

Solutions Manual to Pattern Recognition and Machine Learning

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1 Introduction

1.1

Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y(x_n, \mathbf{w}) - t_n)^2. \quad (1.1)$$

Setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = \sum_{n=1}^N \frac{\partial y(x_n, \mathbf{w})}{\partial \mathbf{w}} (y(x_n, \mathbf{w}) - t_n). \quad (1.2)$$

If

$$y(x_n, \mathbf{w}) = \mathbf{w}^\top \boldsymbol{\phi}(x_n), \quad (1.3)$$

then

$$\mathbf{0} = \sum_{n=1}^N \boldsymbol{\phi}(x_n) (\mathbf{w}^\top \boldsymbol{\phi}(x_n) - t_n). \quad (1.4)$$

Then,

$$\left(\sum_{n=1}^N \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^\top \right) \mathbf{w} = \sum_{n=1}^N t_n \boldsymbol{\phi}(x_n). \quad (1.5)$$

Therefore,

$$\underset{\mathbf{w}}{\operatorname{argmin}} E(\mathbf{w}) = \mathbf{A}^{-1} \mathbf{v}, \quad (1.6)$$

where

$$\begin{aligned} \mathbf{A} &= \sum_{n=1}^N \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^\top, \\ \mathbf{v} &= \sum_{n=1}^N t_n \boldsymbol{\phi}(x_n). \end{aligned} \quad (1.7)$$

If

$$\boldsymbol{\phi}(x_n) = \begin{bmatrix} 1 \\ x_n \\ \vdots \\ x_n^M \end{bmatrix},$$

then

$$\begin{aligned} A_{mm'} &= \sum_{n=1}^N x_n^{m+m'}, \\ v_m &= \sum_{n=1}^N t_n x_n^m. \end{aligned} \tag{1.8}$$

1.2

Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y(x_n, \mathbf{w}) - t_n)^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2. \tag{1.9}$$

Setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = \sum_{n=1}^N \frac{\partial y(x_n, \mathbf{w})}{\partial \mathbf{w}} (y(x_n, \mathbf{w}) - t_n) + \lambda \mathbf{w}. \tag{1.10}$$

If

$$y(x_n, \mathbf{w}) = \mathbf{w}^\top \boldsymbol{\phi}(x_n), \tag{1.11}$$

then

$$\mathbf{0} = \sum_{n=1}^N \boldsymbol{\phi}(x_n) (\mathbf{w}^\top \boldsymbol{\phi}(x_n) - t_n) + \lambda \mathbf{w}. \tag{1.12}$$

Then,

$$\left(\sum_{n=1}^N \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^\top + \lambda \mathbf{I} \right) \mathbf{w} = \sum_{n=1}^N t_n \boldsymbol{\phi}(x_n). \tag{1.13}$$

Therefore,

$$\underset{\mathbf{w}}{\operatorname{argmin}} E(\mathbf{w}) = \mathbf{A}^{-1} \mathbf{v}, \tag{1.14}$$

where

$$\begin{aligned} \mathbf{A} &= \sum_{n=1}^N \boldsymbol{\phi}(x_n) \boldsymbol{\phi}(x_n)^\top + \lambda \mathbf{I}, \\ \mathbf{v} &= \sum_{n=1}^N t_n \boldsymbol{\phi}(x_n). \end{aligned} \tag{1.15}$$

If

$$\phi(x_n) = \begin{bmatrix} 1 \\ x_n \\ \vdots \\ x_n^M \end{bmatrix},$$

then

$$\begin{aligned} A_{mm'} &= \sum_{n=1}^N x_n^{m+m'} + \lambda I_{mm'}, \\ v_m &= \sum_{n=1}^N t_n x_n^m. \end{aligned} \tag{1.16}$$

1.3

Let

$$\begin{aligned} p(a|r) &= \frac{3}{10}, p(o|r) = \frac{2}{5}, p(l|r) = \frac{3}{10}, \\ p(a|b) &= \frac{1}{2}, p(o|b) = \frac{1}{2}, \\ p(a|g) &= \frac{3}{10}, p(o|g) = \frac{3}{10}, p(l|g) = \frac{2}{5}. \end{aligned} \tag{1.17}$$

Let

$$p(r) = \frac{1}{5}, p(b) = \frac{1}{5}, p(g) = \frac{3}{5}. \tag{1.18}$$

(a)

By Bayes' theorem,

$$p(a) = p(a|r)p(r) + p(a|b)p(b) + p(a|g)p(g). \tag{1.19}$$

Therefore,

$$p(a) = \frac{17}{50}. \tag{1.20}$$

(b)

By Bayes' theorem,

$$p(g|o) = \frac{p(g, o)}{p(o)}. \tag{1.21}$$

By Bayes' theorem, the right hand side can be written as

$$\frac{p(o|g)p(g)}{p(o|r)p(r) + p(o|b)p(b) + p(o|g)p(g)}. \quad (1.22)$$

Therefore,

$$p(g|o) = \frac{1}{2}. \quad (1.23)$$

1.4

Let x and y be variables such that

$$x = g(y). \quad (1.24)$$

Let $\hat{x}$ and $\hat{y}$ be the locations of the maximum of p_x and p_y . Let us assume that

$$g'(y) \neq 0, \quad (1.25)$$

in a neighbourhood of $\hat{y}$.

(a)

Since

$$\int p_x(x)dx = \int p_x(g(y)) |g'(y)|dy, \quad (1.26)$$

we have

$$p_y(y) = p_x(g(y)) |g'(y)|. \quad (1.27)$$

Taking the derivative and substituting

$$y = \hat{y} \quad (1.28)$$

gives

$$0 = g'(\hat{y})p'_x(g(\hat{y})) + p_x(g(\hat{y}))g''(\hat{y}). \quad (1.29)$$

Therefore, in general,

$$\hat{x} \neq g(\hat{y}). \quad (1.30)$$

(b)

By (a), if

$$g(y) = ay + b, \quad (1.31)$$

then

$$0 = ap'_x(g(\hat{y})). \quad (1.32)$$

Therefore,

$$\hat{x} = g(\hat{y}). \quad (1.33)$$

1.5

We have

$$\text{var } f(x) = E(f(x) - E f(x))^2. \quad (1.34)$$

The right hand side can be written as

$$E((f(x))^2 - 2f(x) E f(x) + (E f(x))^2) = E(f(x))^2 - (E f(x))^2. \quad (1.35)$$

Therefore,

$$\text{var } f(x) = E(f(x))^2 - (E f(x))^2. \quad (1.36)$$

1.6

We have

$$\text{cov}(x, y) = E((x - E x)(y - E y)). \quad (1.37)$$

The right hand side can be written as

$$E xy - E(x E y) - E(y E x) + E(E x E y) = E xy - E x E y. \quad (1.38)$$

The right hand side can be written as

$$\int xyp(x, y)dxdy - \int xp(x)dx \int yp(y)dy. \quad (1.39)$$

If x and y are independent, then

$$p(x, y) = p(x)p(y). \quad (1.40)$$

Then,

$$\int xyp(x, y)dxdy = \int p(x)dx \int p(y)dy. \quad (1.41)$$

Therefore,

$$\text{cov}(x, y) = 0. \quad (1.42)$$

1.7

(a)

Let

$$I = \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}x^2\right) dx. \quad (1.43)$$

Then,

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x^2 + y^2)\right) dx dy. \quad (1.44)$$

By the transformation

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}, \quad (1.45)$$

the right hand side can be written as

$$\begin{aligned} & \int_0^{\infty} \int_0^{2\pi} \exp\left(-\frac{1}{2\sigma^2}r^2\right) \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} dr d\theta \\ &= 2\pi \int_0^{\infty} r \exp\left(-\frac{1}{2\sigma^2}r^2\right) dr. \end{aligned} \quad (1.46)$$

By the transformation

$$s = \frac{r}{\sigma}. \quad (1.47)$$

the right hand side can be written as

$$2\pi\sigma^2 \int_0^{\infty} s \exp\left(-\frac{1}{2}s^2\right) ds = 2\pi\sigma^2 \left[-\exp\left(-\frac{1}{2}s^2\right)\right]_0^{\infty}. \quad (1.48)$$

Therefore,

$$I = (2\pi\sigma^2)^{\frac{1}{2}}. \quad (1.49)$$

(b)

We have

$$\mathcal{N}(x|\mu, \sigma^2) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right). \quad (1.50)$$

Then,

$$\int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) dx = (2\pi\sigma^2)^{-\frac{1}{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) dx. \quad (1.51)$$

By the transformation $t = x - \mu$, the integral of the right hand side can be written as

$$\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}t^2\right) dt. \quad (1.52)$$

By 1.7(a), it can be written as $(2\pi\sigma^2)^{\frac{1}{2}}$. Therefore,

$$\int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) dx = 1. \quad (1.53)$$

1.8

(a)

Let x be a variable such that

$$p(x) = \mathcal{N}(x|\mu, \sigma^2). \quad (1.54)$$

Then,

$$\mathbb{E} x = \int_{-\infty}^{\infty} x \mathcal{N}(x|\mu, \sigma^2) dx. \quad (1.55)$$

The right hand side can be written as

$$(2\pi\sigma^2)^{-\frac{1}{2}} \int_{-\infty}^{\infty} x \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) dx. \quad (1.56)$$

By the transformation

$$y = x - \mu, \quad (1.57)$$

the integral can be written as

$$\begin{aligned} & \int_{-\infty}^{\infty} (y + \mu) \exp\left(-\frac{1}{2\sigma^2}y^2\right) dy \\ &= \int_{-\infty}^{\infty} y \exp\left(-\frac{1}{2\sigma^2}y^2\right) dy + \mu \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}y^2\right) dy. \end{aligned} \quad (1.58)$$

By 1.7(a), the right hand side can be written as

$$\mu (2\pi\sigma^2)^{\frac{1}{2}}. \quad (1.59)$$

Therefore,

$$\mathbb{E} x = \mu. \quad (1.60)$$

(b)

By 1.7(b),

$$\int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) dx = 1, \quad (1.61)$$

so that

$$(2\pi\sigma^2)^{-\frac{1}{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx = 1. \quad (1.62)$$

Taking the derivative with respect to σ^2 gives

$$\begin{aligned} & (2\pi)^{-\frac{1}{2}} \left(-\frac{1}{2}\right) (\sigma^2)^{-\frac{3}{2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx \\ & + (2\pi\sigma^2)^{-\frac{1}{2}} \int_{-\infty}^{\infty} \frac{1}{2} (\sigma^2)^{-2} (x-\mu)^2 \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right) dx = 0. \end{aligned} \quad (1.63)$$

The left hand side can be written as

$$\begin{aligned} & -\frac{1}{2} (\sigma^2)^{-1} \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) dx \\ & + \frac{1}{2} (\sigma^2)^{-2} \int_{-\infty}^{\infty} (x-\mu)^2 \mathcal{N}(x|\mu, \sigma^2) dx \\ & = -\frac{1}{2} (\sigma^2)^{-1} + \frac{1}{2} (\sigma^2)^{-2} \text{var } x. \end{aligned} \quad (1.64)$$

Therefore,

$$\text{var } x = \sigma^2. \quad (1.65)$$

1.9

(a)

Let x be a variable such that

$$p(x) = \mathcal{N}(x|\mu, \sigma^2). \quad (1.66)$$

Setting the derivative of the right hand side with respect to x to zero gives

$$0 = (2\pi\sigma^2)^{-\frac{1}{2}} \left(-\frac{1}{\sigma^2}(x-\mu)\right) \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right). \quad (1.67)$$

Therefore,

$$\text{mode } x = \mu. \quad (1.68)$$

(b)

Let $\mathbf{x}$ be a variable in D dimensions such that

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}). \quad (1.69)$$

Setting the derivative of the right hand side with respect to $\mathbf{x}$ to zero gives

$$\begin{aligned} \mathbf{0} = & - (2\pi)^{-\frac{D}{2}} (\det \boldsymbol{\Sigma})^{-\frac{1}{2}} (\boldsymbol{\Sigma}^{-1} + (\boldsymbol{\Sigma}^{-1})^\top) (\mathbf{x} - \boldsymbol{\mu}) \\ & \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \end{aligned} \quad (1.70)$$

Therefore,

$$\text{mode } \mathbf{x} = \boldsymbol{\mu}. \quad (1.71)$$

1.10

(a)

We have

$$\mathbb{E}(x + y) = \iint (x + y)p(x, y)dx dy. \quad (1.72)$$

The right hand side can be written as

$$\begin{aligned} & \int x \left(\int p(x, y)dy \right) dx + \int y \left(\int p(x, y)dx \right) dy \\ &= \int xp(x)dx + \int yp(y)dy. \end{aligned} \quad (1.73)$$

The right hand side can be written as

$$\mathbb{E} x + \mathbb{E} y. \quad (1.74)$$

Therefore,

$$\mathbb{E}(x + y) = \mathbb{E} x + \mathbb{E} y. \quad (1.75)$$

(b)

We have

$$\text{var}(x + y) = \mathbb{E}(x + y - \mathbb{E}(x + y))^2 \quad (1.76)$$

The right hand side can be written as

$$\begin{aligned} & \mathbb{E}(x - \mathbb{E}x)^2 + 2\mathbb{E}((x - \mathbb{E}x)(y - \mathbb{E}y)) + \mathbb{E}(y - \mathbb{E}y)^2 \\ &= \text{var } x + 2\text{cov}(x, y) + \text{var } y. \end{aligned} \quad (1.77)$$

By 1.6, if x and y are independent, then

$$\text{cov}(x, y) = 0. \quad (1.78)$$

Therefore,

$$\text{var}(x + y) = \text{var } x + \text{var } y. \quad (1.79)$$

1.11

Let $x_1, \dots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}(x_n | \mu, \sigma^2). \quad (1.80)$$

Then,

$$\ln p(\mathbf{x}) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2. \quad (1.81)$$

Setting the derivatives of $\ln p(\mathbf{x})$ with respect to μ and σ^2 to zero gives

$$\begin{aligned} 0 &= \frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu), \\ 0 &= -\frac{N}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{n=1}^N (x_n - \mu)^2. \end{aligned} \quad (1.82)$$

Therefore, the maximum likelihood solutions to μ and σ^2 are given by

$$\begin{aligned} \mu_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N x_n, \\ \sigma_{\text{ML}}^2 &= \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2. \end{aligned} \quad (1.83)$$

1.12

(a)

Let x_n and $x_{n'}$ be independent variables such that

$$\begin{aligned} p(x_n) &= \mathcal{N}(x_n | \mu, \sigma^2), \\ p(x_{n'}) &= \mathcal{N}(x_{n'} | \mu, \sigma^2). \end{aligned} \tag{1.84}$$

Then,

$$\mathbb{E} x_n x_{n'} = \mu^2. \tag{1.85}$$

By the property

$$\mathbb{E} x_n^2 = \text{var } x_n + (\mathbb{E} x_n)^2, \tag{1.86}$$

we have

$$\mathbb{E} x_n^2 = \sigma^2 + \mu^2. \tag{1.87}$$

Therefore,

$$\mathbb{E} x_n x_{n'} = \mu^2 + I_{nn'} \sigma^2. \tag{1.88}$$

(b)

Let $x_1, \dots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}(x_n | \mu, \sigma^2). \tag{1.89}$$

By 1.11, the maximum likelihood solution to μ is given by

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n. \tag{1.90}$$

Then,

$$\mathbb{E} \mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N \mathbb{E} x_n. \tag{1.91}$$

Therefore,

$$\mathbb{E} \mu_{\text{ML}} = \mu. \tag{1.92}$$

(c)

Let $x_1, \dots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}(x_n | \mu, \sigma^2). \quad (1.93)$$

By 1.11, the maximum likelihood solution to σ^2 is given by

$$\sigma_{\text{ML}}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{\text{ML}})^2. \quad (1.94)$$

Then,

$$\mathbb{E} \sigma_{\text{ML}}^2 = \frac{1}{N} \sum_{n=1}^N \mathbb{E} (x_n - \mu_{\text{ML}})^2. \quad (1.95)$$

The right hand side can be written as

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N \mathbb{E} (x_n^2 - 2\mu_{\text{ML}}x_n + \mu_{\text{ML}}^2) \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E} x_n^2 - \frac{2}{N} \mathbb{E} \left(\mu_{\text{ML}} \left(\sum_{n=1}^N x_n \right) \right) + \mathbb{E} \mu_{\text{ML}}^2. \end{aligned} \quad (1.96)$$

The first term of the right hand side can be written as

$$\frac{1}{N} \sum_{n=1}^N (\mu^2 + \sigma^2) = \mu^2 + \sigma^2, \quad (1.97)$$

while, by 1.11, the second and third terms can be written as

$$-\frac{2}{N} \mathbb{E} \left(N \left(\frac{1}{N} \sum_{n=1}^N x_n \right)^2 \right) + \mathbb{E} \left(\frac{1}{N} \sum_{n=1}^N x_n \right)^2 = -\mathbb{E} \left(\frac{1}{N} \sum_{n=1}^N x_n \right)^2. \quad (1.98)$$

By (a), the right hand side can be written as

$$\begin{aligned} & -\frac{1}{N^2} \sum_{n=1}^N \mathbb{E} x_n^2 - \frac{2}{N^2} \sum_{1 \leq n < n' \leq N} \mathbb{E} x_n x_{n'} \\ &= -\frac{1}{N^2} N (\mu^2 + \sigma^2) - \frac{2}{N^2} \frac{N(N-1)}{2} \mu^2. \end{aligned} \quad (1.99)$$

The right hand side can be written as

$$-\frac{1}{N}(\mu^2 + \sigma^2) - \frac{N-1}{N}\mu^2 = -\mu^2 - \frac{1}{N}\sigma^2. \quad (1.100)$$

Then,

$$\mathbb{E} \sigma_{\text{ML}}^2 = \mu^2 + \sigma^2 - \mu^2 - \frac{1}{N}\sigma^2. \quad (1.101)$$

Therefore,

$$\mathbb{E} \sigma_{\text{ML}}^2 = \frac{N-1}{N}\sigma^2. \quad (1.102)$$

1.13

Let $x_1, \dots, x_N$ be variables such that

$$\begin{aligned} \mathbb{E} x_n &= \mu, \\ \text{var } x_n &= \sigma^2. \end{aligned} \quad (1.103)$$

We have

$$\mathbb{E} \left(\frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2 \right) = \frac{1}{N^2} \sum_{n=1}^N \mathbb{E} (x_n - \mu)^2. \quad (1.104)$$

The right hand side can be written as

$$\frac{1}{N^2} \sum_{n=1}^N \text{var } x_n = \frac{\sigma^2}{N}. \quad (1.105)$$

Therefore,

$$\mathbb{E} \left(\frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2 \right) = \frac{\sigma^2}{N}. \quad (1.106)$$

1.14

Let

$$\begin{aligned} w_{dd'}^{\text{S}} &= \frac{1}{2}(w_{dd'} + w_{d'd}), \\ w_{dd'}^{\text{A}} &= \frac{1}{2}(w_{dd'} - w_{d'd}). \end{aligned} \quad (1.107)$$

(a)

We have

$$\begin{aligned} w_{dd'} &= w_{dd'}^S + w_{dd'}^A, \\ w_{dd'}^S &= w_{d'd}^S, \\ w_{dd'}^A &= -w_{d'd}^A. \end{aligned} \quad (1.108)$$

(b)

We have

$$\sum_{d=1}^D \sum_{d'=1}^D w_{dd'}^A x_d x_{d'} = \frac{1}{2} \sum_{d=1}^D \sum_{d'=1}^D (w_{dd'} - w_{d'd}) x_d x_{d'}. \quad (1.109)$$

The right hand side can be written as

$$\frac{1}{2} \left(\sum_{d=1}^D \sum_{d'=1}^D w_{dd'} x_d x_{d'} - \sum_{d=1}^D \sum_{d'=1}^D w_{d'd} x_d x_{d'} \right) = 0. \quad (1.110)$$

Therefore,

$$\sum_{d=1}^D \sum_{d'=1}^D w_{dd'}^A x_d x_{d'} = 0. \quad (1.111)$$

(c)

We have

$$\sum_{d=1}^D \sum_{d'=1}^D w_{dd'} x_d x_{d'} = \sum_{d=1}^D \sum_{d'=1}^D (w_{dd'}^S + w_{dd'}^A) x_d x_{d'}. \quad (1.112)$$

By (b), the right hand side can be written as

$$\sum_{d=1}^D \sum_{d'=1}^D w_{dd'}^S x_d x_{d'} + \sum_{d=1}^D \sum_{d'=1}^D w_{dd'}^A x_d x_{d'} = \sum_{d=1}^D \sum_{d'=1}^D w_{dd'}^S x_d x_{d'}, \quad (1.113)$$

Therefore,

$$\sum_{d=1}^D \sum_{d'=1}^D w_{dd'} x_d x_{d'} = \sum_{d=1}^D \sum_{d'=1}^D w_{dd'}^S x_d x_{d'}. \quad (1.114)$$

(d)

Since $\mathbf{W}^S$ is a $D \times D$ symmetric matrix, its number of independent parameters is $\frac{D(D+1)}{2}$.

1.15

(a)

Let $n(D, M)$ be the number of independent parameters of a polynomial in D dimensions and M orders. Then

$$n(1, M) = n(1, M - 1) = 1. \quad (1.115)$$

Let us assume that

$$n(D, M) = \sum_{d=1}^D n(d, M - 1). \quad (1.116)$$

The independent terms of a polynomial in $D + 1$ dimensions and M orders can be split into 1. the ones of a polynomial in D dimensions and M orders and 2. the ones generated by multiplying the ones in $D + 1$ dimensions and M orders by the $D + 1$ th variable. Then,

$$n(D + 1, M) = n(D, M) + n(D + 1, M - 1), \quad (1.117)$$

so that

$$n(D + 1, M) = \sum_{d=1}^{D+1} n(d, M - 1). \quad (1.118)$$

Therefore, the assumption is proved by induction on D .

(b)

We have

$$\sum_{d=1}^1 \frac{(d + M - 2)!}{(d - 1)!(M - 1)!} = 1. \quad (1.119)$$

Let us assume that

$$\sum_{d=1}^D \frac{(d + M - 2)!}{(d - 1)!(M - 1)!} = \frac{(D + M - 1)!}{(D - 1)!M!}. \quad (1.120)$$

Then,

$$\sum_{d=1}^{D+1} \frac{(d+M-2)!}{(d-1)!(M-1)!} = \frac{(D+M-1)!}{(D-1)!M!} + \frac{(D+M-1)!}{D!(M-1)!}. \quad (1.121)$$

The right hand side can be written as

$$\frac{D(D+M-1)! + M(D+M-1)!}{D!M!} = \frac{(D+M)!}{D!M!}. \quad (1.122)$$

Therefore, the assumption is proved by induction on D .

(c)

By 1.14(d),

$$n(D, 2) = \frac{D(D+1)}{2}. \quad (1.123)$$

Let us assume that

$$n(D, M) = \frac{(D+M-1)!}{(D-1)!M!}. \quad (1.124)$$

By (a),

$$n(D, M+1) = \sum_{d=1}^D n(d, M). \quad (1.125)$$

By the assumption and (b), the right hand side can be written as

$$\sum_{d=1}^D \frac{(d+M-1)!}{(d-1)!M!} = \frac{(D+M)!}{(D-1)!(M+1)!}. \quad (1.126)$$

Therefore, the assumption is proved by induction on M .

1.16

(a)

Let $N(D, M)$ be the number of independent parameters in all of the terms up to and including the ones of D dimensions and M orders. By 1.15,

$$N(D, M) = \sum_{m=0}^M n(D, m), \quad (1.127)$$

where

$$n(D, m) = \frac{(D+m-1)!}{(D-1)!m!}. \quad (1.128)$$

(b)

By (a),

$$N(D, 0) = 1. \quad (1.129)$$

Let us assume that

$$\sum_{m=0}^M n(D, m) = \frac{(D+M)!}{D!M!}. \quad (1.130)$$

Then,

$$\sum_{m=0}^{M+1} n(D, m) = \frac{(D+M)!}{D!M!} + \frac{(D+M)!}{(D-1)!(M+1)!}. \quad (1.131)$$

The right hand side can be written as

$$\frac{(M+1)(D+M)! + D(D+M)!}{D!(M+1)!} = \frac{(D+M+1)!}{D!(M+1)!}. \quad (1.132)$$

Then, the assumption is proved by induction on M . Therefore,

$$N(D, M) = \frac{(D+M)!}{D!M!}. \quad (1.133)$$

(c)

By the approximation

$$n! \simeq n^n \exp(-n), \quad (1.134)$$

we have

$$\frac{(D+M)!}{D!M!} \simeq \frac{(D+M)^{D+M}}{D^D M^M}. \quad (1.135)$$

The right hand side can be written as

$$D^M \left(1 + \frac{M}{D}\right)^D \left(\frac{1}{M} + \frac{1}{D}\right)^M = M^D \left(1 + \frac{D}{M}\right)^M \left(\frac{1}{D} + \frac{1}{M}\right)^D. \quad (1.136)$$

Therefore,

$$N(D, M) \simeq \begin{cases} D^M, & D \gg M, \\ M^D, & M \gg D. \end{cases} \quad (1.137)$$

(d)

By (b),

$$\begin{aligned}N(10, 3) &= 286, \\N(100, 3) &= 176851, \\N(1000, 3) &= 167668501.\end{aligned}\tag{1.138}$$

1.17

Let

$$\Gamma(x) = \int_0^\infty u^{x-1} \exp(-u) du.\tag{1.139}$$

(a)

We have

$$\Gamma(x+1) = \int_0^\infty u^x \exp(-u) du.\tag{1.140}$$

The right hand side can be written as

$$[-u^x \exp(-u)]_{u=0}^{u=\infty} + \int_0^\infty x u^{x-1} \exp(-u) du = x \Gamma(x).\tag{1.141}$$

Therefore,

$$\Gamma(x+1) = x \Gamma(x).\tag{1.142}$$

(b)

We have

$$\Gamma(1) = \int_0^\infty \exp(-u) du,\tag{1.143}$$

so that

$$\Gamma(1) = 0!.\tag{1.144}$$

For a positive integer x , let us assume that

$$\Gamma(x) = (x-1)!.\tag{1.145}$$

By (a),

$$\Gamma(x+1) = x \Gamma(x),\tag{1.146}$$

so that

$$\Gamma(x+1) = x!. \quad (1.147)$$

Therefore, the assumption is proved by induction on x .

1.18

(a)

Let

$$\prod_{d=1}^D \int_{-\infty}^{\infty} \exp(-x_d^2) dx_d = S_D \int_0^{\infty} \exp(-r^2) r^{D-1} dr, \quad (1.148)$$

where S_D is the surface area of a sphere of unit radius in D dimensions. By 1.7, the left hand side can be written as $\pi^{\frac{D}{2}}$. By the transformation

$$s = r^2, \quad (1.149)$$

the right hand side can be written as

$$\frac{S_D}{2} \int_0^{\infty} \exp(-s) s^{\frac{D-1}{2}} s^{-\frac{1}{2}} ds = \frac{S_D}{2} \Gamma\left(\frac{D}{2}\right). \quad (1.150)$$

Therefore,

$$S_D = \frac{2\pi^{\frac{D}{2}}}{\Gamma\left(\frac{D}{2}\right)}. \quad (1.151)$$

(b)

The volume of the sphere can be written as

$$V_D = S_D \int_0^1 r^{D-1} dr. \quad (1.152)$$

Therefore,

$$V_D = \frac{S_D}{D}. \quad (1.153)$$

(c)

By (a) and (b),

$$\begin{aligned} S_2 &= 2\pi, \\ V_2 &= \pi. \end{aligned} \tag{1.154}$$

Similarly,

$$\begin{aligned} S_3 &= 4\pi, \\ V_3 &= \frac{4}{3}\pi. \end{aligned} \tag{1.155}$$

1.19

(a)

The volume of a cube of side 2 in D dimensions is 2^D . By 1.18, the ratio of the volume of the cocentric sphere of radius 1 divided by the volume of the cube is given by

$$\frac{V_D}{2^D} = \frac{\pi^{\frac{D}{2}}}{D2^{D-1}\Gamma\left(\frac{D}{2}\right)}. \tag{1.156}$$

(b)

By (a) and the Sterling's formula

$$\Gamma(x+1) \simeq (2\pi)^{\frac{1}{2}} \exp(-x)x^{\frac{x+1}{2}}, \tag{1.157}$$

we have

$$\frac{V_D}{2^D} \simeq \frac{\pi^{\frac{D}{2}}}{D2^{D-1}(2\pi)^{\frac{1}{2}} \exp\left(1 - \frac{D}{2}\right) \left(\frac{D}{2} - 1\right)^{\frac{D}{4}}}. \tag{1.158}$$

The right hand side can be written as

$$\frac{1}{2e(2\pi)^{\frac{1}{2}}} \frac{1}{D} \left(\frac{e^2\pi^2}{8D-16} \right)^{\frac{D}{4}}. \tag{1.159}$$

Therefore,

$$\lim_{D \rightarrow \infty} \frac{V_D}{2^D} = 0. \tag{1.160}$$

(c)

The ratio of the distance from the center of the cube to one of the corners divided by the perpendicular distance to one of the sides is given by

$$\frac{\sqrt{\sum_{i=1}^D 1^2}}{1} = \sqrt{D}. \quad (1.161)$$

Therefore, the ratio goes to ∞ as $D \rightarrow \infty$.

1.20

Let $\mathbf{x}$ be a variable in D dimensions such that

$$p(\mathbf{x}) = (2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\right). \quad (1.162)$$

(a)

We have

$$\int_{r \leq \|\mathbf{x}\| \leq r+\epsilon} p(\mathbf{x}) d\mathbf{x} = \int_r^{r+\epsilon} \int (2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{r'^2}{2\sigma^2}\right) J dr' d\phi, \quad (1.163)$$

where ϕ is the vector of the angular components of the polar coordinate and J is the Jacobian of the transformation from the Cartesian to polar coordinate. For a sufficiently small ϵ , the right hand side can be approximated as

$$\begin{aligned} & (2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{r^2}{2\sigma^2}\right) \int_r^{r+\epsilon} \int J dr' d\phi \\ &= (2\pi\sigma^2)^{-\frac{D}{2}} \exp\left(-\frac{r^2}{2\sigma^2}\right) \int_{r \leq \|\mathbf{x}\| \leq r+\epsilon} d\mathbf{x}. \end{aligned} \quad (1.164)$$

Therefore,

$$\int_{r \leq \|\mathbf{x}\| \leq r+\epsilon} p(\mathbf{x}) d\mathbf{x} \simeq p(r)\epsilon, \quad (1.165)$$

where

$$p(r) = (2\pi\sigma^2)^{-\frac{D}{2}} S_D r^{D-1} \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad (1.166)$$

and S_D is the surface area of a unit sphere in D dimensions.

(b)

Setting the derivative of $p(r)$ to zero gives

$$0 = (2\pi\sigma^2)^{-\frac{D}{2}} S_D \left((D-1)r^{D-2} - \frac{r^D}{\sigma^2} \right) \exp \left(-\frac{r^2}{2\sigma^2} \right). \quad (1.167)$$

Therefore, $p(r)$ is maximised at a single stationary point

$$\hat{r} = \sqrt{D-1}\sigma. \quad (1.168)$$

(c)

We have

$$\frac{p(\hat{r} + \epsilon)}{p(\hat{r})} = \left(\frac{\hat{r} + \epsilon}{\hat{r}} \right)^{D-1} \exp \left(-\frac{2\hat{r}\epsilon + \epsilon^2}{2\sigma^2} \right). \quad (1.169)$$

By (b), the right hand side can be written as

$$\begin{aligned} & \exp \left((D-1) \ln \left(1 + \frac{\epsilon}{\hat{r}} \right) - \frac{2\hat{r}\epsilon + \epsilon^2}{2\sigma^2} \right) \\ &= \exp \left(\frac{\hat{r}^2}{\sigma^2} \ln \left(1 + \frac{\epsilon}{\hat{r}} \right) - \frac{2\hat{r}\epsilon + \epsilon^2}{2\sigma^2} \right). \end{aligned} \quad (1.170)$$

By the Taylor series

$$\ln(1+x) = x - \frac{1}{2}x^2 + o(x^3), \quad (1.171)$$

the right hand side can be approximated as

$$\exp \left(\frac{\hat{r}^2}{\sigma^2} \left(\frac{\epsilon}{\hat{r}} - \frac{\epsilon^2}{2\hat{r}^2} \right) - \frac{2\hat{r}\epsilon + \epsilon^2}{2\sigma^2} \right) = \exp \left(-\frac{\epsilon^2}{\sigma^2} \right). \quad (1.172)$$

Therefore,

$$p(\hat{r} + \epsilon) \simeq p(\hat{r}) \exp \left(-\frac{\epsilon^2}{\sigma^2} \right). \quad (1.173)$$

(d)

Let $\hat{\mathbf{r}}$ be a vector of length $\hat{r}$. We have

$$\frac{p(\mathbf{0})}{p(\hat{\mathbf{r}})} = \exp\left(\frac{\hat{r}^2}{2\sigma^2}\right). \quad (1.174)$$

By (b), the right hand side can be written as

$$\exp\left(\frac{D-1}{2}\right). \quad (1.175)$$

Therefore,

$$\frac{p(\mathbf{0})}{p(\hat{\mathbf{r}})} = \exp\left(\frac{D-1}{2}\right). \quad (1.176)$$

1.21

(a)

If $0 \leq a \leq b$, then

$$0 \leq a(b-a). \quad (1.177)$$

Therefore,

$$a \leq (ab)^{\frac{1}{2}}. \quad (1.178)$$

(b)

For a two-class classification problem of $\mathbf{x}$, let the classes be $\mathcal{C}_1$ and $\mathcal{C}_2$ and let the decision regions be $\mathcal{R}_1$ and $\mathcal{R}_2$. Let us choose the decision regions to minimise the probability of misclassification. Then,

$$\begin{aligned} p(\mathbf{x}, \mathcal{C}_1) > p(\mathbf{x}, \mathcal{C}_2) &\Rightarrow \mathbf{x} \in \mathcal{C}_1, \\ p(\mathbf{x}, \mathcal{C}_2) > p(\mathbf{x}, \mathcal{C}_1) &\Rightarrow \mathbf{x} \in \mathcal{C}_2. \end{aligned} \quad (1.179)$$

By (a),

$$\begin{aligned} \int_{\mathcal{R}_1} p(\mathbf{x}, \mathcal{C}_2) d\mathbf{x} &\leq \int_{\mathcal{R}_1} (p(\mathbf{x}, \mathcal{C}_1)p(\mathbf{x}, \mathcal{C}_2))^{\frac{1}{2}} d\mathbf{x}, \\ \int_{\mathcal{R}_2} p(\mathbf{x}, \mathcal{C}_1) d\mathbf{x} &\leq \int_{\mathcal{R}_2} (p(\mathbf{x}, \mathcal{C}_1)p(\mathbf{x}, \mathcal{C}_2))^{\frac{1}{2}} d\mathbf{x}. \end{aligned} \quad (1.180)$$

Therefore,

$$\int_{\mathcal{R}_1} p(\mathbf{x}, \mathcal{C}_2) d\mathbf{x} + \int_{\mathcal{R}_2} p(\mathbf{x}, \mathcal{C}_1) d\mathbf{x} \leq \int (p(\mathbf{x}, \mathcal{C}_1)p(\mathbf{x}, \mathcal{C}_2))^{\frac{1}{2}} d\mathbf{x}. \quad (1.181)$$

1.22

Let

$$E L = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x}, \quad (1.182)$$

where

$$L_{kj} = 1 - I_{kj}. \quad (1.183)$$

The right hand side can be written as

$$\begin{aligned} & \sum_k \sum_j \int_{\mathcal{R}_j} (p(\mathbf{x}, \mathcal{C}_k) - p(\mathbf{x}, \mathcal{C}_j)) d\mathbf{x} \\ &= \sum_j \int_{\mathcal{R}_j} \left(\sum_k p(\mathbf{x}, \mathcal{C}_k) - p(\mathbf{x}, \mathcal{C}_j) \right) d\mathbf{x}. \end{aligned} \quad (1.184)$$

The right hand side can be written as

$$\sum_j \int_{\mathcal{R}_j} (p(\mathbf{x}) - p(\mathbf{x}, \mathcal{C}_j)) d\mathbf{x} = 1 - \sum_j \int_{\mathcal{R}_j} p(\mathbf{x}, \mathcal{C}_j) d\mathbf{x}. \quad (1.185)$$

Then,

$$E L = 1 - \sum_j \int_{\mathcal{R}_j} p(\mathcal{C}_j | \mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \quad (1.186)$$

Therefore, minimising $E L$ reduces to choosing the criterion to maximise the posterior probability $p(\mathcal{C}_j | \mathbf{x})$.

1.23

Let

$$E L = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x}. \quad (1.187)$$

The right hand side can be written as

$$\sum_j \int_{\mathcal{R}_j} \sum_k L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x} = \sum_j \int_{\mathcal{R}_j} \left(\sum_k L_{kj} p(\mathcal{C}_k | \mathbf{x}) \right) p(\mathbf{x}) d\mathbf{x}. \quad (1.188)$$

Then,

$$E L = \sum_j \int_{\mathcal{R}_j} \left(\sum_k L_{kj} p(\mathcal{C}_k | \mathbf{x}) \right) p(\mathbf{x}) d\mathbf{x}. \quad (1.189)$$

Therefore, minimising $E L$ reduces to minimising $\sum_k L_{kj} p(\mathcal{C}_k | \mathbf{x})$.

1.24 (Incomplete)

Let

$$E L = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, \mathcal{C}_k) d\mathbf{x} + \lambda \int_{\forall k p(\mathcal{C}_k|\mathbf{x}) < \theta} p(\mathbf{x}) d\mathbf{x}. \quad (1.190)$$

1.25

Let

$$E L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \iint \|\mathbf{y}(\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}. \quad (1.191)$$

Setting the derivative of $E L(\mathbf{t}, \mathbf{y}(\mathbf{x}))$ with respect to $\mathbf{y}(\mathbf{x})$ to zero gives

$$\mathbf{0} = 2 \int (\mathbf{y}(\mathbf{x}) - \mathbf{t}) p(\mathbf{x}, \mathbf{t}) d\mathbf{t}. \quad (1.192)$$

The integral of the right hand side can be written as

$$\mathbf{y}(\mathbf{x}) \int p(\mathbf{x}, \mathbf{t}) d\mathbf{t} - \int \mathbf{t} p(\mathbf{x}, \mathbf{t}) d\mathbf{t} = \mathbf{y}(\mathbf{x}) p(\mathbf{x}) - p(\mathbf{x}) \int \mathbf{t} p(\mathbf{t}|\mathbf{x}) d\mathbf{t}. \quad (1.193)$$

The integral in the second term of the right hand side can be written as $E_{\mathbf{t}}(\mathbf{t}|\mathbf{x})$. Then, the right hand side can be written as

$$\mathbf{0} = p(\mathbf{x}) (\mathbf{y}(\mathbf{x}) - E_{\mathbf{t}}(\mathbf{t}|\mathbf{x})). \quad (1.194)$$

Therefore,

$$\underset{\mathbf{y}(\mathbf{x})}{\operatorname{argmin}} E L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = E_{\mathbf{t}}(\mathbf{t}|\mathbf{x}). \quad (1.195)$$

For a single target variable t , it reduces to

$$\underset{\mathbf{y}(\mathbf{x})}{\operatorname{argmin}} E L(t, \mathbf{y}(\mathbf{x})) = E_t(t|\mathbf{x}). \quad (1.196)$$

1.26

Let

$$E L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \iint \|\mathbf{y}(\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}. \quad (1.197)$$

The right hand side can be written as

$$\iint \|\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x}) + \mathbf{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t} = I_1 + 2I_2 + I_3, \quad (1.198)$$

where

$$\begin{aligned} I_1 &= \iint \|\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}, \\ I_2 &= \iint (\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x}))^\top (\mathbf{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}) p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}, \\ I_3 &= \iint \|\mathbf{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}\|^2 p(\mathbf{x}, \mathbf{t}) d\mathbf{x} d\mathbf{t}. \end{aligned} \quad (1.199)$$

The expression of I_1 can be written as

$$\begin{aligned} &\int \|\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x})\|^2 \left(\int p(\mathbf{x}, \mathbf{t}) d\mathbf{t} \right) d\mathbf{x} \\ &= \int \|\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x}. \end{aligned} \quad (1.200)$$

The expression of I_2 can be written as

$$\int (\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x}))^\top \left(\int (\mathbf{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}) p(\mathbf{t}|\mathbf{x}) d\mathbf{t} \right) p(\mathbf{x}) d\mathbf{x}. \quad (1.201)$$

We have

$$\int (\mathbf{E}(\mathbf{t}|\mathbf{x}) - \mathbf{t}) p(\mathbf{t}|\mathbf{x}) d\mathbf{t} = \mathbf{E}(\mathbf{t}|\mathbf{x}) \int p(\mathbf{t}|\mathbf{x}) d\mathbf{t} - \int \mathbf{t} p(\mathbf{t}|\mathbf{x}) d\mathbf{t}. \quad (1.202)$$

The right hand side can be written as

$$\mathbf{E}(\mathbf{t}|\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x}) = 0. \quad (1.203)$$

The expression of I_3 can be written as

$$\int \left(\int \|\mathbf{t} - \mathbf{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{t}|\mathbf{x}) d\mathbf{t} \right) p(\mathbf{x}) d\mathbf{x} = \int \text{var}(\mathbf{t}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \quad (1.204)$$

Then,

$$\mathbf{E} L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \int \|\mathbf{y}(\mathbf{x}) - \mathbf{E}(\mathbf{t}|\mathbf{x})\|^2 p(\mathbf{x}) d\mathbf{x} + \int \text{var}(\mathbf{t}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \quad (1.205)$$

Therefore,

$$\underset{\mathbf{y}(\mathbf{x})}{\text{argmin}} \mathbf{E} L(\mathbf{t}, \mathbf{y}(\mathbf{x})) = \mathbf{E}(\mathbf{t}|\mathbf{x}). \quad (1.206)$$

1.27 (Incomplete)

(a)

Let

$$E L_q = \iint |y(\mathbf{x}) - t|^q p(\mathbf{x}, t) d\mathbf{x} dt. \quad (1.207)$$

Setting the derivative of $E L_q$ with respect to $y(\mathbf{x})$ to zero gives

$$0 = qp(\mathbf{x}) \int |y(\mathbf{x}) - t|^{q-1} \text{sign}(y(\mathbf{x}) - t) p(t|\mathbf{x}) dt. \quad (1.208)$$

Therefore,

$$\begin{aligned} & \underset{y(\mathbf{x})}{\operatorname{argmin}} E L_q \\ &= \left\{ y(\mathbf{x}) \mid \int |y(\mathbf{x}) - t|^{q-1} \text{sign}(y(\mathbf{x}) - t) p(t|\mathbf{x}) dt = 0 \right\}. \end{aligned} \quad (1.209)$$

(b)

We have

$$E L_1 = \int \left(\int \text{sign}(y(\mathbf{x}) - t) p(t|\mathbf{x}) dt \right) p(\mathbf{x}) d\mathbf{x}. \quad (1.210)$$

The integral of the right hand side with respect to t can be written as

$$\int_{y(\mathbf{x})}^{\infty} p(t|\mathbf{x}) dt - \int_{-\infty}^{y(\mathbf{x})} p(t|\mathbf{x}) dt. \quad (1.211)$$

Therefore,

$$\underset{y(\mathbf{x})}{\operatorname{argmin}} E L_1 = \text{median}(t|\mathbf{x}). \quad (1.212)$$

(c)

We have

$$\lim_{q \rightarrow 0} \left(\underset{y(\mathbf{x})}{\operatorname{argmin}} E L_q \right) = \text{mode}(t|\mathbf{x})? \quad (1.213)$$

1.28

(a)

Let us assume that

$$p(x, y) = p(x)p(y) \Rightarrow h(x, y) = h(x) + h(y). \quad (1.214)$$

Then,

$$h(p^2) = 2h(p). \quad (1.215)$$

Let us assume that, for a positive integer n ,

$$h(p^n) = nh(p). \quad (1.216)$$

Then, by the first assumption,

$$h(p^{n+1}) = h(p^n) + h(p), \quad (1.217)$$

so that

$$h(p^{n+1}) = (n+1)h(p). \quad (1.218)$$

Therefore, the second assumption is proved by induction on n .

(b)

For positive integers m and n ,

$$h(p^n) = h(p^{\frac{n}{m}m}). \quad (1.219)$$

By the second assumption in (a), the left hand side can be written as $nh(p)$. By the first assumption in (a), the right hand side can be written as $mh(p^{\frac{n}{m}})$. Therefore,

$$h(p^{\frac{n}{m}}) = \frac{n}{m}h(p). \quad (1.220)$$

(c)

By the continuity, for a positive real number a ,

$$h(p^a) = ah(p). \quad (1.221)$$

Taking the derivative with respect to a and substituting $a = 1$ gives

$$(p \ln p)h'(p) = h(p). \quad (1.222)$$

Then,

$$\int \frac{h'(p)}{h(p)} dp = \int \frac{1}{p \ln p} dp + \text{const} . \quad (1.223)$$

Ignoring the constants, the left hand side can be written as $\ln h(p)$ and the right hand side can be written as $\ln(\ln p)$. Therefore,

$$h(p) \propto \ln p. \quad (1.224)$$

1.29

Let x be an M -state discrete random variable. Then, the entropy is given by

$$H(x) = - \sum_{m=1}^M p(x_m) \ln p(x_m), \quad (1.225)$$

where

$$\sum_{m=1}^M p(x_m) = 1. \quad (1.226)$$

By Jensen's inequality,

$$\sum_{m=1}^M p(x_i) \ln \frac{1}{p(x_m)} \leq \ln \left(\sum_{m=1}^M 1 \right). \quad (1.227)$$

Therefore,

$$H(x) \leq \ln M. \quad (1.228)$$

1.30

Let

$$\begin{aligned} p(x) &= \mathcal{N}(x|\mu, \sigma^2), \\ q(x) &= \mathcal{N}(x|m, s^2). \end{aligned} \quad (1.229)$$

Then, the Kullback-Leibler divergence is given by

$$\text{KL}(p||q) = - \int p(x) \ln \frac{q(x)}{p(x)} dx. \quad (1.230)$$

The right hand side can be written as

$$\begin{aligned}
& - \int_{-\infty}^{\infty} p(x) \ln \frac{(2\pi s^2)^{-\frac{1}{2}} \exp\left(-\frac{(x-m)^2}{2s^2}\right)}{(2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)} dx \\
& = - \int_{-\infty}^{\infty} p(x) \left(-\frac{1}{2} \ln \frac{s^2}{\sigma^2} - \frac{(x-m)^2}{2s^2} + \frac{(x-\mu)^2}{2\sigma^2} \right) dx.
\end{aligned} \tag{1.231}$$

The right hand side can be written as

$$\ln \frac{s}{\sigma} \int_{-\infty}^{\infty} p(x) dx + \frac{1}{2s^2} \int_{-\infty}^{\infty} (x-m)^2 p(x) dx - \frac{1}{2\sigma^2} \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx. \tag{1.232}$$

The integral of the second term can be written as

$$\begin{aligned}
& \int_{-\infty}^{\infty} (x-\mu+\mu-m)^2 p(x) dx \\
& = \int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx + 2(\mu-m) \int_{-\infty}^{\infty} (x-\mu) p(x) dx \\
& \quad + (\mu-m)^2 \int_{-\infty}^{\infty} p(x) dx.
\end{aligned} \tag{1.233}$$

Therefore,

$$\text{KL}(p||q) = \ln \frac{s}{\sigma} + \frac{\sigma^2 + (\mu-m)^2}{2s^2} - \frac{1}{2}. \tag{1.234}$$

1.31

Let $\mathbf{x}$ and $\mathbf{y}$ be two variables. We have

$$\begin{aligned}
H(\mathbf{x}) &= - \int p(\mathbf{x}) \ln p(\mathbf{x}) d\mathbf{x}, \\
H(\mathbf{y}) &= - \int p(\mathbf{y}) \ln p(\mathbf{y}) d\mathbf{y}, \\
H(\mathbf{x}, \mathbf{y}) &= - \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y}.
\end{aligned} \tag{1.235}$$

The expressions of $H(\mathbf{x})$ and $H(\mathbf{y})$ can be written as

$$\begin{aligned}
& - \int \left(\int p(\mathbf{x}, \mathbf{y}) d\mathbf{y} \right) \ln p(\mathbf{x}) d\mathbf{x}, \\
& - \int \left(\int p(\mathbf{x}, \mathbf{y}) d\mathbf{x} \right) \ln p(\mathbf{y}) d\mathbf{y}.
\end{aligned} \tag{1.236}$$

Then,

$$H(\mathbf{x}) + H(\mathbf{y}) - H(\mathbf{x}, \mathbf{y}) = - \iint p(\mathbf{x}, \mathbf{y}) \ln \left(\frac{p(\mathbf{x})p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})} \right) d\mathbf{x}d\mathbf{y}. \quad (1.237)$$

We have

$$\iint p(\mathbf{x}, \mathbf{y}) d\mathbf{x}d\mathbf{y} = 1. \quad (1.238)$$

Then, by Jensen's inequality,

$$- \iint p(\mathbf{x}, \mathbf{y}) \ln \left(\frac{p(\mathbf{x})p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})} \right) d\mathbf{x}d\mathbf{y} \geq - \ln \left(\iint p(\mathbf{x})p(\mathbf{y}) d\mathbf{x}d\mathbf{y} \right). \quad (1.239)$$

The right hand side can be written as

$$- \ln \left(\int p(\mathbf{x}) d\mathbf{x} \int p(\mathbf{y}) d\mathbf{y} \right) = 0. \quad (1.240)$$

Therefore,

$$H(\mathbf{x}, \mathbf{y}) \leq H(\mathbf{x}) + H(\mathbf{y}). \quad (1.241)$$

1.32

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\mathbf{y} = \mathbf{A}\mathbf{x}, \quad (1.242)$$

where $\mathbf{A}$ is a nonsingular matrix. We have

$$\int p_x(\mathbf{x}) d\mathbf{x} = \int p_x(\mathbf{A}^{-1}\mathbf{y}) |\det \mathbf{A}^{-1}| d\mathbf{y}. \quad (1.243)$$

Then,

$$p_y(\mathbf{y}) = p_x(\mathbf{A}^{-1}\mathbf{y}) |\det \mathbf{A}^{-1}|. \quad (1.244)$$

We have

$$H(\mathbf{y}) = - \int p_y(\mathbf{y}) \ln p_y(\mathbf{y}) d\mathbf{y}. \quad (1.245)$$

The right hand side can be written as

$$\begin{aligned} & - \int p_y(\mathbf{y}) \ln (p_x(\mathbf{A}^{-1}\mathbf{y}) |\det \mathbf{A}^{-1}|) d\mathbf{y} \\ &= - \int p_y(\mathbf{y}) \ln p_x(\mathbf{A}^{-1}\mathbf{y}) d\mathbf{y} + \ln |\det \mathbf{A}| \int p_y(\mathbf{y}) d\mathbf{y}. \end{aligned} \quad (1.246)$$

The first term can be written as

$$-|\det \mathbf{A}^{-1}| \int p_x(\mathbf{A}^{-1}\mathbf{y}) \ln p_x(\mathbf{A}^{-1}\mathbf{y}) d\mathbf{y}. \quad (1.247)$$

By the transformation

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{y}, \quad (1.248)$$

it can be written as

$$-\int p_x(\mathbf{x}) \ln p_x(\mathbf{x}) d\mathbf{x} = H(\mathbf{x}). \quad (1.249)$$

Therefore,

$$H(\mathbf{y}) = H(\mathbf{x}) + \ln |\det \mathbf{A}|. \quad (1.250)$$

1.33

Let x and y be two discrete variables with K and L states. Then,

$$H(y|x) = -\sum_{k=1}^K \sum_{l=1}^L p(x_k, y_l) \ln p(y_l|x_k). \quad (1.251)$$

If

$$H(y|x) = 0, \quad (1.252)$$

then

$$0 = -\sum_{k=1}^K p(x_k) \left(\sum_{l=1}^L p(y_l|x_k) \ln p(y_l|x_k) \right). \quad (1.253)$$

We have

$$\begin{aligned} p(x_k) &\geq 0, \\ p(y_l|x_k) \ln p(y_l|x_k) &\leq 0. \end{aligned} \quad (1.254)$$

Then, the equation reduces to

$$p(y_l|x_k) \ln p(y_l|x_k) = 0. \quad (1.255)$$

Then,

$$p(y_l|x_k) = \begin{cases} 1, \\ 0. \end{cases} \quad (1.256)$$

We have

$$\sum_{l=1}^L p(y_l|x_k) = 1. \quad (1.257)$$

Therefore,

$$p(y_l|x_k) = \begin{cases} 1, & \text{if } K = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (1.258)$$

1.34

Let x be a variable. We have

$$H(x) = - \int_{-\infty}^{\infty} p(x) \ln p(x) dx. \quad (1.259)$$

In order to maximise $H(x)$ with the constraints

$$\begin{aligned} \int_{-\infty}^{\infty} p(x) dx &= 1, \\ \int_{-\infty}^{\infty} xp(x) dx &= \mu, \\ \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx &= \sigma^2, \end{aligned} \quad (1.260)$$

let

$$\begin{aligned} L(p) = & H(x) + \lambda_1 \left(\int_{-\infty}^{\infty} p(x) dx - 1 \right) + \lambda_2 \left(\int_{-\infty}^{\infty} xp(x) dx - \mu \right) \\ & + \lambda_3 \left(\int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx - \sigma^2 \right). \end{aligned} \quad (1.261)$$

Setting the variation with respect to p to zero gives

$$0 = -\ln p - 1 + \lambda_1 + \lambda_2 x + \lambda_3 (x - \mu)^2. \quad (1.262)$$

Then,

$$p(x) = \exp \left(-1 + \lambda_1 + \lambda_2 x + \lambda_3 (x - \mu)^2 \right), \quad (1.263)$$

so that

$$p(x) = c \exp \left(-q(x) \right), \quad (1.264)$$

where

$$\begin{aligned} c &= \exp \left(-1 + \lambda_1 - \frac{\lambda_2^2}{4\lambda_3} \right), \\ q(x) &= -\lambda_3 \left(x - \left(\mu - \frac{\lambda_2}{2\lambda_3} \right) \right)^2. \end{aligned} \quad (1.265)$$

Substituting it to the constraints gives

$$\begin{aligned} c \int_{-\infty}^{\infty} \exp(-q(x)) dx &= 1, \\ c \int_{-\infty}^{\infty} x \exp(-q(x)) dx &= \mu, \\ c \int_{-\infty}^{\infty} (x - \mu)^2 \exp(-q(x)) dx &= \sigma^2. \end{aligned} \quad (1.266)$$

By the transformation

$$y^2 = q(x), \quad (1.267)$$

they can be written as

$$\begin{aligned} c \int_{-\infty}^{\infty} \exp(-y^2) (-\lambda_3)^{-\frac{1}{2}} dy &= 1, \\ c \int_{-\infty}^{\infty} \left((-\lambda_3)^{-\frac{1}{2}} y + \mu - \frac{\lambda_2}{2\lambda_3} \right) \exp(-y^2) (-\lambda_3)^{-\frac{1}{2}} dy &= \mu, \\ c \int_{-\infty}^{\infty} \left((-\lambda_3)^{-\frac{1}{2}} y - \frac{\lambda_2}{2\lambda_3} \right)^2 \exp(-y^2) (-\lambda_3)^{-\frac{1}{2}} dy &= \sigma^2. \end{aligned} \quad (1.268)$$

We have

$$\begin{aligned} \int_{-\infty}^{\infty} \exp(-y^2) dy &= \Gamma\left(\frac{1}{2}\right), \\ \int_{-\infty}^{\infty} y \exp(-y^2) dy &= 0, \\ \int_{-\infty}^{\infty} y^2 \exp(-y^2) dy &= \Gamma\left(\frac{3}{2}\right), \end{aligned} \quad (1.269)$$

Then,

$$\begin{aligned}
c(-\lambda_3)^{-\frac{1}{2}}\Gamma\left(\frac{1}{2}\right) &= 1, \\
c\left(\mu - \frac{\lambda_2}{2\lambda_3}\right)(-\lambda_3)^{-\frac{1}{2}}\Gamma\left(\frac{1}{2}\right) &= \mu, \\
c\left((-\lambda_3)^{-\frac{3}{2}}\Gamma\left(\frac{3}{2}\right) + (-\lambda_3)^{-\frac{1}{2}}\frac{\lambda_2^2}{4\lambda_3^2}\Gamma\left(\frac{1}{2}\right)\right) &= \sigma^2.
\end{aligned} \tag{1.270}$$

Then,

$$\begin{aligned}
\lambda_1 &= 1 - \frac{1}{2} \ln(2\pi\sigma^2), \\
\lambda_2 &= 0, \\
\lambda_3 &= -\frac{1}{2\sigma^2}.
\end{aligned} \tag{1.271}$$

Therefore,

$$p(x) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right). \tag{1.272}$$

1.35

Let x be a variable such that

$$p(x) = \mathcal{N}(x|\mu, \sigma^2). \tag{1.273}$$

Then,

$$H(x) = - \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) \ln \mathcal{N}(x|\mu, \sigma^2) dx. \tag{1.274}$$

The right hand side can be written as

$$\begin{aligned}
& - \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) \left(-\frac{1}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(x - \mu)^2 \right) dx \\
&= \frac{1}{2} \ln(2\pi\sigma^2) \int_{-\infty}^{\infty} \mathcal{N}(x|\mu, \sigma^2) dx \\
&+ \frac{1}{2\sigma^2} \int_{-\infty}^{\infty} (x - \mu)^2 \mathcal{N}(x|\mu, \sigma^2) dx.
\end{aligned} \tag{1.275}$$

Therefore,

$$H(x) = \frac{1}{2} (1 + \ln(2\pi\sigma^2)). \tag{1.276}$$

1.36 (Incomplete)

Let f be a strictly convex function. Then,

$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b), \quad (1.277)$$

where $a \leq b$ and $0 \leq \lambda \leq 1$. Let

$$x = \lambda a + (1 - \lambda)b. \quad (1.278)$$

Then, the inequality can be written as

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b). \quad (1.279)$$

Let

$$g(x) = \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) - f(x). \quad (1.280)$$

Then,

$$g(x) \geq 0. \quad (1.281)$$

Additionally, for $x > a$,

$$g(x) = (x-a) \left(\frac{f(b)-f(a)}{b-a} - \frac{f(x)-f(a)}{x-a} \right). \quad (1.282)$$

By the mean value theorem, there exists c and y such that $a \leq c \leq b$, $a \leq y \leq x$ and

$$\begin{aligned} f'(c) &= \frac{f(b)-f(a)}{b-a}, \\ f'(y) &= \frac{f(x)-f(a)}{x-a}. \end{aligned} \quad (1.283)$$

Then, for $x > a$, the inequality reduces to

$$f'(y) \leq f'(c). \quad (1.284)$$

1.37

Let $\mathbf{x}$ and $\mathbf{y}$ be two variables. We have

$$H(\mathbf{x}, \mathbf{y}) = - \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y}. \quad (1.285)$$

The right hand side can be written as

$$\begin{aligned}
& - \iint p(\mathbf{x}, \mathbf{y}) (\ln p(\mathbf{y}|\mathbf{x}) + \ln p(\mathbf{x})) d\mathbf{x}d\mathbf{y} \\
& = - \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{y}|\mathbf{x}) d\mathbf{x}d\mathbf{y} - \int \left(\int p(\mathbf{x}, \mathbf{y}) d\mathbf{y} \right) \ln p(\mathbf{x}) d\mathbf{x}.
\end{aligned} \tag{1.286}$$

We have

$$\begin{aligned}
H(\mathbf{y}|\mathbf{x}) & = - \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{y}|\mathbf{x}) d\mathbf{x}d\mathbf{y}, \\
H(\mathbf{x}) & = - \int p(\mathbf{x}) \ln p(\mathbf{x}) d\mathbf{x}.
\end{aligned} \tag{1.287}$$

Therefore,

$$H(\mathbf{x}, \mathbf{y}) = H(\mathbf{y}|\mathbf{x}) + H(\mathbf{x}). \tag{1.288}$$

1.38

Let f be a strictly convex function. Then,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2), \tag{1.289}$$

where

$$0 \leq \lambda \leq 1. \tag{1.290}$$

Let us assume that

$$f\left(\sum_{m=1}^M \lambda_m x_m\right) \leq \sum_{m=1}^M \lambda_m f(x_m), \tag{1.291}$$

where

$$\begin{aligned}
\sum_{m=1}^M \lambda_m & = 1, \\
\lambda_m & \geq 0.
\end{aligned} \tag{1.292}$$

We have

$$\begin{aligned}
& f\left(\sum_{m=1}^{M+1} \lambda_m x_m\right) \\
& \leq \lambda_{M+1} f(x_{M+1}) + (1 - \lambda_{M+1}) f\left(\sum_{m=1}^M \frac{\lambda_m}{1 - \lambda_{M+1}} x_m\right),
\end{aligned} \tag{1.293}$$

where

$$\begin{aligned}\sum_{m=1}^{M+1} \lambda_m &= 1, \\ \lambda_m &\geq 0.\end{aligned}\tag{1.294}$$

By the assumption,

$$f\left(\sum_{m=1}^M \frac{\lambda_m}{1 - \lambda_{M+1}} x_m\right) \leq \sum_{m=1}^M \frac{\lambda_m}{1 - \lambda_{M+1}} f(x_m).\tag{1.295}$$

Then,

$$f\left(\sum_{m=1}^{M+1} \lambda_m x_m\right) \leq \lambda_{M+1} f(x_{M+1}) + (1 - \lambda_{M+1}) \sum_{m=1}^M \frac{\lambda_m}{1 - \lambda_{M+1}} f(x_m),\tag{1.296}$$

so that

$$f\left(\sum_{m=1}^{M+1} \lambda_m x_m\right) \leq \sum_{m=1}^{M+1} \lambda_m f(x_m).\tag{1.297}$$

Therefore, the assumption is proved by induction on M .

1.39

Let x and y be two binary variables where

$$\begin{aligned}p(x=0, y=0) &= \frac{1}{3}, \\ p(x=0, y=1) &= \frac{1}{3}, \\ p(x=1, y=0) &= 0, \\ p(x=1, y=1) &= \frac{1}{3}.\end{aligned}\tag{1.298}$$

(a)

We have

$$H(x) = - \sum_x p(x) \ln p(x).\tag{1.299}$$

We have

$$\begin{aligned} p(x = 0) &= p(x = 0, y = 0) + p(x = 0, y = 1), \\ p(x = 1) &= p(x = 1, y = 0) + p(x = 1, y = 1). \end{aligned} \tag{1.300}$$

Then,

$$\begin{aligned} p(x = 0) &= \frac{2}{3}, \\ p(x = 1) &= \frac{1}{3}. \end{aligned} \tag{1.301}$$

Therefore,

$$H(x) = \ln 3 - \frac{2}{3} \ln 2. \tag{1.302}$$

(b)

We have

$$H(y) = - \sum_y p(y) \ln p(y). \tag{1.303}$$

We have

$$\begin{aligned} p(y = 0) &= p(x = 0, y = 0) + p(x = 1, y = 0), \\ p(y = 1) &= p(x = 0, y = 1) + p(x = 1, y = 1). \end{aligned} \tag{1.304}$$

Then,

$$\begin{aligned} p(y = 0) &= \frac{1}{3}, \\ p(y = 1) &= \frac{2}{3}. \end{aligned} \tag{1.305}$$

Therefore,

$$H(y) = \ln 3 - \frac{2}{3} \ln 2. \tag{1.306}$$

(c)

We have

$$H(y|x) = - \sum_{x,y} p(x, y) \ln p(y|x). \tag{1.307}$$

By Bayes' theorem,

$$\begin{aligned}
p(y = 0|x = 0) &= \frac{p(x = 0, y = 0)}{p(x = 0)}, \\
p(y = 0|x = 1) &= \frac{p(x = 1, y = 0)}{p(x = 1)}, \\
p(y = 1|x = 0) &= \frac{p(x = 0, y = 1)}{p(x = 0)}, \\
p(y = 1|x = 1) &= \frac{p(x = 1, y = 1)}{p(x = 1)}.
\end{aligned} \tag{1.308}$$

Then,

$$\begin{aligned}
p(y = 0|x = 0) &= \frac{1}{2}, \\
p(y = 0|x = 1) &= 0, \\
p(y = 1|x = 0) &= \frac{1}{2}, \\
p(y = 1|x = 1) &= 1.
\end{aligned} \tag{1.309}$$

Therefore,

$$H(y|x) = \frac{2}{3} \ln 2. \tag{1.310}$$

(d)

We have

$$H(x|y) = - \sum_{x,y} p(x, y) \ln p(x|y). \tag{1.311}$$

By Bayes' theorem,

$$\begin{aligned}
p(x = 0|y = 0) &= \frac{p(x = 0, y = 0)}{p(y = 0)}, \\
p(x = 0|y = 1) &= \frac{p(x = 0, y = 1)}{p(y = 1)}, \\
p(x = 1|y = 0) &= \frac{p(x = 1, y = 0)}{p(y = 0)}, \\
p(x = 1|y = 1) &= \frac{p(x = 1, y = 1)}{p(y = 1)}.
\end{aligned} \tag{1.312}$$

Then,

$$\begin{aligned} p(x=0|y=0) &= 1, \\ p(x=0|y=1) &= \frac{1}{2}, \\ p(x=1|y=0) &= 0, \\ p(x=1|y=1) &= \frac{1}{2}. \end{aligned} \tag{1.313}$$

Therefore,

$$H(x|y) = \frac{2}{3} \ln 2. \tag{1.314}$$

(e)

We have

$$H(x, y) = - \sum_{x,y} p(x, y) \ln p(x, y). \tag{1.315}$$

Therefore,

$$H(x, y) = \ln 3. \tag{1.316}$$

(f)

We have

$$I(x, y) = - \sum_{x,y} p(x, y) \ln \frac{p(x)p(y)}{p(x, y)}. \tag{1.317}$$

The right hand side can be written as

$$H(x) + H(y) - H(x, y). \tag{1.318}$$

Therefore,

$$I(x, y) = \ln 3 - \frac{4}{3} \ln 2. \tag{1.319}$$

1.40

Let $\lambda_1, \dots, \lambda_M$ and $x_1, \dots, x_M$ be numbers such that

$$\begin{aligned} \sum_{m=1}^M \lambda_m &= 1, \\ \lambda_m &\geq 0, \\ x_m &> 0. \end{aligned} \tag{1.320}$$

By Jensen's inequality,

$$\sum_{m=1}^M \lambda_m \ln x_m \leq \ln \left(\sum_{m=1}^M \lambda_m x_m \right), \quad (1.321)$$

so that

$$\prod_{m=1}^M x_m^{\lambda_m} \leq \sum_{m=1}^M \lambda_m x_m. \quad (1.322)$$

Substituting

$$\lambda_m = \frac{1}{M} \quad (1.323)$$

to the inequality gives

$$\left(\prod_{m=1}^M x_m \right)^{\frac{1}{M}} \leq \frac{1}{M} \sum_{m=1}^M x_m. \quad (1.324)$$

1.41

Let $\mathbf{x}$ and $\mathbf{y}$ be variables.

(a)

We have

$$I(\mathbf{x}, \mathbf{y}) = - \iint p(\mathbf{x}, \mathbf{y}) \ln \left(\frac{p(\mathbf{x})p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})} \right) d\mathbf{x}d\mathbf{y}. \quad (1.325)$$

The right hand side can be written as

$$\begin{aligned} & - \iint p(\mathbf{x}, \mathbf{y}) \left(\ln p(\mathbf{x}) + \ln \left(\frac{p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})} \right) \right) d\mathbf{x}d\mathbf{y} \\ & = - \iint p(\mathbf{x}, \mathbf{y}) (\ln p(\mathbf{x}) - \ln p(\mathbf{x}|\mathbf{y})) d\mathbf{x}d\mathbf{y}. \end{aligned} \quad (1.326)$$

The right hand side can be written as

$$- \int \left(\int p(\mathbf{x}, \mathbf{y}) d\mathbf{y} \right) \ln p(\mathbf{x}) d\mathbf{x} + \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}|\mathbf{y}) d\mathbf{x}d\mathbf{y}. \quad (1.327)$$

Therefore,

$$I(\mathbf{x}, \mathbf{y}) = H(\mathbf{x}) - H(\mathbf{x}|\mathbf{y}). \quad (1.328)$$

(b)

We have

$$I(\mathbf{x}, \mathbf{y}) = I(\mathbf{y}, \mathbf{x}). \quad (1.329)$$

By (a), the right hand side can be written as

$$H(\mathbf{y}) - H(\mathbf{y}|\mathbf{x}). \quad (1.330)$$

Therefore,

$$I(\mathbf{x}, \mathbf{y}) = H(\mathbf{y}) - H(\mathbf{y}|\mathbf{x}). \quad (1.331)$$

2 Probability Distributions

2.1

Let x be a binary variable such that

$$p(x) = \mu^x(1 - \mu)^{1-x}. \quad (2.1)$$

(a)

We have

$$\sum_x p(x) = 1 - \mu + \mu. \quad (2.2)$$

Therefore,

$$\sum_x p(x) = 1. \quad (2.3)$$

(b)

We have

$$\begin{aligned} \mathbb{E} x &= \sum_x x p(x), \\ \mathbb{E} x^2 &= \sum_x x^2 p(x), \end{aligned} \quad (2.4)$$

Then,

$$\begin{aligned} \mathbb{E} x &= \mu, \\ \mathbb{E} x^2 &= \mu. \end{aligned} \quad (2.5)$$

We have

$$\text{var } x = \mathbb{E} x^2 - (\mathbb{E} x)^2. \quad (2.6)$$

Therefore,

$$\text{var } x = \mu(1 - \mu). \quad (2.7)$$

(c)

We have

$$H(x) = - \sum_x p(x) \ln p(x). \quad (2.8)$$

Therefore,

$$H(x) = -\mu \ln \mu - (1 - \mu) \ln(1 - \mu). \quad (2.9)$$

2.2

Let x be a variable from $\{-1, 1\}$ such that

$$p(x) = \left(\frac{1-\mu}{2}\right)^{\frac{1-x}{2}} \left(\frac{1+\mu}{2}\right)^{\frac{1+x}{2}}. \quad (2.10)$$

(a)

We have

$$\sum_x p(x) = \frac{1-\mu}{2} + \frac{1+\mu}{2}. \quad (2.11)$$

Therefore,

$$\sum_x p(x) = 1. \quad (2.12)$$

(b)

We have

$$\begin{aligned} \mathbb{E} x &= \sum_x x p(x), \\ \mathbb{E} x^2 &= \sum_x x^2 p(x). \end{aligned} \quad (2.13)$$

The right hand sides can be written as

$$\begin{aligned} -\frac{1-\mu}{2} + \frac{1+\mu}{2} &= \mu, \\ \frac{1-\mu}{2} + \frac{1+\mu}{2} &= 1. \end{aligned} \quad (2.14)$$

Then,

$$\begin{aligned} \mathbb{E} x &= \mu, \\ \mathbb{E} x^2 &= 1. \end{aligned} \quad (2.15)$$

We have

$$\text{var } x = \mathbb{E} x^2 - (\mathbb{E} x)^2. \quad (2.16)$$

Therefore,

$$\text{var } x = 1 - \mu^2. \quad (2.17)$$

(c)

We have

$$H(x) = - \sum_x p(x|\mu) \ln p(x|\mu). \quad (2.18)$$

Therefore,

$$H(x) = -\frac{1-\mu}{2} \ln \frac{1-\mu}{2} - \frac{1+\mu}{2} \ln \frac{1+\mu}{2}. \quad (2.19)$$

2.3

(a)

We have

$$\begin{aligned} \binom{N}{n} &= \frac{N!}{n!(N-n)!}, \\ \binom{N}{n-1} &= \frac{N!}{(n-1)!(N-n+1)!} \end{aligned} \quad (2.20)$$

Then,

$$\binom{N}{n} + \binom{N}{n-1} = \frac{(N-n+1)N! + nN!}{n!(N-n+1)!}. \quad (2.21)$$

The right hand side can be written as

$$\frac{(N+1)!}{n!(N+1-n)!} = \binom{N+1}{n}. \quad (2.22)$$

Therefore,

$$\binom{N}{n} + \binom{N}{n-1} = \binom{N+1}{n}. \quad (2.23)$$

(b)

We have

$$1+x = \sum_{n=0}^1 \binom{1}{n} x^n. \quad (2.24)$$

Let us assume that

$$(1+x)^N = \sum_{n=0}^N \binom{N}{n} x^n. \quad (2.25)$$

Then,

$$(1+x)^{N+1} = \sum_{n=0}^N \binom{N}{n} x^n + \sum_{n=0}^N \binom{N}{n} x^{n+1}. \quad (2.26)$$

By (a), the right hand side can be written as

$$\sum_{n=0}^N \binom{N}{n} x^n + \sum_{n=1}^{N+1} \binom{N}{n-1} x^n = 1 + x^{N+1} + \sum_{n=1}^N \binom{N+1}{n} x^n. \quad (2.27)$$

Then,

$$(1+x)^{N+1} = \sum_{n=0}^{N+1} \binom{N+1}{n} x^n. \quad (2.28)$$

Therefore, the assumption is proved by induction on N .

(c)

Let n be a variable such that

$$p(n) = \binom{N}{n} \mu^n (1-\mu)^{N-n}. \quad (2.29)$$

Then,

$$\sum_{n=0}^N p(n) = \sum_{n=0}^N \binom{N}{n} \mu^n (1-\mu)^{N-n}. \quad (2.30)$$

By (b), the right hand side can be written as

$$(1-\mu)^N \sum_{n=0}^N \binom{N}{n} \left(\frac{\mu}{1-\mu} \right)^n = (1-\mu)^N \left(1 + \frac{\mu}{1-\mu} \right)^N. \quad (2.31)$$

Therefore,

$$\sum_{n=0}^N p(n) = 1. \quad (2.32)$$

2.4

Let n be a variable such that

$$p(n) = \binom{N}{n} \mu^n (1-\mu)^{N-n}. \quad (2.33)$$

(a)

We have

$$E n = \sum_{n=0}^N n \binom{N}{n} \mu^n (1 - \mu)^{N-n}. \quad (2.34)$$

By 2.3(c),

$$\sum_{n=0}^N \binom{N}{n} \mu^n (1 - \mu)^{N-n} = 1. \quad (2.35)$$

Taking the derivative with respect to μ gives

$$\sum_{n=0}^N n \binom{N}{n} \mu^{n-1} (1 - \mu)^{N-n} - \sum_{n=0}^N (N - n) \binom{N}{n} \mu^n (1 - \mu)^{N-n-1} = 0. \quad (2.36)$$

The first term of the left hand side can be written as

$$\frac{1}{\mu} \sum_{n=0}^N n p(n) = \frac{1}{\mu} E n. \quad (2.37)$$

Since

$$(N - n) \binom{N}{n} = N \binom{N-1}{n}, \quad (2.38)$$

the second term can be written as

$$-N \sum_{n=0}^{N-1} \binom{N-1}{n} \mu^n (1 - \mu)^{N-n-1} = -N. \quad (2.39)$$

Therefore,

$$E n = N\mu. \quad (2.40)$$

(b)

By 2.3(c),

$$\sum_{n=0}^N \binom{N}{n} \mu^n (1 - \mu)^{N-n} = 1. \quad (2.41)$$

Taking the second derivative with respect to μ gives

$$\begin{aligned}
& \sum_{n=0}^N n(n-1) \binom{N}{n} \mu^{n-2} (1-\mu)^{N-n} \\
& - 2 \sum_{n=0}^N n(N-n) \binom{N}{n} \mu^{n-1} (1-\mu)^{N-n-1} \\
& + \sum_{n=0}^N (N-n)(N-n-1) \binom{N}{n} \mu^n (1-\mu)^{N-n-2} = 0.
\end{aligned} \tag{2.42}$$

The first term of the left hand side can be written as

$$\frac{1}{\mu^2} \sum_{n=0}^N n(n-1) p(n) = \frac{1}{\mu^2} E n(n-1). \tag{2.43}$$

Since

$$\begin{aligned}
n(N-n) \binom{N}{n} &= N(N-1) \binom{N-2}{n-1}, \\
(N-n)(N-n-1) \binom{N}{n} &= N(N-1) \binom{N-2}{n},
\end{aligned} \tag{2.44}$$

the second and third terms can be written as

$$\begin{aligned}
-2N(N-1) \sum_{n=1}^{N-1} \binom{N-2}{n-1} \mu^{n-1} (1-\mu)^{N-n-1} &= -2N(N-1), \\
N(N-1) \sum_{n=0}^N \binom{N-2}{n} \mu^n (1-\mu)^{N-n-2} &= N(N-1).
\end{aligned} \tag{2.45}$$

Then,

$$E n(n-1) = N(N-1) \mu^2. \tag{2.46}$$

We have

$$\text{var } n = E n(n-1) + E n - (E n)^2. \tag{2.47}$$

Therefore,

$$\text{var } n = N\mu(1-\mu). \tag{2.48}$$

2.5

We have

$$\Gamma(a)\Gamma(b) = \int_0^\infty x^{a-1} \exp(-x) dx \int_0^\infty y^{b-1} \exp(-y) dy. \quad (2.49)$$

By the transformation

$$t = x + y, \quad (2.50)$$

the right hand side can be written as

$$\begin{aligned} & \int_0^\infty x^{a-1} \left(\int_x^\infty (t-x)^{b-1} \exp(-t) dt \right) dx \\ &= \int_0^\infty \left(\int_0^t x^{a-1} (t-x)^{b-1} dx \right) \exp(-t) dt. \end{aligned} \quad (2.51)$$

By the transformation

$$x = t\mu, \quad (2.52)$$

the right hand side can be written as

$$\begin{aligned} & \int_0^\infty \left(\int_0^1 (t\mu)^{a-1} t^{b-1} (1-\mu)^{b-1} t d\mu \right) \exp(-t) dt \\ &= \int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu \int_0^\infty t^{a+b-1} \exp(-t) dt. \end{aligned} \quad (2.53)$$

Then,

$$\Gamma(a)\Gamma(b) = \Gamma(a+b) \int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu. \quad (2.54)$$

Therefore,

$$\int_0^1 \mu^{a-1} (1-\mu)^{b-1} d\mu = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \quad (2.55)$$

2.6

Let μ be a variable such that

$$p(\mu) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}. \quad (2.56)$$

(a)

We have

$$\begin{aligned} E\mu &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \mu^a (1-\mu)^{b-1} d\mu, \\ E\mu^2 &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \mu^{a+1} (1-\mu)^{b-1} d\mu. \end{aligned} \quad (2.57)$$

By 2.5,

$$\begin{aligned} \int_0^1 \mu^a (1-\mu)^{b-1} d\mu &= \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+b+1)}, \\ \int_0^1 \mu^{a+1} (1-\mu)^{b-1} d\mu &= \frac{\Gamma(a+2)\Gamma(b)}{\Gamma(a+b+2)}. \end{aligned} \quad (2.58)$$

Therefore,

$$\begin{aligned} E\mu &= \frac{a}{a+b}, \\ E\mu^2 &= \frac{a(a+1)}{(a+b)(a+b+1)}. \end{aligned} \quad (2.59)$$

(b)

We have

$$\text{var } \mu = E\mu^2 - (E\mu)^2. \quad (2.60)$$

By (a), the right hand side can be written as

$$\frac{a(a+1)}{(a+b)(a+b+1)} - \left(\frac{a}{a+b} \right)^2 = \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b)^2(a+b+1)}. \quad (2.61)$$

Therefore,

$$\text{var } \mu = \frac{ab}{(a+b)^2(a+b+1)}. \quad (2.62)$$

(c)

We have

$$\ln p(\mu) = \ln \Gamma(a+b) - \ln \Gamma(a) - \ln \Gamma(b) + (a-1) \ln \mu + (b-1) \ln(1-\mu). \quad (2.63)$$

Setting the derivative of $\ln p$ with respect to μ to zero gives

$$0 = \frac{a-1}{\mu} - \frac{b-1}{1-\mu}. \quad (2.64)$$

Therefore,

$$\text{mode } \mu = \begin{cases} \frac{a-1}{a+b-2}, & \text{if } a \geq 1 \text{ and } b \geq 1, \\ \text{none}, & \text{otherwise.} \end{cases} \quad (2.65)$$

2.7

Let m and l be variables such that

$$\begin{aligned} p(m, l | \mu) &= \binom{m+l}{m} \mu^m (1-\mu)^l, \\ p(\mu) &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}. \end{aligned} \quad (2.66)$$

By 2.6,

$$E \mu = \frac{a}{a+b}. \quad (2.67)$$

Setting the derivative of $p(m, l | \mu)$ with respect to μ to zero gives

$$0 = \binom{m+l}{m} \mu^m (1-\mu)^l \left(\frac{m}{\mu} + \frac{l}{1-\mu} \right). \quad (2.68)$$

Then, the maximum likelihood solution to μ is given by

$$\mu_{\text{ML}} = \frac{m}{m+l}. \quad (2.69)$$

By Bayes' theorem,

$$p(\mu | m, l) = \frac{p(m, l | \mu) p(\mu)}{p(m, l)}. \quad (2.70)$$

By 2.5, the right hand side can be written as

$$p(\mu | m, l) = \frac{\Gamma(m+l+a+b)}{\Gamma(m+a)\Gamma(l+b)} \mu^{m+a-1} (1-\mu)^{l+b-1}. \quad (2.71)$$

By 2.6,

$$E(\mu | m, l) = \frac{m+a}{m+l+a+b}. \quad (2.72)$$

Therefore,

$$E(\mu | m, l) = \lambda \mu_{\text{ML}} + (1-\lambda) E \mu, \quad (2.73)$$

where

$$\lambda = \frac{m+l}{m+l+a+b}. \quad (2.74)$$

2.8

Let x and y be variables.

(a)

We have

$$E x = \int x p(x) dx. \quad (2.75)$$

The right hand side can be written as

$$\int x \left(\int p(x, y) dy \right) dx = \int \left(\int x p(x|y) dx \right) p(y) dy. \quad (2.76)$$

Therefore,

$$E x = E (E(x|y)). \quad (2.77)$$

(b)

We have

$$\text{var } x = E (x - E x)^2. \quad (2.78)$$

By (a), the right hand side can be written as

$$\begin{aligned} & E (E ((x - E x)^2 | y)) \\ &= E (E ((f(x, y) + g(y))^2 | y)), \end{aligned} \quad (2.79)$$

where

$$\begin{aligned} f(x, y) &= x - E(x|y), \\ g(y) &= E(x|y) - E x. \end{aligned} \quad (2.80)$$

The right hand side can be written as $I_1 + 2I_2 + I_3$, where

$$\begin{aligned} I_1 &= E (E (f^2(x, y) | y)), \\ I_2 &= E (E (f(x, y)g(y) | y)), \\ I_3 &= E (g^2(y)). \end{aligned} \quad (2.81)$$

We have

$$\text{var}(x|y) = E (f^2(x, y) | y). \quad (2.82)$$

Then, the expression of I_1 can be written as

$$E (\text{var}(x|y)). \quad (2.83)$$

We have

$$\mathbb{E}(f(x, y)|y) = 0. \quad (2.84)$$

Then, the expression of I_2 can be written as

$$\mathbb{E}(\mathbb{E}(f(x, y)|y) g(y)) = 0. \quad (2.85)$$

By (a),

$$\mathbb{E}(\mathbb{E}(x|y)) = \mathbb{E} x. \quad (2.86)$$

Then, the expression of I_3 can be written as

$$\text{var}(\mathbb{E}(x|y)). \quad (2.87)$$

Therefore,

$$\text{var } x = \mathbb{E}(\text{var}(x|y)) + \text{var}(\mathbb{E}(x|y)). \quad (2.88)$$

2.9

For a vector $\boldsymbol{\mu}$ in 2 dimensions, by 2.5,

$$\int_{\substack{\mu_1 + \mu_2 = 1 \\ \mu_1 \geq 0, \mu_2 \geq 0}} \mu_1^{\alpha_1 - 1} \mu_2^{\alpha_2 - 1} d\boldsymbol{\mu} = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}. \quad (2.89)$$

For a vector $\boldsymbol{\mu}$ in K dimensions, let us assume that

$$\int_{\substack{\sum_{k=1}^K \mu_k = 1 \\ \mu_k \geq 0}} \prod_{k=1}^K \mu_k^{\alpha_k - 1} d\boldsymbol{\mu} = \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K \alpha_k)}. \quad (2.90)$$

For a vector $\boldsymbol{\mu}$ in $K + 1$ dimensions, let

$$\boldsymbol{\mu}' = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_{K-1} \\ \mu_K + \mu_{K+1} \end{bmatrix}. \quad (2.91)$$

Then,

$$\begin{aligned} & \int_{\substack{\sum_{k=1}^{K+1} \mu_k = 1 \\ \mu_k \geq 0}} \prod_{k=1}^{K+1} \mu_k^{\alpha_k - 1} d\boldsymbol{\mu} \\ &= \int_{\substack{\sum_{k=1}^K \mu'_k = 1 \\ \mu'_k \geq 0}} \left(\prod_{k=1}^{K-1} \mu'_k{}^{\alpha_k - 1} \right) I(\mu'_K) d\boldsymbol{\mu}', \end{aligned} \quad (2.92)$$

where

$$I(\mu'_K) = \int_0^{\mu'_K} (\mu'_K - \mu_{K+1})^{\alpha_K-1} \mu_{K+1}^{\alpha_{K+1}-1} d\mu_{K+1}. \quad (2.93)$$

By the transformation

$$x = \frac{\mu_{K+1}}{\mu'_K}, \quad (2.94)$$

the expression of $I(\mu'_K)$ can be written as

$$\begin{aligned} & \int_0^1 (\mu'_K(1-x))^{\alpha_K-1} (\mu'_K x)^{\alpha_{K+1}-1} \mu'_K dx \\ &= \mu_K'^{\alpha_K+\alpha_{K+1}-1} \int_0^1 x^{\alpha_{K+1}-1} (1-x)^{\alpha_K-1} dx. \end{aligned} \quad (2.95)$$

By 2.5, the integral of the right hand side can be written as

$$\frac{\Gamma(\alpha_K)\Gamma(\alpha_{K+1})}{\Gamma(\alpha_K + \alpha_{K+1})}. \quad (2.96)$$

Then,

$$\begin{aligned} & \int_{\sum_{k=1}^{K+1} \mu_k=1} \prod_{\substack{k=1 \\ \mu_k \geq 0}}^{K+1} \mu_k^{\alpha_k-1} d\boldsymbol{\mu} \\ &= \frac{\Gamma(\alpha_K)\Gamma(\alpha_{K+1})}{\Gamma(\alpha_K + \alpha_{K+1})} \int_{\sum_{k=1}^K \mu'_k=1} \left(\prod_{k=1}^{K-1} \mu'_k{}^{\alpha_k-1} \right) \mu_K'^{\alpha_K+\alpha_{K+1}-1} d\boldsymbol{\mu}'. \end{aligned} \quad (2.97)$$

By 2.5, the integral of the right hand side can be written as

$$\frac{(\prod_{k=1}^{K-1} \Gamma(\alpha_k))\Gamma(\alpha_K + \alpha_{K+1})}{\Gamma(\sum_{k=1}^{K+1} \alpha_k)}. \quad (2.98)$$

Then,

$$\int_{\sum_{k=1}^{K+1} \mu_k=1} \prod_{\substack{k=1 \\ \mu_k \geq 0}}^{K+1} \mu_k^{\alpha_k-1} d\boldsymbol{\mu} = \frac{\prod_{k=1}^{K+1} \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^{K+1} \alpha_k)}. \quad (2.99)$$

Therefore, the assumption is proved by induction on K .

2.10

Let $\boldsymbol{\mu}$ be a variable such that

$$p(\boldsymbol{\mu}) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \mu_k^{\alpha_k-1}. \quad (2.100)$$

We have

$$\begin{aligned} \mathbb{E} \mu_k &= \int \mu_k p(\boldsymbol{\mu}) d\boldsymbol{\mu}, \\ \mathbb{E} \mu_k^2 &= \int \mu_k^2 p(\boldsymbol{\mu}) d\boldsymbol{\mu}, \\ \mathbb{E} \mu_k \mu_{k'} &= \int \mu_k \mu_{k'} p(\boldsymbol{\mu}) d\boldsymbol{\mu}. \end{aligned} \quad (2.101)$$

Let $k \neq k'$. By 2.9, the right hand sides can be written as

$$\begin{aligned} \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \frac{\Gamma(\alpha_k+1)}{\Gamma(\alpha_k)} \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K \alpha_k + 1)} &= \frac{\alpha_k}{\sum_{k=1}^K \alpha_k}, \\ \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \frac{\Gamma(\alpha_k+2)}{\Gamma(\alpha_k)} \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K \alpha_k + 2)} &= \frac{\alpha_k(\alpha_k + 1)}{\sum_{k=1}^K \alpha_k (\sum_{k=1}^K \alpha_k + 1)}, \\ \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \frac{\Gamma(\alpha_k+1)\Gamma(\alpha_{k'}+1)}{\Gamma(\alpha_k)\Gamma(\alpha_{k'})} \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K \alpha_k + 2)} &= \frac{\alpha_k \alpha_{k'}}{\sum_{k=1}^K \alpha_k (\sum_{k=1}^K \alpha_k + 1)}. \end{aligned} \quad (2.102)$$

Then,

$$\begin{aligned} \mathbb{E} \mu_k &= \frac{\alpha_k}{\sum_{k=1}^K \alpha_k}, \\ \mathbb{E} \mu_k^2 &= \frac{\alpha_k(\alpha_k + 1)}{\sum_{k=1}^K \alpha_k (\sum_{k=1}^K \alpha_k + 1)}, \\ \mathbb{E} \mu_k \mu_{k'} &= \frac{\alpha_k \alpha_{k'}}{\sum_{k=1}^K \alpha_k (\sum_{k=1}^K \alpha_k + 1)}. \end{aligned} \quad (2.103)$$

We have

$$\begin{aligned} \text{var } \mu_k &= \mathbb{E} \mu_k^2 - (\mathbb{E} \mu_k)^2, \\ \text{cov}(\mu_k, \mu_{k'}) &= \mathbb{E} \mu_k \mu_{k'} - \mathbb{E} \mu_k \mathbb{E} \mu_{k'}. \end{aligned} \quad (2.104)$$

Therefore,

$$\begin{aligned}\text{var } \mu_k &= \frac{\alpha_k \left(\left(\sum_{k=1}^K \alpha_k \right) - \alpha_k \right)}{\left(\sum_{k=1}^K \alpha_k \right)^2 \left(\sum_{k=1}^K \alpha_k + 1 \right)}, \\ \text{cov}(\mu_k, \mu_{k'}) &= -\frac{\alpha_k \alpha_{k'}}{\left(\sum_{k=1}^K \alpha_k \right)^2 \left(\sum_{k=1}^K \alpha_k + 1 \right)}.\end{aligned}\tag{2.105}$$

2.11

Let $\boldsymbol{\mu}$ be a variable such that

$$p(\boldsymbol{\mu}) = \frac{\Gamma \left(\sum_{k=1}^K \alpha_k \right)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K \mu_k^{\alpha_k - 1}.\tag{2.106}$$

We have

$$\mathbb{E} \ln \mu_k = \int (\ln \mu_k) p(\boldsymbol{\mu}) d\boldsymbol{\mu}.\tag{2.107}$$

We have

$$\frac{\partial}{\partial \alpha_k} p(\boldsymbol{\mu}) = \left(\frac{\Gamma' \left(\sum_{k=1}^K \alpha_k \right)}{\Gamma \left(\sum_{k=1}^K \alpha_k \right)} - \frac{\Gamma'(\alpha_k)}{\Gamma(\alpha_k)} + \ln \mu_k \right) p(\boldsymbol{\mu}).\tag{2.108}$$

Then,

$$\mathbb{E} \ln \mu_k = \frac{\partial}{\partial \alpha_k} \int p(\boldsymbol{\mu}) d\boldsymbol{\mu} + \left(\psi(\alpha_k) - \psi \left(\sum_{k=1}^K \alpha_k \right) \right) \int p(\boldsymbol{\mu}) d\boldsymbol{\mu},\tag{2.109}$$

where

$$\psi(a) = \frac{d}{da} \ln \Gamma(a).\tag{2.110}$$

Therefore,

$$\mathbb{E} \ln \mu_k = \psi(\alpha_k) - \psi \left(\sum_{k=1}^K \alpha_k \right).\tag{2.111}$$

2.12

Let x be a variable such that

$$p(x) = \frac{1}{b-a}, \quad (2.112)$$

where $a < b$. Then

$$\int_a^b p(x) dx = 1. \quad (2.113)$$

We have

$$\begin{aligned} \mathbb{E} x &= \frac{1}{b-a} \int_a^b x dx, \\ \mathbb{E} x^2 &= \frac{1}{b-a} \int_a^b x^2 dx. \end{aligned} \quad (2.114)$$

Then,

$$\begin{aligned} \mathbb{E} x &= \frac{1}{2}(a+b), \\ \mathbb{E} x^2 &= \frac{1}{3}(a^2 + ab + b^2). \end{aligned} \quad (2.115)$$

We have

$$\text{var } x = \mathbb{E} x^2 - (\mathbb{E} x)^2. \quad (2.116)$$

Therefore,

$$\text{var } x = \frac{1}{12}(b-a)^2. \quad (2.117)$$

2.13

Let $\mathbf{x}$ be a variable in D dimensions and let

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}), \\ q(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{L}). \end{aligned} \quad (2.118)$$

We have

$$\text{KL}(p||q) = - \int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \ln \frac{\mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{L})}{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})} d\mathbf{x}. \quad (2.119)$$

The logarithm in the right hand side can be written as

$$\begin{aligned} & \ln \frac{(2\pi)^{-\frac{D}{2}} |\det \mathbf{L}|^{-\frac{1}{2}} \exp \left(-\frac{1}{2}(\mathbf{x} - \mathbf{m})^\top \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}) \right)}{(2\pi)^{-\frac{D}{2}} |\det \boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp \left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right)} \\ &= \frac{1}{2} \ln \left| \frac{\det \boldsymbol{\Sigma}}{\det \mathbf{L}} \right| + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) - \frac{1}{2}(\mathbf{x} - \mathbf{m})^\top \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}). \end{aligned} \quad (2.120)$$

Then, the integral can be written as

$$\frac{1}{2} \ln \left| \frac{\det \Sigma}{\det \mathbf{L}} \right| + \frac{1}{2} I_1 - \frac{1}{2} I_2, \quad (2.121)$$

where

$$\begin{aligned} I_1 &= \int (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x}, \\ I_2 &= \int (\mathbf{x} - \mathbf{m})^\top \mathbf{L}^{-1} (\mathbf{x} - \mathbf{m}) \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x}. \end{aligned} \quad (2.122)$$

We have

$$(\mathbf{x} - \mathbf{m})^\top \mathbf{L}^{-1} (\mathbf{x} - \mathbf{m}) = (\mathbf{x} - \boldsymbol{\mu} + \boldsymbol{\mu} - \mathbf{m})^\top \mathbf{L}^{-1} (\mathbf{x} - \boldsymbol{\mu} + \boldsymbol{\mu} - \mathbf{m}). \quad (2.123)$$

Then, the expression of I_2 can be written as

$$\begin{aligned} & \int (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{L}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x} \\ & + 2(\boldsymbol{\mu} - \mathbf{m})^\top \mathbf{L}^{-1} \int (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x} \\ & + (\boldsymbol{\mu} - \mathbf{m})^\top \mathbf{L}^{-1} (\boldsymbol{\mu} - \mathbf{m}) \int \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x} \\ & = I'_2 + (\boldsymbol{\mu} - \mathbf{m})^\top \mathbf{L}^{-1} (\boldsymbol{\mu} - \mathbf{m}), \end{aligned} \quad (2.124)$$

where

$$I'_2 = \int (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{L}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x}. \quad (2.125)$$

We have

$$\int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x} = \Sigma, \quad (2.126)$$

so that

$$\int (\mathbf{x} - \boldsymbol{\mu})^\top (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \Sigma) d\mathbf{x} = \text{tr } \Sigma. \quad (2.127)$$

Then,

$$\begin{aligned} I_1 &= \text{tr } (\Sigma^{-1} \Sigma), \\ I'_2 &= \text{tr } (\mathbf{L}^{-1} \Sigma). \end{aligned} \quad (2.128)$$

Therefore,

$$\begin{aligned} & \text{KL}(p||q) \\ & = \frac{1}{2} \left(\ln \left| \frac{\det \mathbf{L}}{\det \Sigma} \right| - D + \text{tr } (\mathbf{L}^{-1} \Sigma) + (\boldsymbol{\mu} - \mathbf{m})^\top \mathbf{L}^{-1} (\boldsymbol{\mu} - \mathbf{m}) \right). \end{aligned} \quad (2.129)$$

2.14

Let $\mathbf{x}$ be a variable in D dimensions. We have

$$H(\mathbf{x}) = - \int p(\mathbf{x}) \ln p(\mathbf{x}) d\mathbf{x}. \quad (2.130)$$

In order to maximise $H(x)$ with the constraints

$$\begin{aligned} \int p(\mathbf{x}) d\mathbf{x} &= 1, \\ \int \mathbf{x} p(\mathbf{x}) d\mathbf{x} &= \boldsymbol{\mu}, \\ \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top p(\mathbf{x}) d\mathbf{x} &= \boldsymbol{\Sigma}, \end{aligned} \quad (2.131)$$

let

$$\begin{aligned} L(p) = H(\mathbf{x}) + \lambda \left(\int p(\mathbf{x}) d\mathbf{x} - 1 \right) + \mathbf{l}^\top \left(\int \mathbf{x} p(\mathbf{x}) d\mathbf{x} - \boldsymbol{\mu} \right) \\ + \mathbf{m}^\top \left(\int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top p(\mathbf{x}) d\mathbf{x} - \boldsymbol{\Sigma} \right) \mathbf{m}. \end{aligned} \quad (2.132)$$

Setting the variation with respect to p to zero gives

$$0 = -\ln p(\mathbf{x}) - 1 + \lambda + \mathbf{l}^\top \mathbf{x} + \|\mathbf{m}^\top (\mathbf{x} - \boldsymbol{\mu})\|^2. \quad (2.133)$$

Then,

$$p(\mathbf{x}) = \exp \left(-1 + \lambda + \mathbf{l}^\top \mathbf{x} + \|\mathbf{m}^\top (\mathbf{x} - \boldsymbol{\mu})\|^2 \right). \quad (2.134)$$

We have

$$\begin{aligned} \mathbf{l}^\top \mathbf{x} + \|\mathbf{m}^\top (\mathbf{x} - \boldsymbol{\mu})\|^2 \\ = -(\mathbf{x} - \boldsymbol{\mu} - \mathbf{M}\mathbf{l})^\top \mathbf{M}^{-1} (\mathbf{x} - \boldsymbol{\mu} - \mathbf{M}\mathbf{l}) + \mathbf{l}^\top \mathbf{M}\mathbf{l} + \mathbf{l}^\top \boldsymbol{\mu}. \end{aligned} \quad (2.135)$$

where

$$\mathbf{M} = -(\mathbf{m}\mathbf{m}^\top)^{-1}. \quad (2.136)$$

Then,

$$p(\mathbf{x}) = c \exp \left(-(\mathbf{x} - \boldsymbol{\mu} - \mathbf{M}\mathbf{l})^\top \mathbf{M}^{-1} (\mathbf{x} - \boldsymbol{\mu} - \mathbf{M}\mathbf{l}) \right), \quad (2.137)$$

where

$$c = \exp(-1 + \lambda + \mathbf{l}^\top \mathbf{M} \mathbf{l} + \mathbf{l}^\top \boldsymbol{\mu}). \quad (2.138)$$

Substituting it to the constraints and the transformation

$$\mathbf{y} = \mathbf{x} - \boldsymbol{\mu} - \mathbf{M} \mathbf{l} \quad (2.139)$$

gives

$$\begin{aligned} c \int \exp(-\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}) d\mathbf{y} &= 1, \\ c \int (\mathbf{y} + \boldsymbol{\mu} + \mathbf{M} \mathbf{l}) \exp(-\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}) d\mathbf{y} &= \boldsymbol{\mu}, \\ c \int (\mathbf{y} + \mathbf{M} \mathbf{l}) (\mathbf{y} + \mathbf{M} \mathbf{l})^\top \exp(-\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}) d\mathbf{y} &= \boldsymbol{\Sigma}. \end{aligned} \quad (2.140)$$

We have

$$\begin{aligned} \int \exp(-\mathbf{y}^\top \mathbf{y}) d\mathbf{y} &= \left(\Gamma\left(\frac{1}{2}\right) \right)^D, \\ \int \mathbf{y} \exp(-\mathbf{y}^\top \mathbf{y}) d\mathbf{y} &= \mathbf{0}, \\ \int \mathbf{y} \mathbf{y}^\top \exp(-\mathbf{y}^\top \mathbf{y}) d\mathbf{y} &= \Gamma\left(\frac{3}{2}\right) \left(\Gamma\left(\frac{1}{2}\right) \right)^{D-1} \mathbf{I}, \end{aligned} \quad (2.141)$$

so that

$$\begin{aligned} \int \exp(-\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}) d\mathbf{y} &= \pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}}, \\ \int \mathbf{y} \exp(-\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}) d\mathbf{y} &= \mathbf{0}, \\ \int \mathbf{y} \mathbf{y}^\top \exp(-\mathbf{y}^\top \mathbf{M}^{-1} \mathbf{y}) d\mathbf{y} &= \frac{1}{2} \pi^{\frac{D}{2}} (\det \mathbf{M})^{\frac{1}{2}} \mathbf{M}. \end{aligned} \quad (2.142)$$

Then,

$$\begin{aligned} c \pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}} &= 1, \\ c \pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}} (\boldsymbol{\mu} + \mathbf{M} \mathbf{l}) &= \boldsymbol{\mu}, \\ c \pi^{\frac{D}{2}} |\det \mathbf{M}|^{\frac{1}{2}} \left(\frac{1}{2} \mathbf{M} + \mathbf{M} \mathbf{l} (\mathbf{M} \mathbf{l})^\top \right) &= \boldsymbol{\Sigma}, \end{aligned} \quad (2.143)$$

so that

$$\begin{aligned} c &= (2\pi)^{-\frac{D}{2}} |\det \boldsymbol{\Sigma}|^{-\frac{1}{2}}, \\ \mathbf{l} &= \mathbf{0}, \\ \mathbf{M} &= 2\boldsymbol{\Sigma}. \end{aligned} \quad (2.144)$$

Therefore,

$$p(\mathbf{x}) = (2\pi)^{-\frac{D}{2}} |\det \Sigma|^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right). \quad (2.145)$$

2.15

Let $\mathbf{x}$ be a variable in D dimensions such that

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma). \quad (2.146)$$

Then, by the definition, the entropy is given by

$$H(\mathbf{x}) = - \int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) \ln \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) d\mathbf{x}. \quad (2.147)$$

We have

$$\ln \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) = -\frac{D}{2} \ln(2\pi) - \frac{1}{2} \ln |\det \Sigma| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}). \quad (2.148)$$

Then, the expression of $H(\mathbf{x})$ can be written as

$$\begin{aligned} & \left(\frac{D}{2} \ln(2\pi) + \frac{1}{2} \ln |\det \Sigma| \right) \int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) d\mathbf{x} \\ & + \frac{1}{2} \int (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) d\mathbf{x}. \end{aligned} \quad (2.149)$$

We have

$$\int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) d\mathbf{x} = \Sigma, \quad (2.150)$$

so that

$$\int (\mathbf{x} - \boldsymbol{\mu})^\top (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) d\mathbf{x} = \text{tr } \Sigma. \quad (2.151)$$

Then,

$$\int (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) d\mathbf{x} = \text{tr } (\Sigma^{-1} \Sigma), \quad (2.152)$$

where the right hand side can be written as D . Therefore,

$$H(\mathbf{x}) = \frac{D}{2} (1 + \ln(2\pi)) + \frac{1}{2} \ln |\det \Sigma|. \quad (2.153)$$

2.16

Let x be a variable such that

$$x = x_1 + x_2, \quad (2.154)$$

where

$$\begin{aligned} p(x_1) &= \mathcal{N}(x_1 | \mu_1, \tau_1^{-1}), \\ p(x_2) &= \mathcal{N}(x_2 | \mu_2, \tau_2^{-1}). \end{aligned} \quad (2.155)$$

By marginalisation,

$$p(x) = \int_{-\infty}^{\infty} p(x|x_2)p(x_2)dx_2. \quad (2.156)$$

The integrand of the right hand side can be written as

$$\mathcal{N}(x | \mu_1 + x_2, \tau_1^{-1}) \mathcal{N}(x_2 | \mu_2, \tau_2^{-1}). \quad (2.157)$$

The logarithm of the integrand except the terms independent of x and z is given by

$$-\frac{\tau_1}{2}(x - \mu_1 - x_2)^2 - \frac{\tau_2}{2}(x_2 - \mu_2)^2 = -\frac{1}{2}\mathbf{v}^T \mathbf{M} \mathbf{v}, \quad (2.158)$$

where

$$\mathbf{v} = \begin{bmatrix} x_2 - \mu_2 \\ x - \mu_1 - \mu_2 \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \tau_1 + \tau_2 & -\tau_1 \\ -\tau_1 & \tau_1 \end{bmatrix}. \quad (2.159)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \tau_2^{-1} & \tau_2^{-1} \\ \tau_2^{-1} & \tau_1^{-1} + \tau_2^{-1} \end{bmatrix}. \quad (2.160)$$

Then,

$$p(x) = \mathcal{N}(x | \mu_1 + \mu_2, \tau_1^{-1} + \tau_2^{-1}). \quad (2.161)$$

Therefore, by 1.35,

$$H(x) = \frac{1}{2} (1 + \ln(2\pi) + \ln(\tau_1^{-1} + \tau_2^{-1})). \quad (2.162)$$

2.17

Let Σ be a matrix and let

$$\begin{aligned}\mathbf{S} &= \frac{1}{2} (\Sigma^{-1} + (\Sigma^{-1})^\top), \\ \mathbf{A} &= \frac{1}{2} (\Sigma^{-1} - (\Sigma^{-1})^\top).\end{aligned}\tag{2.163}$$

Then,

$$\Sigma^{-1} = \mathbf{S} + \mathbf{A},\tag{2.164}$$

so that

$$(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{S} (\mathbf{x} - \boldsymbol{\mu}) + (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{A} (\mathbf{x} - \boldsymbol{\mu}).\tag{2.165}$$

The second term of the right hand side can be written as

$$\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top (\Sigma^{-1})^\top (\mathbf{x} - \boldsymbol{\mu}).\tag{2.166}$$

The second term of the right hand side can be written as

$$-\frac{1}{2} (\Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}))^\top (\mathbf{x} - \boldsymbol{\mu}) = -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}).\tag{2.167}$$

Then,

$$(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{A} (\mathbf{x} - \boldsymbol{\mu}) = 0.\tag{2.168}$$

Therefore,

$$(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{S} (\mathbf{x} - \boldsymbol{\mu}).\tag{2.169}$$

2.18

(a)

Let Σ be a $D \times D$ real symmetric matrix such that

$$\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d,\tag{2.170}$$

where $\mathbf{u}_1, \dots, \mathbf{u}_D$ are unit eigenvectors. Then,

$$\overline{\mathbf{u}}_d^\top \Sigma \mathbf{u}_d = \lambda_d,\tag{2.171}$$

where $\overline{\mathbf{u}}_d$ is the conjugate of $\mathbf{u}_d$. Since Σ is real and symmetric, the left hand side can be written as

$$\overline{\mathbf{u}}_d^\top \Sigma^\top \mathbf{u}_d = (\overline{\Sigma \mathbf{u}}_d)^\top \mathbf{u}_d. \quad (2.172)$$

The right hand side can be writtet as

$$\overline{\lambda_d} \overline{\mathbf{u}}_d^\top \mathbf{u}_d = \overline{\lambda_d}. \quad (2.173)$$

Therefore,

$$\lambda_d = \overline{\lambda_d}. \quad (2.174)$$

(b)

For $d \neq d'$, taking the inner product with $\mathbf{u}'_d$ on both sides of

$$\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d \quad (2.175)$$

gives

$$\mathbf{u}_{d'}^\top \Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_{d'}^\top \mathbf{u}_d. \quad (2.176)$$

Since Σ is symmetric, the left hand side can be written as

$$\mathbf{u}_{d'}^\top \Sigma^\top \mathbf{u}_d = (\Sigma \mathbf{u}_{d'})^\top \mathbf{u}_d, \quad (2.177)$$

where the right hand side can be written as $\lambda_{d'} \mathbf{u}_{d'}^\top \mathbf{u}_d$. Then,

$$\lambda_d \mathbf{u}_{d'}^\top \mathbf{u}_d = \lambda_{d'} \mathbf{u}_{d'}^\top \mathbf{u}_d. \quad (2.178)$$

Therefore, if $\lambda_d \neq \lambda_{d'}$, then

$$\mathbf{u}_{d'}^\top \mathbf{u}_d = 0. \quad (2.179)$$

2.19

Let Σ be a $D \times D$ real symmetric matrix such that

$$\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d, \quad (2.180)$$

where $\mathbf{u}_1, \dots, \mathbf{u}_D$ are unit eigenvectors. Let

$$\begin{aligned} \Lambda &= \text{diag}(\lambda_1, \dots, \lambda_D), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_D]. \end{aligned} \quad (2.181)$$

Then,

$$\Sigma \mathbf{U} = \mathbf{U} \Lambda. \quad (2.182)$$

We have

$$\mathbf{U}^\top \mathbf{U} = \mathbf{I}. \quad (2.183)$$

Then,

$$\Sigma = \mathbf{U} \Lambda \mathbf{U}^\top, \quad (2.184)$$

so that

$$\Sigma^{-1} = \mathbf{U} \Lambda^{-1} \mathbf{U}^\top. \quad (2.185)$$

Therefore,

$$\begin{aligned} \Sigma &= \sum_{d=1}^D \lambda_d \mathbf{u}_d \mathbf{u}_d^\top, \\ \Sigma^{-1} &= \sum_{d=1}^D \frac{1}{\lambda_d} \mathbf{u}_d \mathbf{u}_d^\top. \end{aligned} \quad (2.186)$$

2.20

Let Σ be a $D \times D$ real symmetric matrix such that

$$\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d, \quad (2.187)$$

where $\mathbf{u}_1, \dots, \mathbf{u}_D$ are unit eigenvectors. Let

$$\begin{aligned} \Lambda &= \text{diag}(\lambda_1, \dots, \lambda_D), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_D]. \end{aligned} \quad (2.188)$$

By 2.19,

$$\Sigma = \mathbf{U} \Lambda \mathbf{U}^\top. \quad (2.189)$$

Then,

$$\mathbf{v}^\top \Sigma \mathbf{v} = \mathbf{w}^\top \Lambda \mathbf{w}, \quad (2.190)$$

where

$$\mathbf{w} = \mathbf{U}^\top \mathbf{v}. \quad (2.191)$$

The right hand side can be written as $\sum_{d=1}^D \lambda_d w_d^2$. Therefore, the necessary and sufficient condition for

$$\mathbf{v}^\top \Sigma \mathbf{v} > 0 \quad (2.192)$$

for any real vector $\mathbf{v}$ is

$$\lambda_d > 0. \quad (2.193)$$

2.21

Let Σ be a $D \times D$ real symmetric matrix. Then, the number of independent parameters is $\frac{D(D+1)}{2}$.

2.22

Let Σ be a $D \times D$ symmetric matrix such that

$$\Sigma \Lambda = \mathbf{I}. \quad (2.194)$$

Taking the transpose of the both sides gives

$$\Lambda^\top \Sigma = \mathbf{I}. \quad (2.195)$$

Therefore,

$$\Lambda^\top = \Lambda. \quad (2.196)$$

2.23

Let Σ be a $D \times D$ real symmetric matrix such that

$$\Sigma \mathbf{u}_d = \lambda_d \mathbf{u}_d, \quad (2.197)$$

where $\mathbf{u}_1, \dots, \mathbf{u}_D$ are unit eigenvectors. Let

$$\begin{aligned} \Lambda' &= \text{diag} \left(\lambda_1^{-\frac{1}{2}}, \dots, \lambda_D^{-\frac{1}{2}} \right), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_D]. \end{aligned} \quad (2.198)$$

By 2.19,

$$\Sigma = \mathbf{U} \Lambda'^{-2} \mathbf{U}^\top. \quad (2.199)$$

Then,

$$\int_{(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu}) = \Delta} d\mathbf{x} = \int_{(\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{U} \Lambda'^2 \mathbf{U}^\top (\mathbf{x}-\boldsymbol{\mu}) = \Delta} d\mathbf{x}. \quad (2.200)$$

By the transformation

$$\mathbf{y} = \Lambda' \mathbf{U}^\top (\mathbf{x} - \boldsymbol{\mu}), \quad (2.201)$$

the right hand side can be written as

$$\int_{\|\mathbf{y}\|^2=\Delta} \left| \det \left(\mathbf{U} \mathbf{\Lambda}'^{-1} \right) \right| d\mathbf{y}. \quad (2.202)$$

We have

$$\left| \det \left(\mathbf{U} \mathbf{\Lambda}'^{-1} \right) \right| = |\det \mathbf{\Sigma}|^{\frac{1}{2}}. \quad (2.203)$$

Therefore,

$$\int_{(\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})=\Delta} d\mathbf{x} = |\det \mathbf{\Sigma}|^{\frac{1}{2}} \Delta^D V_D, \quad (2.204)$$

where

$$V_D = \int_{\|\mathbf{x}\|=1} d\mathbf{x}. \quad (2.205)$$

2.24

Let $\mathbf{A}$ be a square matrix and $\mathbf{D}$ be an invertible matrix. We have

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}. \quad (2.206)$$

The right hand side can be written as

$$\begin{bmatrix} \mathbf{I} & \mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C} & \mathbf{O} \\ \mathbf{O} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{D}^{-1}\mathbf{C} & \mathbf{I} \end{bmatrix}. \quad (2.207)$$

Then,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{I} & \mathbf{O} \\ -\mathbf{D}^{-1}\mathbf{C} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{M} & \mathbf{O} \\ \mathbf{O} & \mathbf{D}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\mathbf{B}\mathbf{D}^{-1} \\ \mathbf{O} & \mathbf{I} \end{bmatrix}, \quad (2.208)$$

where

$$\mathbf{M} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1}. \quad (2.209)$$

Therefore,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{M} & -\mathbf{B}\mathbf{D}^{-1}\mathbf{M} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{M} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{M}\mathbf{B}\mathbf{D}^{-1} \end{bmatrix}. \quad (2.210)$$

2.25

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad (2.211)$$

where

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \\ \mathbf{x}_c \end{bmatrix}, \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \\ \boldsymbol{\mu}_c \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} & \Sigma_{ac} \\ \Sigma_{ba} & \Sigma_{bb} & \Sigma_{bc} \\ \Sigma_{ca} & \Sigma_{cb} & \Sigma_{cc} \end{bmatrix}. \quad (2.212)$$

Let

$$\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}, \quad (2.213)$$

where

$$\boldsymbol{\Lambda} = \begin{bmatrix} \Lambda_{aa} & \Lambda_{ab} & \Lambda_{ac} \\ \Lambda_{ba} & \Lambda_{bb} & \Lambda_{bc} \\ \Lambda_{ca} & \Lambda_{cb} & \Lambda_{cc} \end{bmatrix}. \quad (2.214)$$

By marginalisation,

$$p(\mathbf{x}_a|\mathbf{x}_b) = \int p(\mathbf{x}_a|\mathbf{x}_b, \mathbf{x}_c)p(\mathbf{x}_c)d\mathbf{x}_c. \quad (2.215)$$

The logarithm of $p(\mathbf{x})$ except the terms independent of $\mathbf{x}_a$ can be written as

$$\begin{aligned} & -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_a)^\top \boldsymbol{\Lambda}_{aa}(\mathbf{x}_a - \boldsymbol{\mu}_a) - \frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_a)^\top \boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b) \\ & -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_a)^\top \boldsymbol{\Lambda}_{ac}(\mathbf{x}_c - \boldsymbol{\mu}_c) - \frac{1}{2}(\mathbf{x}_b - \boldsymbol{\mu}_b)^\top \boldsymbol{\Lambda}_{ba}(\mathbf{x}_a - \boldsymbol{\mu}_a) \\ & -\frac{1}{2}(\mathbf{x}_c - \boldsymbol{\mu}_c)^\top \boldsymbol{\Lambda}_{ca}(\mathbf{x}_a - \boldsymbol{\mu}_a) \\ & = -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_{a|b,c})^\top \boldsymbol{\Lambda}_{aa}(\mathbf{x}_a - \boldsymbol{\mu}_{a|b,c}) + \frac{1}{2}\boldsymbol{\mu}_{a|b,c}^\top \boldsymbol{\Lambda}_{aa}\boldsymbol{\mu}_{a|b,c}, \end{aligned} \quad (2.216)$$

where

$$\boldsymbol{\mu}_{a|b,c} = \boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b) - \boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ac}(\mathbf{x}_c - \boldsymbol{\mu}_c). \quad (2.217)$$

Then,

$$\begin{aligned} p(\mathbf{x}_a|\mathbf{x}_b, \mathbf{x}_c) &= \mathcal{N}(\mathbf{x}_a|\boldsymbol{\mu}_{a|b,c}, \boldsymbol{\Lambda}_{aa}^{-1}), \\ p(\mathbf{x}_c) &= \mathcal{N}(\mathbf{x}_c|\boldsymbol{\mu}_c, \boldsymbol{\Lambda}_{cc}^{-1}), \end{aligned} \quad (2.218)$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{x}_a$, $\mathbf{x}_b$ and $\mathbf{x}_c$ can be written as

$$\begin{aligned} & -\frac{1}{2}(\mathbf{x}_a - \boldsymbol{\mu}_{a|b,c})^\top \boldsymbol{\Lambda}_{aa}(\mathbf{x}_a - \boldsymbol{\mu}_{a|b,c}) - \frac{1}{2}(\mathbf{x}_c - \boldsymbol{\mu}_c)^\top \boldsymbol{\Lambda}_{cc}(\mathbf{x}_c - \boldsymbol{\mu}_c) \\ & = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v}, \end{aligned} \quad (2.219)$$

where

$$\begin{aligned} \mathbf{v} &= \begin{bmatrix} \mathbf{x}_c - \boldsymbol{\mu}_c \\ \mathbf{x}_a - \boldsymbol{\mu}_a + \boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b) \end{bmatrix}, \\ \mathbf{M} &= \begin{bmatrix} \boldsymbol{\Lambda}_{ca} \boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ac} + \boldsymbol{\Lambda}_{cc} & \boldsymbol{\Lambda}_{ca} \\ \boldsymbol{\Lambda}_{ac} & \boldsymbol{\Lambda}_{aa} \end{bmatrix}. \end{aligned} \quad (2.220)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \boldsymbol{\Lambda}_{cc}^{-1} & -\boldsymbol{\Lambda}_{cc}^{-1} \boldsymbol{\Lambda}_{ca} \boldsymbol{\Lambda}_{aa}^{-1} \\ -\boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ac} \boldsymbol{\Lambda}_{cc}^{-1} & \boldsymbol{\Lambda}_{aa}^{-1} + \boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ac} \boldsymbol{\Lambda}_{cc}^{-1} \boldsymbol{\Lambda}_{ca} \boldsymbol{\Lambda}_{aa}^{-1} \end{bmatrix}. \quad (2.221)$$

Therefore,

$$p(\mathbf{x}_a | \mathbf{x}_b) = \mathcal{N}(\mathbf{x}_a | \boldsymbol{\mu}_{a|b}, \boldsymbol{\Sigma}_{a|b}), \quad (2.222)$$

where

$$\begin{aligned} \boldsymbol{\mu}_{a|b} &= \boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ab}(\mathbf{x}_b - \boldsymbol{\mu}_b), \\ \boldsymbol{\Sigma}_{a|b} &= \boldsymbol{\Lambda}_{aa}^{-1} + \boldsymbol{\Lambda}_{aa}^{-1} \boldsymbol{\Lambda}_{ac} \boldsymbol{\Lambda}_{cc}^{-1} \boldsymbol{\Lambda}_{ca} \boldsymbol{\Lambda}_{aa}^{-1}. \end{aligned} \quad (2.223)$$

2.26

Let $\mathbf{A}$ be a square matrix and $\mathbf{C}$ be an invertible matrix. By 2.24,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{D} & -\mathbf{C}^{-1} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{M} & \mathbf{MBC} \\ \mathbf{CDM} & -\mathbf{C} + \mathbf{CDMBC} \end{bmatrix}, \quad (2.224)$$

where

$$\mathbf{M} = (\mathbf{A} + \mathbf{BCD})^{-1}. \quad (2.225)$$

By 2.24,

$$\begin{bmatrix} -\mathbf{C}^{-1} & \mathbf{D} \\ \mathbf{B} & \mathbf{A} \end{bmatrix}^{-1} = \begin{bmatrix} -\mathbf{N} & \mathbf{NDA}^{-1} \\ \mathbf{A}^{-1}\mathbf{BN} & \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{BNDA}^{-1} \end{bmatrix}, \quad (2.226)$$

where

$$\mathbf{N} = (\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}. \quad (2.227)$$

Therefore,

$$\mathbf{M} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{BNDA}^{-1}. \quad (2.228)$$

2.27

(a)

Let $\mathbf{x}$ and $\mathbf{z}$ be two variables. We have

$$E(\mathbf{x} + \mathbf{z}) = \iint (\mathbf{x} + \mathbf{z}) p(\mathbf{x}, \mathbf{z}) d\mathbf{x} d\mathbf{z}. \quad (2.229)$$

The right hand side can be written as

$$\begin{aligned} & \int \mathbf{x} \left(\int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \right) d\mathbf{x} + \int \mathbf{z} \left(\int p(\mathbf{x}, \mathbf{z}) d\mathbf{x} \right) d\mathbf{z} \\ &= \int \mathbf{x} p(\mathbf{x}) d\mathbf{x} + \int \mathbf{z} p(\mathbf{z}) d\mathbf{z}. \end{aligned} \quad (2.230)$$

Therefore,

$$E(\mathbf{x} + \mathbf{z}) = E \mathbf{x} + E \mathbf{z}. \quad (2.231)$$

(b)

Let $\mathbf{x}$ and $\mathbf{z}$ be two independent variables in the same dimension. We have

$$\begin{aligned} & \text{var}(\mathbf{x} + \mathbf{z}) \\ &= \iint (\mathbf{x} + \mathbf{z} - E(\mathbf{x} + \mathbf{z})) (\mathbf{x} + \mathbf{z} - E(\mathbf{x} + \mathbf{z}))^\top p(\mathbf{x}, \mathbf{z}) d\mathbf{x} d\mathbf{z}. \end{aligned} \quad (2.232)$$

The right hand side can be written as $I_1 + I_2 + I_3 + I_4$, where

$$\begin{aligned} I_1 &= \iint (\mathbf{x} - E \mathbf{x}) (\mathbf{x} - E \mathbf{x})^\top p(\mathbf{x}, \mathbf{z}) d\mathbf{x} d\mathbf{z}, \\ I_2 &= \iint (\mathbf{x} - E \mathbf{x}) (\mathbf{z} - E \mathbf{z})^\top p(\mathbf{x}, \mathbf{z}) d\mathbf{x} d\mathbf{z}, \\ I_3 &= \iint (\mathbf{z} - E \mathbf{z}) (\mathbf{x} - E \mathbf{x})^\top p(\mathbf{x}, \mathbf{z}) d\mathbf{x} d\mathbf{z}, \\ I_4 &= \iint (\mathbf{z} - E \mathbf{z}) (\mathbf{z} - E \mathbf{z})^\top p(\mathbf{x}, \mathbf{z}) d\mathbf{x} d\mathbf{z}. \end{aligned} \quad (2.233)$$

The expression of I_1 can be written as

$$\begin{aligned} & \int (\mathbf{x} - E \mathbf{x}) (\mathbf{x} - E \mathbf{x})^\top \left(\int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \right) d\mathbf{x} \\ &= \int (\mathbf{x} - E \mathbf{x}) (\mathbf{x} - E \mathbf{x})^\top p(\mathbf{x}) d\mathbf{x}. \end{aligned} \quad (2.234)$$

The expression of I_2 can be written as

$$\int (\mathbf{x} - \mathbf{E} \mathbf{x}) p(\mathbf{x}) d\mathbf{x} \left(\int (\mathbf{z} - \mathbf{E} \mathbf{z}) p(\mathbf{z}) d\mathbf{z} \right)^\top = \mathbf{O}. \quad (2.235)$$

The expression of I_3 can be written as

$$\int (\mathbf{z} - \mathbf{E} \mathbf{z}) p(\mathbf{z}) d\mathbf{z} \left(\int (\mathbf{x} - \mathbf{E} \mathbf{x}) p(\mathbf{x}) d\mathbf{x} \right)^\top = \mathbf{O}. \quad (2.236)$$

The expression of I_4 can be written as

$$\begin{aligned} & \int (\mathbf{z} - \mathbf{E} \mathbf{z}) (\mathbf{z} - \mathbf{E} \mathbf{z})^\top \left(\int p(\mathbf{x}, \mathbf{z}) d\mathbf{x} \right) d\mathbf{z} \\ &= \int (\mathbf{z} - \mathbf{E} \mathbf{z}) (\mathbf{z} - \mathbf{E} \mathbf{z})^\top p(\mathbf{z}) d\mathbf{z}. \end{aligned} \quad (2.237)$$

Therefore,

$$\text{var}(\mathbf{x} + \mathbf{z}) = \text{var} \mathbf{x} + \text{var} \mathbf{z}. \quad (2.238)$$

2.28

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}), \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} | \mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^\top), \\ \text{cov}(\mathbf{x}, \mathbf{y}) &= \boldsymbol{\Lambda}^{-1}\mathbf{A}^\top. \end{aligned} \quad (2.239)$$

Then,

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \mathbf{m}, \mathbf{S}), \quad (2.240)$$

where

$$\mathbf{z} = \begin{bmatrix} \mathbf{y} \\ \mathbf{x} \end{bmatrix}, \mathbf{m} = \begin{bmatrix} \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \\ \boldsymbol{\mu} \end{bmatrix}, \mathbf{S} = \begin{bmatrix} \mathbf{L}^{-1} + \mathbf{A}\boldsymbol{\Lambda}^{-1}\mathbf{A}^\top & \mathbf{A}\boldsymbol{\Lambda}^{-1} \\ \boldsymbol{\Lambda}^{-1}\mathbf{A}^\top & \boldsymbol{\Lambda}^{-1} \end{bmatrix}. \quad (2.241)$$

By 2.24,

$$\mathbf{S}^{-1} = \begin{bmatrix} \mathbf{L} & -\mathbf{L}\mathbf{A} \\ -\mathbf{A}^\top\mathbf{L} & \mathbf{M} \end{bmatrix}, \quad (2.242)$$

where

$$\mathbf{M} = \boldsymbol{\Lambda} + \mathbf{A}^\top\mathbf{L}\mathbf{A}. \quad (2.243)$$

We have

$$\mathbf{A}\boldsymbol{\mu} + \mathbf{b} + \mathbf{A}(\mathbf{x} - \boldsymbol{\mu}) = \mathbf{A}\mathbf{x} + \mathbf{b}, \quad (2.244)$$

and

$$\boldsymbol{\mu} + \mathbf{M}^{-1}\mathbf{A}^\top\mathbf{L}(\mathbf{y} - \mathbf{A}\boldsymbol{\mu} - \mathbf{b}) = \mathbf{M}^{-1}(\Lambda\boldsymbol{\mu} + \mathbf{A}^\top\mathbf{L}(\mathbf{y} - \mathbf{b})). \quad (2.245)$$

Therefore,

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1}), \\ p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x}|\mathbf{M}^{-1}(\Lambda\boldsymbol{\mu} + \mathbf{A}^\top\mathbf{L}(\mathbf{y} - \mathbf{b})), \mathbf{M}^{-1}). \end{aligned} \quad (2.246)$$

2.29

Let

$$\mathbf{R} = \begin{bmatrix} \Lambda + \mathbf{A}^\top\mathbf{L}\mathbf{A} & -\mathbf{A}^\top\mathbf{L} \\ -\mathbf{L}\mathbf{A} & \mathbf{L} \end{bmatrix}. \quad (2.247)$$

By 2.24,

$$\mathbf{R}^{-1} = \begin{bmatrix} \Lambda^{-1} & \Lambda^{-1}\mathbf{A}^\top \\ \mathbf{A}\Lambda^{-1} & \mathbf{L}^{-1} + \mathbf{A}\Lambda^{-1}\mathbf{A}^\top \end{bmatrix}, \quad (2.248)$$

so that

$$\mathbf{R}^{-1} \begin{bmatrix} \Lambda\boldsymbol{\mu} - \mathbf{A}^\top\mathbf{L}\mathbf{b} \\ \mathbf{L}\mathbf{b} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mu} \\ \mathbf{A}\boldsymbol{\mu} + \mathbf{b} \end{bmatrix}. \quad (2.249)$$

2.30

Refer to 2.29.

2.31

Let $\mathbf{y}$ be a variable such that

$$\mathbf{y} = \mathbf{x} + \mathbf{z}, \quad (2.250)$$

where

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_{\mathbf{x}}, \boldsymbol{\Sigma}_{\mathbf{x}}), \\ p(\mathbf{z}) &= \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_{\mathbf{z}}, \boldsymbol{\Sigma}_{\mathbf{z}}). \end{aligned} \quad (2.251)$$

By marginalisation,

$$p(\mathbf{y}) = \int p(\mathbf{y}|\mathbf{x})p(\mathbf{x})d\mathbf{x}. \quad (2.252)$$

The right hand side can be written as

$$\int \mathcal{N}(\mathbf{y}|\mathbf{x} + \boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z) \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x) d\mathbf{x}. \quad (2.253)$$

The logarithm of the integrand except the terms independent of $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\begin{aligned} & -\frac{1}{2}(\mathbf{y} - \mathbf{x} - \boldsymbol{\mu}_z)^\top \boldsymbol{\Sigma}_z^{-1}(\mathbf{y} - \mathbf{x} - \boldsymbol{\mu}_z) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_x)^\top \boldsymbol{\Sigma}_x^{-1}(\mathbf{x} - \boldsymbol{\mu}_x) \\ & = -\frac{1}{2}\mathbf{u}^\top \mathbf{M}\mathbf{u} + \mathbf{u}^\top \mathbf{v} - \frac{1}{2}\boldsymbol{\mu}_z^\top \boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}_z - \frac{1}{2}\boldsymbol{\mu}_x^\top \boldsymbol{\Sigma}_x^{-1}\boldsymbol{\mu}_x, \end{aligned} \quad (2.254)$$

where

$$\mathbf{u} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \boldsymbol{\Sigma}_x^{-1} + \boldsymbol{\Sigma}_z^{-1} & -\boldsymbol{\Sigma}_z^{-1} \\ -\boldsymbol{\Sigma}_z^{-1} & \boldsymbol{\Sigma}_z^{-1} \end{bmatrix}, \mathbf{v} = \begin{bmatrix} \boldsymbol{\Sigma}_x^{-1}\boldsymbol{\mu}_x - \boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}_z \\ \boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}_z \end{bmatrix}. \quad (2.255)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \boldsymbol{\Sigma}_x & \boldsymbol{\Sigma}_x \\ \boldsymbol{\Sigma}_x & \boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_z \end{bmatrix}, \quad (2.256)$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_x + \boldsymbol{\mu}_z \end{bmatrix}. \quad (2.257)$$

Therefore,

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\boldsymbol{\mu}_x + \boldsymbol{\mu}_z, \boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_z). \quad (2.258)$$

2.32

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}), \\ p(\mathbf{y}|\mathbf{x}) &= \mathcal{N}(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1}). \end{aligned} \quad (2.259)$$

By Bayes' theorem,

$$p(\mathbf{x}|\mathbf{y})p(\mathbf{y}) = p(\mathbf{y}|\mathbf{x})p(\mathbf{x}). \quad (2.260)$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$ and $\mathbf{y}$ can be written as

$$\begin{aligned} & -\frac{1}{2}(\mathbf{y} - \mathbf{A}\mathbf{x} - \mathbf{b})^\top \mathbf{L}(\mathbf{y} - \mathbf{A}\mathbf{x} - \mathbf{b}) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Lambda}(\mathbf{x} - \boldsymbol{\mu}) \\ & = -\frac{1}{2}(\mathbf{y} - \mathbf{A}(\mathbf{x} - \boldsymbol{\mu}) - \mathbf{A}\boldsymbol{\mu} - \mathbf{b})^\top \mathbf{L}(\mathbf{y} - \mathbf{A}(\mathbf{x} - \boldsymbol{\mu}) - \mathbf{A}\boldsymbol{\mu} - \mathbf{b}) \\ & \quad - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Lambda}(\mathbf{x} - \boldsymbol{\mu}). \end{aligned} \quad (2.261)$$

The right hand side can be written as

$$\begin{aligned}
& -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top (\boldsymbol{\Lambda} + \mathbf{A}^\top \mathbf{L} \mathbf{A}) (\mathbf{x} - \boldsymbol{\mu}) + (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b}) \\
& -\frac{1}{2}(\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b})^\top \mathbf{L} (\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b}) \\
& = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu} - \mathbf{z})^\top \mathbf{M} (\mathbf{x} - \boldsymbol{\mu} - \mathbf{z}) + \frac{1}{2} \mathbf{z}^\top \mathbf{M} \mathbf{z} \\
& -\frac{1}{2}(\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b})^\top \mathbf{L} (\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b}),
\end{aligned} \tag{2.262}$$

where

$$\begin{aligned}
\mathbf{z} &= \mathbf{M}^{-1} \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b}), \\
\mathbf{M} &= \boldsymbol{\Lambda} + \mathbf{A}^\top \mathbf{L} \mathbf{A}.
\end{aligned} \tag{2.263}$$

The right hand side can be written as

$$-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu} - \mathbf{z})^\top \mathbf{M} (\mathbf{x} - \boldsymbol{\mu} - \mathbf{z}) - \frac{1}{2}(\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b})^\top \mathbf{N} (\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b}), \tag{2.264}$$

where

$$\mathbf{N} = \mathbf{L} - \mathbf{L} \mathbf{A} \mathbf{M}^{-1} \mathbf{A}^\top \mathbf{L}. \tag{2.265}$$

Then,

$$\begin{aligned}
p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x} | \boldsymbol{\mu} + \mathbf{z}, \mathbf{M}^{-1}), \\
p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} | \mathbf{A} \boldsymbol{\mu} + \mathbf{b}, \mathbf{N}^{-1}).
\end{aligned} \tag{2.266}$$

We have

$$\boldsymbol{\mu} + \mathbf{z} = \mathbf{M}^{-1} (\mathbf{M} \boldsymbol{\mu} + \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{A} \boldsymbol{\mu} - \mathbf{b})). \tag{2.267}$$

The right hand side can be written as

$$\mathbf{M}^{-1} (\boldsymbol{\Lambda} \boldsymbol{\mu} + \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{b})). \tag{2.268}$$

By 2.26,

$$\mathbf{N}^{-1} = \mathbf{L}^{-1} + \mathbf{A} \boldsymbol{\Lambda}^{-1} \mathbf{A}^\top. \tag{2.269}$$

Therefore,

$$\begin{aligned}
p(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\mathbf{x} | \mathbf{M}^{-1} (\boldsymbol{\Lambda} \boldsymbol{\mu} + \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{b})), \mathbf{M}^{-1}), \\
p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} | \mathbf{A} \boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A} \boldsymbol{\Lambda}^{-1} \mathbf{A}^\top).
\end{aligned} \tag{2.270}$$

2.33

Refer to 2.32, while a different approach is presented below.

Let $\mathbf{x}$ and $\mathbf{y}$ be variables such that

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}), \\ p(\mathbf{y} | \mathbf{x}) &= \mathcal{N}(\mathbf{y} | \mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1}). \end{aligned} \quad (2.271)$$

By Bayes' theorem,

$$p(\mathbf{x} | \mathbf{y}) p(\mathbf{y}) = p(\mathbf{y} | \mathbf{x}) p(\mathbf{x}). \quad (2.272)$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$ and $\mathbf{y}$ can be written as

$$\begin{aligned} & -\frac{1}{2} (\mathbf{y} - \mathbf{A}\mathbf{x} - \mathbf{b})^\top \mathbf{L} (\mathbf{y} - \mathbf{A}\mathbf{x} - \mathbf{b}) - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Lambda} (\mathbf{x} - \boldsymbol{\mu}) \\ &= -\frac{1}{2} \mathbf{v}^\top \mathbf{S}^{-1} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2} \mathbf{b}^\top \mathbf{L} \mathbf{b} - \frac{1}{2} \boldsymbol{\mu}^\top \boldsymbol{\Lambda} \boldsymbol{\mu}, \end{aligned} \quad (2.273)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}, \mathbf{S}^{-1} = \begin{bmatrix} \mathbf{M} & -\mathbf{A}^\top \mathbf{L} \\ -\mathbf{L} \mathbf{A} & \mathbf{L} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} -\mathbf{A}^\top \mathbf{L} \mathbf{b} + \boldsymbol{\Lambda} \boldsymbol{\mu} \\ \mathbf{L} \mathbf{b} \end{bmatrix}, \quad (2.274)$$

and

$$\mathbf{M} = \boldsymbol{\Lambda} + \mathbf{A}^\top \mathbf{L} \mathbf{A}. \quad (2.275)$$

We have

$$\mathbf{M}^{-1} (-\mathbf{A}^\top \mathbf{L} \mathbf{b} + \boldsymbol{\Lambda} \boldsymbol{\mu} + \mathbf{A}^\top \mathbf{L} \mathbf{y}) = \mathbf{M}^{-1} (\boldsymbol{\Lambda} \boldsymbol{\mu} + \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{b})). \quad (2.276)$$

Then,

$$p(\mathbf{x} | \mathbf{y}) = \mathcal{N}(\mathbf{x} | \mathbf{M}^{-1} (\boldsymbol{\Lambda} \boldsymbol{\mu} + \mathbf{A}^\top \mathbf{L} (\mathbf{y} - \mathbf{b})), \mathbf{M}^{-1}). \quad (2.277)$$

By 2.24,

$$\mathbf{S} = \begin{bmatrix} \boldsymbol{\Lambda}^{-1} & \boldsymbol{\Lambda}^{-1} \mathbf{A}^\top \\ \mathbf{A} \boldsymbol{\Lambda}^{-1} & \mathbf{L}^{-1} + \mathbf{A} \boldsymbol{\Lambda}^{-1} \mathbf{A}^\top \end{bmatrix}, \quad (2.278)$$

so that

$$\mathbf{S} \mathbf{u} = \begin{bmatrix} \boldsymbol{\mu} \\ \mathbf{A} \boldsymbol{\mu} + \mathbf{b} \end{bmatrix}. \quad (2.279)$$

Then,

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y} | \mathbf{A} \boldsymbol{\mu} + \mathbf{b}, \mathbf{L}^{-1} + \mathbf{A} \boldsymbol{\Lambda}^{-1} \mathbf{A}^\top). \quad (2.280)$$

2.34

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be independent variables such that

$$p(\mathbf{x}_n) = \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}, \boldsymbol{\Sigma}). \quad (2.281)$$

Then,

$$\ln p(\mathbf{X}) = -\frac{ND}{2} \ln(2\pi) - \frac{N}{2} \ln |\det \boldsymbol{\Sigma}| - \frac{1}{2} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}). \quad (2.282)$$

By 3.21, setting the derivatives of $\ln p(\mathbf{X})$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ to zero gives

$$\begin{aligned} \mathbf{0} &= \sum_{n=1}^N (\boldsymbol{\Sigma}^{-1} + (\boldsymbol{\Sigma}^{-1})^\top) (\mathbf{x}_n - \boldsymbol{\mu}), \\ \mathbf{O} &= -\frac{N}{2} (\boldsymbol{\Sigma}^{-1})^\top + \frac{1}{2} (\boldsymbol{\Sigma}^{-1})^2 \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\top. \end{aligned} \quad (2.283)$$

Therefore, the maximum likelihood solutions to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are given by

$$\begin{aligned} \boldsymbol{\mu}_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n, \\ \boldsymbol{\Sigma}_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})^\top. \end{aligned} \quad (2.284)$$

2.35

(a)

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}). \quad (2.285)$$

Then,

$$\mathbf{E} \mathbf{x} \mathbf{x}^\top = \mathbf{E}(\mathbf{x} - \boldsymbol{\mu} + \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu} + \boldsymbol{\mu})^\top. \quad (2.286)$$

The right hand side can be written as

$$\mathbf{E}(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top + \boldsymbol{\mu} \mathbf{E}(\mathbf{x} - \boldsymbol{\mu})^\top + \mathbf{E}(\mathbf{x} - \boldsymbol{\mu}) \boldsymbol{\mu}^\top + \boldsymbol{\mu} \boldsymbol{\mu}^\top. \quad (2.287)$$

We have

$$\begin{aligned} \mathbf{E} \mathbf{x} &= \boldsymbol{\mu}, \\ \text{var } \mathbf{x} &= \boldsymbol{\Sigma}. \end{aligned} \quad (2.288)$$

Therefore,

$$\mathbf{E} \mathbf{x} \mathbf{x}^\top = \boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top. \quad (2.289)$$

(b)

Let $\mathbf{x}_n$ and $\mathbf{x}_m$ be independent variables such that

$$\begin{aligned} p(\mathbf{x}_n) &= \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}, \boldsymbol{\Sigma}), \\ p(\mathbf{x}_m) &= \mathcal{N}(\mathbf{x}_m | \boldsymbol{\mu}, \boldsymbol{\Sigma}). \end{aligned} \quad (2.290)$$

If $n \neq m$, then

$$\mathbf{E} \mathbf{x}_n \mathbf{x}_m^\top = \mathbf{E} \mathbf{x}_n \mathbf{E} \mathbf{x}_m^\top. \quad (2.291)$$

Therefore, by (a),

$$\mathbf{E} \mathbf{x}_n \mathbf{x}_m^\top = I_{nm} \boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top. \quad (2.292)$$

(c)

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be independent variables such that

$$p(\mathbf{x}_n) = \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}, \boldsymbol{\Sigma}). \quad (2.293)$$

By 2.34, the maximum likelihood solutions to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are given by

$$\begin{aligned} \boldsymbol{\mu}_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n, \\ \boldsymbol{\Sigma}_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})^\top. \end{aligned} \quad (2.294)$$

Then,

$$\mathbf{E} \boldsymbol{\Sigma}_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N \mathbf{E}(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})^\top. \quad (2.295)$$

The right hand side can be written as

$$\frac{1}{N} S_1 - \frac{1}{N^2} S_2 - \frac{1}{N^2} S_3 + \frac{1}{N^3} S_4, \quad (2.296)$$

where

$$\begin{aligned}
S_1 &= \sum_{n=1}^N \mathbb{E} \mathbf{x}_n \mathbf{x}_n^\top, \\
S_2 &= \sum_{n=1}^N \mathbb{E} \left(\sum_{n'=1}^N \mathbf{x}'_{\mathbf{n}} \right) \mathbf{x}_n^\top, \\
S_3 &= \sum_{n=1}^N \mathbb{E} \mathbf{x}_n \left(\sum_{n'=1}^N \mathbf{x}'_{\mathbf{n}} \right)^\top, \\
S_4 &= \sum_{n=1}^N \mathbb{E} \left(\sum_{n'=1}^N \mathbf{x}'_{\mathbf{n}} \right) \left(\sum_{n''=1}^N \mathbf{x}''_{\mathbf{n}} \right)^\top.
\end{aligned} \tag{2.297}$$

By (b),

$$S_1 = N (\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top). \tag{2.298}$$

By (b),

$$S_2 = S_3 = (\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top) + (N-1) \boldsymbol{\mu} \boldsymbol{\mu}^\top. \tag{2.299}$$

The right hand side can be written as

$$N \boldsymbol{\Sigma} + N^2 \boldsymbol{\mu} \boldsymbol{\mu}^\top. \tag{2.300}$$

By (b),

$$S_4 = N (N (\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top) + N(N-1) \boldsymbol{\mu} \boldsymbol{\mu}^\top). \tag{2.301}$$

The right hand side can be written as

$$N^2 \boldsymbol{\Sigma} + N^3 \boldsymbol{\mu} \boldsymbol{\mu}^\top. \tag{2.302}$$

Then,

$$\mathbb{E} \boldsymbol{\Sigma}_{\text{ML}} = \boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top - 2 \left(\frac{1}{N} \boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top \right) + \frac{1}{N} \boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top. \tag{2.303}$$

Therefore,

$$\mathbb{E} \boldsymbol{\Sigma}_{\text{ML}} = \frac{N-1}{N} \boldsymbol{\Sigma}. \tag{2.304}$$

2.36

Let $x_1, \dots, x_N$ be independent variables such that

$$p(x_n) = \mathcal{N}(x_n | \mu, \sigma^2). \quad (2.305)$$

where μ is known. By 2.34, the maximum likelihood solution to σ^2 is given by

$$\sigma_{\text{ML}}^{2(N)} = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2. \quad (2.306)$$

The right hand side can be written as

$$\frac{1}{N} (x_N - \mu)^2 + \frac{1}{N} \sum_{n=1}^{N-1} (x_n - \mu)^2 = \frac{1}{N} (x_N - \mu)^2 + \frac{N-1}{N} \sigma_{\text{ML}}^{2(N-1)}. \quad (2.307)$$

Then,

$$\sigma_{\text{ML}}^{2(N)} = \sigma_{\text{ML}}^{2(N-1)} + \frac{1}{N} \left((x_N - \mu)^2 - \sigma_{\text{ML}}^{2(N-1)} \right). \quad (2.308)$$

We have

$$\frac{\partial}{\partial \sigma^2} (-\ln p(x_n)) = \frac{1}{2\sigma^2} - \frac{1}{2(\sigma^2)^2} (x_n - \mu)^2. \quad (2.309)$$

Therefore,

$$\sigma_{\text{ML}}^{2(N)} = \sigma_{\text{ML}}^{2(N-1)} - \frac{2 \left(\sigma_{\text{ML}}^{2(N-1)} \right)^2}{N} \delta, \quad (2.310)$$

where

$$\delta = \left(\frac{\partial}{\partial \sigma^2} (-\ln p(x_N)) \right) \Big|_{\sigma^2 = \sigma_{\text{ML}}^{2(N-1)}}. \quad (2.311)$$

2.37

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be independent variables such that

$$p(\mathbf{x}_n) = \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad (2.312)$$

where $\boldsymbol{\mu}$ is known. By 2.34, the maximum likelihood solution to $\boldsymbol{\Sigma}$ is given by

$$\boldsymbol{\Sigma}_{\text{ML}}^{(N)} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\top. \quad (2.313)$$

The right hand side can be written as

$$\begin{aligned} & \frac{1}{N}(\mathbf{x}_N - \boldsymbol{\mu})(\mathbf{x}_N - \boldsymbol{\mu})^\top + \frac{1}{N} \sum_{n=1}^{N-1} (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\top \\ &= \frac{1}{N}(\mathbf{x}_N - \boldsymbol{\mu})(\mathbf{x}_N - \boldsymbol{\mu})^\top + \frac{N-1}{N} \boldsymbol{\Sigma}_{\text{ML}}^{(N-1)}. \end{aligned} \quad (2.314)$$

Then,

$$\boldsymbol{\Sigma}_{\text{ML}}^{(N)} = \boldsymbol{\Sigma}_{\text{ML}}^{(N-1)} + \frac{1}{N} \left((\mathbf{x}_N - \boldsymbol{\mu})(\mathbf{x}_N - \boldsymbol{\mu})^\top - \boldsymbol{\Sigma}_{\text{ML}}^{(N-1)} \right). \quad (2.315)$$

By 3.21, we have

$$\frac{\partial}{\partial \boldsymbol{\Sigma}} (-\ln p(x_n)) = -\frac{1}{2} (\boldsymbol{\Sigma}^{-1})^\top + \frac{1}{2} (\boldsymbol{\Sigma}^{-1})^2 (\mathbf{x}_N - \boldsymbol{\mu})(\mathbf{x}_N - \boldsymbol{\mu})^\top. \quad (2.316)$$

Therefore,

$$\boldsymbol{\Sigma}_{\text{ML}}^{(N)} = \boldsymbol{\Sigma}_{\text{ML}}^{(N-1)} - \frac{\left(\boldsymbol{\Sigma}_{\text{ML}}^{(N-1)} \right)^2}{N} \left(\frac{\partial}{\partial \boldsymbol{\Sigma}} (-\ln p(\mathbf{x}_N)) \right) \Big|_{\boldsymbol{\Sigma}=\boldsymbol{\Sigma}_{\text{ML}}^{(N-1)}}. \quad (2.317)$$

2.38

Let $x_1, \dots, x_N$ be conditionally independent variables such that

$$\begin{aligned} p(x_n|\mu) &= \mathcal{N}(x_n|\mu, \sigma^2), \\ p(\mu) &= \mathcal{N}(\mu|\mu_0, \sigma_0^2). \end{aligned} \quad (2.318)$$

By Bayes' theorem,

$$p(\mu|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|\mu)p(\mu). \quad (2.319)$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$ and μ can be written as

$$-\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 = -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{\mu_0^2}{2\sigma_0^2}, \quad (2.320)$$

where

$$\mathbf{v} = \begin{bmatrix} \mu \\ \mathbf{x} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} s^{-2} & -\sigma^{-2} \mathbf{1}^\top \\ -\sigma^{-2} \mathbf{1} & \sigma^{-2} \mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \sigma_0^{-2} \mu_0 \\ \mathbf{0} \end{bmatrix}, \quad (2.321)$$

and

$$s^2 = (N\sigma^{-2} + \sigma_0^{-2})^{-1}. \quad (2.322)$$

We have

$$s^2 (\sigma_0^{-2} \mu_0 + \sigma^{-2} \mathbf{1}^\top \mathbf{x}) = m, \quad (2.323)$$

where

$$m = s^2 (N\sigma^{-2} \bar{x} + \sigma_0^{-2} \mu_0), \quad (2.324)$$

$$\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n.$$

Therefore,

$$p(\mu|\mathbf{x}) = \mathcal{N}(\mu|m, s^2). \quad (2.325)$$

2.39

Let $x_1, \dots, x_N$ be conditionally independent variables such that

$$\begin{aligned} p(x_n|\mu) &= \mathcal{N}(x_n|\mu, \sigma^2), \\ p(\mu) &= \mathcal{N}(\mu|\mu_0, \sigma_0^2). \end{aligned} \quad (2.326)$$

(a)

By 2.38,

$$p(\mu|\mathbf{x}) = \mathcal{N}(\mu|m, s^2), \quad (2.327)$$

where

$$\begin{aligned} m &= s^2 (N\sigma^{-2} \bar{x} + \sigma_0^{-2} \mu_0), \\ s^2 &= (N\sigma^{-2} + \sigma_0^{-2})^{-1}, \end{aligned} \quad (2.328)$$

where

$$\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n. \quad (2.329)$$

Let

$$p(\mu|\mathbf{x}') = \mathcal{N}(\mu|m', s'^2). \quad (2.330)$$

where

$$\mathbf{x}' = \begin{bmatrix} x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}. \quad (2.331)$$

We have

$$s^{-2}m = \sigma^{-2} \sum_{n=1}^N x_n + \sigma_0^{-2}\mu_0, \quad (2.332)$$

so that

$$s'^{-2}m' = \sigma^{-2} \sum_{n=1}^{N-1} x_n + \sigma_0^{-2}\mu_0. \quad (2.333)$$

Therefore,

$$s^{-2}m = s'^{-2}m' + \sigma^{-2}x_N. \quad (2.334)$$

We have

$$s^{-2} = (N-1)\sigma^{-2} + \sigma_0^{-2} + \sigma^2. \quad (2.335)$$

Therefore,

$$s^{-2} = s'^{-2} + \sigma^{-2}. \quad (2.336)$$

(b)

By Bayes' theorem,

$$p(\mu|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|\mu)p(\mu). \quad (2.337)$$

We have

$$\begin{aligned} p(\mathbf{x}) &= p(x_N)p(\mathbf{x}'), \\ p(\mathbf{x}|\mu) &= p(x_N|\mu)p(\mathbf{x}'|\mu), \end{aligned} \quad (2.338)$$

where

$$\mathbf{x}' = \begin{bmatrix} x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}. \quad (2.339)$$

Then,

$$p(\mu|\mathbf{x})p(x_N)p(\mathbf{x}') = p(x_N|\mu)p(\mathbf{x}'|\mu)p(\mu). \quad (2.340)$$

By Bayes' theorem,

$$p(\mathbf{x}'|\mu)p(\mu) = p(\mu|\mathbf{x}')p(\mathbf{x}'). \quad (2.341)$$

Then,

$$p(\mu|\mathbf{x})p(x_N) = p(\mu|\mathbf{x}')p(x_N|\mu). \quad (2.342)$$

Let

$$p(\mu|\mathbf{x}') = \mathcal{N}(\mu|m', s'^2). \quad (2.343)$$

Then, the logarithm of the right hand side except the terms independent of μ or x_N can be written as

$$-\frac{1}{2s'^2}(\mu - m')^2 - \frac{1}{2\sigma^2}(x_N - \mu)^2 = -\frac{1}{2}\mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{m'^2}{2s'^2}, \quad (2.344)$$

where

$$\mathbf{v} = \begin{bmatrix} \mu \\ x_N \end{bmatrix}, \mathbf{M} = \begin{bmatrix} s^{-2} & -\sigma^{-2} \\ -\sigma^{-2} & \sigma^{-2} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} s'^{-2}m' \\ 0 \end{bmatrix}, \quad (2.345)$$

and

$$s^2 = \left(s'^{-2} + \sigma^{-2} \right)^{-1}. \quad (2.346)$$

Then,

$$p(\mu|\mathbf{x}) = \mathcal{N}(\mu|m, s^2), \quad (2.347)$$

where

$$m = s^2 \left(s'^{-2}m' + \sigma^{-2}x_N \right). \quad (2.348)$$

Therefore,

$$\begin{aligned} s^{-2}m &= s'^{-2}m' + \sigma^{-2}x_N, \\ s^{-2} &= s'^2 + \sigma^{-2}. \end{aligned} \quad (2.349)$$

2.40

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{x}_n|\boldsymbol{\mu}) &= \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Sigma}), \\ p(\boldsymbol{\mu}) &= \mathcal{N}(\boldsymbol{\mu}|\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0). \end{aligned} \quad (2.350)$$

By Bayes' theorem,

$$p(\boldsymbol{\mu}|\mathbf{X})p(\mathbf{X}) = p(\mathbf{X}|\boldsymbol{\mu})p(\boldsymbol{\mu}). \quad (2.351)$$

The logarithm of the right hand side except the terms independent of $\mathbf{X}$ and $\boldsymbol{\mu}$ can be written as

$$-\frac{1}{2} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}) - \frac{1}{2} (\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_0^{-1} (\boldsymbol{\mu} - \boldsymbol{\mu}_0). \quad (2.352)$$

We have

$$\begin{aligned} & \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}) \\ &= \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \bar{\mathbf{x}}) + N(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}). \end{aligned} \quad (2.353)$$

where

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n. \quad (2.354)$$

Then, the logarithm can be written as

$$-\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \frac{1}{2} \mathbf{v}^\top \mathbf{u} - \frac{1}{2} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \bar{\mathbf{x}}) - \frac{1}{2} \boldsymbol{\mu}_0^\top \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0, \quad (2.355)$$

where

$$\mathbf{v} = \begin{bmatrix} \boldsymbol{\mu} \\ \bar{\mathbf{x}} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} & -N\boldsymbol{\Sigma}^{-1} \\ -N\boldsymbol{\Sigma}^{-1} & N\boldsymbol{\Sigma}^{-1} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0 \\ \mathbf{0} \end{bmatrix}, \quad (2.356)$$

and

$$\mathbf{S} = (N\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}_0^{-1})^{-1}. \quad (2.357)$$

Therefore,

$$p(\boldsymbol{\mu} | \mathbf{X}) = \mathcal{N}(\boldsymbol{\mu} | \mathbf{m}, \mathbf{S}), \quad (2.358)$$

where

$$\mathbf{m} = \mathbf{S} (N\boldsymbol{\Sigma}^{-1} \bar{\mathbf{x}} + \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0). \quad (2.359)$$

2.41

We have

$$\text{Gam}(\lambda | a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda). \quad (2.360)$$

Then,

$$\int_0^\infty \text{Gam}(\lambda | a, b) d\lambda = \frac{b^a}{\Gamma(a)} \int_0^\infty \lambda^{a-1} \exp(-b\lambda) d\lambda. \quad (2.361)$$

By the transformation

$$\lambda' = b\lambda, \quad (2.362)$$

the right hand side can be written as

$$\frac{b^a}{\Gamma(a)} \int_0^\infty \left(\frac{\lambda'}{b}\right)^{a-1} \exp(-\lambda') \frac{1}{b} d\lambda' = \frac{1}{\Gamma(a)} \int_0^\infty \lambda'^{a-1} \exp(-\lambda') d\lambda'. \quad (2.363)$$

The integral of the right hand side can be written as $\Gamma(a)$. Therefore,

$$\int_0^\infty \text{Gam}(\lambda|a, b) d\lambda = 1. \quad (2.364)$$

2.42

Let λ be a variable such that

$$p(\lambda) = \text{Gam}(\lambda|a, b). \quad (2.365)$$

We have

$$\text{Gam}(\lambda|a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda). \quad (2.366)$$

(a)

We have

$$E \lambda = \frac{b^a}{\Gamma(a)} \int_0^\infty \lambda^a \exp(-b\lambda) d\lambda. \quad (2.367)$$

By the transformation

$$\lambda' = b\lambda, \quad (2.368)$$

the right hand side can be written as

$$\frac{b^a}{\Gamma(a)} \int_0^\infty \left(\frac{\lambda'}{b}\right)^a \exp(-\lambda') \frac{1}{b} d\lambda' = \frac{1}{b\Gamma(a)} \int_0^\infty \lambda'^a \exp(-\lambda') d\lambda'. \quad (2.369)$$

The integral of the right hand side can be written as $\Gamma(a+1)$. Therefore,

$$E \lambda = \frac{a}{b}. \quad (2.370)$$

(b)

We have

$$E \lambda^2 = \frac{b^a}{\Gamma(a)} \int_0^\infty \lambda^{a+1} \exp(-b\lambda) d\lambda. \quad (2.371)$$

By the transformation

$$\lambda' = b\lambda, \quad (2.372)$$

the right hand side can be written as

$$\frac{b^a}{\Gamma(a)} \int_0^\infty \left(\frac{\lambda'}{b}\right)^{a+1} \exp(-\lambda') \frac{1}{b} d\lambda' = \frac{1}{b^2 \Gamma(a)} \int_0^\infty \lambda'^{a+1} \exp(-\lambda') d\lambda'. \quad (2.373)$$

The integral of the right hand side can be written as $\Gamma(a+2)$. Then,

$$E \lambda^2 = \frac{a(a+1)}{b^2}. \quad (2.374)$$

We have

$$\text{var } \lambda = E \lambda^2 - (E \lambda)^2. \quad (2.375)$$

Therefore, by (a),

$$\text{var } \lambda = \frac{a}{b^2}. \quad (2.376)$$

(c)

Setting the derivative of $\ln \text{Gam}(\lambda|a, b)$ with respect to λ to zero gives

$$0 = \frac{a-1}{\lambda} - b. \quad (2.377)$$

Therefore,

$$\text{mode } \lambda = \frac{a-1}{b}. \quad (2.378)$$

2.43

Let

$$p(x|\sigma^2, q) = \frac{q}{2\Gamma(\frac{1}{q})} (2\sigma^2)^{-\frac{1}{q}} \exp\left(-\frac{|x|^q}{2\sigma^2}\right). \quad (2.379)$$

(a)

We have

$$\int_{-\infty}^{\infty} p(x|\sigma^2, q) dx = \frac{q}{\Gamma(\frac{1}{q})} (2\sigma^2)^{-\frac{1}{q}} \int_0^{\infty} \exp\left(-\frac{x^q}{2\sigma^2}\right) dx. \quad (2.380)$$

By the transformation

$$x' = \frac{x^q}{2\sigma^2}, \quad (2.381)$$

the integral of the right hand side can be written as

$$\begin{aligned} & \int_0^{\infty} \exp(-x') (2\sigma^2)^{\frac{1}{q}} \frac{1}{q} x^{\frac{1}{q}-1} dx' \\ &= (2\sigma^2)^{\frac{1}{q}} \frac{1}{q} \int_0^{\infty} x^{\frac{1}{q}-1} \exp(-x') dx'. \end{aligned} \quad (2.382)$$

The integral of the right hand side can be written as $\Gamma(\frac{1}{q})$. Therefore,

$$\int_{-\infty}^{\infty} p(x|\sigma^2, q) dx = 1. \quad (2.383)$$

(b)

We have

$$p(x|\sigma^2, 2) = \frac{1}{\Gamma(\frac{1}{2})} (2\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{x^2}{2\sigma^2}\right). \quad (2.384)$$

Therefore,

$$p(x|\sigma^2, 2) = \mathcal{N}(x|0, \sigma^2). \quad (2.385)$$

(c)

Let $\mathbf{t}$ be a vector such that

$$t_n = y(\mathbf{x}_n, \mathbf{w}) + \epsilon_n, \quad (2.386)$$

where

$$p(\epsilon_n) = p(\epsilon_n|\sigma^2, q). \quad (2.387)$$

Then, the logarithm of $p(\epsilon_n)$ except the terms independent of $\mathbf{w}$ and σ^2 can be written as

$$-\frac{|\epsilon_n|^q}{2\sigma^2} - \frac{1}{q} \ln(2\sigma^2). \quad (2.388)$$

Therefore, the logarithm of $p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \sigma^2)$ except the terms independent of $\mathbf{w}$ and σ^2 can be written as

$$-\frac{1}{2\sigma^2} \sum_{n=1}^N |y(\mathbf{x}_n, \mathbf{w}) - t_n|^q - \frac{N}{q} \ln(2\sigma^2). \quad (2.389)$$

2.44

Let $x_1, \dots, x_N$ be conditionally independent variables such that

$$\begin{aligned} p(x_n|\mu, \tau) &= \mathcal{N}(x_n|\mu, \tau^{-1}), \\ p(\mu|\tau) &= \mathcal{N}(\mu|\mu_0, (\beta\tau)^{-1}), \\ p(\tau) &= \text{Gam}(\tau|a, b). \end{aligned} \quad (2.390)$$

By Bayes' theorem,

$$p(\mu, \tau|\mathbf{x}) = p(\mu|\tau, \mathbf{x})p(\tau|\mathbf{x}). \quad (2.391)$$

By Bayes' theorem,

$$p(\mu|\tau, \mathbf{x})p(\tau|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|\mu, \tau)p(\mu|\tau)p(\tau). \quad (2.392)$$

The logarithm of the right hand side except the terms independent of $\mathbf{x}$, μ and τ can be written as

$$\begin{aligned} l(\mu, \tau) &= -\frac{N}{2} \ln \tau^{-1} - \frac{\tau}{2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{1}{2} \ln \tau^{-1} - \frac{\beta\tau}{2} (\mu - \mu_0)^2 \\ &\quad + (a-1) \ln \tau - b\tau. \end{aligned} \quad (2.393)$$

We have

$$\sum_{n=1}^N (x_n - \mu)^2 = \sum_{n=1}^N (x_n - \bar{x})^2 + N(\bar{x} - \mu)^2, \quad (2.394)$$

where

$$\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n. \quad (2.395)$$

Then,

$$l(\mu, \tau) = l_1(\tau) + l_2(\mu, \tau), \quad (2.396)$$

where

$$\begin{aligned} l_1(\tau) &= \left(a + \frac{N+1}{2} - 1 \right) \ln \tau - \left(b + \frac{1}{2} \sum_{n=1}^N (x_n - \bar{x})^2 \right) \tau, \\ l_2(\mu, \tau) &= -\frac{N\tau}{2}(\bar{x} - \mu)^2 - \frac{\beta\tau}{2}(\mu - \mu_0)^2. \end{aligned} \quad (2.397)$$

We have

$$l_2(\mu, \tau) = -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{\beta\tau}{2} \mu_0^2, \quad (2.398)$$

where

$$\mathbf{v} = \begin{bmatrix} \mu \\ \bar{x} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} (N+\beta)\tau & -N\tau \\ -N\tau & N\tau \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta\tau\mu_0 \\ 0 \end{bmatrix}. \quad (2.399)$$

Then,

$$p(\mu|\tau, \mathbf{x}) = \mathcal{N}(\mu|m, s^2), \quad (2.400)$$

where

$$\begin{aligned} m &= (N+\beta)^{-1} (N\bar{x} + \beta\mu_0), \\ s^2 &= (N+\beta)^{-1} \tau^{-1}. \end{aligned} \quad (2.401)$$

Then, $\ln p(\mu|\tau, \mathbf{x})$ except the terms independent of τ or dependent on μ can be written as

$$l_3(\tau) = -\frac{1}{2} \ln \tau^{-1}. \quad (2.402)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} (\beta\tau)^{-1} & (\beta\tau)^{-1} \\ (\beta\tau)^{-1} & (N\tau)^{-1} + (\beta\tau)^{-1} \end{bmatrix}, \quad (2.403)$$

so that

$$\mathbf{M}^{-1} \mathbf{u} = \begin{bmatrix} \mu_0 \\ \mu_0 \end{bmatrix}. \quad (2.404)$$

Then,

$$p(\bar{x}|\tau) = \mathcal{N}(\bar{x}|\mu_0, (N^{-1} + \beta^{-1}) \tau^{-1}). \quad (2.405)$$

Then, $\ln p(\bar{x}|\tau)$ except the terms independent of $\mathbf{x}$ can be written as

$$l_4(\tau) = -\frac{\tau(\bar{x} - \mu_0)^2}{2(N^{-1} + \beta^{-1})}. \quad (2.406)$$

Then, $\ln p(\tau|\mathbf{x})$ except the terms independent of τ and $\mathbf{x}$ can be written as

$$l_1(\tau) - l_3(\tau) + l_4(\tau) = (a' - 1) \ln \tau - b' \tau, \quad (2.407)$$

where

$$\begin{aligned} a' &= a + \frac{N}{2}, \\ b' &= b + \frac{1}{2} \sum_{n=1}^N (x_n - \bar{x})^2 + \frac{(\bar{x} - \mu_0)^2}{2(N^{-1} + \beta^{-1})}. \end{aligned} \quad (2.408)$$

Then,

$$p(\tau|\mathbf{x}) = \text{Gam}(\tau|a', b'). \quad (2.409)$$

Therefore,

$$p(\mu, \tau|\mathbf{x}) = \mathcal{N}(\mu|m, s^2) \text{Gam}(\tau|a', b'). \quad (2.410)$$

2.45

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be conditionally independent variables in D dimensions such that

$$\begin{aligned} p(\mathbf{x}_n|\mathbf{\Lambda}) &= \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}, \mathbf{\Lambda}^{-1}), \\ p(\mathbf{\Lambda}) &= \mathcal{W}(\mathbf{\Lambda}|\mathbf{W}, \nu), \end{aligned} \quad (2.411)$$

where

$$\mathcal{W}(\mathbf{\Lambda}|\mathbf{W}, \nu) = B(\mathbf{W}, \nu) |\det \mathbf{\Lambda}|^{\frac{\nu-D-1}{2}} \exp \left(-\frac{1}{2} \text{tr}(\mathbf{W}^{-1} \mathbf{\Lambda}) \right). \quad (2.412)$$

By Bayes' theorem,

$$p(\mathbf{\Lambda}|\mathbf{X})p(\mathbf{X}) = p(\mathbf{X}|\mathbf{\Lambda})p(\mathbf{\Lambda}). \quad (2.413)$$

The logarithm of the right hand side except the terms independent of $\mathbf{\Lambda}$ can be written as

$$\begin{aligned} & -\frac{N}{2} \ln |\det(\mathbf{\Lambda}^{-1})| - \frac{1}{2} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})^\top \mathbf{\Lambda} (\mathbf{x}_n - \boldsymbol{\mu}) \\ & + \frac{\nu - D - 1}{2} \ln |\det \mathbf{\Lambda}| - \frac{1}{2} \text{tr}(\mathbf{W}^{-1} \mathbf{\Lambda}) \\ & = \frac{\nu + N - D - 1}{2} \ln |\det \mathbf{\Lambda}| - \frac{1}{2} \text{tr}((\mathbf{W}^{-1} + \mathbf{S}) \mathbf{\Lambda}), \end{aligned} \quad (2.414)$$

where

$$\mathbf{S} = \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\top. \quad (2.415)$$

Then,

$$p(\boldsymbol{\Lambda}|\mathbf{X}) = \mathcal{W}\left(\boldsymbol{\Lambda} \mid (\mathbf{W}^{-1} + \mathbf{S})^{-1}, \nu + N\right). \quad (2.416)$$

Therefore, $\mathcal{W}$ is a conjugate prior distribution of $\boldsymbol{\Lambda}$.

2.46

Let x be a variable such that

$$\begin{aligned} p(x|\tau) &= \mathcal{N}(x|\mu, \tau^{-1}), \\ p(\tau) &= \text{Gam}(\tau|a, b). \end{aligned} \quad (2.417)$$

By marginalisation,

$$p(x) = \int_0^\infty p(x|\tau)p(\tau)d\tau. \quad (2.418)$$

The right hand side can be written as

$$\begin{aligned} & \int_0^\infty (2\pi\tau^{-1})^{-\frac{1}{2}} \exp\left(-\frac{\tau}{2}(x-\mu)^2\right) \frac{b^a}{\Gamma(a)} \tau^{a-1} \exp(-b\tau) d\tau \\ &= (2\pi)^{-\frac{1}{2}} \frac{b^a}{\Gamma(a)} \int_0^\infty \tau^{a-\frac{1}{2}} \exp\left(-\left(b + \frac{(x-\mu)^2}{2}\right)\tau\right) d\tau. \end{aligned} \quad (2.419)$$

By the transformation

$$\tau' = \left(b + \frac{(x-\mu)^2}{2}\right)\tau, \quad (2.420)$$

the integral of the right hand side can be written as

$$\begin{aligned} & \int_0^\infty \left(\frac{\tau'}{b + \frac{(x-\mu)^2}{2}}\right)^{a-\frac{1}{2}} \exp(-\tau') \frac{1}{b + \frac{(x-\mu)^2}{2}} d\tau' \\ &= \Gamma\left(a + \frac{1}{2}\right) \left(b + \frac{(x-\mu)^2}{2}\right)^{-a-\frac{1}{2}}. \end{aligned} \quad (2.421)$$

Then,

$$p(x) = (2\pi)^{-\frac{1}{2}} \frac{\Gamma(a + \frac{1}{2})}{\Gamma(a)} b^a \left(b + \frac{(x - \mu)^2}{2} \right)^{-a - \frac{1}{2}}. \quad (2.422)$$

The right hand side can be written as

$$(2\pi)^{-\frac{1}{2}} \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} \left(\frac{\nu\sigma^2}{2} \right)^{-\frac{1}{2}} \left(1 + \frac{(x - \mu)^2}{\nu\sigma^2} \right)^{-\frac{\nu+1}{2}}. \quad (2.423)$$

where

$$\begin{aligned} \nu &= 2a, \\ \sigma^2 &= \frac{b}{a}. \end{aligned} \quad (2.424)$$

Therefore,

$$p(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} (\pi\nu\sigma^2)^{-\frac{1}{2}} \left(1 + \frac{(x - \mu)^2}{\nu\sigma^2} \right)^{-\frac{\nu+1}{2}}. \quad (2.425)$$

2.47

We have

$$\text{St}(x|\mu, \sigma^2, \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})} (\pi\nu\sigma^2)^{-\frac{1}{2}} \left(1 + \frac{(x - \mu)^2}{\nu\sigma^2} \right)^{-\frac{\nu+1}{2}}. \quad (2.426)$$

We have

$$\left(1 + \frac{(x - \mu)^2}{\nu\sigma^2} \right)^{-\frac{\nu+1}{2}} = \left(1 + \frac{1}{y} \right)^{-\frac{(x - \mu)^2}{2\sigma^2} y - \frac{1}{2}}, \quad (2.427)$$

where

$$\frac{1}{y} = \frac{(x - \mu)^2}{\nu\sigma^2}, \quad (2.428)$$

We have

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e. \quad (2.429)$$

Then,

$$\lim_{y \rightarrow \infty} \left(1 + \frac{1}{y} \right)^{-\frac{(x - \mu)^2}{2\sigma^2} y - \frac{1}{2}} = \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right). \quad (2.430)$$

Therefore,

$$\lim_{\nu \rightarrow \infty} \text{St}(x|\mu, \sigma^2, \nu) = \mathcal{N}(x|\mu, \sigma^2). \quad (2.431)$$

2.48

Let $\mathbf{x}$ be a variable in D dimensions such that

$$\begin{aligned} p(\mathbf{x}|\eta) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\boldsymbol{\Sigma}), \\ p(\eta) &= \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right). \end{aligned} \quad (2.432)$$

By marginalisation,

$$p(\mathbf{x}) = \int_0^\infty p(\mathbf{x}|\eta)p(\eta)d\eta. \quad (2.433)$$

The logarithm of the integrand can be written as

$$\begin{aligned} & -\frac{D}{2}\ln(2\pi) - \frac{1}{2}\ln|\det(\eta^{-1}\boldsymbol{\Sigma})| - \frac{\eta}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \\ & + \frac{\nu}{2}\ln\frac{\nu}{2} - \ln\Gamma\left(\frac{\nu}{2}\right) + \left(\frac{\nu}{2} - 1\right)\ln\eta - \frac{\nu}{2}\eta \\ = & -\frac{D}{2}\ln(2\pi) - \frac{1}{2}\ln|\det\boldsymbol{\Sigma}| + \frac{\nu}{2}\ln\frac{\nu}{2} - \ln\Gamma\left(\frac{\nu}{2}\right) \\ & + \left(\frac{D+\nu}{2} - 1\right)\ln\eta - \frac{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{2}\eta. \end{aligned} \quad (2.434)$$

Then, the integral can be written as

$$(2\pi)^{-\frac{D}{2}} \frac{\left(\frac{\nu}{2}\right)^{\frac{\nu}{2}}}{\Gamma\left(\frac{\nu}{2}\right)} |\det\boldsymbol{\Sigma}|^{-\frac{1}{2}} I, \quad (2.435)$$

where

$$I = \int_0^\infty \eta^{\frac{D+\nu}{2}-1} \exp\left(-\frac{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{2}\eta\right) d\eta. \quad (2.436)$$

By the transformation

$$\eta' = \frac{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{2}\eta, \quad (2.437)$$

I can be written as

$$\begin{aligned} & \left(\frac{2}{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}\right)^{\frac{D+\nu}{2}} \int_0^\infty \eta'^{\frac{D+\nu}{2}-1} \exp(-\eta') d\eta' \\ = & \Gamma\left(\frac{D+\nu}{2}\right) \left(\frac{2}{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}\right)^{\frac{D+\nu}{2}}. \end{aligned} \quad (2.438)$$

Then, $\ln p(\mathbf{x})$ can be written as

$$\begin{aligned} & -\frac{D}{2} \ln(2\pi) + \frac{\nu}{2} \ln\left(\frac{\nu}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln |\det \Sigma| \\ & + \ln \Gamma\left(\frac{D+\nu}{2}\right) + \frac{D+\nu}{2} \ln\left(\frac{2}{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})}\right). \end{aligned} \quad (2.439)$$

We have

$$\begin{aligned} & \frac{\nu}{2} \ln\left(\frac{\nu}{2}\right) + \frac{D+\nu}{2} \ln\left(\frac{2}{\nu + (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})}\right) \\ & = -\frac{D}{2} \ln\left(\frac{\nu}{2}\right) - \frac{D+\nu}{2} \ln\left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})}{\nu}\right). \end{aligned} \quad (2.440)$$

Then, $\ln p(\mathbf{x})$ can be written as

$$\begin{aligned} & -\frac{D}{2} \ln(\pi\nu) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln |\det \Sigma| + \ln \Gamma\left(\frac{D+\nu}{2}\right) \\ & - \frac{D+\nu}{2} \ln\left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})}{\nu}\right). \end{aligned} \quad (2.441)$$

Therefore,

$$p(\mathbf{x}) = c \left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})}{\nu}\right)^{-\frac{D+\nu}{2}}, \quad (2.442)$$

where

$$c = \frac{\Gamma(\frac{D+\nu}{2})}{\Gamma(\frac{\nu}{2})} (\pi\nu)^{-\frac{D}{2}} |\det \Sigma|^{-\frac{1}{2}}. \quad (2.443)$$

2.49

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}) = \text{St}(\mathbf{x}|\boldsymbol{\mu}, \Sigma, \nu). \quad (2.444)$$

We have

$$\text{St}(\mathbf{x}|\boldsymbol{\mu}, \Sigma, \nu) = \int \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \eta^{-1}\Sigma) \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta. \quad (2.445)$$

(a)

We have

$$\mathbf{E} \mathbf{x} = \int \mathbf{x} \text{St}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) d\mathbf{x}. \quad (2.446)$$

The right hand side can be written as

$$\begin{aligned} & \int \mathbf{x} \left(\int \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \eta^{-1} \boldsymbol{\Sigma}) \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta \right) d\mathbf{x} \\ &= \int \left(\int \mathbf{x} \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \eta^{-1} \boldsymbol{\Sigma}) d\mathbf{x} \right) \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta. \end{aligned} \quad (2.447)$$

The right hand side can be written as

$$\boldsymbol{\mu} \int \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta = \boldsymbol{\mu}. \quad (2.448)$$

Therefore,

$$\mathbf{E} \mathbf{x} = \boldsymbol{\mu}. \quad (2.449)$$

(b)

By (a), we have

$$\text{var} \mathbf{x} = \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \text{St}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) d\mathbf{x}. \quad (2.450)$$

The right hand side can be written as

$$\begin{aligned} & \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \left(\int \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \eta^{-1} \boldsymbol{\Sigma}) \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta \right) d\mathbf{x} \\ &= \int \left(\int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \eta^{-1} \boldsymbol{\Sigma}) d\mathbf{x} \right) \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta. \end{aligned} \quad (2.451)$$

The right hand side can be written as

$$\begin{aligned} & \left(\int \eta^{-1} \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta \right) \boldsymbol{\Sigma} \\ &= \frac{\left(\frac{\nu}{2}\right)^{\frac{\nu}{2}}}{\Gamma\left(\frac{\nu}{2}\right)} \left(\int \eta^{\frac{\nu}{2}-2} \exp\left(-\frac{\nu}{2}\eta\right) d\eta \right) \boldsymbol{\Sigma}. \end{aligned} \quad (2.452)$$

By the transformation

$$\eta' = \frac{\nu}{2}\eta, \quad (2.453)$$

the integral of the right hand side can be written as

$$\int \left(\frac{2}{\nu}\eta'\right)^{\frac{\nu}{2}-2} \exp(-\eta') \frac{2}{\nu} d\eta' = \left(\frac{2}{\nu}\right)^{\frac{\nu}{2}-1} \Gamma\left(\frac{\nu}{2} - 1\right). \quad (2.454)$$

Then,

$$\text{var } \mathbf{x} = \frac{\left(\frac{\nu}{2}\right)^{\frac{\nu}{2}}}{\Gamma\left(\frac{\nu}{2}\right)} \left(\frac{2}{\nu}\right)^{\frac{\nu}{2}-1} \Gamma\left(\frac{\nu}{2} - 1\right) \boldsymbol{\Sigma}. \quad (2.455)$$

Therefore,

$$\text{var } \mathbf{x} = \frac{\nu}{\nu - 2} \boldsymbol{\Sigma}. \quad (2.456)$$

(c)

Setting the derivative of p with respect to $\mathbf{x}$ to zero gives

$$\mathbf{0} = -\frac{1}{2} (\boldsymbol{\Sigma}^{-1} + (\boldsymbol{\Sigma}^{-1})^\top) (\mathbf{x} - \boldsymbol{\mu}) I, \quad (2.457)$$

where

$$I = \int \eta \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \eta^{-1} \boldsymbol{\Sigma}) \text{Gam}\left(\eta \mid \frac{\nu}{2}, \frac{\nu}{2}\right) d\eta. \quad (2.458)$$

Therefore,

$$\text{mode } \mathbf{x} = \boldsymbol{\mu}. \quad (2.459)$$

2.50

We have

$$\text{St}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) = c \left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{\nu} \right)^{-\frac{D+\nu}{2}}, \quad (2.460)$$

where

$$c = \frac{\Gamma\left(\frac{D+\nu}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)} (\pi\nu)^{-\frac{D}{2}} (\det \boldsymbol{\Sigma})^{-\frac{1}{2}}. \quad (2.461)$$

We have

$$\left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{\nu}\right)^{-\frac{D+\nu}{2}} = \left(1 + \frac{1}{y}\right)^{-\frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{2} y - \frac{D}{2}}, \quad (2.462)$$

where

$$\frac{1}{y} = \frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{\nu}. \quad (2.463)$$

We have

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e. \quad (2.464)$$

Then,

$$\begin{aligned} & \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^{-\frac{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})}{2} y - \frac{D}{2}} \\ &= \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \end{aligned} \quad (2.465)$$

Therefore,

$$\lim_{\nu \rightarrow \infty} \text{St}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}). \quad (2.466)$$

2.51

(a)

We have

$$\exp(iA) \exp(-iA) = 1. \quad (2.467)$$

The left hand side can be written as

$$(\cos A + i \sin A)(\cos A - i \sin A) = \cos^2 A + \sin^2 A. \quad (2.468)$$

Therefore,

$$\cos^2 A + \sin^2 A = 1. \quad (2.469)$$

(b)

We have

$$\cos(A - B) = \text{Re}(\exp(i(A - B))). \quad (2.470)$$

The right hand side can be written as

$$\operatorname{Re}(\exp(iA) \exp(-iB)) = \operatorname{Re}((\cos A + i \sin A)(\cos B - i \sin B)). \quad (2.471)$$

Therefore,

$$\cos(A - B) = \cos A \cos B + \sin A \sin B. \quad (2.472)$$

(c)

We have

$$\sin(A - B) = \operatorname{Im}(\exp(i(A - B))). \quad (2.473)$$

The right hand side can be written as

$$\operatorname{Im}(\exp(iA) \exp(-iB)) = ((\cos A + i \sin A)(\cos B - i \sin B)). \quad (2.474)$$

Therefore,

$$\sin(A - B) = \sin A \cos B - \cos A \sin B. \quad (2.475)$$

2.52

Let

$$f(\theta|\theta_0, m) = \frac{1}{2\pi I_0(m)} \exp(m \cos(\theta - \theta_0)), \quad (2.476)$$

where

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp(m \cos \theta) d\theta. \quad (2.477)$$

By the Taylor series

$$\cos \alpha = 1 - \frac{1}{2} \alpha^2 + O(\alpha^4), \quad (2.478)$$

the right hand side can be written as

$$\begin{aligned} & \frac{\exp\left(m\left(1 - \frac{1}{2}(\theta - \theta_0)^2 + O((\theta - \theta_0)^4)\right)\right)}{\int_0^{2\pi} \exp\left(m\left(1 - \frac{1}{2}\theta^2 + O(\theta^4)\right)\right) d\theta} \\ &= \exp\left(-\frac{m}{2}(\theta - \theta_0)^2\right) \frac{\exp(mO((\theta - \theta_0)^4))}{\int_0^{2\pi} \exp\left(m\left(-\frac{1}{2}\theta^2 + O(\theta^4)\right)\right) d\theta}. \end{aligned} \quad (2.479)$$

Therefore,

$$\lim_{m \rightarrow \infty} f(\theta|\theta_0, m) = \mathcal{N}(\theta|\theta_0, m^{-1}). \quad (2.480)$$

2.53

Let

$$\sum_{n=1}^N \sin(\theta_n - \theta_0) = 0. \quad (2.481)$$

The left hand side can be written as

$$\sum_{n=1}^N (\sin \theta_n \cos \theta_0 - \cos \theta_n \sin \theta_0) = \cos \theta_0 \sum_{n=1}^N \sin \theta_n - \sin \theta_0 \sum_{n=1}^N \cos \theta_n. \quad (2.482)$$

Therefore,

$$\theta_0 = \arctan \left(\frac{\sum_{n=1}^N \sin \theta_n}{\sum_{n=1}^N \cos \theta_n} \right). \quad (2.483)$$

2.54

Let

$$f(\theta|\theta_0, m) = \frac{1}{2\pi I_0(m)} \exp(m \cos(\theta - \theta_0)), \quad (2.484)$$

where

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp(m \cos \theta) d\theta. \quad (2.485)$$

Setting the first and second derivatives with respect to θ to zero gives

$$\begin{aligned} 0 &= -m \sin(\theta - \theta_0) f(\theta|\theta_0, m), \\ 0 &= (m^2 \sin^2(\theta - \theta_0) - m \cos(\theta - \theta_0)) f(\theta|\theta_0, m). \end{aligned} \quad (2.486)$$

Therefore,

$$\begin{aligned} \operatorname{argmax}_{\theta} f(\theta|\theta_0, m) &= \theta_0, \\ \operatorname{argmin}_{\theta} f(\theta|\theta_0, m) &= \theta_0 - \pi \operatorname{sgn}(\theta_0 - \pi). \end{aligned} \quad (2.487)$$

2.55

(a)

Let $\theta_1, \dots, \theta_N$ be independent variables such that

$$p(\theta_n) = \frac{1}{2\pi I_0(m)} \exp(m \cos(\theta - \theta_0)), \quad (2.488)$$

where

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp(m \cos \theta) d\theta. \quad (2.489)$$

Then,

$$\ln p(\boldsymbol{\theta}) = -\frac{N}{2} \ln(2\pi I_0(m)) + m \sum_{n=1}^N \cos(\theta_n - \theta_0). \quad (2.490)$$

Setting the derivative of $\ln p$ with respect to θ_0 to zero gives

$$0 = m \sum_{n=1}^N \sin(\theta_n - \theta_0). \quad (2.491)$$

Therefore, by 2.53, the maximum likelihood solution to θ_0 is given by

$$\theta_0^{\text{ML}} = \arctan \left(\frac{\sum_{n=1}^N \sin \theta_n}{\sum_{n=1}^N \cos \theta_n} \right). \quad (2.492)$$

(b)

Let

$$\begin{aligned} \bar{r} \cos \bar{\theta} &= \frac{1}{N} \sum_{n=1}^N \cos \theta_n, \\ \bar{r} \sin \bar{\theta} &= \frac{1}{N} \sum_{n=1}^N \sin \theta_n. \end{aligned} \quad (2.493)$$

By (a),

$$\bar{\theta} = \theta_0^{\text{ML}}. \quad (2.494)$$

We have

$$\begin{aligned} &\frac{1}{N} \sum_{n=1}^N \cos(\theta_n - \theta_0^{\text{ML}}) \\ &= \left(\frac{1}{N} \sum_{n=1}^N \cos \theta_n \right) \cos \theta_0^{\text{ML}} + \left(\frac{1}{N} \sum_{n=1}^N \sin \theta_n \right) \sin \theta_0^{\text{ML}}. \end{aligned} \quad (2.495)$$

The right hand side can be written as

$$\bar{r} \cos^2 \bar{\theta} + \bar{r} \sin^2 \bar{\theta} = \bar{r}. \quad (2.496)$$

Therefore,

$$\frac{1}{N} \sum_{n=1}^N \cos(\theta_n - \theta_0^{\text{ML}}) = \bar{r}. \quad (2.497)$$

2.56

(a)

Let

$$\text{Beta}(\mu|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}. \quad (2.498)$$

The right hand side can be written as

$$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \exp((a-1) \ln \mu + (b-1) \ln(1-\mu)) \quad (2.499)$$

Therefore, the natural parameters are given by

$$\boldsymbol{\eta} = \begin{bmatrix} a-1 \\ b-1 \end{bmatrix}. \quad (2.500)$$

(b)

Let

$$\text{Gam}(\lambda|a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda). \quad (2.501)$$

The right hand side can be written as

$$\frac{b^a}{\Gamma(a)} \exp((a-1) \ln \lambda - b\lambda). \quad (2.502)$$

Therefore, the natural parameters are given by

$$\boldsymbol{\eta} = \begin{bmatrix} a-1 \\ -b \end{bmatrix}. \quad (2.503)$$

(c)

Let

$$f(\theta|\theta_0, m) = \frac{1}{2\pi I_0(m)} \exp(m \cos(\theta - \theta_0)), \quad (2.504)$$

where

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp(m \cos \theta) d\theta, \quad (2.505)$$

the right hand side can be written as

$$\frac{1}{2\pi I_0(m)} \exp(m \cos \theta_0 \cos \theta + m \sin \theta_0 \sin \theta). \quad (2.506)$$

Therefore, the natural parameters are given by

$$\boldsymbol{\eta} = \begin{bmatrix} m \cos \theta_0 \\ m \sin \theta_0 \end{bmatrix}. \quad (2.507)$$

2.57

We have

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-\frac{D}{2}} (\det \boldsymbol{\Sigma})^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right). \quad (2.508)$$

Therefore,

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = h(\mathbf{x}) g(\boldsymbol{\eta}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})), \quad (2.509)$$

where

$$\begin{aligned} h(\mathbf{x}) &= (2\pi)^{-\frac{D}{2}}, \\ g(\boldsymbol{\eta}) &= (\det(-2\boldsymbol{\eta}_2))^{-\frac{1}{2}} \exp \left(\frac{1}{4} \boldsymbol{\eta}_1^\top \boldsymbol{\eta}_2^{-1} \boldsymbol{\eta}_1 \right), \\ \boldsymbol{\eta} &= \begin{bmatrix} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \\ -\frac{1}{2} \boldsymbol{\Sigma}^{-1} \end{bmatrix}, \mathbf{u}(\mathbf{x}) = \begin{bmatrix} \mathbf{x} \\ \mathbf{x} \mathbf{x}^\top \end{bmatrix}. \end{aligned} \quad (2.510)$$

2.58

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}|\boldsymbol{\eta}) = h(\mathbf{x}) g(\boldsymbol{\eta}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})). \quad (2.511)$$

(a)

Setting the first derivative of

$$\int p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} = 1 \quad (2.512)$$

with respect to $\boldsymbol{\eta}$ to zero gives

$$\begin{aligned} \mathbf{0} = & \nabla g(\boldsymbol{\eta}) \int h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})) d\mathbf{x} \\ & + g(\boldsymbol{\eta}) \int \mathbf{u}(\mathbf{x}) h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})) d\mathbf{x}. \end{aligned} \quad (2.513)$$

The right hand side can be written as

$$\frac{\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} \int p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} + \int \mathbf{u}(\mathbf{x})p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} = \frac{\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} + \mathbb{E} \mathbf{u}(\mathbf{x}). \quad (2.514)$$

Then,

$$\mathbb{E} \mathbf{u}(\mathbf{x}) = -\frac{\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})}. \quad (2.515)$$

Therefore,

$$\mathbb{E} \mathbf{u}(\mathbf{x}) = -\nabla \ln g(\boldsymbol{\eta}). \quad (2.516)$$

(b)

Setting the second derivative with respect to $\boldsymbol{\eta}$ to zero gives

$$\begin{aligned} \mathbf{0} = & \nabla \nabla g(\boldsymbol{\eta}) \int h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})) d\mathbf{x} \\ & + 2\nabla g(\boldsymbol{\eta}) \int \mathbf{u}(\mathbf{x})^\top h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})) d\mathbf{x} \\ & + g(\boldsymbol{\eta}) \int \mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\top h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{u}(\mathbf{x})) d\mathbf{x} \end{aligned} \quad (2.517)$$

The right hand side can be written as

$$\begin{aligned} & \frac{\nabla \nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} \int p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} + \frac{2\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} \int \mathbf{u}(\mathbf{x})^\top p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} \\ & + \int \mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\top p(\mathbf{x}|\boldsymbol{\eta})d\mathbf{x} \\ = & \frac{\nabla \nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} - 2\mathbb{E} \mathbf{u}(\mathbf{x}) \mathbb{E} \mathbf{u}(\mathbf{x})^\top + \mathbb{E} (\mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\top). \end{aligned} \quad (2.518)$$

Then,

$$\mathbb{E}(\mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\top) = -\frac{\nabla\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} + \frac{2\nabla g(\boldsymbol{\eta})(\nabla g(\boldsymbol{\eta}))^\top}{g^2(\boldsymbol{\eta})}. \quad (2.519)$$

We have

$$\text{var } \mathbf{u}(\mathbf{x}) = \mathbb{E}(\mathbf{u}(\mathbf{x})\mathbf{u}(\mathbf{x})^\top) - \mathbb{E} \mathbf{u}(\mathbf{x}) \mathbb{E} \mathbf{u}(\mathbf{x})^\top. \quad (2.520)$$

Then,

$$\text{var } \mathbf{u}(\mathbf{x}) = -\frac{\nabla\nabla g(\boldsymbol{\eta})}{g(\boldsymbol{\eta})} + \frac{\nabla g(\boldsymbol{\eta})(\nabla g(\boldsymbol{\eta}))^\top}{g^2(\boldsymbol{\eta})}. \quad (2.521)$$

Therefore,

$$\text{var } \mathbf{u}(\mathbf{x}) = -\nabla\nabla \ln g(\boldsymbol{\eta}). \quad (2.522)$$

2.59

Let

$$p(x|\sigma) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right). \quad (2.523)$$

Then

$$\int p(x|\sigma) dx = \frac{1}{\sigma} \int f\left(\frac{x}{\sigma}\right) dx. \quad (2.524)$$

By the transformation

$$x' = \frac{x}{\sigma}, \quad (2.525)$$

the right hand side can be written as

$$\frac{1}{\sigma} \int f(x') \sigma dx' = \int f(x') dx'. \quad (2.526)$$

Therefore, $p(x|\sigma)$ will be normalised if $f(x)$ is normalised.

2.60

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be variables such that

$$\mathbf{x}_n \in \mathcal{R}_k \Rightarrow p(\mathbf{x}_n) = \pi_k. \quad (2.527)$$

Let n_k be the number of observations which fall in $\mathcal{R}_k$ and let

$$\Delta_k = \int_{\mathcal{R}_k} d\mathbf{x}_n. \quad (2.528)$$

In order to maximise $\ln p(\mathbf{X})$ under the constraint

$$\int p(\mathbf{x}_n) d\mathbf{x}_n = 1, \quad (2.529)$$

let

$$L = \ln p(\mathbf{X}) + \lambda \left(\int p(\mathbf{x}_n) d\mathbf{x}_n - 1 \right). \quad (2.530)$$

We have

$$\begin{aligned} \ln p(\mathbf{X}) &= \sum_{k=1}^K n_k \ln \pi_k, \\ \int p(\mathbf{x}_n) d\mathbf{x}_n &= \sum_{k=1}^K \pi_k \Delta_k. \end{aligned} \quad (2.531)$$

Then,

$$L = \sum_{k=1}^K n_k \ln \pi_k + \lambda \left(\sum_{k=1}^K \pi_k \Delta_k - 1 \right). \quad (2.532)$$

Setting the derivatives of L with respect to π_k and λ to zero gives

$$\begin{aligned} \frac{n_k}{\pi_k} + \lambda \Delta_k &= 0, \\ \sum_{k=1}^K \pi_k \Delta_k - 1 &= 0. \end{aligned} \quad (2.533)$$

We have

$$\sum_{k=1}^K n_k = N. \quad (2.534)$$

Therefore, the maximum likelihood solution to π_k is given by

$$\pi_{k\text{ML}} = \frac{n_k}{N \Delta_k}. \quad (2.535)$$

2.61 (Incomplete)

Let $\mathbf{x}$ be a variable and $\mathbf{x}_1, \dots, \mathbf{x}_N$ be observations. Let

$$p(\mathbf{x}) = \frac{K}{NV(\mathbf{x})}, \quad (2.536)$$

where

$$V(\mathbf{x}) = \int_{\|\mathbf{x}' - \mathbf{x}\| \leq \|\mathbf{x}_{(K)} - \mathbf{x}\|} d\mathbf{x}', \quad (2.537)$$

K is a constant and $\mathbf{x}_{(K)}$ is the K th nearest observation from the point $\mathbf{x}$.

3 Linear Models for Regression

3.1

(a)

We have

$$\tanh a = \frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)}. \quad (3.1)$$

The right hand side can be written as

$$\frac{1 - \exp(-2a)}{1 + \exp(-2a)} = \frac{2}{1 + \exp(-2a)} - 1. \quad (3.2)$$

Therefore,

$$\tanh a = 2\sigma(2a) - 1, \quad (3.3)$$

where

$$\sigma(a) = \frac{1}{1 + \exp(-a)}. \quad (3.4)$$

(b)

Let

$$y(x_n, \mathbf{w}) = w_0 + \sum_{m=1}^M w_m \sigma\left(\frac{x - \mu_m}{s}\right). \quad (3.5)$$

By the result above, the right hand side can be written as

$$\begin{aligned} & w_0 f + \sum_{m=1}^M w_m \frac{1 + \tanh\left(\frac{x - \mu_m}{2s}\right)}{2} \\ &= w_0 + \frac{1}{2} \sum_{m=1}^M w_m + \frac{1}{2} \sum_{m=1}^M w_m \tanh\left(\frac{x - \mu_m}{2s}\right). \end{aligned} \quad (3.6)$$

Therefore, $y(x_n, \mathbf{w})$ is equivalent to

$$y(x_n, \mathbf{u}) = u_0 + \sum_{m=1}^M u_m \tanh\left(\frac{x - \mu_m}{2s}\right), \quad (3.7)$$

where

$$\begin{aligned} u_0 &= w_0 + \frac{1}{2} \sum_{m=1}^M w_m, \\ u_m &= \frac{1}{2} w_m. \end{aligned} \tag{3.8}$$

3.2 (Incomplete)

Let Φ be an $N \times M$ matrix. Then, for any vector $\mathbf{v}$ in N dimensions,

$$\Phi (\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{v} \tag{3.9}$$

is a projection of $\mathbf{v}$ onto the space spanned by the columns of Φ ?

Additionally, for a vector $\mathbf{t}$ in N dimensions,

$$(\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{t} \tag{3.10}$$

is an orthogonal projection of $\mathbf{t}$ onto the space spanned by the columns of Φ ?

3.3

Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N r_n (t_n - \mathbf{w}^\top \phi_n)^2. \tag{3.11}$$

The right hand side can be written as

$$\frac{1}{2} \|\mathbf{t}' - \Phi' \mathbf{w}\|^2, \tag{3.12}$$

where

$$\mathbf{t}' = \begin{bmatrix} \sqrt{r_1} t_1 \\ \vdots \\ \sqrt{r_N} t_N \end{bmatrix}, \Phi' = \begin{bmatrix} \sqrt{r_1} \phi(\mathbf{x}_1)^\top \\ \vdots \\ \sqrt{r_N} \phi(\mathbf{x}_N)^\top \end{bmatrix}. \tag{3.13}$$

Setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = -\Phi'^\top (\mathbf{t}' - \Phi' \mathbf{w}). \tag{3.14}$$

Therefore,

$$\underset{\mathbf{w}}{\operatorname{argmin}} E(\mathbf{w}) = (\Phi'^\top \Phi')^{-1} \Phi'^\top \mathbf{t}'. \tag{3.15}$$

3.4

Let

$$\tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (t_n - \tilde{y}_n)^2, \quad (3.16)$$

where

$$\begin{aligned} \tilde{y}_n &= w_0 + \mathbf{w}^\top (\mathbf{x}_n + \boldsymbol{\epsilon}_n), \\ p(\boldsymbol{\epsilon}_n) &= \mathcal{N}(\boldsymbol{\epsilon}_n | \mathbf{0}, \sigma^2 \mathbf{I}). \end{aligned} \quad (3.17)$$

We have

$$\mathbb{E} \tilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \mathbb{E} (t_n - y_n - \mathbf{w}^\top \boldsymbol{\epsilon}_n)^2, \quad (3.18)$$

where

$$y_n = w_0 + \mathbf{w}^\top \mathbf{x}_n. \quad (3.19)$$

The summand of the right hand side can be written as

$$(t_n - y_n)^2 - 2(t_n - y_n) \mathbf{w}^\top \mathbb{E} \boldsymbol{\epsilon}_n + \mathbf{w}^\top (\mathbb{E} \boldsymbol{\epsilon} \boldsymbol{\epsilon}^\top) \mathbf{w} = (t_n - y_n)^2 + \sigma^2 \|\mathbf{w}\|^2. \quad (3.20)$$

Therefore,

$$\mathbb{E} \tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \frac{N\sigma^2}{2} \|\mathbf{w}\|^2, \quad (3.21)$$

where

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (t_n - y_n)^2. \quad (3.22)$$

3.5

Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^\top \boldsymbol{\phi}(\mathbf{x}_n))^2, \quad (3.23)$$

where

$$\sum_{m=1}^M |w_m|^q \leq \eta. \quad (3.24)$$

In order to minimise $E(\mathbf{w})$, let

$$L(\mathbf{w}, \lambda) = E(\mathbf{w}) + \lambda \left(\sum_{m=1}^M |w_m|^q - \eta \right). \quad (3.25)$$

Therefore,

$$\eta = \sum_{m=1}^M |w_m^*(\lambda)|^q, \quad (3.26)$$

where

$$\mathbf{w}^*(\lambda) = \underset{\mathbf{w}}{\operatorname{argmin}} L(\mathbf{w}, \lambda). \quad (3.27)$$

3.6

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables in D dimensions such that

$$p(\mathbf{t}_n | \mathbf{y}_n) = \mathcal{N}(\mathbf{t}_n | \mathbf{y}_n, \mathbf{\Sigma}), \quad (3.28)$$

where

$$\mathbf{y}_n = \mathbf{W}^\top \phi_n. \quad (3.29)$$

Then,

$$\begin{aligned} \ln p(\mathbf{T} | \mathbf{Y}) &= -\frac{ND}{2} \ln(2\pi) - \frac{N}{2} \ln(\det \mathbf{\Sigma}) \\ &\quad - \frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - \mathbf{W}^\top \phi_n)^\top \mathbf{\Sigma}^{-1} (\mathbf{t}_n - \mathbf{W}^\top \phi_n). \end{aligned} \quad (3.30)$$

By 3.21, setting the derivatives of $\ln p(\mathbf{T} | \mathbf{Y})$ with respect to $\mathbf{W}$ and $\mathbf{\Sigma}$ to zero gives

$$\begin{aligned} \mathbf{O} &= -\frac{1}{2} (\mathbf{\Sigma}^{-1} + (\mathbf{\Sigma}^{-1})^\top) \sum_{n=1}^N (\mathbf{t}_n - \mathbf{W}^\top \phi_n) \phi_n^\top, \\ \mathbf{O} &= -\frac{N}{2} (\mathbf{\Sigma}^{-1})^\top + \frac{1}{2} (\mathbf{\Sigma}^{-1})^2 \sum_{n=1}^N (\mathbf{t}_n - \mathbf{W}^\top \phi_n) (\mathbf{t}_n - \mathbf{W}^\top \phi_n)^\top. \end{aligned} \quad (3.31)$$

Therefore, the maximum likelihood solutions to $\mathbf{W}$ and $\mathbf{\Sigma}$ are given by

$$\begin{aligned} \mathbf{W}_{\text{ML}} &= (\mathbf{\Phi}^\top \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \mathbf{t}, \\ \mathbf{\Sigma}_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N (\mathbf{t}_n - \mathbf{W}_{\text{ML}}^\top \phi_n) (\mathbf{t}_n - \mathbf{W}_{\text{ML}}^\top \phi_n)^\top. \end{aligned} \quad (3.32)$$

3.7

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \quad (3.33)$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \quad (3.34)$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2} \|\mathbf{t} - \boldsymbol{\Phi} \mathbf{w}\|^2 - \frac{1}{2} (\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\mathbf{w} - \mathbf{m}_0) \\ &= -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2} \mathbf{m}_0^\top \mathbf{S}_0^{-1} \mathbf{m}_0, \end{aligned} \quad (3.35)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi} & -\beta \boldsymbol{\Phi}^\top \\ -\beta \boldsymbol{\Phi} & \beta \mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1} \mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}. \quad (3.36)$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3.37)$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S} (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \boldsymbol{\Phi}^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned} \quad (3.38)$$

3.8

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \quad (3.39)$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3.40)$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S} (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \boldsymbol{\Phi}^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned} \quad (3.41)$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t}')p(\mathbf{t}') = p(\mathbf{t}'|\mathbf{w})p(\mathbf{w}), \quad (3.42)$$

where

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}. \quad (3.43)$$

By Bayes' theorem, the right hand side can be written as

$$p(t_{N+1}|\mathbf{w})p(\mathbf{t}|\mathbf{w})p(\mathbf{w}) = p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})p(\mathbf{t}). \quad (3.44)$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}(t_{N+1} - \mathbf{w}^\top \boldsymbol{\phi}_{N+1})^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}) \\ & = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2}\mathbf{m}^\top \mathbf{S}^{-1}\mathbf{m}, \end{aligned} \quad (3.45)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} & -\beta\boldsymbol{\phi}_{N+1} \\ -\beta\boldsymbol{\phi}_{N+1}^\top & \beta \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}^{-1}\mathbf{m} \\ 0 \end{bmatrix}. \quad (3.46)$$

Then,

$$p(\mathbf{w}|\mathbf{t}') = \mathcal{N}(\mathbf{w}|\mathbf{m}', \mathbf{S}'), \quad (3.47)$$

where

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}'(\mathbf{S}^{-1}\mathbf{m} + \beta t_{N+1}\boldsymbol{\phi}_{N+1}), \\ \mathbf{S}' &= (\mathbf{S}^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\top)^{-1}. \end{aligned} \quad (3.48)$$

We have

$$\begin{aligned} \mathbf{S}^{-1}\mathbf{m} + \beta t_{N+1}\boldsymbol{\phi}_{N+1} &= \mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}'^\top \mathbf{t}', \\ \mathbf{S}^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\top &= \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}'^\top \boldsymbol{\Phi}', \end{aligned} \quad (3.49)$$

where

$$\boldsymbol{\Phi}' = \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{\phi}_{N+1}^\top \end{bmatrix}. \quad (3.50)$$

Therefore,

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}'(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}'^\top \mathbf{t}'), \\ \mathbf{S}' &= (\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}'^\top \boldsymbol{\Phi}')^{-1}. \end{aligned} \quad (3.51)$$

3.9

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \quad (3.52)$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3.53)$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S} (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \boldsymbol{\Phi}^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned} \quad (3.54)$$

Let

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}, \boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Phi} \\ \boldsymbol{\phi}_{N+1}^\top \end{bmatrix}. \quad (3.55)$$

Then,

$$p(\mathbf{w}|\mathbf{t}') = \mathcal{N}(\mathbf{w}|\mathbf{m}', \mathbf{t}'), \quad (3.56)$$

where

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}' (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \boldsymbol{\Phi}'^\top \mathbf{t}'), \\ \mathbf{S}' &= (\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}'^\top \boldsymbol{\Phi}')^{-1}. \end{aligned} \quad (3.57)$$

3.10

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \quad (3.58)$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3.59)$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S} (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \boldsymbol{\Phi}^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned} \quad (3.60)$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w}) p(\mathbf{w}|\mathbf{t}) d\mathbf{w}. \quad (3.61)$$

The logarithm of the integrand of the right hand side except the terms independent of t_{N+1} and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}(t_{N+1} - \mathbf{w}^\top \boldsymbol{\phi}_{N+1})^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}) \\ & = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2}\mathbf{m}^\top \mathbf{S}^{-1}\mathbf{m}, \end{aligned} \quad (3.62)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} + \beta \boldsymbol{\phi}_{N+1} \boldsymbol{\phi}_{N+1}^\top & -\beta \boldsymbol{\phi}_{N+1} \\ -\beta \boldsymbol{\phi}_{N+1}^\top & \beta \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}^{-1}\mathbf{m} \\ 0 \end{bmatrix}. \quad (3.63)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S} & \mathbf{S}\boldsymbol{\phi}_{N+1} \\ \boldsymbol{\phi}_{N+1}^\top \mathbf{S} & \beta^{-1} + \boldsymbol{\phi}_{N+1}^\top \mathbf{S}\boldsymbol{\phi}_{N+1} \end{bmatrix}. \quad (3.64)$$

Then,

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\top \boldsymbol{\phi}_{N+1} \end{bmatrix}. \quad (3.65)$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1}|\mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \sigma^2(\boldsymbol{\phi}_{N+1})), \quad (3.66)$$

where

$$\sigma^2(\boldsymbol{\phi}) = \beta^{-1} + \boldsymbol{\phi}^\top \mathbf{S}\boldsymbol{\phi}. \quad (3.67)$$

3.11

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned} \quad (3.68)$$

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3.69)$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta \boldsymbol{\Phi}^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned} \quad (3.70)$$

By 3.10,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1}|\mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \sigma^2(\boldsymbol{\phi}_{N+1})), \quad (3.71)$$

where

$$\sigma^2(\phi) = \beta^{-1} + \phi^\top \mathbf{S} \phi. \quad (3.72)$$

Then,

$$p(t_{N+2}|\mathbf{t}') = \mathcal{N}\left(t_{N+2}|\mathbf{m}'^\top \phi_{N+2}, \sigma'^2(\phi_{N+2})\right), \quad (3.73)$$

where

$$\begin{aligned} \mathbf{m}' &= \mathbf{S}' (\mathbf{S}_0^{-1} \mathbf{m}_0 + \beta \Phi'^\top \mathbf{t}'), \\ \mathbf{S}' &= (\mathbf{S}_0^{-1} + \beta \Phi'^\top \Phi')^{-1}, \\ \sigma'^2(\phi) &= \beta^{-1} + \phi^\top \mathbf{S}' \phi, \end{aligned} \quad (3.74)$$

and

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}, \Phi = \begin{bmatrix} \Phi \\ \phi_{N+1}^\top \end{bmatrix}. \quad (3.75)$$

We have

$$\sigma^2(\phi) - \sigma'^2(\phi) = \phi^\top (\mathbf{S} - \mathbf{S}') \phi, \quad (3.76)$$

and

$$\mathbf{S}' = (\mathbf{S}^{-1} + \beta \phi_{N+1} \phi_{N+1}^\top)^{-1}. \quad (3.77)$$

By 2.26,

$$\mathbf{S}' = \mathbf{S} - c \beta^2 \mathbf{S} \phi_{N+1} \phi_{N+1}^\top \mathbf{S}, \quad (3.78)$$

where

$$c = (\beta + \beta^2 \phi_{N+1}^\top \mathbf{S} \phi_{N+1})^{-1}. \quad (3.79)$$

Then,

$$\sigma^2(\phi) - \sigma'^2(\phi) = c \beta^2 (\phi^\top \mathbf{S} \phi_{N+1})^2. \quad (3.80)$$

Therefore,

$$\sigma^2(\phi) \geq \sigma'^2(\phi). \quad (3.81)$$

3.12

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}, \beta) &= \mathcal{N}(t_n|\mathbf{w}^\top \phi_n, \beta^{-1}), \\ p(\mathbf{w}|\beta) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \beta^{-1} \mathbf{S}_0), \\ p(\beta) &= \text{Gam}(\beta|a_0, b_0). \end{aligned} \quad (3.82)$$

where $\mathbf{w}$ is a vector in M dimensions.

(a)

By Bayes' theorem,

$$p(\mathbf{w}|\beta, \mathbf{t})p(\mathbf{t}|\beta) = p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta). \quad (3.83)$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \Phi\mathbf{w}\|^2 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ & = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{\beta}{2}\mathbf{m}_0^\top \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned} \quad (3.84)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta(\mathbf{S}_0^{-1} + \Phi^\top \Phi) & -\beta\Phi^\top \\ -\beta\Phi & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta\mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}. \quad (3.85)$$

Therefore,

$$p(\mathbf{w}|\beta, \mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \beta^{-1}\mathbf{S}), \quad (3.86)$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}(\mathbf{S}_0^{-1}\mathbf{m}_0 + \Phi^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \Phi^\top \Phi)^{-1}. \end{aligned} \quad (3.87)$$

(b)

By Bayes' theorem,

$$p(\beta|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\beta)p(\beta). \quad (3.88)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\Phi^\top \\ \beta^{-1}\Phi\mathbf{S}_0 & \beta^{-1}(\mathbf{I} + \Phi\mathbf{S}_0\Phi^\top) \end{bmatrix}, \quad (3.89)$$

so that

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mathbf{m}_0 \\ \Phi\mathbf{m}_0 \end{bmatrix}. \quad (3.90)$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}(\mathbf{t}|\Phi\mathbf{m}_0, \beta^{-1}(\mathbf{I} + \Phi\mathbf{S}_0\Phi^\top)). \quad (3.91)$$

Then, the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of β can be written as

$$\begin{aligned} & -\frac{N}{2} \ln \beta^{-1} - \frac{\beta}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0) \\ & + (a_0 - 1) \ln \beta - b_0 \beta \\ & = (a - 1) \ln \beta - b \beta, \end{aligned} \quad (3.92)$$

where

$$\begin{aligned} a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0). \end{aligned} \quad (3.93)$$

Therefore,

$$p(\beta|\mathbf{t}) = \text{Gam}(\beta|a, b). \quad (3.94)$$

(c)

By Bayes' theorem,

$$p(\mathbf{w}|\beta, \mathbf{t})p(\beta|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta)p(\beta). \quad (3.95)$$

Then, by (a) and (b),

$$p(\mathbf{t}) = \frac{\mathcal{N}(\mathbf{t}|\Phi \mathbf{w}, \beta^{-1} \mathbf{I}) \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \beta^{-1} \mathbf{S}_0) \text{Gam}(\beta|a_0, b_0)}{\mathcal{N}(\mathbf{w}|\mathbf{m}, \beta^{-1} \mathbf{S}) \text{Gam}(\beta|a, b)}. \quad (3.96)$$

The logarithm of the right hand side can be written as

$$\begin{aligned} & -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln \beta^{-1} - \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 \\ & -\frac{M}{2} \ln(2\pi) - \frac{M}{2} \ln \beta^{-1} - \frac{1}{2} \det \mathbf{S}_0 - \frac{\beta}{2} (\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\mathbf{w} - \mathbf{m}_0) \\ & + a_0 \ln b_0 - \ln \Gamma(a_0) + (a_0 - 1) \ln \beta - b_0 \beta \\ & + \frac{M}{2} \ln(2\pi) + \frac{M}{2} \ln \beta^{-1} + \frac{1}{2} \det \mathbf{S} + \frac{\beta}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}) \\ & - a \ln b + \ln \Gamma(a) - (a - 1) \ln \beta + b \beta \\ & = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \det \mathbf{S}_0 + a_0 \ln b_0 - \ln \Gamma(a_0) + \frac{1}{2} \det \mathbf{S} \\ & - a \ln b + \ln \Gamma(a) + a' \ln \beta - b' \beta, \end{aligned} \quad (3.97)$$

where

$$\begin{aligned} a' &= a_0 + \frac{N}{2} - a, \\ b' &= b_0 + \frac{1}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{1}{2} (\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\mathbf{w} - \mathbf{m}_0) \\ &\quad - b - \frac{1}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}). \end{aligned} \quad (3.98)$$

By (a), a' and b' can be written as zero. Then, $p(\mathbf{t})$ except the terms independent of $\mathbf{t}$ can be written as

$$b^{-a} = \left(1 + (\mathbf{t} - \boldsymbol{\mu})^\top (\nu \boldsymbol{\Sigma})^{-1} (\mathbf{t} - \boldsymbol{\mu})\right)^{-\frac{\nu+N}{2}} b_0^{-a}, \quad (3.99)$$

where

$$\begin{aligned} \boldsymbol{\mu} &= \Phi \mathbf{m}_0, \\ \boldsymbol{\Sigma} &= \frac{b_0}{a_0} (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top), \\ \nu &= 2a_0. \end{aligned} \quad (3.100)$$

Therefore, by 2.48,

$$p(\mathbf{t}) = \text{St}(\mathbf{t} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu). \quad (3.101)$$

3.13

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | \mathbf{w}, \beta) &= \mathcal{N}(t_n | \mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w} | \beta) &= \mathcal{N}(\mathbf{w} | \mathbf{m}_0, \beta^{-1} \mathbf{S}_0), \\ p(\beta) &= \text{Gam}(\beta | a_0, b_0). \end{aligned} \quad (3.102)$$

where $\mathbf{w}$ is a vector in M dimensions. By marginalisation,

$$\begin{aligned} p(t_{N+1} | \mathbf{t}) &= \int_0^\infty p(t_{N+1} | \beta, \mathbf{t}) p(\beta | \mathbf{t}) d\beta, \\ p(t_{N+1} | \beta, \mathbf{t}) &= \int p(t_{N+1} | \mathbf{w}, \beta) p(\mathbf{w} | \beta, \mathbf{t}) d\mathbf{w}. \end{aligned} \quad (3.103)$$

By 3.12(a),

$$p(\mathbf{w}, \beta | \mathbf{t}) = \mathcal{N}(\mathbf{w} | \mathbf{m}, \beta^{-1} \mathbf{S}), \quad (3.104)$$

where

$$\begin{aligned}\mathbf{m} &= \mathbf{S} (\mathbf{S}_0^{-1} \mathbf{m}_0 + \Phi^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \Phi^\top \Phi)^{-1}.\end{aligned}\tag{3.105}$$

Then, the logarithm of $p(t_{N+1}|\mathbf{w}, \beta)p(\mathbf{w}|\beta, \mathbf{t})$ except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned}& -\frac{\beta}{2} (t_{N+1} - \mathbf{w}^\top \phi_{N+1})^2 - \frac{\beta}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}) \\ &= -\frac{\beta}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \beta \mathbf{v}^\top \mathbf{u} - \frac{\beta}{2} \mathbf{m}^\top \mathbf{S}^{-1} \mathbf{m},\end{aligned}\tag{3.106}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} + \phi_{N+1} \phi_{N+1}^\top & -\phi_{N+1} \\ -\phi_{N+1}^\top & 1 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}^{-1} \mathbf{m} \\ 0 \end{bmatrix}.\tag{3.107}$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S} & \mathbf{S} \phi_{N+1} \\ \phi_{N+1}^\top \mathbf{S} & 1 + \phi_{N+1}^\top \mathbf{S} \phi_{N+1} \end{bmatrix},\tag{3.108}$$

so that

$$\mathbf{M}^{-1} \mathbf{u} = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\top \phi_{N+1} \end{bmatrix}.\tag{3.109}$$

Then,

$$p(t_{N+1}|\beta, \mathbf{t}) = \mathcal{N}(t_{N+1} | \mathbf{m}^\top \phi_{N+1}, \beta^{-1} (1 + \phi_{N+1}^\top \mathbf{S} \phi_{N+1})).\tag{3.110}$$

By 3.12(b),

$$p(\beta|\mathbf{t}) = \text{Gam}(\beta|a, b),\tag{3.111}$$

where

$$\begin{aligned}a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0).\end{aligned}\tag{3.112}$$

Then, the logarithm of $p(t_{N+1}|\beta, \mathbf{t})p(\beta|\mathbf{t})$ except the terms independent of β can be written as

$$\begin{aligned}& -\frac{1}{2} \ln \beta^{-1} - \frac{(t_{N+1} - \mathbf{m}^\top \phi_{N+1})^2}{2 (1 + \phi_{N+1}^\top \mathbf{S} \phi_{N+1})} \beta + (a - 1) \ln \beta - b \beta \\ &= (a' - 1) \ln \beta - b' \beta,\end{aligned}\tag{3.113}$$

where

$$\begin{aligned} a' &= a + \frac{1}{2}, \\ b' &= b + \frac{(t_{N+1} - \mathbf{m}^\top \boldsymbol{\phi}_{N+1})^2}{2(1 + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1})}. \end{aligned} \quad (3.114)$$

We have

$$\int_0^\infty \beta^{a'-1} \exp(-b'\beta) d\beta = \Gamma(a') b'^{-a'}. \quad (3.115)$$

Then, $p(t_{N+1}|\mathbf{t})$ except the terms independent of t_{N+1} can be written as

$$b'^{-a'} = \left(1 + \frac{(t_{N+1} - \mu)^2}{\nu \sigma^2}\right)^{-\frac{\nu+1}{2}} b^{-a'}, \quad (3.116)$$

where

$$\begin{aligned} \mu &= \mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \\ \sigma^2 &= \frac{b}{a} (1 + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1}), \\ \nu &= 2a. \end{aligned} \quad (3.117)$$

Therefore, by 2.46,

$$p(t_{N+1}|\mathbf{t}) = \text{St}(t_{N+1}|\mu, \sigma^2, \nu). \quad (3.118)$$

3.14 (Incomplete)

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \end{aligned} \quad (3.119)$$

where $\mathbf{w}$ is a vector in M dimensions.

(a)

By 3.7,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (3.120)$$

where

$$\begin{aligned} \mathbf{m} &= \beta \mathbf{S} \boldsymbol{\Phi}^\top \mathbf{t}, \\ \mathbf{S} &= (\alpha \mathbf{I} + \beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned} \quad (3.121)$$

Let

$$y(\boldsymbol{\phi}, \mathbf{w}) = \mathbf{w}^\top \boldsymbol{\phi}. \quad (3.122)$$

Then,

$$y(\boldsymbol{\phi}, \mathbf{m}) = \beta \boldsymbol{\phi}^\top \mathbf{S} \boldsymbol{\Phi}^\top \mathbf{t}. \quad (3.123)$$

The right hand side can be written as

$$\sum_{n=1}^N k(\boldsymbol{\phi}, \boldsymbol{\phi}_n) t_n, \quad (3.124)$$

where

$$k(\boldsymbol{\phi}, \boldsymbol{\phi}') = \beta \boldsymbol{\phi}^\top \mathbf{S} \boldsymbol{\phi}'. \quad (3.125)$$

Let us suppose that $\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_M$ are linearly independent, $N > M$ and

$$\phi_1 = 1. \quad (3.126)$$

Then, we can construct a new basis set $\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_M$ such that

$$\boldsymbol{\Psi}^\top \boldsymbol{\Psi} = \mathbf{I} \quad (3.127)$$

where

$$\psi_1 = 1. \quad (3.128)$$

(b)

3.15

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | \mathbf{w}) &= \mathcal{N}(t_n | \mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}), \end{aligned} \quad (3.129)$$

where $\mathbf{w}$ is a vector in M dimensions. By 3.19,

$$\begin{aligned} \ln p(\mathbf{t}) &= \frac{M}{2} \ln \alpha + \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) + \frac{1}{2} \ln(\det \mathbf{S}) \\ &\quad - E(\mathbf{m}, \alpha, \beta), \end{aligned} \quad (3.130)$$

where

$$\begin{aligned} E(\mathbf{w}, \alpha, \beta) &= \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2, \\ \mathbf{m} &= \beta \mathbf{S} \Phi^\top \mathbf{t}, \\ \mathbf{S}^{-1} &= \alpha \mathbf{I} + \beta \Phi^\top \Phi. \end{aligned} \quad (3.131)$$

By 3.20, the maximum likelihood solutions to α and β are given by

$$\begin{aligned} \alpha_{\text{ML}} &= \frac{\gamma}{\|\mathbf{m}\|^2} \\ \beta_{\text{ML}} &= \frac{N - \gamma}{\|\mathbf{t} - \Phi \mathbf{m}\|^2} \end{aligned} \quad (3.132)$$

where

$$\gamma = \sum_{m=1}^M \frac{\beta \lambda_m}{\alpha + \beta \lambda_m}, \quad (3.133)$$

and $\lambda_1, \dots, \lambda_M$ are the eigenvalues of $\Phi^\top \Phi$. Therefore,

$$E(\mathbf{m}, \alpha_{\text{ML}}, \beta_{\text{ML}}) = \frac{N}{2}. \quad (3.134)$$

3.16

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | \mathbf{w}) &= \mathcal{N}(t_n | \mathbf{w}^\top \phi_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}), \end{aligned} \quad (3.135)$$

where $\mathbf{w}$ is a vector in M dimensions. By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t} | \mathbf{w}) p(\mathbf{w}) d\mathbf{w}. \quad (3.136)$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$-\frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 - \frac{\alpha}{2} \mathbf{w}^\top \mathbf{w} = -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v}, \quad (3.137)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \alpha \mathbf{I} + \beta \Phi^\top \Phi & -\beta \Phi^\top \\ -\beta \Phi & \beta \mathbf{I} \end{bmatrix}. \quad (3.138)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \alpha^{-1}\mathbf{I} & \alpha^{-1}\mathbf{\Phi}^\top \\ \alpha^{-1}\mathbf{\Phi} & \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^\top + \beta^{-1}\mathbf{I} \end{bmatrix}. \quad (3.139)$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^\top + \beta^{-1}\mathbf{I}). \quad (3.140)$$

3.17

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top\boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \end{aligned} \quad (3.141)$$

where $\mathbf{w}$ is a vector in M dimensions. By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \quad (3.142)$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} & -\frac{N}{2} \ln(2\pi\beta^{-1}) - \frac{\beta}{2} \|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 \\ & -\frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\alpha^{-1}\mathbf{I})) - \frac{\alpha}{2} \|\mathbf{w}\|^2. \end{aligned} \quad (3.143)$$

Therefore,

$$p(\mathbf{t}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \int \exp(-E(\mathbf{w})) d\mathbf{w}, \quad (3.144)$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{t} - \mathbf{\Phi}\mathbf{w}\|^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (3.145)$$

3.18

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top\boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \end{aligned} \quad (3.146)$$

where $\mathbf{w}$ is a vector in M dimensions. By 3.17,

$$p(\mathbf{t}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \int \exp(-E(\mathbf{w})) d\mathbf{w}, \quad (3.147)$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (3.148)$$

We have

$$\nabla E(\mathbf{w}) = -\beta \Phi^\top (\mathbf{t} - \Phi \mathbf{w}) + \alpha \mathbf{w}. \quad (3.149)$$

Then,

$$\begin{aligned} \nabla E(\mathbf{w}) &= \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}), \\ \nabla \nabla E(\mathbf{w}) &= \mathbf{S}^{-1}. \end{aligned} \quad (3.150)$$

where

$$\begin{aligned} \mathbf{m} &= \beta \mathbf{S} \Phi^\top \mathbf{t}, \\ \mathbf{S} &= (\beta \Phi^\top \Phi + \alpha \mathbf{I})^{-1}. \end{aligned} \quad (3.151)$$

Then,

$$E(\mathbf{w}) = E(\mathbf{m}) + \frac{1}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}). \quad (3.152)$$

3.19

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | \mathbf{w}) &= \mathcal{N}(t_n | \mathbf{w}^\top \phi_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}), \end{aligned} \quad (3.153)$$

where $\mathbf{w}$ is a vector in M dimensions. By 3.17,

$$p(\mathbf{t}) = \left(\frac{\beta}{2\pi}\right)^{\frac{N}{2}} \left(\frac{\alpha}{2\pi}\right)^{\frac{M}{2}} \int \exp(-E(\mathbf{w})) d\mathbf{w}, \quad (3.154)$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (3.155)$$

By 3.18,

$$E(\mathbf{w}) = E(\mathbf{m}) + \frac{1}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}), \quad (3.156)$$

where

$$\begin{aligned}\mathbf{m} &= \beta \mathbf{S} \Phi^\top \mathbf{t}, \\ \mathbf{S} &= (\beta \Phi^\top \Phi + \alpha \mathbf{I})^{-1}.\end{aligned}\tag{3.157}$$

Then, the integral can be written as

$$\begin{aligned}& \exp(-E(\mathbf{m})) \int \exp\left(-\frac{1}{2}(\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m})\right) d\mathbf{w} \\ &= (2\pi)^{\frac{M}{2}} (\det \mathbf{S})^{\frac{1}{2}} \exp(-E(\mathbf{m})).\end{aligned}\tag{3.158}$$

Therefore,

$$\ln p(\mathbf{t}) = \frac{M}{2} \ln \alpha + \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) + \frac{1}{2} \ln(\det \mathbf{S}) - E(\mathbf{m}).\tag{3.159}$$

3.20

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n | \mathbf{w}) &= \mathcal{N}(t_n | \mathbf{w}^\top \phi_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}),\end{aligned}\tag{3.160}$$

where $\mathbf{w}$ is a vector in M dimensions. By 3.19,

$$\ln p(\mathbf{t}) = \frac{M}{2} \ln \alpha + \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) + \frac{1}{2} \ln(\det \mathbf{S}) - E(\mathbf{m}),\tag{3.161}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2,\tag{3.162}$$

and

$$\begin{aligned}\mathbf{m} &= \beta \mathbf{S} \Phi^\top \mathbf{t}, \\ \mathbf{S} &= (\beta \Phi^\top \Phi + \alpha \mathbf{I})^{-1}.\end{aligned}\tag{3.163}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{t})$ with respect to α and β to zero gives

$$\begin{aligned}0 &= \frac{M}{2\alpha} + \frac{1}{2} \text{tr} \left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha} \right) - \frac{\partial E(\mathbf{m})}{\partial \alpha}, \\ 0 &= \frac{N}{2\beta} + \frac{1}{2} \text{tr} \left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} \right) - \frac{\partial E(\mathbf{m})}{\partial \beta}.\end{aligned}\tag{3.164}$$

We have

$$\begin{aligned}\mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \alpha}, \\ \mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \beta}.\end{aligned}\tag{3.165}$$

The right hand sides can be written as

$$\begin{aligned}\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha} + \frac{\partial \mathbf{S}^{-1}}{\partial \alpha} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha} + \mathbf{S}, \\ \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \frac{\partial \mathbf{S}^{-1}}{\partial \beta} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \mathbf{\Phi}^\top \mathbf{\Phi} \mathbf{S}.\end{aligned}\tag{3.166}$$

Then,

$$\begin{aligned}\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha} &= -\mathbf{S}, \\ \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} &= -\mathbf{\Phi}^\top \mathbf{\Phi} \mathbf{S}.\end{aligned}\tag{3.167}$$

We have

$$\begin{aligned}\text{tr } \mathbf{S} &= \sum_{m=1}^M (\beta \lambda_m + \alpha)^{-1}, \\ \text{tr } (\mathbf{\Phi}^\top \mathbf{\Phi} \mathbf{S}) &= \sum_{m=1}^M \lambda_m (\beta \lambda_m + \alpha)^{-1},\end{aligned}\tag{3.168}$$

where $\lambda_1, \dots, \lambda_M$ are the eigenvalues of $\mathbf{\Phi}^\top \mathbf{\Phi}$. The right hand sides can be written as $\alpha^{-1}(M - \gamma)$ and $\beta^{-1}\gamma$, where

$$\gamma = \sum_{m=1}^M \frac{\beta \lambda_m}{\alpha + \beta \lambda_m}.\tag{3.169}$$

We have

$$\begin{aligned}\frac{\partial E(\mathbf{m})}{\partial \alpha} &= \nabla E(\mathbf{m}) \frac{\partial \mathbf{m}}{\partial \alpha} + \frac{1}{2} \|\mathbf{m}\|^2, \\ \frac{\partial E(\mathbf{m})}{\partial \beta} &= \nabla E(\mathbf{m}) \frac{\partial \mathbf{m}}{\partial \beta} + \frac{1}{2} \|\mathbf{t} - \mathbf{\Phi} \mathbf{m}\|^2.\end{aligned}\tag{3.170}$$

We have

$$\nabla E(\mathbf{m}) = \mathbf{0}.\tag{3.171}$$

Then,

$$\begin{aligned} 0 &= \frac{M}{2\alpha} - \frac{M - \gamma}{2\alpha} - \frac{1}{2} \|\mathbf{m}\|^2, \\ 0 &= \frac{N}{2\beta} - \frac{\gamma}{2\beta} - \frac{1}{2} \|\mathbf{t} - \Phi \mathbf{m}\|^2. \end{aligned} \quad (3.172)$$

Therefore, the maximum likelihood solutions to α and β are given by

$$\begin{aligned} \alpha_{\text{ML}} &= \frac{\gamma}{\|\mathbf{m}\|^2}, \\ \beta_{\text{ML}} &= \frac{N - \gamma}{\|\mathbf{t} - \Phi \mathbf{m}\|^2}. \end{aligned} \quad (3.173)$$

3.21

Let Σ be a $M \times M$ real symmetric matrix such that

$$\Sigma \mathbf{u}_m = \lambda_m \mathbf{u}_m, \quad (3.174)$$

where $\mathbf{u}_1, \dots, \mathbf{u}_M$ are unit vectors. Let

$$\begin{aligned} \Lambda &= \text{diag}(\lambda_1, \dots, \lambda_M), \\ \mathbf{U} &= [\mathbf{u}_1 \cdots \mathbf{u}_M]. \end{aligned} \quad (3.175)$$

By 2.19,

$$\begin{aligned} \Sigma &= \mathbf{U} \Lambda \mathbf{U}^\top, \\ \mathbf{U}^\top \mathbf{U} &= \mathbf{I}. \end{aligned} \quad (3.176)$$

Then,

$$\det \Sigma = \prod_{m=1}^M \lambda_m, \quad (3.177)$$

so that

$$\ln(\det \Sigma) = \sum_{m=1}^M \ln \lambda_m. \quad (3.178)$$

Then,

$$\frac{\partial}{\partial \alpha} \ln(\det \Sigma) = \sum_{m=1}^M \frac{\partial \lambda_m}{\partial \alpha} \frac{1}{\lambda_m}. \quad (3.179)$$

Then,

$$\frac{\partial}{\partial \alpha} \ln(\det \boldsymbol{\Sigma}) = \text{tr} \left(\boldsymbol{\Lambda}^{-1} \frac{\partial \boldsymbol{\Lambda}}{\partial \alpha} \right). \quad (3.180)$$

The right hand side can be written as

$$\text{tr} \left(\mathbf{U} \boldsymbol{\Lambda}^{-1} \mathbf{U}^\top \frac{\partial \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^\top}{\partial \alpha} \right) = \text{tr} \left(\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha} \right). \quad (3.181)$$

Therefore,

$$\frac{\partial}{\partial \alpha} \ln(\det \boldsymbol{\Sigma}) = \text{tr} \left(\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha} \right). \quad (3.182)$$

3.22

Refer to 3.20.

3.23

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | \mathbf{w}, \beta) &= \mathcal{N}(t_n | \mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w} | \beta) &= \mathcal{N}(\mathbf{w} | \mathbf{m}_0, \beta^{-1} \mathbf{S}_0), \\ p(\beta) &= \text{Gam}(\beta | a_0, b_0). \end{aligned} \quad (3.183)$$

where $\mathbf{w}$ is a vector in M dimensions. By marginalisation,

$$\begin{aligned} p(\mathbf{t}) &= \int_0^\infty p(\mathbf{t} | \beta) p(\beta) d\beta, \\ p(\mathbf{t} | \beta) &= \int p(\mathbf{t} | \mathbf{w}, \beta) p(\mathbf{w} | \beta) d\mathbf{w}. \end{aligned} \quad (3.184)$$

The logarithm of $p(\mathbf{t} | \mathbf{w}, \beta) p(\mathbf{w} | \beta)$ except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2} \|\mathbf{t} - \boldsymbol{\Phi} \mathbf{w}\|^2 - \frac{\beta}{2} (\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\mathbf{w} - \mathbf{m}_0) \\ &= -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{\beta}{2} \mathbf{m}_0^\top \mathbf{S}_0^{-1} \mathbf{m}_0, \end{aligned} \quad (3.185)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta (\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\top \boldsymbol{\Phi}) & -\beta \boldsymbol{\Phi}^\top \\ -\beta \boldsymbol{\Phi} & \beta \mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta \mathbf{S}_0^{-1} \mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}. \quad (3.186)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \beta^{-1} \mathbf{S}_0 & \beta^{-1} \mathbf{S}_0 \Phi^\top \\ \beta^{-1} \Phi \mathbf{S}_0 & \beta^{-1} (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top) \end{bmatrix}, \quad (3.187)$$

so that

$$\mathbf{M}^{-1} \mathbf{u} = \begin{bmatrix} \mathbf{m}_0 \\ \Phi \mathbf{m}_0 \end{bmatrix}. \quad (3.188)$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}(\mathbf{t}|\Phi \mathbf{m}_0, \beta^{-1} (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)). \quad (3.189)$$

Then, the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of β can be written as

$$\begin{aligned} & -\frac{N}{2} \ln \beta^{-1} - \frac{\beta}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0) \\ & + (a_0 - 1) \ln \beta - b_0 \beta \\ & = (a - 1) \ln \beta - b \beta, \end{aligned} \quad (3.190)$$

where

$$\begin{aligned} a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0). \end{aligned} \quad (3.191)$$

We have

$$\int_0^\infty \beta^{a_0-1} \exp(-b\beta) d\beta = b^{-a} \Gamma(a). \quad (3.192)$$

We have

$$b^{-a} = (1 + (\mathbf{t} - \boldsymbol{\mu})^\top (\nu \boldsymbol{\Sigma})^{-1} (\mathbf{t} - \boldsymbol{\mu}))^{-\frac{\nu+N}{2}} b_0^{-a}, \quad (3.193)$$

where

$$\begin{aligned} \boldsymbol{\mu} &= \Phi \mathbf{m}_0, \\ \boldsymbol{\Sigma} &= \frac{b_0}{a_0} (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top), \\ \nu &= 2a_0. \end{aligned} \quad (3.194)$$

Therefore, by 2.48,

$$p(\mathbf{t}) = \text{St}(\mathbf{t}|\boldsymbol{\mu}, \boldsymbol{\Lambda}, \nu), \quad (3.195)$$

3.24

Refer to 3.12.

4 Linear Models for Classification

4.1

Let $x_1, \dots, x_M$ and $y_1, \dots, y_N$ be two sets of data points. Then, the corresponding convex hulls are defined as the sets of all points $\mathbf{x}$ and $\mathbf{y}$ such that

$$\begin{aligned}\mathbf{x} &= \sum_{m=1}^M \alpha_m \mathbf{x}_m, \\ \mathbf{y} &= \sum_{n=1}^N \beta_n \mathbf{y}_n,\end{aligned}\tag{4.1}$$

where

$$\begin{aligned}\sum_{m=1}^M \alpha_m &= \sum_{n=1}^N \beta_n = 1, \\ \alpha_m &\geq 0, \beta_n \geq 0.\end{aligned}\tag{4.2}$$

Let us assume that $\alpha_1, \dots, \alpha_M$ and $\beta_1, \dots, \beta_N$ below are subject to the constraints above.

(a)

If the convex hulls intersect, then there exist $\alpha_1, \dots, \alpha_M$ and $\beta_1, \dots, \beta_N$ such that

$$\sum_{m=1}^M \alpha_m \mathbf{x}_m = \sum_{n=1}^N \beta_n \mathbf{y}_n.\tag{4.3}$$

Then,

$$\sum_{m=1}^M \alpha_m (\hat{\mathbf{w}}^\top \mathbf{x}_m + w_0) = \hat{\mathbf{w}}^\top \sum_{m=1}^M \alpha_m \mathbf{x}_m + w_0 \sum_m \alpha_m,\tag{4.4}$$

for any $\hat{\mathbf{w}}$ and w_0 . The right hand side can be written as

$$\hat{\mathbf{w}}^\top \sum_{n=1}^N \beta_n \mathbf{y}_n + w_0 \sum_{n=1}^N \beta_n = \sum_{n=1}^N \beta_n (\hat{\mathbf{w}}^\top \mathbf{y}_n + w_0).\tag{4.5}$$

Therefore, there do not exist $\hat{\mathbf{w}}$ and w_0 such that

$$\begin{aligned}\hat{\mathbf{w}}^\top \mathbf{x}_m + w_0 &> 0, \\ \hat{\mathbf{w}}^\top \mathbf{y}_n + w_0 &< 0.\end{aligned}\tag{4.6}$$

(b)

If there exist $\hat{\mathbf{w}}$ and w_0 such that

$$\begin{aligned}\hat{\mathbf{w}}^\top \mathbf{x}_m + w_0 &> 0, \\ \hat{\mathbf{w}}^\top \mathbf{y}_n + w_0 &< 0,\end{aligned}\tag{4.7}$$

then

$$\begin{aligned}\sum_{m=1}^M \alpha_m (\hat{\mathbf{w}}^\top \mathbf{x}_m + w_0) &> 0, \\ \sum_{n=1}^N \beta_n (\hat{\mathbf{w}}^\top \mathbf{y}_n + w_0) &< 0.\end{aligned}\tag{4.8}$$

The left hand sides can be written as

$$\begin{aligned}\hat{\mathbf{w}}^\top \sum_{m=1}^M \alpha_m \mathbf{x}_m + w_0 \sum_{m=1}^M \alpha_m &= \hat{\mathbf{w}}^\top \sum_{m=1}^M \alpha_m \mathbf{x}_m + w_0, \\ \hat{\mathbf{w}}^\top \sum_{n=1}^N \beta_n \mathbf{y}_n + w_0 \sum_{n=1}^N \beta_n &= \hat{\mathbf{w}}^\top \sum_{n=1}^N \beta_n \mathbf{y}_n + w_0.\end{aligned}\tag{4.9}$$

Then, there do not exist $\alpha_1, \dots, \alpha_M$ and $\beta_1, \dots, \beta_N$ such that

$$\sum_{m=1}^M \alpha_m \mathbf{x}_m = \sum_{n=1}^N \beta_n \mathbf{y}_n.\tag{4.10}$$

Therefore, the convex hulls do not intersect.

4.2 (Incomplete)

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ and $\mathbf{w}_1, \dots, \mathbf{w}_K$ be variables in M dimensions and $\mathbf{t}_1, \dots, \mathbf{t}_N$ be variables in K dimensions. Let

$$E(\tilde{\mathbf{W}}) = \frac{1}{2} \text{tr} \left((\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T})^\top (\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T}) \right),\tag{4.11}$$

where

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & \mathbf{x}_1^\top \\ \vdots & \vdots \\ 1 & \mathbf{x}_N^\top \end{bmatrix}, \tilde{\mathbf{W}} = \begin{bmatrix} w_{10} & \cdots & w_{K0} \\ \mathbf{w}_1 & \cdots & \mathbf{w}_K \end{bmatrix}, \mathbf{T} = \begin{bmatrix} \mathbf{t}_1^\top \\ \vdots \\ \mathbf{t}_N^\top \end{bmatrix}.\tag{4.12}$$

Setting the derivative of $E(\tilde{\mathbf{W}})$ with respect to $\tilde{\mathbf{W}}$ to zero gives

$$\mathbf{0} = \tilde{\mathbf{X}}^\top (\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T}). \quad (4.13)$$

Then,

$$\underset{\tilde{\mathbf{W}}}{\operatorname{argmin}} E(\tilde{\mathbf{W}}) = \tilde{\mathbf{W}}^*, \quad (4.14)$$

where

$$\tilde{\mathbf{W}}^* = \left(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \right)^{-1} \tilde{\mathbf{X}}^\top \mathbf{T}. \quad (4.15)$$

Then,

$$(\tilde{\mathbf{W}}^*)^\top \tilde{\mathbf{x}} - \mathbf{t}_n = \mathbf{T}^\top \tilde{\mathbf{X}} \left(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \right)^{-1} \tilde{\mathbf{x}} - \mathbf{t}_n, \quad (4.16)$$

where $\tilde{\mathbf{x}}$ is a vector in $M + 1$ dimensions whose first element is 1. The right hand side can be written as

$$\mathbf{T}^\top \left(\tilde{\mathbf{X}} \left(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \right)^{-1} \tilde{\mathbf{x}} - \mathbf{v}_n \right) = \mathbf{0} \quad (4.17)$$

where $\mathbf{v}_n$ is a vector in N dimensions whose n th element is 1 and other elements are zero. Then,

$$(\tilde{\mathbf{W}}^*)^\top \tilde{\mathbf{x}} - \mathbf{t}_n = \mathbf{0}. \quad (4.18)$$

Therefore, if

$$\mathbf{a}^\top \mathbf{t}_n + b = 0, \quad (4.19)$$

then

$$\mathbf{a}^\top (\tilde{\mathbf{W}}^*)^\top \tilde{\mathbf{x}} + b = 0. \quad (4.20)$$

4.3 (Incomplete)

4.4

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be variables and let

$$\mathbf{m}_k = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n, \quad (4.21)$$

where

$$N_k = \sum_{n \in \mathcal{C}_k} 1. \quad (4.22)$$

In order to maximise $\mathbf{w}^\top(\mathbf{m}_2 - \mathbf{m}_1)$ under the constraint

$$\|\mathbf{w}\|^2 = 1, \quad (4.23)$$

let

$$L = \mathbf{w}^\top(\mathbf{m}_2 - \mathbf{m}_1) + \lambda (\|\mathbf{w}\|^2 - 1). \quad (4.24)$$

Setting the derivatives of L with respect to $\mathbf{w}$ and λ to zero gives

$$\begin{aligned} \mathbf{m}_2 - \mathbf{m}_1 + 2\lambda\mathbf{w} &= \mathbf{0}, \\ \|\mathbf{w}\|^2 - 1 &= 0. \end{aligned} \quad (4.25)$$

Therefore, the optimal solution to $\mathbf{w}$ satisfy

$$\mathbf{w}^* = c(\mathbf{m}_2 - \mathbf{m}_1), \quad (4.26)$$

where c is a constant.

4.5

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be variables and let

$$\mathbf{m}_k = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n, \quad (4.27)$$

where

$$N_k = \sum_{n \in \mathcal{C}_k} 1. \quad (4.28)$$

Let

$$J(\mathbf{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2}, \quad (4.29)$$

where

$$\begin{aligned} s_k^2 &= \sum_{n \in \mathcal{C}_k} (y_n - m_k)^2, \\ y_n &= \mathbf{w}^\top \mathbf{x}_n, \\ m_k &= \mathbf{w}^\top \mathbf{m}_k. \end{aligned} \quad (4.30)$$

Then, $J(\mathbf{w})$ can be written as

$$\frac{(\mathbf{w}^\top(\mathbf{m}_2 - \mathbf{m}_1))^2}{\sum_{n \in \mathcal{C}_1} (\mathbf{w}^\top(\mathbf{x}_n - \mathbf{m}_1))^2 + \sum_{n \in \mathcal{C}_2} (\mathbf{w}^\top(\mathbf{x}_n - \mathbf{m}_2))^2} = \frac{\mathbf{w}^\top \mathbf{S}_B \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \mathbf{w}}, \quad (4.31)$$

where

$$\begin{aligned}\mathbf{S}_B &= (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^\top, \\ \mathbf{S}_W &= \sum_{k=1,2} \sum_{n \in \mathcal{C}_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^\top.\end{aligned}\tag{4.32}$$

4.6

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be variables and let

$$\mathbf{m}_k = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n,\tag{4.33}$$

where $k \in \{1, 2\}$ and

$$N_k = \sum_{n \in \mathcal{C}_k} 1.\tag{4.34}$$

Let

$$E = \frac{1}{2} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}_n + w_0 - t_n)^2,\tag{4.35}$$

where

$$t_n = \begin{cases} \frac{N}{N_1}, & n \in \mathcal{C}_1, \\ -\frac{N}{N_2}, & n \in \mathcal{C}_2. \end{cases}\tag{4.36}$$

Setting the derivative of E with respect to $\mathbf{w}$ and w_0 gives

$$\begin{aligned}0 &= \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}_n + w_0 - t_n), \\ \mathbf{0} &= \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}_n + w_0 - t_n) \mathbf{x}_n.\end{aligned}\tag{4.37}$$

The right hand side of the first equation can be written as

$$\mathbf{w}^\top \sum_{n=1}^N \mathbf{x}_n + Nw_0 - \sum_{n=1}^N t_n = N(\mathbf{w}^\top \mathbf{m} + w_0),\tag{4.38}$$

where

$$\mathbf{m} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n.\tag{4.39}$$

Then,

$$w_0 = -\mathbf{w}^\top \mathbf{m}. \quad (4.40)$$

Then, the right hand side of the second equation above can be written as

$$\sum_{n=1}^N (\mathbf{w}^\top (\mathbf{x}_n - \mathbf{m}) - t_n) \mathbf{x}_n = \mathbf{v}_1 + \mathbf{v}_2, \quad (4.41)$$

where

$$\begin{aligned} \mathbf{v}_1 &= \sum_{n \in \mathcal{C}_1} \left(\mathbf{w}^\top (\mathbf{x}_n - \mathbf{m}) - \frac{N}{N_1} \right) \mathbf{x}_n, \\ \mathbf{v}_2 &= \sum_{n \in \mathcal{C}_2} \left(\mathbf{w}^\top (\mathbf{x}_n - \mathbf{m}) + \frac{N}{N_2} \right) \mathbf{x}_n. \end{aligned} \quad (4.42)$$

We have

$$\begin{aligned} \mathbf{m} &= \frac{N_1}{N} \mathbf{m}_1 + \frac{N_2}{N} \mathbf{m}_2, \\ \sum_{n \in \mathcal{C}_1} (\mathbf{x}_n - \mathbf{m}_1) &= \mathbf{0}, \\ \sum_{n \in \mathcal{C}_2} (\mathbf{x}_n - \mathbf{m}_2) &= \mathbf{0}. \end{aligned} \quad (4.43)$$

Then, $\mathbf{v}_1$ and $\mathbf{v}_2$ can be written as

$$\begin{aligned} &\sum_{n \in \mathcal{C}_1} \left(\mathbf{w}^\top \left(\mathbf{x}_n - \mathbf{m}_1 + \frac{N_2}{N} (\mathbf{m}_1 - \mathbf{m}_2) \right) - \frac{N}{N_1} \right) (\mathbf{x}_n - \mathbf{m}_1 + \mathbf{m}_1) \\ &= \left(\sum_{n \in \mathcal{C}_1} (\mathbf{x}_n - \mathbf{m}_1) (\mathbf{x}_n - \mathbf{m}_1)^\top \right) \mathbf{w} + \frac{N_1 N_2}{N} (\mathbf{m}_1 - \mathbf{m}_2) \mathbf{m}_1^\top \mathbf{w} - N \mathbf{m}_1, \end{aligned} \quad (4.44)$$

and

$$\begin{aligned} &\sum_{n \in \mathcal{C}_2} \left(\mathbf{w}^\top \left(\mathbf{x}_n - \mathbf{m}_2 + \frac{N_1}{N} (\mathbf{m}_2 - \mathbf{m}_1) \right) + \frac{N}{N_2} \right) (\mathbf{x}_n - \mathbf{m}_2 + \mathbf{m}_2) \\ &= \left(\sum_{n \in \mathcal{C}_2} (\mathbf{x}_n - \mathbf{m}_2) (\mathbf{x}_n - \mathbf{m}_2)^\top \right) \mathbf{w} + \frac{N_1 N_2}{N} (\mathbf{m}_2 - \mathbf{m}_1) \mathbf{m}_2^\top \mathbf{w} + N \mathbf{m}_2. \end{aligned} \quad (4.45)$$

Then,

$$\begin{aligned} \mathbf{0} = & \left(\sum_{k=1,2} \sum_{n \in \mathcal{C}_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^\top \right) \mathbf{w} \\ & + \frac{N_1 N_2}{N} (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^\top \mathbf{w} - N(\mathbf{m}_1 - \mathbf{m}_2). \end{aligned} \quad (4.46)$$

Therefore, the optimal solution to $\mathbf{w}$ satisfies

$$\left(\mathbf{S}_W + \frac{N_1 N_2}{N} \mathbf{S}_B \right) \mathbf{w}^* = N(\mathbf{m}_1 - \mathbf{m}_2). \quad (4.47)$$

where

$$\begin{aligned} \mathbf{S}_W &= \sum_{k=1,2} \sum_{n \in \mathcal{C}_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^\top, \\ \mathbf{S}_B &= (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^\top. \end{aligned} \quad (4.48)$$

4.7

Let

$$\sigma(a) = \frac{1}{1 + \exp(-a)}. \quad (4.49)$$

(a)

We have

$$\sigma(-a) = \frac{1}{1 + \exp(a)}. \quad (4.50)$$

The right hand side can be written as

$$1 - \frac{\exp(a)}{1 + \exp(a)} = 1 - \frac{1}{1 + \exp(-a)}. \quad (4.51)$$

Therefore,

$$\sigma(-a) = 1 - \sigma(a). \quad (4.52)$$

(b)

We have

$$\exp(-a) = \frac{1}{\sigma(a)} - 1. \quad (4.53)$$

Then,

$$a = -\ln \left(\frac{1}{\sigma(a)} - 1 \right). \quad (4.54)$$

Therefore,

$$\sigma^{-1}(y) = \ln \left(\frac{y}{1-y} \right). \quad (4.55)$$

4.8

Let $\mathbf{x}$ be a variable in D dimensions such that

$$p(\mathbf{x}|\mathcal{C}_k) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}), \quad (4.56)$$

where

$$p(\mathcal{C}_1) + p(\mathcal{C}_2) = 1. \quad (4.57)$$

By Bayes' theorem,

$$p(\mathcal{C}_1|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}. \quad (4.58)$$

The right hand side can be written as

$$\sigma(a) = \frac{1}{1 + \exp(-a)}, \quad (4.59)$$

where

$$a = \ln \left(\frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} \right). \quad (4.60)$$

Substituting the expressions above of $p(\mathbf{x}|\mathcal{C}_k)$, we have

$$\begin{aligned} a = & -\frac{D}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \boldsymbol{\Sigma}) - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \ln p(\mathcal{C}_1) \\ & + \frac{D}{2} \ln(2\pi) + \frac{1}{2} \ln(\det \boldsymbol{\Sigma}) + \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) - \ln p(\mathcal{C}_2). \end{aligned} \quad (4.61)$$

Therefore,

$$p(\mathcal{C}_1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + w_0), \quad (4.62)$$

where

$$\begin{aligned} \mathbf{w} &= \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2), \\ w_0 &= -\frac{1}{2}\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 + \ln p(\mathcal{C}_1) - \ln p(\mathcal{C}_2). \end{aligned} \quad (4.63)$$

4.9

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n, \boldsymbol{\phi}_n) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \quad (4.64)$$

where

$$\begin{aligned} y_{nk} &= p(\boldsymbol{\phi}_n, \mathcal{C}_k), \\ p(\mathcal{C}_k) &= \pi_k, \end{aligned} \quad (4.65)$$

By Bayes' theorem,

$$y_{nk} = \pi_k p(\boldsymbol{\phi}_n | \mathcal{C}_k). \quad (4.66)$$

Then,

$$\ln p(\mathbf{T}, \boldsymbol{\Phi}) = \sum_{n=1}^N \sum_{k=1}^K t_{nk} (\ln \pi_k + \ln p(\boldsymbol{\phi}_n | \mathcal{C}_k)). \quad (4.67)$$

In order to maximise $\ln p(\mathbf{T}, \boldsymbol{\Phi})$ under the constraint

$$\sum_{k=1}^K \pi_k = 1, \quad (4.68)$$

let

$$L = \ln p(\mathbf{T}, \boldsymbol{\Phi}) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right). \quad (4.69)$$

Setting the derivatives of L with respect to π_k and λ to zero gives

$$\begin{aligned} 0 &= \frac{N_k}{\pi_k} + \lambda, \\ 0 &= \sum_{k=1}^K \pi_k - 1, \end{aligned} \quad (4.70)$$

where

$$N_k = \sum_{n=1}^N t_{nk}. \quad (4.71)$$

Then,

$$\lambda = - \sum_{k=1}^K N_k. \quad (4.72)$$

The right hand side can be written as $-N$. Therefore, the maximum likelihood solution to π_k is given by

$$\pi_{k\text{ML}} = \frac{N_k}{N}. \quad (4.73)$$

4.10

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n, \phi_n) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \quad (4.74)$$

where

$$\begin{aligned} y_{nk} &= p(\phi_n, \mathcal{C}_k), \\ p(\phi_n | \mathcal{C}_k) &= \mathcal{N}(\phi_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}), \\ p(\mathcal{C}_k) &= \pi_k. \end{aligned} \quad (4.75)$$

By Bayes' theorem,

$$y_{nk} = \pi_k \mathcal{N}(\phi_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}). \quad (4.76)$$

Then,

$$\ln p(\mathbf{T}, \boldsymbol{\Phi}) = \sum_{n=1}^N \sum_{k=1}^K t_{nk} (\ln \mathcal{N}(\phi_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}) + \ln \pi_k). \quad (4.77)$$

The right hand side except the terms independent of $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}$ can be written as

$$l = \sum_{n=1}^N \sum_{k=1}^K t_{nk} \left(-\frac{1}{2} \ln(\det \boldsymbol{\Sigma}) - \frac{1}{2} (\phi_n - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1} (\phi_n - \boldsymbol{\mu}_k) \right). \quad (4.78)$$

By 3.21, setting the derivatives of l with respect to $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}$ to zero gives

$$\begin{aligned}\mathbf{0} &= \frac{1}{2} \sum_{n=1}^N t_{nk} (\boldsymbol{\Sigma}^{-1} + (\boldsymbol{\Sigma}^{-1})^\top) (\boldsymbol{\phi}_n - \boldsymbol{\mu}_k), \\ \mathbf{O} &= -\frac{1}{2} \sum_{n=1}^N \sum_{k=1}^K t_{nk} \left((\boldsymbol{\Sigma}^{-1})^\top - (\boldsymbol{\Sigma}^{-1})^2 (\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)^\top \right).\end{aligned}\tag{4.79}$$

Therefore, the maximum likelihood solutions to $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}$ are given by

$$\begin{aligned}\boldsymbol{\mu}_{k\text{ML}} &= \frac{1}{N_k} \sum_{n=1}^N t_{nk} \boldsymbol{\phi}_n, \\ \boldsymbol{\Sigma}_{\text{ML}} &= \frac{1}{N} \sum_{k=1}^K N_k \mathbf{S}_k,\end{aligned}\tag{4.80}$$

where

$$\begin{aligned}N_k &= \sum_{n=1}^N t_{nk}, \\ \mathbf{S}_k &= \frac{1}{N_k} \sum_{n=1}^N t_{nk} (\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)(\boldsymbol{\phi}_n - \boldsymbol{\mu}_k)^\top.\end{aligned}\tag{4.81}$$

4.11

Let $\boldsymbol{\phi}$ be a variable in M dimensions each of whose component is a binary code with length L such that

$$p(\boldsymbol{\phi}|\mathcal{C}_k) = \prod_{m=1}^M \prod_{l=1}^L \mu_{kml}^{\phi_{ml}}.\tag{4.82}$$

By Bayes' theorem,

$$p(\mathcal{C}_k|\boldsymbol{\phi}) = \frac{p(\boldsymbol{\phi}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_{k'=1}^K p(\boldsymbol{\phi}|\mathcal{C}_{k'})p(\mathcal{C}_{k'})}.\tag{4.83}$$

Therefore,

$$p(\mathcal{C}_k|\boldsymbol{\phi}) = \frac{\exp(a_k(\boldsymbol{\phi}))}{\sum_{k'=1}^K \exp(a_{k'}(\boldsymbol{\phi}))},\tag{4.84}$$

where

$$a_k(\boldsymbol{\phi}) = \left(\sum_{m=1}^M \sum_{l=1}^L \phi_{ml} \ln \mu_{kml} \right) + \ln p(\mathcal{C}_k).\tag{4.85}$$

4.12

Let

$$\sigma(a) = \frac{1}{1 + \exp(-a)}. \quad (4.86)$$

Then,

$$\frac{d\sigma(a)}{da} = \frac{\exp(-a)}{(1 + \exp(-a))^2}. \quad (4.87)$$

The right hand side can be written as

$$\frac{1}{1 + \exp(-a)} - \frac{1}{(1 + \exp(-a))^2} = \sigma(a) - (\sigma(a))^2. \quad (4.88)$$

Therefore,

$$\frac{d\sigma(a)}{da} = \sigma(a) (1 - \sigma(a)). \quad (4.89)$$

4.13

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n | \mathbf{w}) = y_n^{t_n} (1 - y_n)^{1-t_n}, \quad (4.90)$$

where

$$\begin{aligned} y_n &= \sigma(\mathbf{w}^\top \boldsymbol{\phi}_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}. \end{aligned} \quad (4.91)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t} | \mathbf{w}). \quad (4.92)$$

The right hand side can be written as

$$-\sum_{n=1}^N (t_n \ln y_n + (1 - t_n) \ln(1 - y_n)). \quad (4.93)$$

By 4.12,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^N \left(\frac{t_n}{y_n} y_n (1 - y_n) \boldsymbol{\phi}_n - \frac{1 - t_n}{1 - y_n} y_n (1 - y_n) \boldsymbol{\phi}_n \right). \quad (4.94)$$

The right hand side can be written as

$$-\sum_{n=1}^N (t_n(1 - y_n)\phi_n - (1 - t_n)y_n\phi_n) = \sum_{n=1}^N (y_n - t_n)\phi_n. \quad (4.95)$$

Therefore,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N (y_n - t_n)\phi_n. \quad (4.96)$$

4.14

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1 - y_n)^{1-t_n}, \quad (4.97)$$

where

$$\begin{aligned} y_n &= \sigma(\mathbf{w}^\top \phi_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}. \end{aligned} \quad (4.98)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \quad (4.99)$$

By 4.13, setting the derivative of $E(\mathbf{w})$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = \sum_{n=1}^N (y_n - t_n) \phi_n. \quad (4.100)$$

If $\phi_1, \dots, \phi_N$ are linearly independent, then

$$y_n = t_n. \quad (4.101)$$

Then,

$$\sigma(\mathbf{w}^\top \phi_n) = \begin{cases} 1, & t_n = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (4.102)$$

Therefore,

$$\mathbf{w}^\top \phi_n = \begin{cases} \infty, & t_n = 1, \\ -\infty, & \text{otherwise.} \end{cases} \quad (4.103)$$

4.15

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1 - y_n)^{1-t_n}, \quad (4.104)$$

where

$$\begin{aligned} y_n &= \sigma(\mathbf{w}^\top \boldsymbol{\phi}_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}. \end{aligned} \quad (4.105)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \quad (4.106)$$

By 4.13,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N (y_n - t_n) \boldsymbol{\phi}_n. \quad (4.107)$$

By 4.12,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N y_n(1 - y_n) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^\top. \quad (4.108)$$

The right hand side can be written as

$$\mathbf{H} = \boldsymbol{\Phi}^\top \mathbf{R} \boldsymbol{\Phi}, \quad (4.109)$$

where

$$R_{nn'} = y_n(1 - y_n) I_{nn'}, \quad (4.110)$$

Then,

$$\mathbf{u}^\top \mathbf{H} \mathbf{u} = \mathbf{v}^\top \mathbf{R} \mathbf{v}, \quad (4.111)$$

where

$$\mathbf{v} = \boldsymbol{\Phi} \mathbf{u}. \quad (4.112)$$

The right hand side can be written as

$$\sum_{n=1}^N \sum_{n'=1}^N v_n y_n(1 - y_n) I_{nn'} v_{n'} = \sum_{n=1}^N y_n(1 - y_n) v_n^2. \quad (4.113)$$

We have

$$y_n(1 - y_n) > 0. \quad (4.114)$$

Then,

$$\mathbf{u}^\top \mathbf{H} \mathbf{u} > 0, \quad (4.115)$$

so that $\mathbf{H}$ is positive definite. Therefore, E is a convex function of $\mathbf{w}$ and it has a unique minimum.

4.16

Let $t_1, \dots, t_N$ be independent binary variables such that

$$p(t_n = 1) = \pi_n. \quad (4.116)$$

Then,

$$p(t_n) = \pi_n^{t_n} (1 - \pi_n)^{1-t_n}, \quad (4.117)$$

so that

$$p(\mathbf{t}) = \prod_{n=1}^N \pi_n^{t_n} (1 - \pi_n)^{1-t_n}. \quad (4.118)$$

Therefore,

$$-\ln p(\mathbf{t}) = -\sum_{n=1}^N (t_n \ln \pi_n + (1 - t_n) \ln(1 - \pi_n)). \quad (4.119)$$

4.17

Let

$$y_k = \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})}. \quad (4.120)$$

Then,

$$\frac{\partial y_k}{\partial a_k} = \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})} - \frac{\exp(2a_k)}{\left(\sum_{k'=1}^K \exp(a_{k'})\right)^2}. \quad (4.121)$$

The right hand side can be written as $y_k(1 - y_k)$. If $k \neq k'$, then

$$\frac{\partial y_k}{\partial a_{k'}} = -\frac{\exp(a_k + a_{k'})}{\left(\sum_{k''=1}^K \exp(a_{k''})\right)^2}. \quad (4.122)$$

The right hand side can be written as $-y_k y_{k'}$. Therefore,

$$\frac{\partial y_k}{\partial a_{k'}} = y_k(I_{kk'} - y_{k'}). \quad (4.123)$$

4.18

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n | \mathbf{W}) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \quad (4.124)$$

where

$$y_{nk} = \frac{\exp(a_{nk})}{\sum_{k'=1}^K \exp(a_{nk'})}, \quad (4.125)$$

$$a_{nk} = \mathbf{w}_k^\top \phi_n.$$

Let

$$E(\mathbf{W}) = -\ln p(\mathbf{T} | \mathbf{W}). \quad (4.126)$$

The right hand side can be written as

$$-\sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk}. \quad (4.127)$$

By 4.17,

$$\nabla_{\mathbf{w}_{k'}} E(\mathbf{W}) = -\sum_{n=1}^N \sum_{k=1}^K (I_{kk'} - y_{nk'}) t_{nk} \phi_n. \quad (4.128)$$

The right hand side can be written as

$$-\sum_{n=1}^N \left(\sum_{k=1}^K (I_{kk'} - y_{nk'}) t_{nk} \right) \phi_n = -\sum_{n=1}^N (t_{nk'} - y_{nk'}) \phi_n. \quad (4.129)$$

Therefore,

$$\nabla_{\mathbf{w}_k} E(\mathbf{W}) = \sum_{n=1}^N (y_{nk} - t_{nk}) \phi_n. \quad (4.130)$$

4.19

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n = 1 | \phi_n) = \Phi(a_n), \quad (4.131)$$

where

$$\Phi(a) = \int_{-\infty}^a \mathcal{N}(\theta | 0, 1) d\theta, \quad (4.132)$$

$$a_n = \mathbf{w}^\top \phi_n.$$

(a)

We have

$$p(t_n|\phi_n) = (\Phi(a_n))^{t_n} (1 - \Phi(a_n))^{1-t_n}. \quad (4.133)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\Phi). \quad (4.134)$$

The right hand side can be written as

$$-\sum_{n=1}^N (t_n \ln \Phi(a_n) + (1 - t_n) \ln (1 - \Phi(a_n))). \quad (4.135)$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^N \left(t_n \frac{\mathcal{N}(a_n|0, 1)}{\Phi(a_n)} - (1 - t_n) \frac{\mathcal{N}(a_n|0, 1)}{1 - \Phi(a_n)} \right) \phi_n. \quad (4.136)$$

The right hand side can be written as

$$\begin{aligned} & -\sum_{n=1}^N \left(\frac{t_n}{\Phi(a_n)} - \frac{1 - t_n}{1 - \Phi(a_n)} \right) \mathcal{N}(a_n|0, 1) \phi_n \\ &= \sum_{n=1}^N \frac{\mathcal{N}(a_n|0, 1)}{\Phi(a_n) (1 - \Phi(a_n))} (\Phi(a_n) - t_n) \phi_n. \end{aligned} \quad (4.137)$$

Therefore,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N \frac{\mathcal{N}(a_n|0, 1)}{\Phi(a_n) (1 - \Phi(a_n))} (\Phi(a_n) - t_n) \phi_n. \quad (4.138)$$

(b)

We have

$$\begin{aligned}
\nabla \nabla E(\mathbf{w}) = & \sum_{n=1}^N \frac{-a_n \mathcal{N}(a_n|0,1)}{\Phi(a_n)(1-\Phi(a_n))} (\Phi(a_n) - t_n) \phi_n \phi_n^\top \\
& - \sum_{n=1}^N \frac{(\mathcal{N}(a_n|0,1))^2}{(\Phi(a_n))^2(1-\Phi(a_n))} (\Phi(a_n) - t_n) \phi_n \phi_n^\top \\
& + \sum_{n=1}^N \frac{(\mathcal{N}(a_n|0,1))^2}{\Phi(a_n)(1-\Phi(a_n))^2} (\Phi(a_n) - t_n) \phi_n \phi_n^\top \\
& + \sum_{n=1}^N \frac{(\mathcal{N}(a_n|0,1))^2}{\Phi(a_n)(1-\Phi(a_n))} \phi_n \phi_n^\top.
\end{aligned} \tag{4.139}$$

Therefore,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N b_n \phi_n \phi_n^\top, \tag{4.140}$$

where

$$\begin{aligned}
b_n = & \left(\frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)(1-\Phi(a_n))} \right)^2 ((\Phi(a_n))^2 - 2t_n\Phi(a_n) + t_n) \\
& - \frac{\mathcal{N}(a_n|0,1)}{\Phi(a_n)(1-\Phi(a_n))} a_n (\Phi(a_n) - t_n).
\end{aligned} \tag{4.141}$$

4.20

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n | \mathbf{W}) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \tag{4.142}$$

where

$$\begin{aligned}
y_{nk} &= \frac{\exp(a_{nk})}{\sum_{k'=1}^K \exp(a_{nk'})}, \\
a_{nk} &= \mathbf{w}_k^\top \phi_n.
\end{aligned} \tag{4.143}$$

Let

$$E(\mathbf{W}) = -\ln p(\mathbf{T}|\mathbf{W}). \quad (4.144)$$

By 4.18,

$$\nabla_{\mathbf{w}_k} E(\mathbf{W}) = \sum_{n=1}^N (y_{nk} - t_{nk}) \boldsymbol{\phi}_n. \quad (4.145)$$

By 4.17,

$$\nabla_{\mathbf{w}_k} \nabla_{\mathbf{w}_{k'}} E(\mathbf{W}) = \sum_{n=1}^N y_{nk} (I_{kk'} - y_{nk'}) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^\top. \quad (4.146)$$

The right hand side can be written as

$$\mathbf{H}_{kk'} = \boldsymbol{\Phi}^\top \mathbf{R}_{kk'} \boldsymbol{\Phi}, \quad (4.147)$$

where

$$R_{kk'nn'} = y_{nk} (I_{kk'} - y_{nk'}) I_{nn'}. \quad (4.148)$$

Let

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \cdots & \mathbf{H}_{1K} \\ \vdots & \ddots & \vdots \\ \mathbf{H}_{K1} & \cdots & \mathbf{H}_{KK} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_K \end{bmatrix}, \quad (4.149)$$

where $\mathbf{u}_1, \dots, \mathbf{u}_K$ are vectors in the same dimension as $\mathbf{w}$. Then,

$$\mathbf{u}^\top \mathbf{H} \mathbf{u} = \sum_{k=1}^K \sum_{k'=1}^K \mathbf{v}_k^\top \mathbf{R}_{kk'} \mathbf{v}_{k'}, \quad (4.150)$$

where

$$\mathbf{v}_k = \boldsymbol{\Phi} \mathbf{u}_k. \quad (4.151)$$

The right hand side can be written as

$$\begin{aligned} & \sum_{k=1}^K \sum_{k'=1}^K \sum_{n=1}^N \sum_{n'=1}^N v_{kn} y_{nk} (I_{kk'} - y_{nk'}) I_{nn'} v_{k'n'} \\ &= \sum_{k=1}^K \sum_{k'=1}^K \sum_{n=1}^N v_{kn} y_{nk} (I_{kk'} - y_{nk'}) v_{k'n}. \end{aligned} \quad (4.152)$$

The right hand side can be written as

$$\sum_{n=1}^N \sum_{k=1}^K y_{nk} v_{kn} \left(v_{kn} - \sum_{k'=1}^K y_{nk'} v_{k'n} \right). \quad (4.153)$$

By Jensen's inequality,

$$\begin{aligned} & \sum_{k=1}^K y_{nk} v_{kn} \left(v_{kn} - \sum_{k'=1}^K y_{nk'} v_{k'n} \right) \\ & \geq \left(\sum_{k=1}^K y_{nk} v_{kn} \right) \left(\sum_{k=1}^K y_{nk} v_{kn} - \sum_{k'=1}^K y_{nk'} v_{k'n} \right), \end{aligned} \quad (4.154)$$

where the right hand side is zero. Then,

$$\mathbf{u}^\top \mathbf{H} \mathbf{u} \geq 0. \quad (4.155)$$

Therefore, $\mathbf{H}$ is positive semidifinite.

4.21

Let

$$\Phi(a) = \int_{-\infty}^a \mathcal{N}(\theta|0, 1) d\theta. \quad (4.156)$$

The right hand side can be written as

$$\int_{-\infty}^0 \mathcal{N}(\theta|0, 1) d\theta + \int_0^a \mathcal{N}(\theta|0, 1) d\theta = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^a \exp\left(-\frac{\theta^2}{2}\right) d\theta. \quad (4.157)$$

The second term of the right hand side can be written as

$$\frac{1}{\sqrt{2\pi}} \int_0^{\frac{a}{\sqrt{2}}} \exp(-t^2) \sqrt{2} dt = \frac{1}{2} \operatorname{erf}\left(\frac{a}{\sqrt{2}}\right), \quad (4.158)$$

where

$$\operatorname{erf}(a) = \frac{2}{\sqrt{\pi}} \int_0^a \exp(-t^2) dt. \quad (4.159)$$

Therefore,

$$\Phi(a) = \frac{1}{2} \left(1 + \operatorname{erf}\left(\frac{a}{\sqrt{2}}\right) \right). \quad (4.160)$$

4.22

Let $\mathcal{D}$ be a set of conditionally independent variables dependent on $\boldsymbol{\theta}$ in M dimensions. By marginalisation,

$$p(\mathcal{D}) = \int p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}. \quad (4.161)$$

Let the logarithm of the integrand be $-E(\boldsymbol{\theta})$. We have the Taylor series

$$\begin{aligned} E(\boldsymbol{\theta}) = & E(\boldsymbol{\theta}_{\text{MAP}}) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}}) \\ & + O(\|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}}\|^3), \end{aligned} \quad (4.162)$$

where $\boldsymbol{\theta}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\boldsymbol{\theta}_{\text{MAP}}). \quad (4.163)$$

Then,

$$\int p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta} \simeq \exp(-E(\boldsymbol{\theta}_{\text{MAP}})) I, \quad (4.164)$$

where

$$I = \int \exp\left(-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{MAP}})\right) d\boldsymbol{\theta}. \quad (4.165)$$

Therefore,

$$p(\mathcal{D}) \simeq (2\pi)^{\frac{M}{2}} (\det \mathbf{H}_{\text{MAP}})^{-\frac{1}{2}} \exp(-E(\boldsymbol{\theta}_{\text{MAP}})), \quad (4.166)$$

so that

$$\ln p(\mathcal{D}) \simeq \frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{H}_{\text{MAP}}) - E(\boldsymbol{\theta}_{\text{MAP}}). \quad (4.167)$$

4.23

Let $\mathcal{D}$ be a set of $t_1, \dots, t_N$ which are conditionally independent and identically distributed variables dependent on $\boldsymbol{\theta}$ in M dimensions such that

$$p(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\theta}|\mathbf{m}_0, \mathbf{S}_0). \quad (4.168)$$

By 4.22,

$$\ln p(\mathcal{D}) \simeq \frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{H}_{\text{MAP}}) - E(\boldsymbol{\theta}_{\text{MAP}}), \quad (4.169)$$

where

$$E(\boldsymbol{\theta}) = -\ln p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}), \quad (4.170)$$

$\boldsymbol{\theta}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\boldsymbol{\theta}_{\text{MAP}}). \quad (4.171)$$

We have

$$\mathbf{H}_{\text{MAP}} = \mathbf{H}_{\mathcal{D}\text{MAP}} + \mathbf{S}_0^{-1}, \quad (4.172)$$

where

$$\mathbf{H}_{\mathcal{D}\text{MAP}} = -\nabla \nabla \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}). \quad (4.173)$$

The right hand side of the approximation can be written as

$$\begin{aligned} & \frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\mathbf{H}_{\mathcal{D}\text{MAP}} + \mathbf{S}_0^{-1})) \\ & + \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}) + \ln p(\boldsymbol{\theta}_{\text{MAP}}). \end{aligned} \quad (4.174)$$

We have

$$\begin{aligned} \ln p(\boldsymbol{\theta}_{\text{MAP}}) &= -\frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{S}_0) \\ & - \frac{1}{2} (\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0). \end{aligned} \quad (4.175)$$

Then,

$$\begin{aligned} \ln p(\mathcal{D}) &\simeq -\frac{1}{2} \ln(\det(\mathbf{H}_{\mathcal{D}\text{MAP}} \mathbf{S}_0 + \mathbf{I})) + \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}) \\ & - \frac{1}{2} (\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\boldsymbol{\theta}_{\text{MAP}} - \mathbf{m}_0). \end{aligned} \quad (4.176)$$

Then, for a large $\mathbf{S}_0$,

$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}) - \frac{1}{2} \ln(\det(\mathbf{H}_{\mathcal{D}\text{MAP}} \mathbf{S}_0)). \quad (4.177)$$

We have

$$\mathbf{H}_{\mathcal{D}\text{MAP}} = N \bar{\mathbf{H}}_{\text{MAP}}, \quad (4.178)$$

where

$$\begin{aligned} \bar{\mathbf{H}}_{\text{MAP}} &= \frac{1}{N} \sum_{n=1}^N \mathbf{H}_{n\text{MAP}}, \\ \mathbf{H}_{n\text{MAP}} &= -\nabla \nabla \ln p(t_n|\boldsymbol{\theta}_{\text{MAP}}). \end{aligned} \quad (4.179)$$

Then,

$$\det(\mathbf{H}_{\mathcal{D}_{\text{MAP}}}\mathbf{S}_0) = N^M \det(\bar{\mathbf{H}}_{\text{MAP}}\mathbf{S}_0), \quad (4.180)$$

so that

$$\ln(\det(\mathbf{H}_{\mathcal{D}_{\text{MAP}}}\mathbf{S}_0)) \simeq M \ln N. \quad (4.181)$$

Therefore,

$$\ln p(\mathcal{D}) \simeq \ln p(\mathcal{D}|\boldsymbol{\theta}_{\text{MAP}}) - \frac{M}{2} \ln N. \quad (4.182)$$

4.24

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= y_n^{t_n}(1 - y_n)^{1-t_n}, \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0), \end{aligned} \quad (4.183)$$

where

$$\begin{aligned} y_n &= \sigma(\mathbf{w}^\top \boldsymbol{\phi}_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}. \end{aligned} \quad (4.184)$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}. \quad (4.185)$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \quad (4.186)$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$\begin{aligned} E(\mathbf{w}) &= \sum_{n=1}^N (t_n \ln y_n + (1 - t_n) \ln(1 - y_n)) \\ &\quad - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0). \end{aligned} \quad (4.187)$$

Let $\mathbf{w}_{\text{MAP}}$ be a stationary point of E . Then, we have the Taylor series

$$\begin{aligned} E(\mathbf{w}) &= E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ &\quad + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3), \end{aligned} \quad (4.188)$$

where

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}). \quad (4.189)$$

Then,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}). \quad (4.190)$$

Then,

$$\begin{aligned} & \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w} \\ & \simeq \int \sigma(a_{N+1})^{t_{N+1}} (1 - \sigma(a_{N+1}))^{1-t_{N+1}} p(a_{N+1}) da_{N+1}, \end{aligned} \quad (4.191)$$

where

$$p(a_{N+1}) = \int \delta(a_{N+1} - \mathbf{w}^\top \boldsymbol{\phi}_{N+1}) \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}) d\mathbf{w}, \quad (4.192)$$

and δ is the Dirac delta function. Since t_{N+1} is a binary variable, the right hand side of the approximation can be written as

$$v_{N+1}^{t_{N+1}} (1 - v_{N+1})^{1-t_{N+1}}, \quad (4.193)$$

where

$$v_{N+1} = \int \sigma(a_{N+1}) p(a_{N+1}) da_{N+1}. \quad (4.194)$$

We have

$$\mathbb{E} a_{N+1} = \int a_{N+1} p(a_{N+1}) da_{N+1}. \quad (4.195)$$

The right hand side can be written as

$$\int \mathbf{w}^\top \boldsymbol{\phi}_{N+1} \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}) d\mathbf{w} = \mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}_{N+1}. \quad (4.196)$$

Then,

$$\text{var } a_{N+1} = \int (a_{N+1} - \mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}_{N+1})^2 p(a_{N+1}) da_{N+1}. \quad (4.197)$$

The right hand side can be written as

$$\int ((\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \boldsymbol{\phi}_{N+1})^2 p(a_{N+1}) da_{N+1} = \boldsymbol{\phi}_{N+1}^\top \mathbf{H}_{\text{MAP}}^{-1} \boldsymbol{\phi}_{N+1}. \quad (4.198)$$

Then,

$$p(a_{N+1}) = \mathcal{N}(a_{N+1} | \mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}_{N+1}, \boldsymbol{\phi}_{N+1}^\top \mathbf{H}_{\text{MAP}}^{-1} \boldsymbol{\phi}_{N+1}). \quad (4.199)$$

Therefore,

$$p(t_{N+1} | \mathbf{t}) = v_{N+1}^{t_{N+1}} (1 - v_{N+1})^{1-t_{N+1}}, \quad (4.200)$$

where

$$v_{N+1} = \int \sigma(a) \mathcal{N}(a | \mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}_{N+1}, \boldsymbol{\phi}_{N+1}^\top \mathbf{H}_{\text{MAP}}^{-1} \boldsymbol{\phi}_{N+1}) da. \quad (4.201)$$

4.25

Let λ be a constant such that

$$\left. \frac{d\sigma(a)}{da} \right|_{a=0} = \left. \frac{d\Phi(\lambda a)}{da} \right|_{a=0}, \quad (4.202)$$

where

$$\begin{aligned} \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ \Phi(a) &= \int_{-\infty}^a \mathcal{N}(\theta | 0, 1) d\theta. \end{aligned} \quad (4.203)$$

By 4.12,

$$\frac{d\sigma(a)}{da} = \sigma(a) (1 - \sigma(a)). \quad (4.204)$$

We have

$$\frac{d\Phi(\lambda a)}{da} = \lambda \mathcal{N}(\lambda a | 0, 1). \quad (4.205)$$

Then,

$$\frac{1}{4} = \lambda (2\pi)^{-\frac{1}{2}}. \quad (4.206)$$

Therefore,

$$\lambda^2 = \frac{\pi}{8}. \quad (4.207)$$

4.26

Let

$$I(\mu) = \int \Phi(\lambda a) \mathcal{N}(a|\mu, \sigma^2) da, \quad (4.208)$$

where

$$\Phi(a) = \int_{-\infty}^a \mathcal{N}(\theta|0, 1) d\theta. \quad (4.209)$$

By the transformation

$$z = \frac{\theta - \mu}{\sigma}, \quad (4.210)$$

the right hand side can be written as

$$\begin{aligned} & \int \Phi(\lambda(\mu + \sigma z)) \mathcal{N}(\mu + \sigma z|\mu, \sigma^2) \sigma dz \\ &= \int \Phi(\lambda(\mu + \sigma z)) \mathcal{N}(z|0, 1) dz. \end{aligned} \quad (4.211)$$

Then,

$$\frac{\partial}{\partial \mu} I(\mu) = \lambda \int \mathcal{N}(\lambda(\mu + \sigma z)|0, 1) \mathcal{N}(z|0, 1) dz. \quad (4.212)$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} & -\frac{1}{2} \ln(2\pi) - \frac{\lambda^2(\mu + \sigma z)^2}{2} - \frac{1}{2} \ln(2\pi) - \frac{z^2}{2} \\ &= -\ln(2\pi) - \frac{1 + \sigma^2 \lambda^2}{2} \left(z + \frac{\mu \sigma \lambda^2}{1 + \sigma^2 \lambda^2} \right)^2 + \frac{\mu^2 \sigma^2 \lambda^4}{2(1 + \sigma^2 \lambda^2)} - \frac{\mu^2 \lambda^2}{2}. \end{aligned} \quad (4.213)$$

The right hand side can be written as

$$\begin{aligned} & -\ln(2\pi) - \frac{1 + \sigma^2 \lambda^2}{2} \left(z + \frac{\mu \sigma \lambda^2}{1 + \sigma^2 \lambda^2} \right)^2 - \frac{\mu^2 \lambda^2}{2(1 + \sigma^2 \lambda^2)} \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(1 + \sigma^2 \lambda^2)^{-1} - \frac{1 + \sigma^2 \lambda^2}{2} \left(z + \frac{\mu \sigma \lambda^2}{1 + \sigma^2 \lambda^2} \right)^2 \\ & \quad - \ln \lambda - \frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\lambda^{-2} + \sigma^2) - \frac{\mu^2}{2(\lambda^{-2} + \sigma^2)}. \end{aligned} \quad (4.214)$$

Then, the integral can be written as

$$\begin{aligned} & \frac{1}{\lambda} \mathcal{N}(\mu|0, \lambda^{-2} + \sigma^2) \int \mathcal{N}\left(z \middle| -\frac{\mu\sigma\lambda^2}{1 + \sigma^2\lambda^2}, (1 + \sigma^2\lambda^2)^{-1}\right) dz \\ &= \frac{1}{\lambda} \mathcal{N}(\mu|0, \lambda^{-2} + \sigma^2). \end{aligned} \quad (4.215)$$

Then,

$$\frac{\partial}{\partial \mu} I(\mu) = \mathcal{N}(\mu|0, \lambda^{-2} + \sigma^2), \quad (4.216)$$

so that

$$I(\mu) = \int_{-\infty}^{\mu} \mathcal{N}(m|0, \lambda^{-2} + \sigma^2) dm. \quad (4.217)$$

By the transformation

$$m' = (\lambda^{-2} + \sigma^2)^{-\frac{1}{2}} m, \quad (4.218)$$

the right hand side can be written as

$$\begin{aligned} & \int_{-\infty}^{(\lambda^{-2} + \sigma^2)^{-\frac{1}{2}} \mu} (\lambda^{-2} + \sigma^2)^{-\frac{1}{2}} \mathcal{N}(m'|0, 1) (\lambda^{-2} + \sigma^2)^{\frac{1}{2}} dm' \\ &= \int_{-\infty}^{(\lambda^{-2} + \sigma^2)^{-\frac{1}{2}} \mu} \mathcal{N}(m'|0, 1) dm'. \end{aligned} \quad (4.219)$$

Therefore,

$$I(\mu) = \Phi\left((\lambda^{-2} + \sigma^2)^{-\frac{1}{2}} \mu\right). \quad (4.220)$$

5 Neural Networks

5.1

Let

$$y_k(\mathbf{x}, \mathbf{w}) = \sigma \left(\sum_{m=1}^M w_{km}^{(2)} \sigma \left(\sum_{d=1}^D w_{md}^{(1)} x_d + w_{m0}^{(1)} \right) + w_{k0}^{(2)} \right), \quad (5.1)$$

where

$$\sigma(a) = \frac{1}{1 + \exp(-a)}. \quad (5.2)$$

We have

$$\sigma(a) = \frac{\exp\left(\frac{a}{2}\right)}{\exp\left(\frac{a}{2}\right) + \exp\left(-\frac{a}{2}\right)}. \quad (5.3)$$

The right hand side can be written as

$$\frac{\exp\left(\frac{a}{2}\right) - \exp\left(-\frac{a}{2}\right)}{\exp\left(\frac{a}{2}\right) + \exp\left(-\frac{a}{2}\right)} + \frac{\exp\left(-\frac{a}{2}\right)}{\exp\left(\frac{a}{2}\right) + \exp\left(-\frac{a}{2}\right)} = \tanh\left(\frac{a}{2}\right) + 1 - \sigma(a). \quad (5.4)$$

Then,

$$\sigma(a) = \frac{1}{2} \left(1 + \tanh\left(\frac{a}{2}\right) \right). \quad (5.5)$$

Then, the argument of the expression of $y_k(\mathbf{x}, \mathbf{w})$ can be written as

$$\begin{aligned} & \sum_{m=1}^M w_{km}^{(2)} \left(\frac{1}{2} \left(1 + \tanh \left(\frac{1}{2} \left(\sum_{d=1}^D w_{md}^{(1)} x_d + w_{m0}^{(1)} \right) \right) \right) \right) + w_{k0}^{(2)} \\ &= \frac{1}{2} \sum_{m=1}^M w_{km}^{(2)} \tanh \left(\frac{1}{2} \sum_{d=1}^D w_{md}^{(1)} x_d + \frac{1}{2} w_{m0}^{(1)} \right) + \frac{1}{2} \sum_{m=1}^M w_{km}^{(2)} + w_{k0}^{(2)}. \end{aligned} \quad (5.6)$$

Therefore,

$$\begin{aligned} & y_k(\mathbf{x}, \mathbf{w}) \\ &= \sigma \left(\frac{1}{2} \sum_{m=1}^M w_{km}^{(2)} \tanh \left(\frac{1}{2} \sum_{d=1}^D w_{md}^{(1)} x_d + \frac{1}{2} w_{m0}^{(1)} \right) + \frac{1}{2} \sum_{m=1}^M w_{km}^{(2)} + w_{k0}^{(2)} \right). \end{aligned} \quad (5.7)$$

5.2

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{w}) = \mathcal{N}(\mathbf{t}_n|\mathbf{y}_n, \beta^{-1}\mathbf{I}), \quad (5.8)$$

where

$$\mathbf{y}_n = \mathbf{y}(\mathbf{x}_n, \mathbf{w}). \quad (5.9)$$

Then, the logarithm of $p(\mathbf{T}|\mathbf{w})$ except the terms independent of $\mathbf{w}$ can be written as

$$-\frac{\beta}{2} \sum_{n=1}^N \|\mathbf{t}_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\|^2. \quad (5.10)$$

Therefore, maximising $p(\mathbf{T}|\mathbf{w})$ is equivalent to minimising $E(\mathbf{w})$, where

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \|\mathbf{y}(\mathbf{x}_n, \mathbf{w}) - \mathbf{t}_n\|^2. \quad (5.11)$$

5.3

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n|\mathbf{w}) = \mathcal{N}(\mathbf{t}_n|\mathbf{y}_n, \mathbf{\Sigma}), \quad (5.12)$$

where

$$\mathbf{y}_n = \mathbf{y}(\mathbf{x}_n, \mathbf{w}). \quad (5.13)$$

(a)

The logarithm of $\ln p(\mathbf{T}|\mathbf{w})$ except the terms independent of $\mathbf{\Sigma}$ can be written as

$$l = -\frac{N}{2} \ln(\det \mathbf{\Sigma}) - \frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - \mathbf{y}_n)^\top \mathbf{\Sigma}^{-1} (\mathbf{t}_n - \mathbf{y}_n). \quad (5.14)$$

By 3.21, setting the derivative of l with respect to $\mathbf{\Sigma}$ to zero gives

$$\mathbf{O} = -\frac{N}{2} \mathbf{\Sigma}^{-1} + \frac{1}{2} (\mathbf{\Sigma}^{-1})^2 \sum_{n=1}^N (\mathbf{t}_n - \mathbf{y}_n) (\mathbf{t}_n - \mathbf{y}_n)^\top. \quad (5.15)$$

Therefore, the maximum likelihood solution to $\mathbf{\Sigma}$ is given by

$$\mathbf{\Sigma}_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N (\mathbf{t}_n - \mathbf{y}_n) (\mathbf{t}_n - \mathbf{y}_n)^\top. \quad (5.16)$$

(b)

If Σ is fixed and known, then the maximum likelihood solution to $\mathbf{w}$ is given by minimising $E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - \mathbf{y}_n)^\top \Sigma^{-1} (\mathbf{t}_n - \mathbf{y}_n). \quad (5.17)$$

5.4

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n = 1 | \mathbf{w}) = (1 - \epsilon)y_n + \epsilon(1 - y_n), \quad (5.18)$$

where

$$y_n = y(\mathbf{x}_n, \mathbf{w}). \quad (5.19)$$

Then,

$$p(t_n | \mathbf{w}) = ((1 - \epsilon)y_n + \epsilon(1 - y_n))^{t_n} (\epsilon y_n + (1 - \epsilon)(1 - y_n))^{1-t_n}. \quad (5.20)$$

Therefore,

$$\begin{aligned} -\ln p(\mathbf{t} | \mathbf{w}) &= -\sum_{n=1}^N t_n \ln((1 - \epsilon)y_n + \epsilon(1 - y_n)) \\ &\quad -\sum_{n=1}^N (1 - t_n) \ln(\epsilon y_n + (1 - \epsilon)(1 - y_n)). \end{aligned} \quad (5.21)$$

5.5

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n | \mathbf{w}) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \quad (5.22)$$

where

$$y_{nk} = y_k(\mathbf{x}_n, \mathbf{w}). \quad (5.23)$$

Therefore,

$$\ln p(\mathbf{T} | \mathbf{w}) = \sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk}. \quad (5.24)$$

5.6

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n|\mathbf{w}) = y_n^{t_n}(1 - y_n)^{1-t_n}, \quad (5.25)$$

where

$$\begin{aligned} y_n &= \sigma(a_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ a_n &= a(\mathbf{x}_n, \mathbf{w}). \end{aligned} \quad (5.26)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \quad (5.27)$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^N (t_n \ln y_n + (1 - t_n) \ln(1 - y_n)). \quad (5.28)$$

Then,

$$\frac{\partial E(\mathbf{w})}{\partial a_n} = -y_n(1 - y_n) \left(\frac{t_n}{y_n} - \frac{1 - t_n}{1 - y_n} \right). \quad (5.29)$$

Therefore,

$$\frac{\partial E(\mathbf{w})}{\partial a_n} = y_n - t_n. \quad (5.30)$$

5.7

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \quad (5.31)$$

where

$$\begin{aligned} y_{nk} &= y_k(\mathbf{a}_n), \\ y_k(\mathbf{a}) &= \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})}, \\ \mathbf{a}_n &= \mathbf{a}(\mathbf{x}_n, \mathbf{w}). \end{aligned} \quad (5.32)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{T}|\mathbf{w}). \quad (5.33)$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk}. \quad (5.34)$$

Then,

$$\frac{\partial E(\mathbf{w})}{\partial a_{nk'}} = -\sum_{k=1}^K \frac{t_{nk}}{y_{nk}} \frac{\partial y_{nk}}{\partial a_{nk'}}. \quad (5.35)$$

By 4.17,

$$\frac{\partial y_{nk}}{\partial a_{nk'}} = y_{nk}(I_{kk'} - y_{nk'}). \quad (5.36)$$

Then,

$$\frac{\partial E(\mathbf{w})}{\partial a_{nk'}} = \sum_{k=1}^K t_{nk}(y_{nk'} - I_{kk'}). \quad (5.37)$$

The right hand side can be written as

$$y_{nk'} \sum_{k=1}^K t_{nk} - t_{nk'} = y_{nk'} - t_{nk'}. \quad (5.38)$$

Therefore,

$$\frac{\partial E(\mathbf{w})}{\partial a_{nk}} = y_{nk} - t_{nk}. \quad (5.39)$$

5.8

We have

$$\tanh a = \frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)}. \quad (5.40)$$

Then,

$$\frac{d}{da} \tanh a = 1 - \left(\frac{\exp(a) - \exp(-a)}{\exp(a) + \exp(-a)} \right)^2. \quad (5.41)$$

Therefore,

$$\frac{d}{da} \tanh a = 1 - (\tanh a)^2. \quad (5.42)$$

5.9

Let $t_1, \dots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n|\mathbf{w}) = \left(\frac{1+y_n}{2}\right)^{\frac{1+t_n}{2}} \left(\frac{1-y_n}{2}\right)^{\frac{1-t_n}{2}}, \quad (5.43)$$

where

$$\begin{aligned} y_n &= y(\mathbf{x}_n, \mathbf{w}), \\ -1 &\leq y(\mathbf{x}_n, \mathbf{w}) \leq 1. \end{aligned} \quad (5.44)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}). \quad (5.45)$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^N \left(\frac{1+t_n}{2} \ln \frac{1+y_n}{2} + \frac{1-t_n}{2} \ln \frac{1-y_n}{2} \right). \quad (5.46)$$

The appropriate choice of y is $\tanh$.

5.10

Let

$$\mathbf{H} = \nabla \nabla E(\mathbf{w}), \quad (5.47)$$

where E is a real function of real vectors. Then, $\mathbf{H}$ is a real symmetric matrix. Therefore, by 2.20, $\mathbf{H}$ is positive definite if and only if all of its eigenvalues are positive.

5.11

Let $\mathbf{w}$ be a real vector in M dimensions. We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2} (\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}^* (\mathbf{w} - \mathbf{w}^*) + O(\|\mathbf{w} - \mathbf{w}^*\|^3), \quad (5.48)$$

where $\mathbf{w}^*$ is a stationary point of E and

$$\mathbf{H}^* = \nabla \nabla E(\mathbf{w}^*). \quad (5.49)$$

Let $\mathbf{u}_1, \dots, \mathbf{u}_M$ be eigenvectors of $\mathbf{H}^*$ such that

$$\mathbf{H}^* \mathbf{u}_m = \lambda_m \mathbf{u}_m. \quad (5.50)$$

Then, there exists $\alpha_1, \dots, \alpha_M$ such that

$$\mathbf{w} - \mathbf{w}^* = \sum_{m=1}^M \alpha_m \mathbf{u}_m. \quad (5.51)$$

By 2.19,

$$\mathbf{u}_m^\top \mathbf{u}_{m'} = I_{mm'}. \quad (5.52)$$

Then,

$$E(\mathbf{w}) \simeq E(\mathbf{w}^*) + \frac{1}{2} \sum_{m=1}^M \lambda_m \alpha_m^2. \quad (5.53)$$

Therefore, the contours of constant E are ellipses whose axes are aligned with $\mathbf{u}_1, \dots, \mathbf{u}_M$ with lengths which are proportional to $\lambda_1^{-\frac{1}{2}}, \dots, \lambda_M^{-\frac{1}{2}}$.

5.12

Let $\mathbf{w}$ be a real vector. We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2} (\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}^* (\mathbf{w} - \mathbf{w}^*) + O(\|\mathbf{w} - \mathbf{w}^*\|^3), \quad (5.54)$$

where $\mathbf{w}^*$ is a stationary point of E and

$$\mathbf{H}^* = \nabla \nabla E(\mathbf{w}^*). \quad (5.55)$$

If $\mathbf{H}^*$ is positive definite, then the second term of the right hand will be non-negative. Then, $\mathbf{w}^*$ is a local minimum of E . On the other hand, if $\mathbf{w}^*$ is a local minimum of E , then the second term of the right hand side will be non-negative. Then, $\mathbf{H}^*$ is positive definite. Therefore, the necessary and sufficient condition for $\mathbf{w}^*$ to be a local minimum is that $\mathbf{H}^*$ is positive definite.

5.13

Let $\mathbf{w}$ be a vector in M dimensions. We have the Taylor series

$$E(\mathbf{w}) = E(\hat{\mathbf{w}}) + \hat{\mathbf{g}}^\top (\mathbf{w} - \hat{\mathbf{w}}) + \frac{1}{2} (\mathbf{w} - \hat{\mathbf{w}})^\top \hat{\mathbf{H}} (\mathbf{w} - \hat{\mathbf{w}}) + O(\|\mathbf{w} - \hat{\mathbf{w}}\|^3), \quad (5.56)$$

where

$$\begin{aligned} \hat{\mathbf{g}} &= \nabla E(\hat{\mathbf{w}}), \\ \hat{\mathbf{H}} &= \nabla \nabla E(\hat{\mathbf{w}}). \end{aligned} \quad (5.57)$$

Since $\hat{\mathbf{g}}$ is a vector in M dimensions and $\hat{\mathbf{H}}$ is a $M \times M$ symmetric matrix, the number of independent elements of the right hand side is

$$M + \frac{M(M+1)}{2} = \frac{M(M+3)}{2}. \quad (5.58)$$

5.14

We have the Taylor series

$$\begin{aligned} E_n(w_{mm'} + \epsilon) &= E_n(w_{mm'}) + \left. \frac{\partial E_n}{\partial w} \right|_{w=w_{mm'}} \epsilon + O(\epsilon^2), \\ E_n(w_{mm'} - \epsilon) &= E_n(w_{mm'}) - \left. \frac{\partial E_n}{\partial w} \right|_{w=w_{mm'}} \epsilon + O(\epsilon^2). \end{aligned} \quad (5.59)$$

Therefore,

$$\left. \frac{\partial E_n}{\partial w} \right|_{w=w_{mm'}} = \frac{E_n(w_{mm'} + \epsilon) - E_n(w_{mm'} - \epsilon)}{2\epsilon} + O(\epsilon^2). \quad (5.60)$$

5.15 (Incomplete)

Let

$$J_{km} = \frac{\partial y_k}{\partial x_m}. \quad (5.61)$$

The right hand side can be written as

$$\sum_{m'=1}^M W_{mm'} \frac{\partial y_k}{\partial a_{m'}}, \quad (5.62)$$

where

$$W_{mm'} = \frac{\partial a_{m'}}{\partial x_m}. \quad (5.63)$$

We have

$$\frac{\partial y_k}{\partial a_{m'}} = \sum_{m''=1}^M \frac{\partial y_k}{\partial a_{m''}} \frac{\partial a_{m''}}{\partial a_{m'}}. \quad (5.64)$$

5.16

Let $\mathbf{y}_1, \dots, \mathbf{y}_N$ be vectors dependent on $\mathbf{w}$. Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \|\mathbf{y}_n - \mathbf{t}_n\|^2. \quad (5.65)$$

Then,

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N (\nabla \mathbf{y}_n)^\top (\mathbf{y}_n - \mathbf{t}_n). \quad (5.66)$$

Therefore,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N (\nabla \text{vec}(\nabla \mathbf{y}_n)^\top)^\top (\mathbf{y}_n - \mathbf{t}_n) + \sum_{n=1}^N (\nabla \mathbf{y}_n)^\top (\nabla \mathbf{y}_n). \quad (5.67)$$

5.17

Let y be a scalar dependent on $\mathbf{x}$ and $\mathbf{w}$. Let

$$E(\mathbf{w}) = \frac{1}{2} \iint (y - t)^2 p(\mathbf{x}, t) d\mathbf{x} dt. \quad (5.68)$$

Then,

$$\nabla E(\mathbf{w}) = \iint (y - t) p(\mathbf{x}, t) \nabla y d\mathbf{x} dt. \quad (5.69)$$

The right hand side can be written as

$$\begin{aligned} & \int y \nabla y \left(\int p(\mathbf{x}, t) dt \right) d\mathbf{x} - \int \nabla y \left(\int t p(t|\mathbf{x}) dt \right) p(\mathbf{x}) d\mathbf{x} \\ &= \int y \nabla y p(\mathbf{x}) d\mathbf{x} - \int \nabla y E(t|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \end{aligned} \quad (5.70)$$

Then,

$$\begin{aligned}\nabla\nabla E(\mathbf{w}) &= \int \nabla y (\nabla y)^\top p(\mathbf{x}) d\mathbf{x} + \int y \nabla\nabla y p(\mathbf{x}) d\mathbf{x} \\ &\quad - \int \nabla\nabla y E(t|\mathbf{x}) p(\mathbf{x}) d\mathbf{x}.\end{aligned}\tag{5.71}$$

The second and the third terms of the right hand side can be written as

$$E(y \nabla\nabla y) - E(\nabla\nabla y E(t|\mathbf{x})) = E((y - E(t|\mathbf{x})) \nabla\nabla y).\tag{5.72}$$

Therefore, if

$$y = E(t|\mathbf{x}),\tag{5.73}$$

then

$$\nabla\nabla E(\mathbf{w}) = \int \nabla y (\nabla y)^\top p(\mathbf{x}) d\mathbf{x}.\tag{5.74}$$

5.18

Let $y_1, \dots, y_N$ be scalars such that

$$\begin{aligned}y_n &= \mathbf{w}_n^{(2)\top} \mathbf{z} + \mathbf{w}_n^{(0)\top} \mathbf{x}, \\ z_m &= \tanh\left(\mathbf{w}_m^{(1)\top} \mathbf{x}\right).\end{aligned}\tag{5.75}$$

Let

$$E = \frac{1}{2} \sum_{n=1}^N (y_n - t_n)^2.\tag{5.76}$$

Then,

$$\begin{aligned}\frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= \frac{\partial E}{\partial y_n} \frac{\partial y_n}{\partial \mathbf{w}_n^{(0)}}, \\ \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= \frac{\partial E}{\partial y_n} \frac{\partial y_n}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial \mathbf{w}_m^{(1)}}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial y_n} \frac{\partial y_n}{\partial \mathbf{w}_n^{(2)}}.\end{aligned}\tag{5.77}$$

Therefore,

$$\begin{aligned}\frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= (y_n - t_n) \mathbf{x}, \\ \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= (y_n - t_n) \mathbf{A} \mathbf{w}_n^{(2)}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= (y_n - t_n) \mathbf{z},\end{aligned}\tag{5.78}$$

where

$$A_{mm'} = (1 - z_m^2) x_{m'}. \quad (5.79)$$

5.19

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$p(t_n | \mathbf{w}) = y_n^{t_n} (1 - y_n)^{1-t_n}, \quad (5.80)$$

where

$$\begin{aligned} y_n &= \sigma(a_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ a_n &= a(\mathbf{x}_n, \mathbf{w}). \end{aligned} \quad (5.81)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{t} | \mathbf{w}). \quad (5.82)$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^N (t_n \ln y_n + (1 - t_n) \ln(1 - y_n)). \quad (5.83)$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^N \left(\frac{t_n}{y_n} - \frac{1 - t_n}{1 - y_n} \right) \nabla y_n. \quad (5.84)$$

We have

$$\nabla y_n = \frac{\partial y_n}{\partial a_n} \nabla a_n. \quad (5.85)$$

Then,

$$\nabla y_n = y_n(1 - y_n) \nabla a_n. \quad (5.86)$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^N (t_n(1 - y_n) - (1 - t_n)y_n) \nabla a_n, \quad (5.87)$$

so that

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N (y_n - t_n) \nabla a_n. \quad (5.88)$$

Then,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N \nabla a_n (\nabla y_n)^\top + \sum_{n=1}^N (y_n - t_n) \nabla \nabla a_n. \quad (5.89)$$

Therefore,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N y_n (1 - y_n) \nabla a_n (\nabla a_n)^\top + \sum_{n=1}^N (y_n - t_n) \nabla \nabla a_n. \quad (5.90)$$

5.20

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n | \mathbf{w}) = \prod_{k=1}^K y_{nk}^{t_{nk}}, \quad (5.91)$$

where

$$\begin{aligned} y_{nk} &= y_k(\mathbf{a}_n), \\ y_k(\mathbf{a}) &= \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})}, \\ \mathbf{a}_n &= \mathbf{a}(\mathbf{x}_n, \mathbf{w}). \end{aligned} \quad (5.92)$$

Let

$$E(\mathbf{w}) = -\ln p(\mathbf{T} | \mathbf{w}). \quad (5.93)$$

Then,

$$E(\mathbf{w}) = -\sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk}. \quad (5.94)$$

Then,

$$\nabla E(\mathbf{w}) = -\sum_{n=1}^N \sum_{k=1}^K \frac{t_{nk}}{y_{nk}} \nabla y_{nk}. \quad (5.95)$$

We have

$$\nabla y_{nk} = \sum_{k'=1}^K \frac{\partial y_{nk}}{\partial a_{nk'}} \nabla a_{nk'}. \quad (5.96)$$

By 4.17,

$$\frac{\partial y_{nk}}{\partial a_{nk'}} = y_{nk} (I_{kk'} - y_{nk'}). \quad (5.97)$$

Then,

$$\nabla y_{nk} = y_{nk} \sum_{k'=1}^K (I_{kk'} - y_{nk'}) \nabla a_{nk'}. \quad (5.98)$$

Then,

$$\nabla E(\mathbf{w}) = - \sum_{n=1}^N \sum_{k=1}^K t_{nk} \sum_{k'=1}^K (I_{kk'} - y_{nk'}) \nabla a_{nk}, \quad (5.99)$$

so that

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N \sum_{k=1}^K (y_{nk} - t_{nk}) \nabla a_{nk}. \quad (5.100)$$

Then,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N \sum_{k=1}^K \nabla a_{nk} (\nabla y_{nk})^\top + \sum_{n=1}^N \sum_{k=1}^K (y_{nk} - t_{nk}) \nabla \nabla a_{nk}. \quad (5.101)$$

Therefore,

$$\begin{aligned} \nabla \nabla E(\mathbf{w}) &= \sum_{n=1}^N \sum_{k=1}^K y_{nk} \nabla a_{nk} \sum_{k'=1}^K (I_{kk'} - y_{nk'}) (\nabla a_{nk'})^\top \\ &\quad + \sum_{n=1}^N \sum_{k=1}^K (y_{nk} - t_{nk}) \nabla \nabla a_{nk}. \end{aligned} \quad (5.102)$$

5.21

Let $\mathbf{y}_1, \dots, \mathbf{y}_N$ be vectors dependent on $\mathbf{w}$. Let

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \|\mathbf{y}_n - \mathbf{t}_n\|^2. \quad (5.103)$$

By 5.16,

$$\nabla \nabla E(\mathbf{w}) = \sum_{n=1}^N (\nabla \text{vec}(\nabla \mathbf{y}_n)^\top)^\top (\mathbf{y}_n - \mathbf{t}_n) + \mathbf{S}, \quad (5.104)$$

where

$$\mathbf{S} = \sum_{n=1}^N (\nabla \mathbf{y}_n)^\top (\nabla \mathbf{y}_n). \quad (5.105)$$

Let

$$\mathbf{S}' = \sum_{n=1}^{N+1} (\nabla \mathbf{y}_n)^\top (\nabla \mathbf{y}_n). \quad (5.106)$$

Then,

$$\mathbf{S}' = \mathbf{S} + \mathbf{B}_{N+1}^\top \mathbf{B}_{N+1}, \quad (5.107)$$

where

$$\mathbf{B}_{N+1} = \nabla \mathbf{y}_{N+1}. \quad (5.108)$$

Therefore, by 2.26,

$$\mathbf{S}'^{-1} = \mathbf{S}^{-1} - \mathbf{S}^{-1} \mathbf{B}_{N+1}^\top \mathbf{M} \mathbf{B}_{N+1} \mathbf{S}^{-1}, \quad (5.109)$$

where

$$\mathbf{M} = (\mathbf{I} + \mathbf{B}_{N+1} \mathbf{S}^{-1} \mathbf{B}_{N+1}^\top)^{-1}. \quad (5.110)$$

5.22

Let $a_1, \dots, a_N$ be variables such that

$$\begin{aligned} a_n &= \mathbf{w}_n^{(2)\top} \mathbf{z}, \\ z_m &= h(b_m), \\ b_m &= \mathbf{w}_m^{(1)\top} \mathbf{x}. \end{aligned} \quad (5.111)$$

Let E be a function of $a_1, \dots, a_N$. Then,

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= \sum_{n=1}^N \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial z_m} \frac{\partial z_m}{\partial b_m} \frac{\partial b_m}{\partial \mathbf{w}_m^{(1)}}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial \mathbf{w}_n^{(2)}}. \end{aligned} \quad (5.112)$$

Then,

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= h'(b_m) \mathbf{x} \sum_{n=1}^N w_{nm}^{(2)} \frac{\partial E}{\partial a_n}, \\ \frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \mathbf{z}. \end{aligned} \quad (5.113)$$

Therefore,

$$\begin{aligned}
\frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)} \partial \mathbf{w}_{m'}^{(1)}} &= C_{mm'}^{(11)} \mathbf{x} \mathbf{x}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)} \partial \mathbf{w}_n^{(2)}} &= C_{mn}^{(12)} \mathbf{x} \mathbf{z}^\top + h'(b_m) \frac{\partial E}{\partial a_n} \mathbf{x} \mathbf{v}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_n^{(2)} \partial \mathbf{w}_{n'}^{(2)}} &= C_{nn'}^{(22)} \mathbf{z} \mathbf{z}^\top,
\end{aligned} \tag{5.114}$$

where $\mathbf{v}$ is a vector from the standard basis in M dimensions such that

$$v_m = 1, \tag{5.115}$$

and

$$\begin{aligned}
C_{mm'}^{(11)} &= h''(b_m) I_{mm'} \sum_{n=1}^N w_{nm}^{(2)} \frac{\partial E}{\partial a_n} \\
&\quad + h'(b_m) h'(b_{m'}) \sum_{n=1}^N w_{nm}^{(2)} \sum_{n'=1}^N w_{n'm'}^{(2)} \frac{\partial^2 E}{\partial y_n \partial a_{n'}}, \\
C_{mn}^{(12)} &= h'(b_m) \sum_{n'=1}^N w_{n'm}^{(2)} \frac{\partial^2 E}{\partial a_n \partial a_{n'}}, \\
C_{nn'}^{(22)} &= \frac{\partial^2 E}{\partial a_n \partial a_{n'}}.
\end{aligned} \tag{5.116}$$

5.23

Let $a_1, \dots, a_N$ be variables such that

$$\begin{aligned}
a_n &= \mathbf{w}_n^{(2)\top} \mathbf{z} + \mathbf{w}_n^{(0)\top} \mathbf{x}, \\
z_m &= h(b_m), \\
b_m &= \mathbf{w}_m^{(1)\top} \mathbf{x}.
\end{aligned} \tag{5.117}$$

Let E be a function of $a_1, \dots, a_N$. Then,

$$\begin{aligned}
\frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial \mathbf{w}_n^{(0)}}, \\
\frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= \sum_{n=1}^N \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial z_m} \frac{\partial z_m}{\partial b_m} \frac{\partial b_m}{\partial \mathbf{w}_m^{(1)}}, \\
\frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \frac{\partial a_n}{\partial \mathbf{w}_n^{(2)}}.
\end{aligned} \tag{5.118}$$

Then,

$$\begin{aligned}
\frac{\partial E}{\partial \mathbf{w}_n^{(0)}} &= \frac{\partial E}{\partial a_n} \mathbf{x}, \\
\frac{\partial E}{\partial \mathbf{w}_m^{(1)}} &= h'(b_m) \mathbf{x} \sum_{n=1}^N w_{nm}^{(2)} \frac{\partial E}{\partial a_n}, \\
\frac{\partial E}{\partial \mathbf{w}_n^{(2)}} &= \frac{\partial E}{\partial a_n} \mathbf{z}.
\end{aligned} \tag{5.119}$$

Therefore,

$$\begin{aligned}
\frac{\partial^2 E}{\partial \mathbf{w}_n^{(0)} \partial \mathbf{w}_{n'}^{(0)}} &= C_{nn'}^{(22)} \mathbf{x} \mathbf{x}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_n^{(0)} \partial \mathbf{w}_m^{(1)}} &= C_{nm}^{(12)} \mathbf{x} \mathbf{x}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_n^{(0)} \partial \mathbf{w}_{n'}^{(2)}} &= C_{nn'}^{(22)} \mathbf{x} \mathbf{z}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)} \partial \mathbf{w}_{m'}^{(1)}} &= C_{mm'}^{(11)} \mathbf{x} \mathbf{x}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_m^{(1)} \partial \mathbf{w}_n^{(2)}} &= C_{mn}^{(12)} \mathbf{x} \mathbf{z}^\top + h'(b_m) \frac{\partial E}{\partial a_n} \mathbf{x} \mathbf{v}^\top, \\
\frac{\partial^2 E}{\partial \mathbf{w}_n^{(2)} \partial \mathbf{w}_{n'}^{(2)}} &= C_{nn'}^{(22)} \mathbf{z} \mathbf{z}^\top,
\end{aligned} \tag{5.120}$$

where $\mathbf{v}$ is a vector from the standard basis in M dimensions such that

$$v_m = 1, \tag{5.121}$$

and

$$\begin{aligned}
C_{mm'}^{(11)} &= h''(b_m) I_{mm'} \sum_{n=1}^N w_{nm}^{(2)} \frac{\partial E}{\partial a_n} \\
&\quad + h'(b_m) h'(b_{m'}) \sum_{n=1}^N w_{nm}^{(2)} \sum_{n'=1}^N w_{n'm'}^{(2)} \frac{\partial^2 E}{\partial a_n \partial a_{n'}}, \\
C_{nm}^{(12)} &= h'(b_m) \sum_{n'=1}^N w_{n'm}^{(2)} \frac{\partial^2 E}{\partial a_n \partial a_{n'}}, \\
C_{nn'}^{(22)} &= \frac{\partial^2 E}{\partial a_n \partial a_{n'}}.
\end{aligned} \tag{5.122}$$

5.24

Let $y_1, \dots, y_N$ be variables such that

$$\begin{aligned} y_n &= \mathbf{w}_n^\top \mathbf{z} + w_{n0}, \\ z_m &= h(\mathbf{w}_m^\top \mathbf{x} + w_{m0}). \end{aligned} \quad (5.123)$$

(a)

Let

$$\tilde{\mathbf{x}} = a\mathbf{x} + b\mathbf{v}, \quad (5.124)$$

where

$$\mathbf{v} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Then,

$$z_m = h\left(\frac{1}{a}\mathbf{w}_m^\top (\tilde{\mathbf{x}} - b\mathbf{v}) + w_{m0}\right). \quad (5.125)$$

Therefore,

$$\tilde{z}_m = h(\tilde{\mathbf{w}}_m^\top \mathbf{x} + \tilde{w}_{m0}), \quad (5.126)$$

where

$$\begin{aligned} \tilde{\mathbf{w}}_m &= \frac{1}{a}\mathbf{w}_m, \\ \tilde{w}_{m0} &= w_{m0} - \frac{b}{a}\mathbf{w}_m^\top \mathbf{v}. \end{aligned} \quad (5.127)$$

(b)

Let

$$\tilde{y}_n = cy_n + d. \quad (5.128)$$

Then,

$$\frac{\tilde{y}_n - d}{c} = \mathbf{w}_n^\top \mathbf{z} + w_{n0}. \quad (5.129)$$

Therefore,

$$\tilde{y}_n = \tilde{\mathbf{w}}_n^\top \mathbf{z} + \tilde{w}_{n0}, \quad (5.130)$$

where

$$\begin{aligned} \tilde{\mathbf{w}}_n &= c\mathbf{w}_n, \\ \tilde{w}_{n0} &= cw_{n0} + d. \end{aligned} \quad (5.131)$$

5.25

Let E be a quadratic error function such that

$$E(\mathbf{w}) = E_0 + \frac{1}{2} (\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}^* (\mathbf{w} - \mathbf{w}^*), \quad (5.132)$$

where $\mathbf{w}^*$ is a stationary point of E and $\mathbf{H}^*$ is a positive definite and constant matrix whose eigenvectors and eigenvalues are $\mathbf{u}_1, \dots, \mathbf{u}_M$ and $\eta_1, \dots, \eta_M$. Let

$$\begin{aligned} \mathbf{w}^{(\tau)} &= \mathbf{w}^{(\tau-1)} - \rho \nabla E, \\ \mathbf{w}^{(0)} &= \mathbf{0}. \end{aligned} \quad (5.133)$$

(a)

We have

$$\nabla E = \mathbf{H}^* (\mathbf{w} - \mathbf{w}^*). \quad (5.134)$$

Then,

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \rho \mathbf{H}^* (\mathbf{w}^{(\tau-1)} - \mathbf{w}^*). \quad (5.135)$$

so that

$$\mathbf{u}_m^\top \mathbf{w}^{(\tau)} = \mathbf{u}_m^\top \mathbf{w}^{(\tau-1)} - \rho \mathbf{u}_m^\top \mathbf{H}^* (\mathbf{w}^{(\tau-1)} - \mathbf{w}^*). \quad (5.136)$$

Since $\mathbf{H}^*$ is symmetric,

$$\mathbf{u}_m^\top \mathbf{H}^* \mathbf{w} = \mathbf{w}^\top \mathbf{H}^* \mathbf{u}_m. \quad (5.137)$$

The right hand side can be written as $\eta_m w_m$, where

$$w_m = \mathbf{w}^\top \mathbf{u}_m. \quad (5.138)$$

Then,

$$w_m^{(\tau)} = w_m^{(\tau-1)} - \rho \eta_m (w_m^{(\tau-1)} - w_m^*), \quad (5.139)$$

so that

$$w_m^{(\tau)} - w_m^* = (1 - \rho \eta_m) (w_m^{(\tau-1)} - w_m^*). \quad (5.140)$$

Therefore,

$$w_m^{(\tau)} = w_m^* + (1 - \rho \eta_m)^\tau (w_m^{(0)} - w_m^*), \quad (5.141)$$

so that

$$w_m^{(\tau)} = (1 - (1 - \rho \eta_m)^\tau) w_m^*. \quad (5.142)$$

(b)

By (a), if

$$|1 - \rho\eta_m| < 1, \quad (5.143)$$

then

$$\lim_{\tau \rightarrow \infty} w_m^{(\tau)} = w_m^*. \quad (5.144)$$

(c)

By (a) and the Taylor series

$$(1 - \rho\eta_m)^\tau = 1 - \rho\tau\eta_m + \rho^2\tau^2 O(\eta_m^2), \quad (5.145)$$

we have

$$w_m^{(\tau)} = (\rho\tau\eta_m - \rho^2\tau^2 O(\eta_m^2)) w_m^*, \quad (5.146)$$

so that

$$w_m^{(\tau)} = \rho\tau\eta_m (1 - \rho\tau\eta_m O(\eta_m)) w_m^*. \quad (5.147)$$

Therefore,

$$|w_m^{(\tau)}| \ll |w_m^|, \quad \text{if } \eta_m \ll (\rho\tau)^{-1}, \quad (5.148)$$

while

$$w_m^{(\tau)} \simeq w_m^*, \quad \text{if } \eta_m \gg (\rho\tau)^{-1}. \quad (5.149)$$

5.26 (Incomplete)

Let

$$\tilde{E} = E + \lambda\Omega, \quad (5.150)$$

where

$$\Omega = \frac{1}{2} \sum_{n=1}^N \sum_{k=1}^K \left(\frac{\partial y_{nk}}{\partial \xi} \right)^2. \quad (5.151)$$

(a)

We have

$$\frac{\partial y_{nk}}{\partial \xi} = \mathcal{G}y_{nk}, \quad (5.152)$$

where

$$\begin{aligned}\mathcal{G} &= \sum_{d=1}^D \tau_d \frac{\partial}{\partial x_d}, \\ \tau_d &= \frac{\partial x_d}{\partial \xi}.\end{aligned}\tag{5.153}$$

Therefore,

$$\Omega = \sum_{n=1}^N \Omega_n,\tag{5.154}$$

where

$$\Omega_n = \frac{1}{2} \sum_{k=1}^K (\mathcal{G} y_{nk})^2.\tag{5.155}$$

(b)

$$\begin{aligned}\alpha_j &= \mathcal{G} z_j = h'(a_j) \beta_j? \\ \beta_j &= \mathcal{G} a_j = \sum_{d=1}^D w_{jd} \alpha_d?\end{aligned}\tag{5.156}$$

5.27

Let

$$\tilde{E} = \frac{1}{2} \iiint (y(\mathbf{x} + \boldsymbol{\xi}) - t)^2 p(t|\mathbf{x}) p(\mathbf{x}) p(\boldsymbol{\xi}) d\mathbf{x} dt d\boldsymbol{\xi},\tag{5.157}$$

where

$$\begin{aligned}\mathbb{E} \boldsymbol{\xi} &= 0, \\ \text{var } \boldsymbol{\xi} &= \lambda \mathbf{I}.\end{aligned}\tag{5.158}$$

(a)

We have the Taylor series

$$y(\mathbf{x} + \boldsymbol{\xi}) = y(\mathbf{x}) + \rho(\mathbf{x}, \boldsymbol{\xi}) + O(\|\boldsymbol{\xi}\|^3),\tag{5.159}$$

where

$$\rho(\mathbf{x}, \boldsymbol{\xi}) = \boldsymbol{\xi}^\top \nabla y(\mathbf{x}) + \frac{1}{2} \boldsymbol{\xi}^\top \nabla \nabla y(\mathbf{x}) \boldsymbol{\xi}.\tag{5.160}$$

Then,

$$(y(\mathbf{x} + \boldsymbol{\xi}) - t)^2 \simeq (y(\mathbf{x}) - t)^2 + 2(y(\mathbf{x}) - t)\rho(\mathbf{x}, \boldsymbol{\xi}) + \rho(\mathbf{x}, \boldsymbol{\xi})^2. \quad (5.161)$$

Then,

$$\tilde{E} \simeq E + I_1 + \frac{1}{2}I_2, \quad (5.162)$$

where

$$\begin{aligned} E &= \frac{1}{2} \iint (y(\mathbf{x}) - t)^2 p(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}dt, \\ I_1 &= \iiint (y(\mathbf{x}) - t)\rho(\mathbf{x}, \boldsymbol{\xi})p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}dtd\boldsymbol{\xi}, \\ I_2 &= \iint \rho(\mathbf{x}, \boldsymbol{\xi})^2 p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}d\boldsymbol{\xi}. \end{aligned} \quad (5.163)$$

I_1 can be written as

$$\begin{aligned} &\iiint (y(\mathbf{x}) - t)\boldsymbol{\xi}^\top \nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}dtd\boldsymbol{\xi} \\ &+ \frac{1}{2} \int (y(\mathbf{x}) - t)\boldsymbol{\xi}^\top \nabla \nabla y(\mathbf{x})\boldsymbol{\xi}p(t|\mathbf{x})p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}dtd\boldsymbol{\xi} \\ &= \left(\int \boldsymbol{\xi}p(\boldsymbol{\xi})d\boldsymbol{\xi} \right)^\top \iint (y(\mathbf{x}) - t)\nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}dt \\ &+ \frac{1}{2} \int \boldsymbol{\xi}^\top \mathbf{M}_1 \boldsymbol{\xi}p(\boldsymbol{\xi})d\boldsymbol{\xi}, \end{aligned} \quad (5.164)$$

where

$$\mathbf{M}_1 = \iint (y(\mathbf{x}) - t)\nabla \nabla y(\mathbf{x})p(t|\mathbf{x})p(\mathbf{x})d\mathbf{x}dt. \quad (5.165)$$

Then,

$$I_1 = \frac{\lambda}{2} \text{tr } \mathbf{M}_1. \quad (5.166)$$

I_2 can be written as

$$\begin{aligned} &\iint \left(\boldsymbol{\xi}^\top \nabla y(\mathbf{x}) + \frac{1}{2}\boldsymbol{\xi}^\top \nabla \nabla y(\mathbf{x})\boldsymbol{\xi} \right)^2 p(\mathbf{x})p(\boldsymbol{\xi})d\mathbf{x}d\boldsymbol{\xi} \\ &\simeq \int \boldsymbol{\xi}^\top \mathbf{M}_2 \boldsymbol{\xi}p(\boldsymbol{\xi})d\boldsymbol{\xi}, \end{aligned} \quad (5.167)$$

where

$$\mathbf{M}_2 = \int (\nabla y(\mathbf{x})) (\nabla y(\mathbf{x}))^\top p(\mathbf{x}) d\mathbf{x}. \quad (5.168)$$

Then,

$$I_2 \simeq \lambda \operatorname{tr} \mathbf{M}_2. \quad (5.169)$$

Therefore,

$$\tilde{E} \simeq E + \frac{\lambda}{2} \operatorname{tr}(\mathbf{M}_1 + \mathbf{M}_2). \quad (5.170)$$

(b)

We have

$$\mathbf{M}_1 = \int \left(y(\mathbf{x}) \int p(t|\mathbf{x}) dt - \int t p(t|\mathbf{x}) dt \right) \nabla \nabla y(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \quad (5.171)$$

The right hand side can be written as

$$\int (y(\mathbf{x}) - E(t|\mathbf{x})) \nabla \nabla y(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}. \quad (5.172)$$

Therefore, if

$$y(\mathbf{x}) = E(t|\mathbf{x}), \quad (5.173)$$

then

$$\tilde{E} \simeq E + \frac{\lambda}{2} \operatorname{tr} \mathbf{M}_2. \quad (5.174)$$

5.28 (Incomplete)

5.29

Let $\mathbf{w}$ be a variable in M dimensions such that

$$p(w_m) = \sum_{k=1}^K \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2). \quad (5.175)$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \quad (5.176)$$

where

$$\Omega(\mathbf{w}) = - \sum_{m=1}^M \ln p(w_m). \quad (5.177)$$

We have

$$\frac{\partial \Omega}{\partial w_m} = \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial w_m}, \quad (5.178)$$

where

$$p_m = p(w_m). \quad (5.179)$$

We have

$$\begin{aligned} \frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial w_m} &= -\sum_{k=1}^K \frac{w_m - \mu_k}{\sigma_k^2} p_{mk}, \end{aligned} \quad (5.180)$$

where

$$p_{mk} = \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2). \quad (5.181)$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial w_m} = \frac{\partial E}{\partial w_m} + \lambda \sum_{k=1}^K \gamma_{mk} \frac{w_m - \mu_k}{\sigma_k^2}, \quad (5.182)$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \quad (5.183)$$

5.30

Let $\mathbf{w}$ be a variable in M dimensions such that

$$p(w_m) = \sum_{k=1}^K \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2). \quad (5.184)$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \quad (5.185)$$

where

$$\Omega(\mathbf{w}) = -\sum_{m=1}^M \ln p(w_m). \quad (5.186)$$

We have

$$\frac{\partial \Omega}{\partial \mu_k} = \sum_{m=1}^M \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial \mu_k}, \quad (5.187)$$

where

$$p_m = p(w_m). \quad (5.188)$$

We have

$$\begin{aligned} \frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial \mu_k} &= -\frac{\mu_k - w_m}{\sigma_k^2} p_{mk}, \end{aligned} \quad (5.189)$$

where

$$p_{mk} = \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2). \quad (5.190)$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial \mu_k} = \lambda \sum_{m=1}^M \gamma_{mk} \frac{\mu_k - w_m}{\sigma_k^2}, \quad (5.191)$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \quad (5.192)$$

5.31

Let $\mathbf{w}$ be a variable in M dimensions such that

$$p(w_m) = \sum_{k=1}^K \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2). \quad (5.193)$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}), \quad (5.194)$$

where

$$\Omega(\mathbf{w}) = -\sum_{m=1}^M \ln p(w_m). \quad (5.195)$$

We have

$$\frac{\partial \Omega}{\partial \sigma_k} = \sum_{m=1}^M \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial \sigma_k}, \quad (5.196)$$

where

$$p_m = p(w_m). \quad (5.197)$$

We have

$$\begin{aligned}\frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial \sigma_k} &= \left(-\frac{1}{\sigma_k} + \frac{(w_m - \mu_k)^2}{\sigma_k^3} \right) p_{mk},\end{aligned}\tag{5.198}$$

where

$$p_{mk} = \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2).\tag{5.199}$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial \sigma_k} = \lambda \sum_{m=1}^M \gamma_{mk} \left(\frac{1}{\sigma_k} - \frac{(w_m - \mu_k)^2}{\sigma_k^3} \right),\tag{5.200}$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}.\tag{5.201}$$

5.32

Let $\mathbf{w}$ be variable in M dimensions such that

$$p(w_m) = \sum_{k=1}^K \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2),\tag{5.202}$$

where

$$\pi_k = \frac{\exp(\eta_k)}{\sum_{k'=1}^K \exp(\eta_{k'})}.\tag{5.203}$$

Let

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \lambda \Omega(\mathbf{w}),\tag{5.204}$$

where

$$\Omega(\mathbf{w}) = - \sum_{m=1}^M \ln p(w_m).\tag{5.205}$$

(a)

If $k \neq k'$, then

$$\frac{\partial \pi_k}{\partial \eta_{k'}} = -\pi_k \pi_{k'}.\tag{5.206}$$

We have

$$\frac{\partial \pi_k}{\partial \eta_k} = \pi_k - \pi_k^2.\tag{5.207}$$

Therefore,

$$\frac{\partial \pi_k}{\partial \eta_{k'}} = I_{kk'} \pi_k - \pi_k \pi_{k'}. \quad (5.208)$$

(b)

We have

$$\frac{\partial \Omega}{\partial \eta_k} = \sum_{m=1}^M \sum_{k'=1}^K \frac{\partial \Omega}{\partial p_m} \frac{\partial p_m}{\partial \pi_{k'}} \frac{\partial \pi_{k'}}{\partial \eta_k}, \quad (5.209)$$

where

$$p_m = p(w_m). \quad (5.210)$$

We have

$$\begin{aligned} \frac{\partial \Omega}{\partial p_m} &= -\frac{1}{p_m}, \\ \frac{\partial p_m}{\partial \pi_{k'}} &= \frac{p_{mk'}}{\pi_{k'}}, \end{aligned} \quad (5.211)$$

where

$$p_{mk} = \pi_k \mathcal{N}(w_m | \mu_k, \sigma_k^2). \quad (5.212)$$

By (a),

$$\frac{\partial \pi_{k'}}{\partial \eta_k} = I_{k'k} \pi_{k'} - \pi_{k'} \pi_k. \quad (5.213)$$

Then,

$$\frac{\partial \Omega}{\partial \eta_k} = - \sum_{m=1}^M \sum_{k'=1}^K \frac{\gamma_{mk'}}{\pi_{k'}} (I_{k'k} \pi_{k'} - \pi_{k'} \pi_k), \quad (5.214)$$

where

$$\gamma_{mk} = \frac{p_{mk}}{p_m}. \quad (5.215)$$

The right hand side can be written as

$$- \sum_{m=1}^M \sum_{k'=1}^K \gamma_{mk'} (I_{k'k} - \pi_k) = \sum_{m=1}^M (\pi_k - \gamma_{mk}). \quad (5.216)$$

Therefore,

$$\frac{\partial \tilde{E}}{\partial \eta_k} = \lambda \sum_{m=1}^M (\pi_k - \gamma_{mk}). \quad (5.217)$$

5.33

Let (x_1, x_2) be the Cartesian coordinates of the end-effector of a two-link robot arm whose joint angles are θ_1 and θ_2 and whose arm lengths are L_1 and L_2 . Let us assume that the origin of the coordinate system is given by the attachment point of the lower arm. We have

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos \theta_1 \\ L_1 \sin \theta_1 \end{bmatrix} + \begin{bmatrix} L_2 \cos (\theta_2 - (\pi - \theta_1)) \\ L_2 \sin (\theta_2 - (\pi - \theta_1)) \end{bmatrix}. \quad (5.218)$$

The second term of the right hand side can be written as

$$\begin{bmatrix} L_2 \cos (\theta_1 + \theta_2 - \pi) \\ L_2 \sin (\theta_1 + \theta_2 - \pi) \end{bmatrix} = \begin{bmatrix} -L_2 \cos (\theta_1 + \theta_2) \\ -L_2 \sin (\theta_1 + \theta_2) \end{bmatrix}. \quad (5.219)$$

Therefore,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos \theta_1 - L_2 \cos (\theta_1 + \theta_2) \\ L_1 \sin \theta_1 - L_2 \sin (\theta_1 + \theta_2) \end{bmatrix}. \quad (5.220)$$

5.34

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n | \mathbf{w}) = \sum_{k=1}^K \pi_k \mathcal{N}_{nk}, \quad (5.221)$$

where

$$\begin{aligned} \pi_k &= \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})}, \\ \mathcal{N}_{nk} &= \mathcal{N}(\mathbf{t}_n | \boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w}) \mathbf{I}). \end{aligned} \quad (5.222)$$

Let

$$E(\mathbf{w}) = \sum_{n=1}^N E_n(\mathbf{w}), \quad (5.223)$$

where

$$E_n(\mathbf{w}) = -\ln p(\mathbf{t}_n | \mathbf{w}). \quad (5.224)$$

We have

$$\frac{\partial E_n}{\partial a_k} = \sum_{k'=1}^K \frac{\partial E_n}{\partial p_n} \frac{\partial p_n}{\partial \pi_{k'}} \frac{\partial \pi_{k'}}{\partial a_k}, \quad (5.225)$$

where

$$p_n = p(\mathbf{t}_n | \mathbf{w}). \quad (5.226)$$

We have

$$\begin{aligned} \frac{\partial E_n}{\partial p_n} &= -\frac{1}{p_n}, \\ \frac{\partial p_n}{\partial \pi_{k'}} &= \frac{p_{nk'}}{\pi_{k'}}, \end{aligned} \quad (5.227)$$

where

$$p_{nk} = \pi_k \mathcal{N}_{nk}. \quad (5.228)$$

By 5.32(a),

$$\frac{\partial \pi_{k'}}{\partial a_k} = I_{k'k} \pi_{k'} - \pi_k \pi_{k'}. \quad (5.229)$$

Then,

$$\frac{\partial E_n}{\partial a_k} = -\frac{1}{p_n} \sum_{k'=1}^K \frac{p_{nk'}}{\pi_{k'}} (I_{k'k} \pi_{k'} - \pi_k \pi_{k'}). \quad (5.230)$$

The right hand side can be written as

$$-\sum_{k'=1}^K \gamma_{nk'} (I_{kk'} - \pi_k) = \pi_k - \gamma_{nk}, \quad (5.231)$$

where

$$\gamma_{nk} = \frac{p_{nk}}{p_n}, \quad (5.232)$$

Therefore,

$$\frac{\partial E_n}{\partial a_k} = \pi_k - \gamma_{nk}. \quad (5.233)$$

5.35

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables such that

$$p(\mathbf{t}_n | \mathbf{w}) = \sum_{k=1}^K \pi_k \mathcal{N}_{nk}, \quad (5.234)$$

where

$$\mathcal{N}_{nk} = \mathcal{N}(\mathbf{t}_n | \boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w}) \mathbf{I}). \quad (5.235)$$

Let

$$E(\mathbf{w}) = \sum_{n=1}^N E_n(\mathbf{w}), \quad (5.236)$$

where

$$E_n(\mathbf{w}) = -\ln p(\mathbf{t}_n|\mathbf{w}). \quad (5.237)$$

Then,

$$\frac{\partial E_n}{\partial \boldsymbol{\mu}_k} = \frac{\partial E_n}{\partial p_n} \frac{\partial p_n}{\partial \boldsymbol{\mu}_k}, \quad (5.238)$$

where

$$p_n = p(t_n|\mathbf{w}). \quad (5.239)$$

We have

$$\begin{aligned} \frac{\partial E_n}{\partial p_n} &= -\frac{1}{p_n}, \\ \frac{\partial p_n}{\partial \boldsymbol{\mu}_k} &= -\sigma_k^{-2}(\boldsymbol{\mu}_k - \mathbf{t}_n)p_{nk}, \end{aligned} \quad (5.240)$$

where

$$p_{nk} = \pi_k \mathcal{N}_{nk}. \quad (5.241)$$

Therefore,

$$\frac{\partial E_n}{\partial \boldsymbol{\mu}_k} = \gamma_{nk} \sigma_k^{-2}(\boldsymbol{\mu}_k - \mathbf{t}_n), \quad (5.242)$$

where

$$\gamma_{nk} = \frac{p_{nk}}{p_n}. \quad (5.243)$$

5.36

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables in M dimensions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \sum_{k=1}^K \pi_k \mathcal{N}_{nk}, \quad (5.244)$$

where

$$\begin{aligned} \mathcal{N}_{nk} &= \mathcal{N}(\mathbf{t}_n | \boldsymbol{\mu}_k(\mathbf{x}_n, \mathbf{w}), \sigma_k^2(\mathbf{x}_n, \mathbf{w}) \mathbf{I}), \\ \sigma_k &= \exp(a_k^\sigma). \end{aligned} \quad (5.245)$$

Let

$$E(\mathbf{w}) = \sum_{n=1}^N E_n(\mathbf{w}), \quad (5.246)$$

where

$$E_n(\mathbf{w}) = -\ln p(\mathbf{t}_n|\mathbf{w}). \quad (5.247)$$

Then,

$$\frac{\partial E_n}{\partial a_k^\sigma} = \frac{\partial E_n}{\partial p_n} \frac{\partial p_n}{\partial \sigma_k} \frac{\partial \sigma_k}{\partial a_k^\sigma}, \quad (5.248)$$

where

$$p_n = p(\mathbf{t}_n|\mathbf{w}). \quad (5.249)$$

We have

$$\begin{aligned} \frac{\partial E_n}{\partial p_n} &= -\frac{1}{p_n}, \\ \frac{\partial p_n}{\partial \sigma_k} &= \left(-\frac{M}{\sigma_k} + \frac{\|\mathbf{t}_n - \boldsymbol{\mu}_k\|^2}{\sigma_k^3} \right) p_{nk}, \\ \frac{\partial \sigma_k}{\partial a_k^\sigma} &= \sigma_k, \end{aligned} \quad (5.250)$$

where

$$p_{nk} = \pi_k \mathcal{N}_{nk}. \quad (5.251)$$

Therefore,

$$\frac{\partial E_n}{\partial a_k^\sigma} = \gamma_{nk} \left(M - \frac{\|\mathbf{t}_n - \boldsymbol{\mu}_k\|^2}{\sigma_k^2} \right), \quad (5.252)$$

where

$$\gamma_{nk} = \frac{p_{nk}}{p_n}. \quad (5.253)$$

5.37

Let $\mathbf{t}$ be a variable such that

$$p(\mathbf{t}|\mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) \mathcal{N}(\mathbf{t}|\boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x})\mathbf{I}). \quad (5.254)$$

(a)

We have

$$\mathbf{E}(\mathbf{t}|\mathbf{x}) = \int \mathbf{t} p(\mathbf{t}|\mathbf{x}) d\mathbf{t}. \quad (5.255)$$

The right hand side can be written as

$$\sum_{k=1}^K \pi_k(\mathbf{x}) \int \mathbf{t} \mathcal{N}(\mathbf{t} | \boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t} = \sum_{k=1}^K \pi_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x}). \quad (5.256)$$

Therefore,

$$\mathbf{E}(\mathbf{t} | \mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x}). \quad (5.257)$$

(b)

We have

$$\text{var}(\mathbf{t} | \mathbf{x}) = \int (\mathbf{t} - \mathbf{E}(\mathbf{t} | \mathbf{x})) (\mathbf{t} - \mathbf{E}(\mathbf{t} | \mathbf{x}))^\top p(\mathbf{t} | \mathbf{x}) d\mathbf{t}. \quad (5.258)$$

The right hand side can be written as

$$\begin{aligned} & \mathbf{M}_1 - \mathbf{E}(\mathbf{t} | \mathbf{x}) \left(\int \mathbf{t} p(\mathbf{t} | \mathbf{x}) d\mathbf{t} \right)^\top - \left(\int \mathbf{t} p(\mathbf{t} | \mathbf{x}) d\mathbf{t} \right) \mathbf{E}(\mathbf{t} | \mathbf{x})^\top \\ & + \mathbf{E}(\mathbf{t} | \mathbf{x}) \mathbf{E}(\mathbf{t} | \mathbf{x})^\top \int p(\mathbf{t} | \mathbf{x}) d\mathbf{t} \\ & = \mathbf{M}_1 - \mathbf{E}(\mathbf{t} | \mathbf{x}) \mathbf{E}(\mathbf{t} | \mathbf{x})^\top, \end{aligned} \quad (5.259)$$

where

$$\mathbf{M}_1 = \int \mathbf{t} \mathbf{t}^\top p(\mathbf{t} | \mathbf{x}) d\mathbf{t}. \quad (5.260)$$

$\mathbf{M}_1$ can be written as

$$\sum_{k=1}^K \pi_k(\mathbf{x}) \int \mathbf{t} \mathbf{t}^\top \mathcal{N}(\mathbf{t} | \boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t}. \quad (5.261)$$

The integral can be written as

$$\begin{aligned} & \int (\mathbf{t}' + \boldsymbol{\mu}_k(\mathbf{x})) (\mathbf{t}' + \boldsymbol{\mu}_k(\mathbf{x}))^\top \mathcal{N}(\mathbf{t} | \boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t} \\ & = \mathbf{M}_2 + \boldsymbol{\mu}_k(\mathbf{x}) \mathbf{v}^\top + \mathbf{v} \boldsymbol{\mu}_k(\mathbf{x})^\top + \boldsymbol{\mu}_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x})^\top I, \end{aligned} \quad (5.262)$$

where

$$\begin{aligned}
\mathbf{t}' &= \mathbf{t} - \boldsymbol{\mu}_k(\mathbf{x}), \\
\mathbf{M}_2 &= \int \mathbf{t}' \mathbf{t}'^\top \mathcal{N}(\mathbf{t} | \boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t}, \\
\mathbf{v} &= \int \mathbf{t}' \mathcal{N}(\mathbf{t} | \boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t}, \\
I &= \int \mathcal{N}(\mathbf{t} | \boldsymbol{\mu}_k(\mathbf{x}), \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t}.
\end{aligned} \tag{5.263}$$

We have

$$\begin{aligned}
\mathbf{M}_2 &= \int \mathbf{t}' \mathbf{t}'^\top \mathcal{N}(\mathbf{t}' | \mathbf{0}, \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t}', \\
\mathbf{v} &= \int \mathbf{t}' \mathcal{N}(\mathbf{t}' | \mathbf{0}, \sigma_k^2(\mathbf{x}) \mathbf{I}) d\mathbf{t}'.
\end{aligned} \tag{5.264}$$

Then,

$$\mathbf{M}_1 = \sigma_k^2(\mathbf{x}) \mathbf{I} + \boldsymbol{\mu}_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x})^\top. \tag{5.265}$$

Therefore, by (a),

$$\begin{aligned}
\text{var}(\mathbf{t} | \mathbf{x}) &= \sum_{k=1}^K \pi_k(\mathbf{x}) (\sigma_k^2(\mathbf{x}) \mathbf{I} + \boldsymbol{\mu}_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x})^\top) \\
&\quad - \left(\sum_{k=1}^K \pi_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x}) \right) \left(\sum_{k=1}^K \pi_k(\mathbf{x}) \boldsymbol{\mu}_k(\mathbf{x}) \right)^\top.
\end{aligned} \tag{5.266}$$

5.38

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned}
p(t_n | \mathbf{w}) &= \mathcal{N}(t_n | y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}), \\
p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}).
\end{aligned} \tag{5.267}$$

By marginalisation,

$$p(t_{N+1} | \mathbf{t}) = \int p(t_{N+1} | \mathbf{w}) p(\mathbf{w} | \mathbf{t}) d\mathbf{w}. \tag{5.268}$$

By Bayes' theorem,

$$p(\mathbf{w} | \mathbf{t}) p(\mathbf{t}) = p(\mathbf{t} | \mathbf{w}) p(\mathbf{w}). \tag{5.269}$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^N (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (5.270)$$

We have the Taylor series

$$\begin{aligned} E(\mathbf{w}) = & E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ & + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3), \end{aligned} \quad (5.271)$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}). \quad (5.272)$$

Then,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}). \quad (5.273)$$

We have the Taylor series

$$\begin{aligned} y(\mathbf{x}_{N+1}, \mathbf{w}) = & y_{\text{MAP}N+1} + \mathbf{g}_{\text{MAP}N+1}^\top (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ & + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^2), \end{aligned} \quad (5.274)$$

where

$$\begin{aligned} y_{\text{MAP}N+1} &= y(\mathbf{x}_{N+1}, \mathbf{w}_{\text{MAP}}), \\ \mathbf{g}_{\text{MAP}N+1} &= \nabla_{\mathbf{w}} y(\mathbf{x}_{N+1}, \mathbf{w}_{\text{MAP}}). \end{aligned} \quad (5.275)$$

Then,

$$p(t_{N+1}|\mathbf{w}) \simeq \mathcal{N}(t_{N+1}|y_{\text{MAP}N+1} + \mathbf{g}_{\text{MAP}N+1}^\top (\mathbf{w} - \mathbf{w}_{\text{MAP}}), \beta^{-1}). \quad (5.276)$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{w}$ can be approximated as

$$\begin{aligned} & -\frac{\beta}{2} (t_{N+1} - y_{\text{MAP}N+1} - \mathbf{g}_{\text{MAP}N+1}^\top (\mathbf{w} - \mathbf{w}_{\text{MAP}}))^2 \\ & -\frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ = & -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v}, \end{aligned} \quad (5.277)$$

where

$$\begin{aligned} \mathbf{v} &= \begin{bmatrix} \mathbf{w} - \mathbf{w}_{\text{MAP}} \\ t_{N+1} - y_{\text{MAP}_{N+1}} \end{bmatrix}, \\ \mathbf{M} &= \begin{bmatrix} \mathbf{H}_{\text{MAP}} + \beta \mathbf{g}_{\text{MAP}_{N+1}} \mathbf{g}_{\text{MAP}_{N+1}}^\top & -\beta \mathbf{g}_{\text{MAP}_{N+1}} \\ -\beta \mathbf{g}_{\text{MAP}_{N+1}}^\top & \beta \end{bmatrix}. \end{aligned} \quad (5.278)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{H}_{\text{MAP}}^{-1} & \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}_{N+1}} \\ \mathbf{g}_{\text{MAP}_{N+1}}^\top \mathbf{H}_{\text{MAP}}^{-1} & \beta^{-1} + \mathbf{g}_{\text{MAP}_{N+1}}^\top \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}_{N+1}} \end{bmatrix}. \quad (5.279)$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) \simeq \mathcal{N}(t_{N+1}|y_{\text{MAP}_{N+1}}, \beta^{-1} + \mathbf{g}_{\text{MAP}_{N+1}}^\top \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}_{N+1}}). \quad (5.280)$$

5.39

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \end{aligned} \quad (5.281)$$

where $\mathbf{w}$ is in M dimensions. By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \quad (5.282)$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} & -\frac{N}{2} \ln(2\pi\beta^{-1}) - \frac{\beta}{2} \sum_{n=1}^N (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 \\ & -\frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\alpha^{-1}\mathbf{I})) - \frac{\alpha}{2} \|\mathbf{w}\|^2 \\ & = -E(\mathbf{w}) - \frac{N+M}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{M}{2} \ln \alpha, \end{aligned} \quad (5.283)$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^N (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (5.284)$$

We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^{\text{T}} \mathbf{H}_{\text{MAP}} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3), \quad (5.285)$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}). \quad (5.286)$$

Then, the logarithm of the integrand can be approximated as

$$\begin{aligned} & -E(\mathbf{w}_{\text{MAP}}) - \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^{\text{T}} \mathbf{H}_{\text{MAP}} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ & - \frac{N+M}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{M}{2} \ln \alpha. \end{aligned} \quad (5.287)$$

We have

$$\begin{aligned} & \int \exp \left(-\frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^{\text{T}} \mathbf{H}_{\text{MAP}} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \right) d\mathbf{w} \\ & = (2\pi)^{\frac{M}{2}} (\det \mathbf{H}_{\text{MAP}}^{-1})^{\frac{1}{2}}. \end{aligned} \quad (5.288)$$

Therefore,

$$\ln p(\mathbf{t}) \simeq -E(\mathbf{w}_{\text{MAP}}) - \frac{1}{2} \ln |\det \mathbf{H}_{\text{MAP}}| - \frac{N}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{M}{2} \ln \alpha. \quad (5.289)$$

5.40

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$\begin{aligned} p(\mathbf{t}_n | \mathbf{w}) &= \prod_{k=1}^K y_{nk}^{t_{nk}}, \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}), \end{aligned} \quad (5.290)$$

where

$$\begin{aligned} y_{nk} &= y_k(\mathbf{a}_n), \\ y_k(\mathbf{a}) &= \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})}, \\ \mathbf{a}_n &= \mathbf{a}(\mathbf{w}, \mathbf{x}_n). \end{aligned} \quad (5.291)$$

(a)

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{T})p(\mathbf{T}) = p(\mathbf{T}|\mathbf{w})p(\mathbf{w}). \quad (5.292)$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = -\sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk} + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (5.293)$$

We have the Taylor series

$$\begin{aligned} E(\mathbf{w}) = & E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^T \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ & + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3), \end{aligned} \quad (5.294)$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}). \quad (5.295)$$

Then,

$$p(\mathbf{w}|\mathbf{T}) \simeq \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}). \quad (5.296)$$

(b)

We have

$$p(\mathbf{t}_{N+1}|\mathbf{w}) = \prod_{k=1}^K y_k(\mathbf{a}_{N+1})^{t_{N+1,k}}. \quad (5.297)$$

We have the Taylor series

$$\begin{aligned} \mathbf{a}(\mathbf{w}, \mathbf{x}_{N+1}) = & \mathbf{a}_{\text{MAP}N+1} + \mathbf{J}_{\text{MAP}N+1}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ & + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^2), \end{aligned} \quad (5.298)$$

where

$$\begin{aligned} \mathbf{a}_{\text{MAP}N+1} &= \mathbf{a}(\mathbf{w}_{\text{MAP}}, \mathbf{x}_{N+1}), \\ \mathbf{J}_{\text{MAP}N+1} &= \nabla_{\mathbf{w}} \mathbf{a}(\mathbf{w}_{\text{MAP}}, \mathbf{x}_{N+1}). \end{aligned} \quad (5.299)$$

Then,

$$p(\mathbf{t}_{N+1}|\mathbf{a}_{N+1}) \simeq \prod_{k=1}^K y_k(\boldsymbol{\alpha}_{N+1})^{t_{N+1,k}}, \quad (5.300)$$

where

$$\boldsymbol{\alpha}_{N+1} = \mathbf{a}_{\text{MAP}N+1} + \mathbf{J}_{\text{MAP}N+1}(\mathbf{w} - \mathbf{w}_{\text{MAP}}). \quad (5.301)$$

(c)

By (a) and (b),

$$\begin{aligned} \mathbf{E}(\boldsymbol{\alpha}_{N+1}|\mathbf{T}) &= \mathbf{a}_{\text{MAP}N+1}, \\ \text{var}(\boldsymbol{\alpha}_{N+1}|\mathbf{T}) &= \mathbf{J}_{\text{MAP}N+1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{J}_{\text{MAP}N+1}^{\top}. \end{aligned} \quad (5.302)$$

Therefore,

$$p(\mathbf{a}_{N+1}|\mathbf{T}) \simeq \mathcal{N}(\mathbf{a}_{N+1}|\mathbf{a}_{\text{MAP}N+1}, \mathbf{J}_{\text{MAP}N+1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{J}_{\text{MAP}N+1}^{\top}). \quad (5.303)$$

(d)

By marginalisation,

$$p(\mathbf{t}_{N+1}|\mathbf{T}) = \int p(\mathbf{t}_{N+1}|\mathbf{a}_{N+1})p(\mathbf{a}_{N+1}|\mathbf{T})d\mathbf{a}_{N+1}. \quad (5.304)$$

Therefore, by (b) and (c),

$$p(\mathbf{t}_{N+1}|\mathbf{T}) \simeq \prod_{k=1}^K v_{N+1,k}^{t_{N+1,k}}, \quad (5.305)$$

where

$$v_{N+1,k} = \int y_k(\mathbf{a}) \mathcal{N}(\mathbf{a}|\mathbf{a}_{\text{MAP}N+1}, \mathbf{J}_{\text{MAP}N+1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{J}_{\text{MAP}N+1}^{\top}) d\mathbf{a}, \quad (5.306)$$

since $\mathbf{t}_{N+1}$ is from the standard basis in K dimensions.

5.41

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$\begin{aligned} p(\mathbf{t}_n|\mathbf{w}) &= \prod_{k=1}^K y_{nk}^{t_{nk}}, \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \end{aligned} \quad (5.307)$$

where $\mathbf{w}$ is in M dimensions and

$$y_{nk} = y_k(\mathbf{x}_n, \mathbf{w}). \quad (5.308)$$

By marginalisation,

$$p(\mathbf{T}) = \int p(\mathbf{T}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \quad (5.309)$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} & \sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk} - \frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\alpha^{-1}\mathbf{I})) - \frac{\alpha}{2} \|\mathbf{w}\|^2 \\ &= -E(\mathbf{w}) - \frac{M}{2} \ln(2\pi) + \frac{M}{2} \ln \alpha, \end{aligned} \quad (5.310)$$

where

$$E(\mathbf{w}) = - \sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk} + \frac{\alpha}{2} \|\mathbf{w}\|^2. \quad (5.311)$$

We have the Taylor series

$$\begin{aligned} E(\mathbf{w}) &= E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ &\quad + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3), \end{aligned} \quad (5.312)$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}). \quad (5.313)$$

Then, the logarithm of the integrand can be approximated as

$$-E(\mathbf{w}_{\text{MAP}}) - \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) - \frac{M}{2} \ln(2\pi) + \frac{M}{2} \ln \alpha. \quad (5.314)$$

We have

$$\begin{aligned} & \int \exp\left(-\frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}})\right) d\mathbf{w} \\ &= (2\pi)^{\frac{M}{2}} (\det \mathbf{H}_{\text{MAP}}^{-1})^{\frac{1}{2}}. \end{aligned} \quad (5.315)$$

Therefore,

$$\ln p(\mathbf{T}) \simeq -E(\mathbf{w}_{\text{MAP}}) - \frac{1}{2} \ln |\det \mathbf{H}_{\text{MAP}}| + \frac{M}{2} \ln \alpha. \quad (5.316)$$

6 Kernel Methods

6.1

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, 1), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \lambda^{-1}\mathbf{I}). \end{aligned} \quad (6.1)$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \quad (6.2)$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E_{\mathbf{w}}(\mathbf{w})$ where

$$E_{\mathbf{w}}(\mathbf{w}) = \frac{1}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2. \quad (6.3)$$

(a)

Let $\mathbf{w}_{\text{MAP}}$ be a stationary point of $E_{\mathbf{w}}$. Setting the derivative of $E_{\mathbf{w}}$ with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = -\boldsymbol{\Phi}^\top(\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}) + \lambda\mathbf{w}. \quad (6.4)$$

Therefore,

$$\mathbf{w}_{\text{MAP}} = \boldsymbol{\Phi}^\top \mathbf{a}_{\text{MAP}}, \quad (6.5)$$

where

$$\mathbf{a}_{\text{MAP}} = \frac{1}{\lambda}(\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}_{\text{MAP}}). \quad (6.6)$$

(b)

Let

$$\mathbf{w} = \boldsymbol{\Phi}^\top \mathbf{a}. \quad (6.7)$$

Then,

$$E_{\mathbf{w}}(\mathbf{w}) = E_{\mathbf{a}}(\mathbf{a}), \quad (6.8)$$

where

$$E_{\mathbf{a}}(\mathbf{a}) = \frac{1}{2}\|\mathbf{t} - \boldsymbol{\Phi}\boldsymbol{\Phi}^\top \mathbf{a}\|^2 + \frac{\lambda}{2}\mathbf{a}^\top \boldsymbol{\Phi}\boldsymbol{\Phi}^\top \mathbf{a}. \quad (6.9)$$

The right hand side can be written as

$$\frac{1}{2}\|\mathbf{t}\|^2 - \mathbf{t}^\top \mathbf{K} \mathbf{a} + \frac{1}{2} \mathbf{a}^\top \mathbf{K} \mathbf{K} \mathbf{a} + \frac{\lambda}{2} \mathbf{a}^\top \mathbf{K} \mathbf{a}, \quad (6.10)$$

where

$$\mathbf{K} = \Phi \Phi^\top. \quad (6.11)$$

Let $\mathbf{a}_{\text{MAP}}$ be a stationary point of $E_{\mathbf{a}}$. Setting the derivative of $E_{\mathbf{a}}$ with respect to $\mathbf{a}$ to zero gives

$$\mathbf{0} = -\mathbf{K} \mathbf{t} + \mathbf{K} \mathbf{K} \mathbf{a} + \lambda \mathbf{K} \mathbf{a}. \quad (6.12)$$

Therefore,

$$\mathbf{a}_{\text{MAP}} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{t}. \quad (6.13)$$

(c)

By (a) and (b),

$$\frac{1}{\lambda} (\mathbf{t} - \Phi \mathbf{w}_{\text{MAP}}) = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{t}, \quad (6.14)$$

so that

$$\mathbf{t} - \Phi \mathbf{w}_{\text{MAP}} = \mathbf{N} \mathbf{t}, \quad (6.15)$$

where

$$\mathbf{N} = (\mathbf{I} + \lambda^{-1} \mathbf{K})^{-1}. \quad (6.16)$$

By 2.26, the right hand side can be written as

$$\mathbf{I} - \Phi \mathbf{M} \Phi^\top, \quad (6.17)$$

where

$$\mathbf{M} = (\lambda \mathbf{I} + \Phi^\top \Phi)^{-1}. \quad (6.18)$$

Then,

$$\mathbf{t} - \Phi \mathbf{w}_{\text{MAP}} = (\mathbf{I} - \Phi \mathbf{M} \Phi^\top) \mathbf{t}, \quad (6.19)$$

so that

$$\Phi (\mathbf{w}_{\text{MAP}} - \Phi \mathbf{M} \Phi^\top \mathbf{t}) = \mathbf{0}. \quad (6.20)$$

Therefore,

$$\mathbf{w}_{\text{MAP}} = \mathbf{M} \Phi^\top \mathbf{t}. \quad (6.21)$$

6.2 (Incomplete)

Let $t_1, \dots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n|\mathbf{w}) = \left(\frac{1+y_n}{2}\right)^{\frac{1+t_n}{2}} \left(\frac{1-y_n}{2}\right)^{\frac{1-t_n}{2}}, \quad (6.22)$$

where

$$y_n = f(\mathbf{w}^\top \phi_n),$$

$$f(a) = \begin{cases} 1, & a \geq 0, \\ -1, & \text{otherwise.} \end{cases} \quad (6.23)$$

Let

$$E(\mathbf{w}) = - \sum_{n \in \mathcal{M}} \mathbf{w}^\top \phi_n t_n, \quad (6.24)$$

where $\mathcal{M}$ is the set of all misclassification. We have the Taylor series

$$E(\mathbf{w}^{(\tau)}) = E(\mathbf{w}^{(\tau-1)}) + \mathbf{b}^\top (\mathbf{w}^{(\tau)} - \mathbf{w}^{(\tau-1)}) + O\left(\|\mathbf{w}^{(\tau)} - \mathbf{w}^{(\tau-1)}\|^2\right), \quad (6.25)$$

where

$$\mathbf{b} = - \sum_{n \in \mathcal{M}} t_n \phi_n. \quad (6.26)$$

6.3

We have

$$\|\mathbf{x} - \mathbf{x}_n\|^2 = \|\mathbf{x}\|^2 - 2\mathbf{x}^\top \mathbf{x}_n + \|\mathbf{x}_n\|^2. \quad (6.27)$$

The right hand side can be written as

$$k(\mathbf{x}, \mathbf{x}) - 2k(\mathbf{x}, \mathbf{x}_n) + k(\mathbf{x}_n, \mathbf{x}_n), \quad (6.28)$$

where

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'. \quad (6.29)$$

6.4

Let $\mathbf{A}$ be a 2×2 matrix with positive eigenvalues and with at least one negative element. We have

$$\det(\lambda \mathbf{I} - \mathbf{A}) = (\lambda - A_{11})(\lambda - A_{22}) - A_{12}A_{21}. \quad (6.30)$$

Seeing the left hand side to zero gives

$$0 = \lambda^2 - (A_{11} + A_{22})\lambda + A_{11}A_{22} - A_{12}A_{21}. \quad (6.31)$$

For the equation to have two positive solutions, we have the conditions

$$\begin{aligned} (A_{11} + A_{22})^2 - 4(A_{11}A_{22} - A_{12}A_{21}) &> 0, \\ A_{11} + A_{22} &> 0, \\ A_{11}A_{22} - A_{12}A_{21} &> 0. \end{aligned} \quad (6.32)$$

Therefore, an example of A is given by

$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}. \quad (6.33)$$

6.5

Let

$$k_1(\mathbf{x}, \mathbf{x}') = \phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}'). \quad (6.34)$$

(a)

Let

$$k(\mathbf{x}, \mathbf{x}') = ck_1(\mathbf{x}, \mathbf{x}'), \quad (6.35)$$

where c is a positive constant. The right hand side can be written as

$$c\phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}') = \phi(\mathbf{x})^\top \phi(\mathbf{x}'), \quad (6.36)$$

where

$$\phi = \sqrt{c}\phi_1. \quad (6.37)$$

Therefore, k is a valid kernel.

(b)

Let

$$k(\mathbf{x}, \mathbf{x}') = f(\mathbf{x})k_1(\mathbf{x}, \mathbf{x}')f(\mathbf{x}'). \quad (6.38)$$

The right hand side can be written as

$$f(\mathbf{x})\phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}')f(\mathbf{x}') = \phi(\mathbf{x})^\top \phi(\mathbf{x}'), \quad (6.39)$$

where

$$\phi = f\phi_1. \quad (6.40)$$

Therefore, k is a valid kernel.

6.6

Let

$$k_1(\mathbf{x}, \mathbf{x}') = \phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}'), \quad (6.41)$$

where ϕ_1 is a vector in M dimensions.

(a)

Let

$$k(\mathbf{x}, \mathbf{x}') = \sum_{j=0}^J c_j q_j(\mathbf{x}, \mathbf{x}'), \quad (6.42)$$

where $c_0, \dots, c_J$ are nonnegative constants and

$$q_j(\mathbf{x}, \mathbf{x}') = (k_1(\mathbf{x}, \mathbf{x}'))^j. \quad (6.43)$$

We have

$$q_0(\mathbf{x}, \mathbf{x}') = 1. \quad (6.44)$$

Let us assume that q_j is a valid kernel. Then,

$$q_{j+1}(\mathbf{x}, \mathbf{x}') = q_j(\mathbf{x}, \mathbf{x}') \phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}'). \quad (6.45)$$

The right hand side can be written as

$$q_j(\mathbf{x}, \mathbf{x}') \sum_{m=1}^M \phi_{1m}(\mathbf{x}) \phi_{1m}(\mathbf{x}') = \sum_{m=1}^M \phi_{1m}(\mathbf{x}) q_j(\mathbf{x}, \mathbf{x}') \phi_{1m}(\mathbf{x}'). \quad (6.46)$$

By 6.5(b) and 6.7(a), the right hand side is a valid kernel. Then, q_{j+1} is a valid kernel. Then, the assumption is proved by induction on j . Therefore, by 6.5(a) and 6.7(a), k is a valid kernel.

(b)

Let

$$k(\mathbf{x}, \mathbf{x}') = \exp(k_1(\mathbf{x}, \mathbf{x}')). \quad (6.47)$$

The right hand side can be written as

$$\sum_{j=0}^{\infty} \frac{1}{j!} (k_1(\mathbf{x}, \mathbf{x}'))^j. \quad (6.48)$$

Therefore, by (a), k is a valid kernel.

6.7

Let

$$\begin{aligned}k_1(\mathbf{x}, \mathbf{x}') &= \boldsymbol{\phi}_1(\mathbf{x})^\top \boldsymbol{\phi}_1(\mathbf{x}'), \\k_2(\mathbf{x}, \mathbf{x}') &= \boldsymbol{\phi}_2(\mathbf{x})^\top \boldsymbol{\phi}_2(\mathbf{x}').\end{aligned}\tag{6.49}$$

(a)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}').\tag{6.50}$$

The right hand side can be written as

$$\boldsymbol{\phi}_1(\mathbf{x})^\top \boldsymbol{\phi}_1(\mathbf{x}') + \boldsymbol{\phi}_2(\mathbf{x})^\top \boldsymbol{\phi}_2(\mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{x}'),\tag{6.51}$$

where

$$\boldsymbol{\phi} = \begin{bmatrix} \boldsymbol{\phi}_1 \\ \boldsymbol{\phi}_2 \end{bmatrix}.\tag{6.52}$$

Therefore, k is a valid kernel.

(b)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') k_2(\mathbf{x}, \mathbf{x}').\tag{6.53}$$

Then,

$$\ln k(\mathbf{x}, \mathbf{x}') = \ln k_1(\mathbf{x}, \mathbf{x}') + \ln k_2(\mathbf{x}, \mathbf{x}').\tag{6.54}$$

By (a) and 6.6(b), the right hand side is a valid kernel. Then, $\ln k$ is a valid kernel. Therefore, by 6.6(b), k is a valid kernel.

6.8

(a)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_1(\boldsymbol{\phi}(\mathbf{x}), \boldsymbol{\phi}(\mathbf{x}')), \tag{6.55}$$

where k_1 is a valid kernel. Therefore, k is a valid kernel.

(b)

Let

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{A} \mathbf{x}', \quad (6.56)$$

where $\mathbf{A}$ is a symmetric positive semidefinite matrix. Then, there exists $\mathbf{M}$ such that

$$\mathbf{A} = \mathbf{M}^\top \mathbf{M}. \quad (6.57)$$

Then,

$$k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{x}'), \quad (6.58)$$

where

$$\boldsymbol{\phi}(\mathbf{x}) = \mathbf{M} \mathbf{x}. \quad (6.59)$$

Therefore, k is a valid kernel.

6.9

Let

$$\begin{aligned} k_a(\mathbf{x}_a, \mathbf{x}'_a) &= \boldsymbol{\phi}_a(\mathbf{x}_a)^\top \boldsymbol{\phi}_a(\mathbf{x}'_a), \\ k_b(\mathbf{x}_b, \mathbf{x}'_b) &= \boldsymbol{\phi}_b(\mathbf{x}_b)^\top \boldsymbol{\phi}_b(\mathbf{x}'_b). \end{aligned} \quad (6.60)$$

(a)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a) + k_b(\mathbf{x}_b, \mathbf{x}'_b), \quad (6.61)$$

where

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{bmatrix}. \quad (6.62)$$

The right hand side can be written as

$$\boldsymbol{\phi}_a(\mathbf{x}_a)^\top \boldsymbol{\phi}_a(\mathbf{x}'_a) + \boldsymbol{\phi}_b(\mathbf{x}_b)^\top \boldsymbol{\phi}_b(\mathbf{x}'_b) = \boldsymbol{\phi}(\mathbf{x})^\top \boldsymbol{\phi}(\mathbf{x}'), \quad (6.63)$$

where

$$\boldsymbol{\phi}(\mathbf{x}) = \begin{bmatrix} \boldsymbol{\phi}_a(\mathbf{x}_a) \\ \boldsymbol{\phi}_b(\mathbf{x}_b) \end{bmatrix}. \quad (6.64)$$

Therefore, k is a valid kernel.

(b)

Let

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a)k_b(\mathbf{x}_b, \mathbf{x}'_b), \quad (6.65)$$

where

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{bmatrix}. \quad (6.66)$$

Then,

$$\ln k(\mathbf{x}, \mathbf{x}') = \ln k_a(\mathbf{x}_a, \mathbf{x}'_a) + \ln k_b(\mathbf{x}_b, \mathbf{x}'_b). \quad (6.67)$$

By (a) and 6.6(b), the right hand side is a valid kernel. Then, $\ln k$ is a valid kernel. Therefore, by 6.6(b), k is a valid kernel.

6.10 (Incomplete)

Let

$$k(\mathbf{x}, \mathbf{x}') = f(\mathbf{x})f(\mathbf{x}'). \quad (6.68)$$

6.11

Let

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right). \quad (6.69)$$

The right hand side can be written as

$$\exp\left(-\frac{\mathbf{x}^\top \mathbf{x}}{2\sigma^2}\right) \exp\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\sigma^2}\right) \exp\left(-\frac{\mathbf{x}'^\top \mathbf{x}'}{2\sigma^2}\right). \quad (6.70)$$

We have

$$\exp\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\sigma^2}\right) = \sum_{j=0}^{\infty} q_j(\mathbf{x}, \mathbf{x}'), \quad (6.71)$$

where

$$q_j(\mathbf{x}, \mathbf{x}') = \frac{1}{j!} \left(\frac{\mathbf{x}^\top \mathbf{x}'}{\sigma^2}\right)^j. \quad (6.72)$$

By 6.6(a), q_j is a valid kernel. Then, there exists ψ_j such that

$$q_j(\mathbf{x}, \mathbf{x}') = \psi_j(\mathbf{x})^\top \psi_j(\mathbf{x}'). \quad (6.73)$$

Then,

$$\exp\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\sigma^2}\right) = \sum_{j=0}^{\infty} \psi_j(\mathbf{x})^\top \psi_j(\mathbf{x}'). \quad (6.74)$$

The right hand side can be written as $\phi(\mathbf{x})^\top \phi(\mathbf{x}')$, where

$$\phi = \begin{bmatrix} \psi_0 \\ \vdots \\ \psi_j \\ \vdots \end{bmatrix}. \quad (6.75)$$

Therefore, k can be written as the inner product of an infinite-dimensional feature vector.

6.12

Let D be a set. Let $A_1, \dots, A_{2^{|D|}}$ be all the subsets of D where $|\cdot|$ is the number of elements in $(\cdot)$. Let

$$k(A_m, A_{m'}) = 2^{|A_m \cap A_{m'}|}. \quad (6.76)$$

Let ϕ be a vector such that

$$\phi_m(A) = \begin{cases} 1, & \text{if } A_m \subseteq A, \\ 0, & \text{otherwise.} \end{cases} \quad (6.77)$$

If

$$|A_m \cap A_{m'}| = 0, \quad (6.78)$$

then

$$k(A_m, A_{m'}) = 1. \quad (6.79)$$

The right hand side can be written as

$$\phi(A_m)^\top \phi(A_{m'}). \quad (6.80)$$

Let us assume that if

$$|A_m \cap A_{m'}| = n, \quad (6.81)$$

then

$$k(A_m, A_{m'}) = \phi(A_m)^\top \phi(A_{m'}). \quad (6.82)$$

Let

$$A_{m''} = A_{m'} \cup \{x\}, \quad (6.83)$$

where x is an element such that

$$x \in A_m - A_{m'}. \quad (6.84)$$

Then,

$$|A_m \cap A_{m''}| = n + 1, \quad (6.85)$$

so that

$$k(A_m, A_{m''}) = 2^{n+1}. \quad (6.86)$$

We have

$$\phi(A_m)^\top \phi(A_{m''}) = \phi(A_m)^\top \phi(A_{m'}) + \phi(A_m)^\top (\phi(A_{m''}) - \phi(A_{m'})). \quad (6.87)$$

By the assumption, the first term of the right hand side is 2^n . By the assumption, $\phi(A_m)$ and $\phi(A_{m'})$ have a set of 2^n 1s in common. Then, $\phi(A_m)$ and $\phi(A_{m''}) - \phi(A_{m'})$ have another set of 2^n 1s in common. Then, the second term of the right hand side is 2^n . Then,

$$\phi(A_m)^\top \phi(A_{m''}) = 2^{n+1}, \quad (6.88)$$

so that

$$k(A_m, A_{m''}) = \phi(A_m)^\top \phi(A_{m''}). \quad (6.89)$$

Therefore, the assumption is proved by induction on n .

6.13

Let

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\top \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}'), \quad (6.90)$$

where

$$\begin{aligned} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) &= \nabla_{\boldsymbol{\theta}} \ln p(\mathbf{x}|\boldsymbol{\theta}), \\ \mathbf{F} &= \mathbf{E}_{\mathbf{x}} (\mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\top). \end{aligned} \quad (6.91)$$

Let ψ be an invertible and differentiable function of $\boldsymbol{\theta}$. Let

$$k_\psi(\mathbf{x}, \mathbf{x}') = \mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x})^\top \mathbf{F}_\psi^{-1} \mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x}'), \quad (6.92)$$

where

$$\begin{aligned}\mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x}) &= \nabla_\psi \ln p(\mathbf{x}|\boldsymbol{\theta}), \\ \mathbf{F}_\psi &= \mathbb{E}_{\mathbf{x}} (\mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x})^\top). \end{aligned} \quad (6.93)$$

We have

$$\mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x}) = \frac{\partial \boldsymbol{\theta}}{\partial \psi} \nabla_{\boldsymbol{\theta}} \ln p(\mathbf{x}|\boldsymbol{\theta}). \quad (6.94)$$

Then,

$$\mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x}) = \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}). \quad (6.95)$$

We have

$$\mathbf{F}_\psi = \int \mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}_\psi(\boldsymbol{\theta}, \mathbf{x})^\top p(\mathbf{x}) d\mathbf{x}. \quad (6.96)$$

The right hand side can be written as

$$\begin{aligned} & \int \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\top \left(\left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\top p(\mathbf{x}) d\mathbf{x} \\ &= \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \left(\int \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}) \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\top p(\mathbf{x}) d\mathbf{x} \right) \left(\left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\top. \end{aligned} \quad (6.97)$$

Then,

$$\mathbf{F}_\psi = \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{F} \left(\left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\top. \quad (6.98)$$

Then,

$$\begin{aligned} & k_\psi(\mathbf{x}, \mathbf{x}') \\ &= \mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\top \left(\left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \right)^\top \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^\top \mathbf{F}^{-1} \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right) \left(\frac{\partial \psi}{\partial \boldsymbol{\theta}} \right)^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}'). \end{aligned} \quad (6.99)$$

The right hand side can be written as

$$\mathbf{g}(\boldsymbol{\theta}, \mathbf{x})^\top \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\theta}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}'). \quad (6.100)$$

Therefore,

$$k_\psi(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}'). \quad (6.101)$$

6.14

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}|\boldsymbol{\mu}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{S}). \quad (6.102)$$

Let

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{g}(\boldsymbol{\mu}, \mathbf{x})^\top \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}'), \quad (6.103)$$

where

$$\begin{aligned} \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}) &= \nabla_{\boldsymbol{\mu}} \ln p(\mathbf{x}|\boldsymbol{\mu}), \\ \mathbf{F} &= \mathbb{E}_{\mathbf{x}} (\mathbf{g}(\boldsymbol{\mu}, \mathbf{x}) \mathbf{g}(\boldsymbol{\mu}, \mathbf{x})^\top). \end{aligned} \quad (6.104)$$

We have

$$\mathbf{g}(\boldsymbol{\mu}, \mathbf{x}) = -\mathbf{S}^{-1}(\mathbf{x} - \boldsymbol{\mu}). \quad (6.105)$$

Then,

$$\mathbf{F} = \int \mathbf{S}^{-1}(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{S}) d\mathbf{x}. \quad (6.106)$$

The right hand side can be written as

$$\mathbf{S}^{-1} \left(\int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{S}) d\mathbf{x} \right) \mathbf{S}^{-1} = \mathbf{S}^{-1}. \quad (6.107)$$

Then,

$$\mathbf{F} = \mathbf{S}^{-1}, \quad (6.108)$$

so that

$$\mathbf{g}(\boldsymbol{\mu}, \mathbf{x})^\top \mathbf{F}^{-1} \mathbf{g}(\boldsymbol{\mu}, \mathbf{x}') = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu}). \quad (6.109)$$

Therefore,

$$k(\mathbf{x}, \mathbf{x}') = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu}). \quad (6.110)$$

6.15

Let k be a positive definite kernel function. Then,

$$\sum_{n=1}^N \sum_{n'=1}^N c_n c_{n'} k(x_n, x_{n'}) > 0. \quad (6.111)$$

Then,

$$\mathbf{c}^\top \mathbf{K} \mathbf{c} \geq 0, \quad (6.112)$$

where

$$\mathbf{c} = \begin{bmatrix} c \\ c' \end{bmatrix}, \mathbf{K} = \begin{bmatrix} k(x, x) & k(x, x') \\ k(x, x') & k(x', x') \end{bmatrix}. \quad (6.113)$$

Then,

$$\det \mathbf{K} \geq 0. \quad (6.114)$$

Therefore,

$$k(x, x')^2 \leq k(x, x)k(x', x'). \quad (6.115)$$

6.16

Let

$$J(\mathbf{w}) = f(\mathbf{w}^\top \phi(\mathbf{x}_1), \dots, \mathbf{w}^\top \phi(\mathbf{x}_N)) + g(\|\mathbf{w}\|^2), \quad (6.116)$$

where g is a monotonically increasing function. Setting the derivative with respect to $\mathbf{w}$ to zero gives

$$\mathbf{0} = \sum_{n=1}^N (\nabla_n f) \phi(\mathbf{x}_n) + 2g'(\|\mathbf{w}\|^2) \mathbf{w}, \quad (6.117)$$

where ∇_n is the partial derivative with respect to the n th argument. Then,

$$\mathbf{w} = -\frac{1}{2g'(\|\mathbf{w}\|^2)} \sum_{n=1}^N (\nabla_n f) \phi(\mathbf{x}_n). \quad (6.118)$$

Therefore, J is minimised by $\mathbf{w}$ in the form of a linear combination of $\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)$.

6.17

Let

$$E = \frac{1}{2} \sum_{n=1}^N \int (y(\mathbf{x}_n + \boldsymbol{\xi}) - t_n)^2 p(\boldsymbol{\xi}) d\boldsymbol{\xi}. \quad (6.119)$$

By the transformation

$$\mathbf{z}_n = \mathbf{x}_n + \boldsymbol{\xi}, \quad (6.120)$$

we have

$$E = \frac{1}{2} \sum_{n=1}^N \int (y(\mathbf{z}_n) - t_n)^2 p(\mathbf{z}_n - \mathbf{x}_n) d\mathbf{z}_n. \quad (6.121)$$

Setting the variation of E with respect to y to zero gives

$$0 = \sum_{n=1}^N \int (y(\mathbf{z}_n) - t_n) p(\mathbf{z}_n - \mathbf{x}_n) \delta y(\mathbf{z}_n) d\mathbf{z}_n. \quad (6.122)$$

Then,

$$\sum_{n=1}^N (y(\mathbf{z}_n) - t_n) p(\mathbf{z}_n - \mathbf{x}_n) = 0, \quad (6.123)$$

so that

$$\sum_{n=1}^N y(\mathbf{z}_n) p(\mathbf{z}_n - \mathbf{x}_n) = \sum_{n=1}^N t_n p(\mathbf{z}_n - \mathbf{x}_n). \quad (6.124)$$

Therefore,

$$y(\mathbf{x}) = \sum_{n=1}^N t_n k(\mathbf{x}, \mathbf{x}_n), \quad (6.125)$$

where

$$k(\mathbf{x}, \mathbf{x}_n) = \frac{p(\mathbf{x} - \mathbf{x}_n)}{\sum_{n'=1}^N p(\mathbf{x} - \mathbf{x}_{n'})}. \quad (6.126)$$

6.18

Let x be a variable such that

$$p(x, t) = \frac{1}{N} \sum_{n=1}^N f(x - x_n, t - t_n), \quad (6.127)$$

where

$$f(x, t) = \mathcal{N}(x|0, \sigma^2) \mathcal{N}(t|0, \sigma^2). \quad (6.128)$$

(a)

By Bayes' theorem,

$$p(t|x) = \frac{p(x, t)}{p(x)}. \quad (6.129)$$

By marginalisation,

$$p(x) = \int p(x, t) dt. \quad (6.130)$$

The right hand side can be written as

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N \mathcal{N}(x - x_n | 0, \sigma^2) \int \mathcal{N}(t - t_n | 0, \sigma^2) dt \\ &= \frac{1}{N} \sum_{n=1}^N \mathcal{N}(x - x_n | 0, \sigma^2). \end{aligned} \quad (6.131)$$

Then,

$$p(t|x) = \frac{\sum_{n=1}^N \mathcal{N}(x - x_n | 0, \sigma^2) \mathcal{N}(t - t_n | 0, \sigma^2)}{\sum_{n=1}^N \mathcal{N}(x - x_n | 0, \sigma^2)}. \quad (6.132)$$

Therefore,

$$p(t|x) = \sum_{n=1}^N k(x, x_n) \mathcal{N}(t - t_n | 0, \sigma^2), \quad (6.133)$$

where

$$k(x, x_n) = \frac{\mathcal{N}(x - x_n | 0, \sigma^2)}{\sum_{n=1}^N \mathcal{N}(x - x_n | 0, \sigma^2)}. \quad (6.134)$$

(b)

We have

$$\mathbb{E}(t|x) = \int t p(t|x) dt. \quad (6.135)$$

By (a), the right hand side can be written as

$$\sum_{n=1}^N k(x, x_n) \int t \mathcal{N}(t - t_n | 0, \sigma^2) dt. \quad (6.136)$$

By the transformation

$$t' = t - t_n, \quad (6.137)$$

the integral can be written as

$$\int (t' + t_n) \mathcal{N}(t' | 0, \sigma^2) dt' = \int t' \mathcal{N}(t' | 0, \sigma^2) dt' + t_n \int \mathcal{N}(t' | 0, \sigma^2) dt'. \quad (6.138)$$

The right hand side can be written as t_n . Therefore,

$$\mathbb{E}(t|x) = \sum_{n=1}^N k(x, x_n) t_n. \quad (6.139)$$

(c)

We have

$$\text{var}(t|x) = \int (t - \mathbb{E}(t|x))^2 p(t|x) dt. \quad (6.140)$$

By (a) and (b), the right hand side can be written as

$$\sum_{n=1}^N k(x, x_n) \int \left(t - \sum_{n=1}^N k(x, x_n) t_n \right)^2 \mathcal{N}(t - t_n | 0, \sigma^2) dt. \quad (6.141)$$

By the transformation

$$t' = t - t_n, \quad (6.142)$$

the integral can be written as

$$\begin{aligned} & \int \left(t' + t_n - \sum_{n=1}^N k(x, x_n) t_n \right)^2 \mathcal{N}(t' | 0, \sigma^2) dt' \\ &= \int t'^2 \mathcal{N}(t' | 0, \sigma^2) dt' + 2 \left(t_n - \sum_{n=1}^N k(x, x_n) t_n \right) \int t' \mathcal{N}(t' | 0, \sigma^2) dt' \\ & \quad + \left(t_n - \sum_{n=1}^N k(x, x_n) t_n \right)^2 \int \mathcal{N}(t' | 0, \sigma^2) dt'. \end{aligned} \quad (6.143)$$

The right hand side can be written as

$$\sigma^2 + \left(t_n - \sum_{n=1}^N k(x, x_n) t_n \right)^2. \quad (6.144)$$

Therefore,

$$\text{var}(t|x) = \sum_{n=1}^N k(x, x_n) \left(\sigma^2 + \left(t_n - \sum_{n'=1}^N k(x, x_{n'}) t_{n'} \right)^2 \right), \quad (6.145)$$

so that

$$\text{var}(t|x) = \sigma^2 + \sum_{n=1}^N k(x, x_n) \left(t_n - \sum_{n'=1}^N k(x, x_{n'}) t_{n'} \right)^2. \quad (6.146)$$

6.19

Let

$$E = \frac{1}{2} \sum_{n=1}^N \int (y(\mathbf{x}_n - \boldsymbol{\xi}_n) - t_n)^2 p(\boldsymbol{\xi}_n) d\boldsymbol{\xi}_n. \quad (6.147)$$

By the transformation

$$\mathbf{z}_n = \mathbf{x}_n - \boldsymbol{\xi}_n, \quad (6.148)$$

we have

$$E = \frac{1}{2} \sum_{n=1}^N \int (y(\mathbf{z}_n) - t_n)^2 p(\mathbf{x}_n - \mathbf{z}_n) d\mathbf{z}_n. \quad (6.149)$$

Setting the variation of E with respect to y to zero gives

$$0 = \sum_{n=1}^N \int (y(\mathbf{z}_n) - t_n) p(\mathbf{x}_n - \mathbf{z}_n) \delta y(\mathbf{z}_n) d\mathbf{z}_n. \quad (6.150)$$

Then,

$$0 = \sum_{n=1}^N (y(\mathbf{z}_n) - t_n) p(\mathbf{x}_n - \mathbf{z}_n), \quad (6.151)$$

so that

$$\sum_{n=1}^N y(\mathbf{z}_n) p(\mathbf{x}_n - \mathbf{z}_n) = \sum_{n=1}^N t_n p(\mathbf{x}_n - \mathbf{z}_n). \quad (6.152)$$

Therefore,

$$y(\mathbf{x}) = \sum_{n=1}^N t_n k(\mathbf{x}, \mathbf{x}_n), \quad (6.153)$$

where

$$k(\mathbf{x}, \mathbf{x}_n) = \frac{p(\mathbf{x}_n - \mathbf{x})}{\sum_{n'=1}^N p(\mathbf{x}_{n'} - \mathbf{x})}. \quad (6.154)$$

6.20

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{t}|\mathbf{y}) &= \mathcal{N}(\mathbf{t}|\mathbf{y}, \beta^{-1}\mathbf{I}), \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{K}), \end{aligned} \quad (6.155)$$

where

$$K_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}). \quad (6.156)$$

By Bayes' theorem,

$$p(t_{N+1}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}'), \quad (6.157)$$

where

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \end{bmatrix}. \quad (6.158)$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{y})p(\mathbf{y})d\mathbf{y}. \quad (6.159)$$

The logarighm of the integrand of the right hand side except the terms independent of $\mathbf{y}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \mathbf{y}\|^2 - \frac{1}{2}\mathbf{y}^\top \mathbf{K}^{-1}\mathbf{y} = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v}, \quad (6.160)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{y} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\mathbf{I} + \mathbf{K}^{-1} & -\beta\mathbf{I} \\ -\beta\mathbf{I} & \beta\mathbf{I} \end{bmatrix}. \quad (6.161)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{K} & \mathbf{K} \\ \mathbf{K} & \mathbf{C} \end{bmatrix}, \quad (6.162)$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{K}. \quad (6.163)$$

Then,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \mathbf{C}). \quad (6.164)$$

Then,

$$p(\mathbf{t}') = \mathcal{N}(\mathbf{t}'|\mathbf{0}, \mathbf{C}'), \quad (6.165)$$

where

$$\mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{k} \\ \mathbf{k}^\top & c \end{bmatrix}, \quad (6.166)$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= \beta^{-1} + k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}). \end{aligned} \quad (6.167)$$

By 2.24,

$$\begin{bmatrix} c & \mathbf{k}^\top \\ \mathbf{k} & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} s^{-1} & -s^{-1}\mathbf{k}^\top\mathbf{C}^{-1} \\ -s^{-1}\mathbf{C}^{-1}\mathbf{k} & \mathbf{C}^{-1} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\top\mathbf{C}^{-1} \end{bmatrix}, \quad (6.168)$$

where

$$s = c - \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{k}. \quad (6.169)$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1}|m, s), \quad (6.170)$$

where

$$m = \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{t}. \quad (6.171)$$

6.21

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top\boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{0}, \mathbf{I}). \end{aligned} \quad (6.172)$$

Then,

$$\begin{aligned} p(\mathbf{t}|\mathbf{y}) &= \mathcal{N}(\mathbf{t}|\mathbf{y}, \beta^{-1}\mathbf{I}), \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{K}), \end{aligned} \quad (6.173)$$

where

$$\mathbf{K} = \boldsymbol{\Phi}\boldsymbol{\Phi}^\top. \quad (6.174)$$

By 6.20,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1}|m, s), \quad (6.175)$$

where

$$\begin{aligned} m &= \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{t}, \\ s &= c - \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{k}, \end{aligned} \quad (6.176)$$

and

$$\begin{aligned} \mathbf{k} &= \boldsymbol{\Phi}\boldsymbol{\phi}_{N+1}, \\ \mathbf{C} &= \beta^{-1}\mathbf{I} + \mathbf{K}, \\ c &= \beta^{-1} + \boldsymbol{\phi}_{N+1}^\top\boldsymbol{\phi}_{N+1}. \end{aligned} \quad (6.177)$$

By 2.26,

$$\mathbf{C}^{-1} = \beta\mathbf{I} - \beta^2\boldsymbol{\Phi}\mathbf{S}\boldsymbol{\Phi}^\top, \quad (6.178)$$

where

$$\mathbf{S} = (\mathbf{I} + \beta \mathbf{\Phi}^\top \mathbf{\Phi})^{-1}. \quad (6.179)$$

Then,

$$\begin{aligned} m &= \phi_{N+1}^\top \mathbf{\Phi}^\top (\beta \mathbf{I} - \beta^2 \mathbf{\Phi} \mathbf{S} \mathbf{\Phi}^\top) \mathbf{t}, \\ s &= \beta^{-1} + \phi_{N+1}^\top \phi_{N+1} - \phi_{N+1}^\top \mathbf{\Phi}^\top (\beta \mathbf{I} - \beta^2 \mathbf{\Phi} \mathbf{S} \mathbf{\Phi}^\top) \mathbf{\Phi} \phi_{N+1}. \end{aligned} \quad (6.180)$$

We have

$$\beta \mathbf{\Phi}^\top \mathbf{\Phi} = \mathbf{S}^{-1} - \mathbf{I}. \quad (6.181)$$

Then,

$$\begin{aligned} m &= \beta \phi_{N+1}^\top (\mathbf{I} - (\mathbf{S}^{-1} - \mathbf{I}) \mathbf{S}) \mathbf{\Phi}^\top \mathbf{t}, \\ s &= \beta^{-1} + \phi_{N+1}^\top (\mathbf{I} - (\mathbf{S}^{-1} - \mathbf{I}) + (\mathbf{S}^{-1} - \mathbf{I}) \mathbf{S} (\mathbf{S}^{-1} - \mathbf{I})) \phi_{N+1}. \end{aligned} \quad (6.182)$$

Therefore,

$$\begin{aligned} m &= \beta \phi_{N+1}^\top \mathbf{S} \mathbf{\Phi}^\top \mathbf{t}, \\ s &= \beta^{-1} + \phi_{N+1}^\top \mathbf{S} \phi_{N+1}. \end{aligned} \quad (6.183)$$

6.22

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{t}|\mathbf{y}) &= \mathcal{N}(\mathbf{t}|\mathbf{y}, \beta^{-1} \mathbf{I}), \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{K}), \end{aligned} \quad (6.184)$$

where

$$K_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}). \quad (6.185)$$

(a)

By Bayes' theorem,

$$p(t_{N+1}, \dots, t_{N+L}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}'), \quad (6.186)$$

where

$$\mathbf{t}' = \begin{bmatrix} \mathbf{t} \\ t_{N+1} \\ \vdots \\ t_{N+L} \end{bmatrix}. \quad (6.187)$$

By 6.20,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \mathbf{C}), \quad (6.188)$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{K}. \quad (6.189)$$

Let $\mathbf{K}'$ and $\mathbf{\Gamma}$ be matrices where

$$\begin{aligned} K'_{nl} &= k(\mathbf{x}_n, \mathbf{x}_{N+l}), \\ \Gamma_{ll'} &= \beta^{-1}I_{ll'} + k(\mathbf{x}_{N+l}, \mathbf{x}_{N+l'}). \end{aligned} \quad (6.190)$$

Then,

$$p(\mathbf{t}') = \mathcal{N}(\mathbf{t}'|\mathbf{0}, \mathbf{C}'), \quad (6.191)$$

where

$$\mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{K}' \\ \mathbf{K}'^\top & \mathbf{\Gamma} \end{bmatrix}. \quad (6.192)$$

By 2.24,

$$\begin{bmatrix} \mathbf{\Gamma} & \mathbf{K}'^\top \\ \mathbf{K}' & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{S}^{-1} & -\mathbf{S}^{-1}\mathbf{K}'^\top\mathbf{C}^{-1} \\ -\mathbf{C}^{-1}\mathbf{K}'\mathbf{S}^{-1} & \mathbf{C}^{-1} + \mathbf{C}^{-1}\mathbf{K}'\mathbf{S}^{-1}\mathbf{K}'^\top\mathbf{C}^{-1} \end{bmatrix}, \quad (6.193)$$

where

$$\mathbf{S} = \mathbf{\Gamma} - \mathbf{K}'^\top\mathbf{C}^{-1}\mathbf{K}'. \quad (6.194)$$

Therefore,

$$p(t_{N+1}, \dots, t_{N+L}|\mathbf{t}) = \mathcal{N}(t_{N+1}, \dots, t_{N+L}|\mathbf{m}, \mathbf{S}), \quad (6.195)$$

where

$$\mathbf{m} = \mathbf{K}'^\top\mathbf{C}^{-1}\mathbf{t}. \quad (6.196)$$

(b)

By (a),

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1}|m, s), \quad (6.197)$$

where

$$\begin{aligned} m &= \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{t}, \\ s &= c - \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{k}, \end{aligned} \quad (6.198)$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= \beta^{-1} + k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}). \end{aligned} \quad (6.199)$$

6.23

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables such that

$$\begin{aligned} p(\mathbf{t}_n|\mathbf{y}_n) &= \mathcal{N}(\mathbf{t}_n|\mathbf{y}_n, \beta^{-1}\mathbf{I}), \\ p(\mathbf{Y}) &= \mathcal{N}(\mathbf{Y}|\mathbf{O}, \mathbf{K}), \end{aligned} \quad (6.200)$$

where

$$K_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}). \quad (6.201)$$

By Bayes' theorem,

$$p(\mathbf{t}_{N+1}|\mathbf{T})p(\mathbf{T}) = p(\mathbf{T}'), \quad (6.202)$$

where

$$\mathbf{T}' = \begin{bmatrix} \mathbf{T} \\ \mathbf{t}_{N+1}^\top \end{bmatrix}. \quad (6.203)$$

By marginalisation,

$$p(\mathbf{T}) = \int p(\mathbf{T}|\mathbf{Y})p(\mathbf{Y})d\mathbf{Y}. \quad (6.204)$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{T}$ and $\mathbf{Y}$ can be written as

$$-\frac{\beta}{2} \sum_{n=1}^N \|\mathbf{t}_n - \mathbf{y}_n\|^2 - \frac{1}{2} \text{tr}(\mathbf{Y}^\top \mathbf{K}^{-1} \mathbf{Y}) = -\frac{1}{2} \text{tr} \mathbf{M}, \quad (6.205)$$

where

$$\mathbf{M} = \begin{bmatrix} \mathbf{Y} \\ \mathbf{T} \end{bmatrix}^\top \begin{bmatrix} \beta\mathbf{I} + \mathbf{K}^{-1} & -\beta\mathbf{I} \\ -\beta\mathbf{I} & \beta\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{Y} \\ \mathbf{T} \end{bmatrix}. \quad (6.206)$$

By 2.24,

$$\begin{bmatrix} \beta\mathbf{I} + \mathbf{K}^{-1} & -\beta\mathbf{I} \\ -\beta\mathbf{I} & \beta\mathbf{I} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{K} & \mathbf{K} \\ \mathbf{K} & \mathbf{C} \end{bmatrix}, \quad (6.207)$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \mathbf{K}. \quad (6.208)$$

Then,

$$p(\mathbf{T}) = \mathcal{N}(\mathbf{T}|\mathbf{O}, \mathbf{C}). \quad (6.209)$$

Then,

$$p(\mathbf{T}') = \mathcal{N}(\mathbf{T}'|\mathbf{O}, \mathbf{C}'), \quad (6.210)$$

where

$$\mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{k} \\ \mathbf{k}^\top & c \end{bmatrix}, \quad (6.211)$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= \beta^{-1} + k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}). \end{aligned} \quad (6.212)$$

By 2.24,

$$\begin{bmatrix} c & \mathbf{k}^\top \\ \mathbf{k} & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} s^{-1} & -s^{-1}\mathbf{k}^\top\mathbf{C}^{-1} \\ -s^{-1}\mathbf{C}^{-1}\mathbf{k} & \mathbf{C}^{-1} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\top\mathbf{C}^{-1} \end{bmatrix}, \quad (6.213)$$

where

$$s = c - \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{k}. \quad (6.214)$$

Therefore,

$$p(\mathbf{t}_{N+1}|\mathbf{T}) = \mathcal{N}(\mathbf{t}_{N+1}|\mathbf{m}, s\mathbf{I}), \quad (6.215)$$

where

$$\mathbf{m} = \mathbf{T}^\top\mathbf{C}^{-1}\mathbf{k}. \quad (6.216)$$

6.24

(a)

Let $\mathbf{M}$ be a $D \times D$ diagonal matrix such that $0 < M_{dd} < 1$. For any vector $\mathbf{v}$ in D dimensions,

$$\mathbf{v}^\top\mathbf{M}\mathbf{v} = \sum_{d=1}^D \sum_{d'=1}^D v_d M_{dd'} v_{d'}. \quad (6.217)$$

The right hand side can be written as

$$\sum_{d=1}^D M_{dd} v_d^2 > 0. \quad (6.218)$$

Therefore, $\mathbf{M}$ is positive definite.

(b)

Let $\mathbf{M}_1$ and $\mathbf{M}_2$ be $D \times D$ positive definite matrices. Then, for any vector $\mathbf{v}$ in D dimensions,

$$\begin{aligned}\mathbf{v}^\top \mathbf{M}_1 \mathbf{v} &> 0, \\ \mathbf{v}^\top \mathbf{M}_2 \mathbf{v} &> 0.\end{aligned}\tag{6.219}$$

Then,

$$\mathbf{v}^\top (\mathbf{M}_1 + \mathbf{M}_2) \mathbf{v} > 0.\tag{6.220}$$

Therefore, $\mathbf{M}_1 + \mathbf{M}_2$ is positive definite.

6.25

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n|a_n) &= \sigma(a_n)^{t_n} (1 - \sigma(a_n))^{1-t_n}, \\ p(\mathbf{a}) &= \mathcal{N}(\mathbf{a}|\mathbf{0}, \mathbf{C}),\end{aligned}\tag{6.221}$$

where

$$\begin{aligned}\sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ C_{nn'} &= k(\mathbf{x}_n, \mathbf{x}_{n'}) + \nu I_{nn'}.\end{aligned}\tag{6.222}$$

(a)

By Bayes' theorem,

$$p(\mathbf{a}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{a})p(\mathbf{a}).\tag{6.223}$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{a}$ can be written as $-E(\mathbf{a})$ where

$$E(\mathbf{a}) = -\sum_{n=1}^N (t_n \ln \sigma(a_n) + (1 - t_n) \ln (1 - \sigma(a_n))) + \frac{1}{2} \mathbf{a}^\top \mathbf{C}^{-1} \mathbf{a}.\tag{6.224}$$

We have the Taylor series

$$\begin{aligned}E(\mathbf{a}) &= E(\mathbf{a}_{\text{MAP}}) + \frac{1}{2}(\mathbf{a} - \mathbf{a}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{a} - \mathbf{a}_{\text{MAP}}) \\ &\quad + O(\|\mathbf{a} - \mathbf{a}_{\text{MAP}}\|^3),\end{aligned}\tag{6.225}$$

where $\mathbf{a}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{a}_{\text{MAP}}). \quad (6.226)$$

Therefore,

$$p(\mathbf{a}|\mathbf{t}) \simeq \mathcal{N}(\mathbf{a}|\mathbf{a}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}). \quad (6.227)$$

(b)

We have the Taylor series

$$\nabla E(\mathbf{a}_{\text{MAP}}) = \nabla E(\mathbf{a}) + \nabla \nabla E(\mathbf{a})(\mathbf{a}_{\text{MAP}} - \mathbf{a}) + O(\|\mathbf{a}_{\text{MAP}} - \mathbf{a}\|^2). \quad (6.228)$$

We have

$$(\nabla E(\mathbf{a}))_n = -(t_n(1 - \sigma(a_n)) - (1 - t_n)\sigma(a_n)) + (\mathbf{C}^{-1}\mathbf{a})_n, \quad (6.229)$$

so that

$$\nabla E(\mathbf{a}) = \mathbf{C}^{-1}\mathbf{a} + \boldsymbol{\sigma} - \mathbf{t}, \quad (6.230)$$

where

$$\sigma_n = \sigma(a_n). \quad (6.231)$$

Then,

$$\nabla \nabla E(\mathbf{a}) = \mathbf{C}^{-1} + \mathbf{W}, \quad (6.232)$$

where

$$W_{nn'} = \sigma(a_n)(1 - \sigma(a_n))I_{nn'}. \quad (6.233)$$

Then,

$$\mathbf{0} \simeq \mathbf{C}^{-1}\mathbf{a} + \boldsymbol{\sigma} - \mathbf{t} + (\mathbf{C}^{-1} + \mathbf{W})(\mathbf{a}_{\text{MAP}} - \mathbf{a}). \quad (6.234)$$

Then,

$$(\mathbf{C}^{-1} + \mathbf{W})\mathbf{a}_{\text{MAP}} \simeq \mathbf{t} - \boldsymbol{\sigma} + \mathbf{W}\mathbf{a}. \quad (6.235)$$

Therefore,

$$\mathbf{a}_{\text{MAP}} \simeq (\mathbf{C}^{-1} + \mathbf{W})^{-1}(\mathbf{t} - \boldsymbol{\sigma} + \mathbf{W}\mathbf{a}). \quad (6.236)$$

(c)

By (b),

$$\mathbf{a}_{\text{MAP}} = (\mathbf{C}^{-1} + \mathbf{W}_{\text{MAP}})^{-1} (\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}} + \mathbf{W}_{\text{MAP}} \mathbf{a}_{\text{MAP}}), \quad (6.237)$$

where

$$\begin{aligned} W_{\text{MAP}nn'} &= \sigma(a_{\text{MAP}n}) (1 - \sigma(a_{\text{MAP}n})) I_{nn'}, \\ \sigma_{\text{MAP}n} &= \sigma(a_{\text{MAP}n}). \end{aligned} \quad (6.238)$$

Then,

$$(\mathbf{C}^{-1} + \mathbf{W}_{\text{MAP}}) \mathbf{a}_{\text{MAP}} = \mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}} + \mathbf{W}_{\text{MAP}} \mathbf{a}_{\text{MAP}}. \quad (6.239)$$

Therefore,

$$\mathbf{a}_{\text{MAP}} = \mathbf{C}(\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}). \quad (6.240)$$

6.26

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | a_n) &= \sigma(a_n)^{t_n} (1 - \sigma(a_n))^{1-t_n}, \\ p(\mathbf{a}) &= \mathcal{N}(\mathbf{a} | \mathbf{0}, \mathbf{C}), \end{aligned} \quad (6.241)$$

where

$$\begin{aligned} \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ C_{nn'} &= k(\mathbf{x}_n, \mathbf{x}_{n'}) + \nu I_{nn'}. \end{aligned} \quad (6.242)$$

By marginalisation,

$$p(a_{N+1} | \mathbf{t}) = \int p(a_{N+1} | \mathbf{a}) p(\mathbf{a} | \mathbf{t}) d\mathbf{a}. \quad (6.243)$$

We have

$$p(\mathbf{a}') = \mathcal{N}(\mathbf{a}' | \mathbf{0}, \mathbf{C}'), \quad (6.244)$$

where

$$\mathbf{a}' = \begin{bmatrix} \mathbf{a} \\ a_{N+1} \end{bmatrix}, \mathbf{C}' = \begin{bmatrix} \mathbf{C} & \mathbf{k} \\ \mathbf{k}^\top & c \end{bmatrix}, \quad (6.245)$$

and

$$\begin{aligned} k_n &= k(\mathbf{x}_n, \mathbf{x}_{N+1}), \\ c &= k(\mathbf{x}_{N+1}, \mathbf{x}_{N+1}) + \nu. \end{aligned} \quad (6.246)$$

By 2.24,

$$\begin{bmatrix} c & \mathbf{k}^\top \\ \mathbf{k} & \mathbf{C} \end{bmatrix}^{-1} = \begin{bmatrix} s^{-1} & -s^{-1}\mathbf{k}^\top\mathbf{C}^{-1} \\ -s^{-1}\mathbf{C}^{-1}\mathbf{k} & \mathbf{C}^{-1} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\top\mathbf{C}^{-1} \end{bmatrix}, \quad (6.247)$$

where

$$s = c - \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{k}. \quad (6.248)$$

Then,

$$p(a_{N+1}|\mathbf{a}) = \mathcal{N}(a_{N+1}|m, s), \quad (6.249)$$

where

$$m = \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{a}. \quad (6.250)$$

By 6.25(a),

$$p(\mathbf{a}|\mathbf{t}) \simeq \mathcal{N}(\mathbf{a}|\mathbf{a}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}), \quad (6.251)$$

where $\mathbf{a}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla\nabla E(\mathbf{a}_{\text{MAP}}), \quad (6.252)$$

where

$$E(\mathbf{a}) = -\sum_{n=1}^N (t_n \ln \sigma(a_n) + (1 - t_n) \ln (1 - \sigma(a_n))) + \frac{1}{2}\mathbf{a}^\top\mathbf{C}^{-1}\mathbf{a}. \quad (6.253)$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{t}$ and $\mathbf{a}$ can be approximated as

$$\begin{aligned} & -\frac{1}{2s} (a_{N+1} - \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{a})^2 - \frac{1}{2}(\mathbf{a} - \mathbf{a}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}} (\mathbf{a} - \mathbf{a}_{\text{MAP}}) \\ & = -\frac{1}{2}\mathbf{a}'^\top \mathbf{M} \mathbf{a}' + \mathbf{a}'^\top \mathbf{v} - \frac{1}{2}\mathbf{a}_{\text{MAP}}^\top \mathbf{H}_{\text{MAP}} \mathbf{a}_{\text{MAP}}. \end{aligned} \quad (6.254)$$

where

$$\begin{aligned} \mathbf{M} &= \begin{bmatrix} \mathbf{H}_{\text{MAP}} + s^{-1}\mathbf{C}^{-1}\mathbf{k}\mathbf{k}^\top\mathbf{C}^{-1} & -s^{-1}\mathbf{C}^{-1}\mathbf{k} \\ -s^{-1}\mathbf{k}^\top\mathbf{C}^{-1} & s^{-1} \end{bmatrix}, \\ \mathbf{v} &= \begin{bmatrix} \mathbf{H}_{\text{MAP}}\mathbf{a}_{\text{MAP}} \\ 0 \end{bmatrix}. \end{aligned} \quad (6.255)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{H}_{\text{MAP}}^{-1} & \mathbf{H}_{\text{MAP}}^{-1}\mathbf{C}^{-1}\mathbf{k} \\ \mathbf{k}^\top\mathbf{C}^{-1}\mathbf{H}_{\text{MAP}}^{-1} & \sigma^2 \end{bmatrix}, \quad (6.256)$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix} \mathbf{a}_{\text{MAP}} \\ \mu \end{bmatrix}, \quad (6.257)$$

where

$$\begin{aligned} \mu &= \mathbf{k}^\top \mathbf{C}^{-1} \mathbf{a}_{\text{MAP}}, \\ \sigma^2 &= s + \mathbf{k}^\top \mathbf{C}^{-1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{C}^{-1} \mathbf{k}. \end{aligned} \quad (6.258)$$

Then,

$$p(a_{N+1}|\mathbf{t}) \simeq \mathcal{N}(a_{N+1}|\mu, \sigma^2). \quad (6.259)$$

By 6.25(c),

$$\mathbf{a}_{\text{MAP}} = \mathbf{C}(\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}), \quad (6.260)$$

where

$$\sigma_{\text{MAP}n} = \sigma(a_{\text{MAP}n}). \quad (6.261)$$

Therefore,

$$\begin{aligned} \mu &= \mathbf{k}^\top (\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}), \\ \sigma^2 &= c - \mathbf{k}^\top \mathbf{C}^{-1} (\mathbf{C} - \mathbf{H}_{\text{MAP}}^{-1}) \mathbf{C}^{-1} \mathbf{k}. \end{aligned} \quad (6.262)$$

6.27

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|a_n) &= \sigma(a_n)^{t_n} (1 - \sigma(a_n))^{1-t_n}, \\ p(\mathbf{a}|\boldsymbol{\theta}) &= \mathcal{N}(\mathbf{a}|\mathbf{0}, \mathbf{C}), \end{aligned} \quad (6.263)$$

where

$$\sigma(a) = \frac{1}{1 + \exp(-a)} \quad (6.264)$$

and $\mathbf{C}$ is dependent on $\boldsymbol{\theta}$ in M dimensions.

(a)

By marginalisation,

$$p(\mathbf{t}|\boldsymbol{\theta}) = \int p(\mathbf{t}|\mathbf{a})p(\mathbf{a}|\boldsymbol{\theta})d\mathbf{a}. \quad (6.265)$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{a}$ can be written as $-E(\mathbf{a})$ where

$$E(\mathbf{a}) = -\sum_{n=1}^N (t_n \ln \sigma(a_n) - (1 - t_n) \ln (1 - \sigma(a_n))) + \frac{1}{2} \mathbf{a}^\top \mathbf{C}^{-1} \mathbf{a}. \quad (6.266)$$

We have the Taylor series

$$E(\mathbf{a}) = E(\mathbf{a}_{\text{MAP}}) + \frac{1}{2} (\mathbf{a} - \mathbf{a}_{\text{MAP}})^{\text{T}} \mathbf{H}_{\text{MAP}} (\mathbf{a} - \mathbf{a}_{\text{MAP}}) + O(\|\mathbf{a} - \mathbf{a}_{\text{MAP}}\|^3), \quad (6.267)$$

where $\mathbf{a}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{a}_{\text{MAP}}). \quad (6.268)$$

Then,

$$p(\mathbf{t}|\boldsymbol{\theta}) \simeq \exp(-E(\mathbf{a}_{\text{MAP}})) I, \quad (6.269)$$

where

$$I = \int \exp\left(-\frac{1}{2} (\mathbf{a} - \mathbf{a}_{\text{MAP}})^{\text{T}} \mathbf{H}_{\text{MAP}} (\mathbf{a} - \mathbf{a}_{\text{MAP}})\right) d\mathbf{a}. \quad (6.270)$$

We have

$$I = (2\pi)^{\frac{N}{2}} (\det \mathbf{H}_{\text{MAP}}^{-1})^{\frac{1}{2}}. \quad (6.271)$$

Therefore,

$$\ln p(\mathbf{t}|\boldsymbol{\theta}) \simeq -E(\mathbf{a}_{\text{MAP}}) + \frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{H}_{\text{MAP}}). \quad (6.272)$$

(b)

By (a),

$$(\nabla E(\mathbf{a}))_n = -t_n (1 - \sigma(a_n)) + (1 - t_n) \sigma(a_n) + (\mathbf{C}^{-1} \mathbf{a})_n. \quad (6.273)$$

Then,

$$\nabla E(\mathbf{a}) = \boldsymbol{\sigma} - \mathbf{t} + \mathbf{C}^{-1} \mathbf{a}, \quad (6.274)$$

where

$$\sigma_n = \sigma(a_n). \quad (6.275)$$

Then,

$$\nabla \nabla E(\mathbf{a}) = \mathbf{W} + \mathbf{C}^{-1}, \quad (6.276)$$

where

$$W_{nn'} = \sigma(a_n) (1 - \sigma(a_n)) I_{nn'}. \quad (6.277)$$

Therefore,

$$\mathbf{H}_{\text{MAP}} = \mathbf{W}_{\text{MAP}} + \mathbf{C}^{-1}, \quad (6.278)$$

where

$$W_{\text{MAP}nn'} = \sigma(a_{\text{MAP}n}) (1 - \sigma(a_{\text{MAP}n})) I_{nn'}. \quad (6.279)$$

(c)

By (a),

$$\frac{\partial \ln p(\mathbf{t}|\boldsymbol{\theta})}{\partial \theta_m} \simeq - \left(\frac{\partial E(\mathbf{a}_{\text{MAP}})}{\partial \theta_m} + \frac{1}{2} \frac{\partial \ln(\det \mathbf{H}_{\text{MAP}})}{\partial \theta_m} \right). \quad (6.280)$$

We have

$$\frac{\partial E(\mathbf{a}_{\text{MAP}})}{\partial \theta_m} = \text{tr} \left(\frac{\partial E(\mathbf{a}_{\text{MAP}})}{\partial \mathbf{C}} \frac{\partial \mathbf{C}}{\partial \theta_m} \right). \quad (6.281)$$

We have

$$\frac{\partial E(\mathbf{a}_{\text{MAP}})}{\partial \mathbf{C}} = \nabla E(\mathbf{a}_{\text{MAP}}) \cdot \frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \mathbf{C}} - \frac{1}{2} \mathbf{a}_{\text{MAP}} (\mathbf{C}^{-1})^2 \mathbf{a}_{\text{MAP}}^{\text{T}}, \quad (6.282)$$

where $(\cdot)$ denotes contraction. We have

$$\nabla E(\mathbf{a}_{\text{MAP}}) = \mathbf{0}. \quad (6.283)$$

Then,

$$\frac{\partial E(\mathbf{a}_{\text{MAP}})}{\partial \theta_m} = -\frac{1}{2} \text{tr} \left(\mathbf{a}_{\text{MAP}} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_m} \mathbf{C}^{-1} \mathbf{a}_{\text{MAP}}^{\text{T}} \right). \quad (6.284)$$

We have

$$\frac{\partial \ln(\det \mathbf{H}_{\text{MAP}})}{\partial \theta_m} = \text{tr} \left(\frac{\partial \ln(\det \mathbf{H}_{\text{MAP}})}{\partial \mathbf{H}_{\text{MAP}}} \frac{\partial \mathbf{H}_{\text{MAP}}}{\partial \theta_m} \right). \quad (6.285)$$

By 3.21 and (b),

$$\frac{\partial \ln(\det \mathbf{H}_{\text{MAP}})}{\partial \mathbf{H}_{\text{MAP}}} \frac{\partial \mathbf{H}_{\text{MAP}}}{\partial \theta_m} = \mathbf{H}_{\text{MAP}}^{-1} \left(\frac{\partial \mathbf{W}_{\text{MAP}}}{\partial \theta_m} + \frac{\partial \mathbf{C}^{-1}}{\partial \theta_m} \right). \quad (6.286)$$

We have

$$\frac{\partial \mathbf{W}_{\text{MAP}}}{\partial \theta_m} = \mathbf{T}_{\text{MAP}} \cdot \frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \theta_m}, \quad (6.287)$$

where

$$T_{\text{MAP}nn'n''} = \sigma(a_{\text{MAP}n}) (1 - \sigma(a_{\text{MAP}n})) (1 - 2\sigma(a_{\text{MAP}n})) I_{nn'} I_{n'n''}. \quad (6.288)$$

Then,

$$\frac{\partial \ln(\det \mathbf{H}_{\text{MAP}})}{\partial \theta_m} = \text{tr} \left(\mathbf{H}_{\text{MAP}}^{-1} \left(\mathbf{T}_{\text{MAP}} \cdot \frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \theta_m} + \frac{\partial \mathbf{C}^{-1}}{\partial \theta_m} \right) \right). \quad (6.289)$$

Therefore,

$$\frac{\partial \ln p(\mathbf{t}|\boldsymbol{\theta})}{\partial \theta_m} \simeq \frac{1}{2} \text{tr } \mathbf{M}, \quad (6.290)$$

where

$$\mathbf{M} = \mathbf{a}_{\text{MAP}} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_m} \mathbf{C}^{-1} \mathbf{a}_{\text{MAP}}^\top - \mathbf{H}_{\text{MAP}}^{-1} \left(\mathbf{T}_{\text{MAP}} \cdot \frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \theta_m} + \frac{\partial \mathbf{C}^{-1}}{\partial \theta_m} \right). \quad (6.291)$$

(d)

By 6.25(c),

$$\mathbf{a}_{\text{MAP}} = \mathbf{C}(\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}). \quad (6.292)$$

Then,

$$\frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \theta_m} = \frac{\partial \mathbf{C}}{\partial \theta_m} (\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}) - \mathbf{C} \frac{\partial \boldsymbol{\sigma}_{\text{MAP}}}{\partial \theta_m}. \quad (6.293)$$

We have

$$\frac{\partial \boldsymbol{\sigma}_{\text{MAP}}}{\partial \theta_m} = \mathbf{W}_{\text{MAP}} \frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \theta_m}. \quad (6.294)$$

Therefore,

$$\frac{\partial \mathbf{a}_{\text{MAP}}}{\partial \theta_m} = (\mathbf{I} + \mathbf{C} \mathbf{W}_{\text{MAP}})^{-1} \frac{\partial \mathbf{C}}{\partial \theta_m} (\mathbf{t} - \boldsymbol{\sigma}_{\text{MAP}}). \quad (6.295)$$

7 Sparse Kernel Machines

7.1 (Incomplete)

7.2

Let $t_1, \dots, t_N$ be variables such that

$$t_n (\mathbf{w}^\top \phi_n + b) \geq \gamma, \quad (7.1)$$

for some positive γ . In order to minimise $\|\mathbf{w}\|$, let

$$L = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{n=1}^N a_n (t_n (\mathbf{w}^\top \phi_n + b) - \gamma), \quad (7.2)$$

where $a_n > 0$. Setting the derivatives of L with respect to $\mathbf{w}$ and b to zero gives

$$\begin{aligned} \mathbf{0} &= \mathbf{w} - \sum_{n=1}^N a_n t_n \phi_n, \\ 0 &= - \sum_{n=1}^N a_n t_n. \end{aligned} \quad (7.3)$$

Therefore,

$$\begin{aligned} \mathbf{w}^* &= \sum_{n=1}^N a_n t_n \phi_n, \\ 0 &= \sum_{n=1}^N a_n t_n, \end{aligned} \quad (7.4)$$

where $\mathbf{w}^*$ is an optimal solution to $\mathbf{w}$, which is independent of γ .

7.3 (Incomplete)

7.4

Let $t_1, \dots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n | \mathbf{w}) = \sigma(y_n)^{\frac{1+t_n}{2}} (1 - \sigma(y_n))^{\frac{1-t_n}{2}}, \quad (7.5)$$

where

$$\begin{aligned} y_n &= \mathbf{w}^\top \phi_n + b, \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}. \end{aligned} \quad (7.6)$$

(a)

Let

$$\rho = \max_{\mathbf{w}, b} \left(\min_n d_n \right), \quad (7.7)$$

where

$$d_n = \|\phi_n - \phi_{n\perp}\|, \quad (7.8)$$

and $\phi_{n\perp}$ is a vector such that

$$\begin{aligned} \mathbf{w}^\top \phi_{n\perp} + b &= 0, \\ \phi_n - \phi_{n\perp} &= c\mathbf{w}, \end{aligned} \quad (7.9)$$

for some c . Then,

$$y_n = c\|\mathbf{w}\|^2, \quad (7.10)$$

so that

$$d_n = \frac{|y_n|}{\|\mathbf{w}\|}. \quad (7.11)$$

Focusing on the correctly classified points gives

$$t_n y_n = |y_n|. \quad (7.12)$$

Then,

$$d_n = \frac{t_n y_n}{\|\mathbf{w}\|}, \quad (7.13)$$

so that

$$\rho = \max_{\mathbf{w}, b} \left(\|\mathbf{w}\|^{-1} \min_n t_n y_n \right). \quad (7.14)$$

Rescaling $\mathbf{w}$ and b gives

$$\rho = \max \|\mathbf{w}\|^{-1}, \quad (7.15)$$

under the constraint

$$\min_n t_n y_n = 1. \quad (7.16)$$

Therefore,

$$\rho = \|\mathbf{w}^*\|^{-1}, \quad (7.17)$$

where $\mathbf{w}^*$ is an optimal solution to $\mathbf{w}$.

(b)

Let

$$L(\mathbf{w}, b, \mathbf{a}) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{n=1}^N a_n (t_n y_n - 1), \quad (7.18)$$

where $a_n > 0$. Setting the derivatives of L with respect to $\mathbf{w}$ and b to zero gives

$$\begin{aligned} \mathbf{0} &= \mathbf{w} - \sum_{n=1}^N a_n t_n \phi_n, \\ 0 &= - \sum_{n=1}^N a_n t_n. \end{aligned} \quad (7.19)$$

We have

$$L(\mathbf{w}^*, b^*, \mathbf{a}) = \frac{1}{2} \|\mathbf{w}^*\|^2 - \sum_{n=1}^N a_n (t_n (\mathbf{w}^{*\top} \phi_n + b^*) - 1), \quad (7.20)$$

where b^* is an optimal solutions to b . The right hand side can be written as

$$\begin{aligned} & \frac{1}{2} \|\mathbf{w}^*\|^2 - \mathbf{w}^{*\top} \sum_{n=1}^N a_n t_n \phi_n - b^* \sum_{n=1}^N a_n t_n + \sum_{n=1}^N a_n \\ &= \frac{1}{2} \left\| \sum_{n=1}^N a_n t_n \phi_n \right\|^2 - \left(\sum_{n=1}^N a_n t_n \phi_n \right)^\top \sum_{n=1}^N a_n t_n \phi_n + \sum_{n=1}^N a_n. \end{aligned} \quad (7.21)$$

The right hand side can be written as

$$\tilde{L}(\mathbf{a}) = \sum_{n=1}^N a_n - \frac{1}{2} \left\| \sum_{n=1}^N a_n t_n \phi_n \right\|^2. \quad (7.22)$$

Setting the derivative of $\tilde{L}$ with respect to a_n to zero gives

$$0 = 1 - t_n \phi_n^\top \sum_{n'=1}^N a_{n'} t_{n'} \phi_{n'}. \quad (7.23)$$

Then,

$$\sum_{n=1}^N a_n^* = \left\| \sum_{n=1}^N a_n^* t_n \phi_n \right\|^2, \quad (7.24)$$

where $\mathbf{a}^*$ is an optimal solution to $\mathbf{a}$. By (a), the right hand side can be written as

$$\|\mathbf{w}^*\|^2 = \rho^{-2}. \quad (7.25)$$

Therefore,

$$\rho = \left(\sum_{n=1}^N a_n^* \right)^{-\frac{1}{2}}. \quad (7.26)$$

(c)

By (b),

$$\tilde{L}(\mathbf{a}^*) = \sum_{n=1}^N a_n^* - \frac{1}{2} \left\| \sum_{n=1}^N a_n^* t_n \phi_n \right\|^2. \quad (7.27)$$

By (b), the right hand side can be written as

$$\frac{1}{2} \sum_{n=1}^N a_n^* = \frac{1}{2} \rho^{-2}. \quad (7.28)$$

Therefore,

$$\rho = \left(2\tilde{L}(\mathbf{a}^*) \right)^{-\frac{1}{2}}. \quad (7.29)$$

7.5

Refer to 7.4.

7.6

Let $t_1, \dots, t_N$ be conditionally independent variables from $\{-1, 1\}$ such that

$$p(t_n | \mathbf{w}) = \sigma(y_n)^{\frac{1+t_n}{2}} (1 - \sigma(y_n))^{\frac{1-t_n}{2}}, \quad (7.30)$$

where

$$\begin{aligned} \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ y_n &= \mathbf{w}^\top \phi_n + b. \end{aligned} \quad (7.31)$$

We have

$$1 - \sigma(a) = \sigma(-a). \quad (7.32)$$

Then,

$$p(t_n|\mathbf{w}) = \sigma(y_n t_n). \quad (7.33)$$

Then,

$$p(\mathbf{t}|\mathbf{w}) = \prod_{n=1}^N \sigma(y_n t_n), \quad (7.34)$$

so that

$$-\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^N \ln \sigma(y_n t_n). \quad (7.35)$$

Therefore,

$$-\ln p(\mathbf{t}|\mathbf{y}) = \sum_{n=1}^N \ln (1 + \exp(-y_n t_n)). \quad (7.36)$$

7.7

Let $t_1, \dots, t_N$ be variables. Let

$$\begin{aligned} \xi_n &= \begin{cases} 0, & t_n - y_n < \epsilon, \\ t_n - y_n - \epsilon, & \text{otherwise,} \end{cases} \\ \hat{\xi}_n &= \begin{cases} 0, & y_n - t_n < \epsilon, \\ y_n - t_n - \epsilon, & \text{otherwise,} \end{cases} \end{aligned} \quad (7.37)$$

where

$$\begin{aligned} y_n &= \mathbf{w}^\top \boldsymbol{\phi}_n + b, \\ \boldsymbol{\phi}_n &= \boldsymbol{\phi}(\mathbf{x}_n), \end{aligned} \quad (7.38)$$

and $\epsilon > 0$. Let

$$E = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N E_\epsilon(y_n - t_n), \quad (7.39)$$

where

$$E_\epsilon(a) = \begin{cases} 0, & |a| < \epsilon, \\ |a| - \epsilon, & \text{otherwise.} \end{cases} \quad (7.40)$$

(a)

We have

$$E = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{n=1}^N (\xi_n + \hat{\xi}_n). \quad (7.41)$$

In order to minimise E , let

$$\begin{aligned} L(\mathbf{w}, b, \boldsymbol{\xi}, \hat{\boldsymbol{\xi}}, \boldsymbol{\mu}, \hat{\boldsymbol{\mu}}, \mathbf{a}, \hat{\mathbf{a}}) = & E - \sum_{n=1}^N \mu_n \xi_n - \sum_{n=1}^N \hat{\mu}_n \hat{\xi}_n \\ & - \sum_{n=1}^N a_n (\xi_n - t_n + y_n + \epsilon) \\ & - \sum_{n=1}^N \hat{a}_n (\hat{\xi}_n - y_n + t_n + \epsilon), \end{aligned} \quad (7.42)$$

where $\mu_n \geq 0$, $\hat{\mu}_n \geq 0$, $a_n \geq 0$ and $\hat{a}_n \geq 0$, . Setting the derivatives of L with respect to $\mathbf{w}$, b , ξ_n and $\hat{\xi}_n$ to zero gives

$$\begin{aligned} \mathbf{0} &= \mathbf{w} - \sum_{n=1}^N a_n \phi_n + \sum_{n=1}^N \hat{a}_n \phi_n, \\ 0 &= - \sum_{n=1}^N a_n + \sum_{n=1}^N \hat{a}_n, \\ 0 &= C - \mu_n - a_n, \\ 0 &= C - \hat{\mu}_n - \hat{a}_n. \end{aligned} \quad (7.43)$$

We have

$$\begin{aligned} L(\mathbf{w}^*, b^*, \boldsymbol{\xi}^*, \hat{\boldsymbol{\xi}}^*, \boldsymbol{\mu}, \hat{\boldsymbol{\mu}}, \mathbf{a}, \hat{\mathbf{a}}) = & \frac{1}{2} \|\mathbf{w}^*\|^2 + C \sum_{n=1}^N (\xi_n^* + \hat{\xi}_n^*) \\ & - \sum_{n=1}^N \mu_n \xi_n^* - \sum_{n=1}^N \hat{\mu}_n \hat{\xi}_n^* \\ & - \sum_{n=1}^N a_n (\xi_n^* - t_n + \mathbf{w}^{*\top} \phi_n + b^* + \epsilon) \\ & - \sum_{n=1}^N \hat{a}_n (\hat{\xi}_n^* - \mathbf{w}^{*\top} \phi_n - b^* + t_n + \epsilon), \end{aligned} \quad (7.44)$$

where $\mathbf{w}^*$, b^* , $\boldsymbol{\xi}^*$ and $\hat{\boldsymbol{\xi}}^*$ are optimal solutions to $\mathbf{w}$, b , $\boldsymbol{\xi}$ and $\hat{\boldsymbol{\xi}}$. The right hand side can be written as

$$\begin{aligned}
& \frac{1}{2} \|\mathbf{w}^*\|^2 - \sum_{n=1}^N a_n (-t_n + \mathbf{w}^{*\top} \boldsymbol{\phi}_n + b^* + \epsilon) \\
& - \sum_{n=1}^N \hat{a}_n (-\mathbf{w}^{*\top} \boldsymbol{\phi}_n - b^* + t_n + \epsilon) \\
& = \frac{1}{2} \left\| \sum_{n=1}^N (a_n - \hat{a}_n) \boldsymbol{\phi}_n \right\|^2 + \sum_{n=1}^N (a_n - \hat{a}_n) t_n \\
& - \left(\sum_{n=1}^N (a_n - \hat{a}_n) \boldsymbol{\phi}_n \right)^\top \sum_{n=1}^N (a_n - \hat{a}_n) \boldsymbol{\phi}_n - \epsilon \sum_{n=1}^N (a_n + \hat{a}_n).
\end{aligned} \tag{7.45}$$

The right hand side can be written as

$$\begin{aligned}
\tilde{L}(\mathbf{a}, \hat{\mathbf{a}}) = & -\frac{1}{2} \sum_{n=1}^N \sum_{n'=1}^N (a_n - \hat{a}_n)(a_{n'} - \hat{a}_{n'}) k(\mathbf{x}_n, \mathbf{x}_{n'}) \\
& + \sum_{n=1}^N (a_n - \hat{a}_n) t_n - \epsilon \sum_{n=1}^N (a_n + \hat{a}_n),
\end{aligned} \tag{7.46}$$

where

$$k(\mathbf{x}_n, \mathbf{x}_{n'}) = \boldsymbol{\phi}_n^\top \boldsymbol{\phi}_{n'}. \tag{7.47}$$

Therefore, the minimisation of E is equivalent to the maximisation of $\tilde{L}$.

(b)

By (a), the corresponding KKT conditions are given by

$$\begin{aligned}
(C - a_n) \xi_n &= 0, \\
(C - \hat{a}_n) \hat{\xi}_n &= 0, \\
a_n (\xi_n - t_n + y_n + \epsilon) &= 0, \\
\hat{a}_n (\hat{\xi}_n - y_n + t_n + \epsilon) &= 0.
\end{aligned} \tag{7.48}$$

Therefore,

$$C = \begin{cases} a_n, & \xi_n > 0, \\ \hat{a}_n, & \hat{\xi}_n > 0. \end{cases} \tag{7.49}$$

7.8

Refer to 7.7.

7.9

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}). \end{aligned} \quad (7.50)$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \quad (7.51)$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}\mathbf{w}^\top \mathbf{A}\mathbf{w} = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v}, \quad (7.52)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \mathbf{A} & -\beta\boldsymbol{\Phi}^\top \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}. \quad (7.53)$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (7.54)$$

where

$$\begin{aligned} \mathbf{m} &= \beta\mathbf{S}\boldsymbol{\Phi}^\top \mathbf{t}, \\ \mathbf{S} &= (\beta\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \mathbf{A})^{-1}. \end{aligned} \quad (7.55)$$

7.10

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}). \end{aligned} \quad (7.56)$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \quad (7.57)$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \Phi\mathbf{w}\|^2 - \frac{1}{2}\mathbf{w}^\top \mathbf{A} \mathbf{w} = -\frac{1}{2}\mathbf{v}^\top \mathbf{M} \mathbf{v}, \quad (7.58)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} & -\beta\Phi^\top \\ -\beta\Phi & \beta\mathbf{I} \end{bmatrix}, \quad (7.59)$$

and

$$\mathbf{S} = (\beta\Phi^\top\Phi + \mathbf{A})^{-1}. \quad (7.60)$$

The right hand side can be written as

$$-\frac{1}{2}\mathbf{u}^\top \mathbf{S}^{-1} \mathbf{u} - \frac{1}{2}\mathbf{t}^\top \mathbf{N} \mathbf{t}, \quad (7.61)$$

where

$$\begin{aligned} \mathbf{u} &= \mathbf{w} - \beta\mathbf{S}\Phi^\top\mathbf{t}, \\ \mathbf{N} &= \beta\mathbf{I} - \beta^2\Phi\mathbf{S}\Phi^\top. \end{aligned} \quad (7.62)$$

By 2.26,

$$\mathbf{N}^{-1} = \beta^{-1}\mathbf{I} + \Phi\mathbf{A}^{-1}\Phi^\top. \quad (7.63)$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \beta^{-1}\mathbf{I} + \Phi\mathbf{A}^{-1}\Phi^\top). \quad (7.64)$$

7.11

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top\phi_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}). \end{aligned} \quad (7.65)$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \quad (7.66)$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \Phi\mathbf{w}\|^2 - \frac{1}{2}\mathbf{w}^\top \mathbf{A} \mathbf{w} = -\frac{1}{2}\mathbf{v}^\top \mathbf{M} \mathbf{v}, \quad (7.67)$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta \Phi^\top \Phi + \mathbf{A} & -\beta \Phi^\top \\ -\beta \Phi & \beta \mathbf{I} \end{bmatrix}. \quad (7.68)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{A}^{-1} & \mathbf{A}^{-1} \Phi^\top \\ \Phi \mathbf{A}^{-1} & \beta^{-1} \mathbf{I} + \Phi \mathbf{A}^{-1} \Phi^\top \end{bmatrix}. \quad (7.69)$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t} | \mathbf{0}, \beta^{-1} \mathbf{I} + \Phi \mathbf{A}^{-1} \Phi^\top). \quad (7.70)$$

7.12

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n | \mathbf{w}) &= \mathcal{N}(t_n | \mathbf{w}^\top \phi_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \mathbf{A}^{-1}), \end{aligned} \quad (7.71)$$

where

$$\mathbf{A} = \text{diag}(\alpha_m). \quad (7.72)$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t} | \mathbf{w}) p(\mathbf{w}) d\mathbf{w}. \quad (7.73)$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned} & -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln(\det(\beta^{-1} \mathbf{I})) \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 \\ & -\frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{A}^{-1}) - \frac{1}{2} \mathbf{w}^\top \mathbf{A} \mathbf{w} \\ & = -\frac{N+M}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{1}{2} \ln(\det \mathbf{A}) - E(\mathbf{w}), \end{aligned} \quad (7.74)$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{1}{2} \mathbf{w}^\top \mathbf{A} \mathbf{w}. \quad (7.75)$$

Then,

$$p(\mathbf{t}) = (2\pi)^{-\frac{N+M}{2}} \beta^{\frac{N}{2}} (\det \mathbf{A})^{\frac{1}{2}} \int \exp(-E(\mathbf{w})) d\mathbf{w}. \quad (7.76)$$

We have

$$\begin{aligned} \nabla E(\mathbf{w}) &= \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}), \\ \nabla \nabla E(\mathbf{w}) &= \mathbf{S}^{-1}, \end{aligned} \quad (7.77)$$

where

$$\begin{aligned}\mathbf{m} &= \beta \mathbf{S} \Phi^\top \mathbf{t}, \\ \mathbf{S} &= (\beta \Phi^\top \Phi + \mathbf{A})^{-1}.\end{aligned}\tag{7.78}$$

Then,

$$E(\mathbf{w}) = E(\mathbf{m}) + \frac{1}{2}(\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}),\tag{7.79}$$

so that

$$\int \exp(-E(\mathbf{w})) d\mathbf{w} = (2\pi)^{\frac{M}{2}} (\det \mathbf{S})^{\frac{1}{2}} \exp(-E(\mathbf{m})).\tag{7.80}$$

Then,

$$\ln p(\mathbf{t}) = -\frac{N}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{1}{2} \ln(\det \mathbf{A}) + \frac{1}{2} \ln(\det \mathbf{S}) - E(\mathbf{m}).\tag{7.81}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{t})$ with respect to α_m and β to zero gives

$$\begin{aligned}0 &= \frac{1}{2\alpha_m} + \frac{1}{2} \operatorname{tr} \left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} \right) - \frac{\partial E(\mathbf{m})}{\partial \alpha}, \\ 0 &= \frac{N}{2\beta} + \frac{1}{2} \operatorname{tr} \left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} \right) - \frac{\partial E(\mathbf{m})}{\partial \beta}.\end{aligned}\tag{7.82}$$

We have

$$\begin{aligned}\mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \alpha_m}, \\ \mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \beta}.\end{aligned}\tag{7.83}$$

The right hand sides can be written as

$$\begin{aligned}\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} + \frac{\partial \mathbf{S}^{-1}}{\partial \alpha_m} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} + \mathbf{e}_m \mathbf{e}_m^\top \mathbf{S}, \\ \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \frac{\partial \mathbf{S}^{-1}}{\partial \beta} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \Phi^\top \Phi \mathbf{S},\end{aligned}\tag{7.84}$$

where $\mathbf{e}_m$ is the m -th vector of the standard basis in M dimensions. Then,

$$\begin{aligned}\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} &= -\mathbf{e}_m \mathbf{e}_m^\top \mathbf{S}, \\ \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} &= -\Phi^\top \Phi \mathbf{S}.\end{aligned}\tag{7.85}$$

We have

$$\begin{aligned}\frac{\partial E(\mathbf{m})}{\partial \alpha} &= \nabla E(\mathbf{m}) \frac{\partial \mathbf{m}}{\partial \alpha} + \frac{m_m^2}{2}, \\ \frac{\partial E(\mathbf{m})}{\partial \beta} &= \nabla E(\mathbf{m}) \frac{\partial \mathbf{m}}{\partial \beta} + \frac{\|\mathbf{t} - \Phi \mathbf{m}\|^2}{2}.\end{aligned}\tag{7.86}$$

Then,

$$\begin{aligned}0 &= \frac{1}{2\alpha_m} - \frac{1 - \gamma_m}{2\alpha_m} - \frac{m_m^2}{2}, \\ 0 &= \frac{N}{2\beta} - \frac{\sum_{m=1}^M \gamma_m}{2\beta} - \frac{\|\mathbf{t} - \Phi \mathbf{m}\|^2}{2},\end{aligned}\tag{7.87}$$

where

$$\gamma_m = 1 - S_{mm}\alpha_m.\tag{7.88}$$

Therefore, the maximum likelihood solutions to α_m and β satisfy

$$\begin{aligned}(\alpha_m)_{\text{ML}} &= \frac{\gamma_m}{m_m^2}, \\ \beta_{\text{ML}} &= \frac{N - \sum_{m=1}^M \gamma_m}{\|\mathbf{t} - \Phi \mathbf{m}\|^2}.\end{aligned}\tag{7.89}$$

7.13

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned}p(t_n | \mathbf{w}, \beta) &= \mathcal{N}(t_n | \mathbf{w}^\top \phi_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{0}, \mathbf{A}^{-1}), \\ p(\beta) &= \text{Gam}(\beta | a_0, b_0),\end{aligned}\tag{7.90}$$

where

$$\mathbf{A} = \text{diag}(\alpha_m).\tag{7.91}$$

By Bayes' theorem,

$$p(\mathbf{t}, \beta) = p(\mathbf{t} | \beta) p(\beta).\tag{7.92}$$

By marginalisation,

$$p(\mathbf{t} | \beta) = \int p(\mathbf{t} | \mathbf{w}, \beta) p(\mathbf{w}) d\mathbf{w}.\tag{7.93}$$

The logarithm of the integrand of the right hand side can be written as

$$\begin{aligned}
& -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln \beta^{-1} - \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 \\
& - \frac{M}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{A}^{-1}) - \frac{1}{2} \mathbf{w}^\top \mathbf{A} \mathbf{w} \\
& = -\frac{N+M}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{1}{2} \ln(\det \mathbf{A}) - E(\mathbf{w}),
\end{aligned} \tag{7.94}$$

where

$$E(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{1}{2} \mathbf{w}^\top \mathbf{A} \mathbf{w}. \tag{7.95}$$

We have

$$\begin{aligned}
\nabla E(\mathbf{w}) &= \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}), \\
\nabla \nabla E(\mathbf{w}) &= \mathbf{S}^{-1},
\end{aligned} \tag{7.96}$$

where

$$\begin{aligned}
\mathbf{S} &= (\beta \Phi^\top \Phi + \mathbf{A})^{-1}, \\
\mathbf{m} &= \beta \mathbf{S} \Phi^\top \mathbf{t}.
\end{aligned} \tag{7.97}$$

Then,

$$E(\mathbf{w}) = E(\boldsymbol{\mu}) + \frac{1}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}). \tag{7.98}$$

Then,

$$\int \exp(-E(\mathbf{w})) d\mathbf{w} = (2\pi)^{\frac{M}{2}} (\det \mathbf{S})^{\frac{1}{2}} \exp(-E(\mathbf{m})). \tag{7.99}$$

Then,

$$\begin{aligned}
\ln p(\mathbf{t}|\beta) &= -\frac{N}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{1}{2} \ln(\det \mathbf{A}) + \frac{1}{2} \ln(\det \mathbf{S}) \\
&\quad - E(\mathbf{m}),
\end{aligned} \tag{7.100}$$

so that

$$\begin{aligned}
\ln p(\mathbf{t}, \beta) &= -\frac{N}{2} \ln(2\pi) + \frac{N}{2} \ln \beta + \frac{1}{2} \ln(\det \mathbf{A}) + \frac{1}{2} \ln(\det \mathbf{S}) \\
&\quad - E(\mathbf{m}) + (a_0 - 1) \ln \beta - b_0 \beta,
\end{aligned} \tag{7.101}$$

By 3.21, setting the derivatives of $\ln p(\mathbf{t}, \beta)$ with respect to α_m and β to zero gives

$$\begin{aligned}
0 &= \frac{1}{2\alpha_m} + \frac{1}{2} \operatorname{tr} \left(\boldsymbol{\Sigma}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} \right) - \frac{\partial E(\mathbf{m})}{\partial \alpha}, \\
0 &= \frac{N}{2\beta} + \frac{1}{2} \operatorname{tr} \left(\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} \right) - \frac{\partial E(\mathbf{m})}{\partial \beta} + \frac{a_0 - 1}{\beta} - b_0.
\end{aligned} \tag{7.102}$$

We have

$$\begin{aligned}\mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \alpha_m}, \\ \mathbf{O} &= \frac{\partial \mathbf{S}^{-1} \mathbf{S}}{\partial \beta}.\end{aligned}\tag{7.103}$$

The right hand side can be written as

$$\begin{aligned}\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} + \frac{\partial \mathbf{S}^{-1}}{\partial \alpha_m} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} + \mathbf{e}_m \mathbf{e}_m^\top \mathbf{S}, \\ \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \frac{\partial \mathbf{S}^{-1}}{\partial \beta} \mathbf{S} &= \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} + \Phi^\top \Phi \mathbf{S},\end{aligned}\tag{7.104}$$

where $\mathbf{e}_m$ is the m -th vector of the standard basis in M dimensions. Then,

$$\begin{aligned}\mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \alpha_m} &= -\mathbf{e}_m \mathbf{e}_m^\top \mathbf{S}, \\ \mathbf{S}^{-1} \frac{\partial \mathbf{S}}{\partial \beta} &= -\Phi^\top \Phi \mathbf{S}.\end{aligned}\tag{7.105}$$

We have

$$\begin{aligned}\frac{\partial E(\mathbf{m})}{\partial \alpha_m} &= \nabla E(\mathbf{m}) \frac{\partial \mathbf{m}}{\partial \alpha_m} + \frac{m_m^2}{2}, \\ \frac{\partial E(\mathbf{m})}{\partial \beta} &= \nabla E(\mathbf{m}) \frac{\partial \mathbf{m}}{\partial \beta} + \frac{\|\mathbf{t} - \Phi \mathbf{m}\|^2}{2}.\end{aligned}\tag{7.106}$$

Then,

$$\begin{aligned}0 &= \frac{1}{2\alpha_m} - \frac{1 - \gamma_m}{2\alpha_m} - \frac{m_m^2}{2}, \\ 0 &= \frac{N}{2\beta} - \frac{\sum_{m=1}^M \gamma_m}{2\beta} - \frac{\|\mathbf{t} - \Phi \mathbf{m}\|^2}{2} + \frac{a_0 - 1}{\beta} - b_0.\end{aligned}\tag{7.107}$$

where

$$\gamma_m = 1 - S_{mm} \alpha_m.\tag{7.108}$$

Therefore, the maximum likelihood solutions to α_m and β satisfy

$$\begin{aligned}(\alpha_m)_{\text{ML}} &= \frac{\gamma_m}{m_m^2}, \\ \beta_{\text{ML}} &= \frac{N - \sum_{m=1}^M \gamma_m + 2a_0 - 2}{\|\mathbf{t} - \Phi \mathbf{m}\|^2 + 2b_0}.\end{aligned}\tag{7.109}$$

7.14

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}). \end{aligned} \quad (7.110)$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}. \quad (7.111)$$

By 7.9,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (7.112)$$

where

$$\begin{aligned} \mathbf{m} &= \beta \mathbf{S} \boldsymbol{\Phi}^\top \mathbf{t}, \\ \mathbf{S} &= (\beta \boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \mathbf{A})^{-1}. \end{aligned} \quad (7.113)$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2} (t_{N+1} - \mathbf{w}^\top \boldsymbol{\phi}_{N+1})^2 - \frac{1}{2} (\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1} (\mathbf{w} - \mathbf{m}) \\ &= -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2} \mathbf{m}^\top \mathbf{S}^{-1} \mathbf{m}, \end{aligned} \quad (7.114)$$

where

$$\begin{aligned} \mathbf{v} &= \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}^{-1} + \beta \boldsymbol{\phi}_{N+1} \boldsymbol{\phi}_{N+1}^\top & -\beta \boldsymbol{\phi}_{N+1} \\ -\beta \boldsymbol{\phi}_{N+1}^\top & \beta \end{bmatrix}, \\ \mathbf{u} &= \begin{bmatrix} \mathbf{S}^{-1} \mathbf{m} \\ 0 \end{bmatrix}. \end{aligned} \quad (7.115)$$

By 2.24,

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S} & \mathbf{S} \boldsymbol{\phi}_{N+1} \\ \boldsymbol{\phi}_{N+1}^\top \mathbf{S} & \beta^{-1} + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1} \end{bmatrix}, \quad (7.116)$$

so that

$$\mathbf{M}^{-1} \mathbf{u} = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\top \boldsymbol{\phi}_{N+1} \end{bmatrix}. \quad (7.117)$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1}|\mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \beta^{-1} + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1}). \quad (7.118)$$

7.15

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}), \end{aligned} \quad (7.119)$$

where

$$\mathbf{A} = \text{diag}(\alpha_m). \quad (7.120)$$

By 7.11,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \mathbf{C}), \quad (7.121)$$

where

$$\mathbf{C} = \beta^{-1} \mathbf{I} + \boldsymbol{\Phi} \mathbf{A}^{-1} \boldsymbol{\Phi}^\top. \quad (7.122)$$

Then,

$$\ln p(\mathbf{t}) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln(\det \mathbf{C}) - \frac{1}{2} \mathbf{t}^\top \mathbf{C}^{-1} \mathbf{t}. \quad (7.123)$$

We have

$$\mathbf{C} = \mathbf{C}_{-m} + \alpha_m^{-1} \boldsymbol{\varphi}_m \boldsymbol{\varphi}_m^\top, \quad (7.124)$$

where $\mathbf{C}_{-m}$ is independent of α_m and $\boldsymbol{\varphi}_m$ is the m^{th} column of $\boldsymbol{\Phi}$. By 2.26,

$$\mathbf{C}^{-1} = \mathbf{C}_{-m}^{-1} - (\alpha_m + s_m)^{-1} \mathbf{C}_{-m}^{-1} \boldsymbol{\varphi}_m \boldsymbol{\varphi}_m^\top \mathbf{C}_{-m}^{-1}, \quad (7.125)$$

where

$$s_m = \boldsymbol{\varphi}_m^\top \mathbf{C}_{-m}^{-1} \boldsymbol{\varphi}_m. \quad (7.126)$$

Then, $\ln p(\mathbf{t})$ except the terms independent of α_m can be written as

$$\lambda(\alpha_m) = \frac{1}{2} \left(\ln \alpha_m - \ln(\alpha_m + s_m) + (\alpha_m + s_m)^{-1} q_m^2 \right), \quad (7.127)$$

where

$$q_m = \boldsymbol{\varphi}_m^\top \mathbf{C}_{-m}^{-1} \mathbf{t}. \quad (7.128)$$

7.16

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}), \end{aligned} \quad (7.129)$$

where

$$\mathbf{A} = \text{diag}(\alpha_m). \quad (7.130)$$

By 7.11,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \mathbf{C}), \quad (7.131)$$

where

$$\mathbf{C} = \beta^{-1}\mathbf{I} + \Phi\mathbf{A}^{-1}\Phi^\top. \quad (7.132)$$

By 7.15, $\ln p(\mathbf{t})$ except the terms independent of α_m can be written as

$$\lambda(\alpha_m) = \frac{1}{2} \left(\ln \alpha_m - \ln(\alpha_m + s_m) + (\alpha_m + s_m)^{-1} q_m^2 \right), \quad (7.133)$$

where

$$\begin{aligned} s_m &= \boldsymbol{\varphi}_m^\top \mathbf{C}_{-m}^{-1} \boldsymbol{\varphi}_m, \\ q_m &= \boldsymbol{\varphi}_m^\top \mathbf{C}_{-m}^{-1} \mathbf{t}, \\ \mathbf{C}_{-m} &= \mathbf{C} - \alpha_m^{-1} \boldsymbol{\varphi}_m \boldsymbol{\varphi}_m^\top, \end{aligned} \quad (7.134)$$

and $\boldsymbol{\varphi}_m$ is the m^{th} column of Φ . Setting the derivative of $\lambda(\alpha_m)$ to zero gives

$$0 = \frac{1}{2} \left(\alpha_m^{-1} - (\alpha_m + s_m)^{-1} - (\alpha_m + s_m)^{-2} q_m^2 \right). \quad (7.135)$$

Then,

$$0 = (\alpha_m + s_m)^2 - (\alpha_m + s_m)\alpha_m - \alpha_m q_m^2. \quad (7.136)$$

Therefore, the maximum likelihood solution to α_m is given by

$$(\alpha_m)_{\text{ML}} = \frac{s_m^2}{q_m^2 - s_m}. \quad (7.137)$$

7.17

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}), \end{aligned} \quad (7.138)$$

where

$$\mathbf{A} = \text{diag}(\alpha_m). \quad (7.139)$$

(a)

By 7.11,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t} | \mathbf{0}, \mathbf{C}), \quad (7.140)$$

where

$$\mathbf{C} = \beta^{-1} \mathbf{I} + \Phi \mathbf{A}^{-1} \Phi^\top. \quad (7.141)$$

By 7.15, $\ln p(\mathbf{t})$ except the terms independent of α_m can be written as

$$\lambda(\alpha_m) = \frac{1}{2} (\ln \alpha_m - \ln (\alpha_m + s_m) + (\alpha_m + s_m)^{-1} q_m^2), \quad (7.142)$$

where

$$\begin{aligned} s_m &= \varphi_m^\top \mathbf{C}_{-m}^{-1} \varphi_m, \\ q_m &= \varphi_m^\top \mathbf{C}_{-m}^{-1} \mathbf{t}, \\ \mathbf{C}_{-m} &= \mathbf{C} - \alpha_m^{-1} \varphi_m \varphi_m^\top, \end{aligned} \quad (7.143)$$

and φ_m is the m^{th} column of Φ . By 2.26,

$$\mathbf{C}_{-m}^{-1} = \mathbf{C}^{-1} + (\alpha_m - S_m)^{-1} \mathbf{C}^{-1} \varphi_m \varphi_m^\top \mathbf{C}^{-1}, \quad (7.144)$$

where

$$S_m = \varphi_m^\top \mathbf{C}^{-1} \varphi_m. \quad (7.145)$$

Then,

$$\begin{aligned} s_m &= S_m + (\alpha_m - S_m)^{-1} S_m^2, \\ q_m &= Q_m + (\alpha_m - S_m)^{-1} S_m Q_m, \end{aligned} \quad (7.146)$$

where

$$Q_m = \varphi_m^\top \mathbf{C}^{-1} \mathbf{t}. \quad (7.147)$$

Therefore,

$$\begin{aligned} s_m &= (\alpha_m - S_m)^{-1} \alpha_m S_m, \\ q_m &= (\alpha_m - S_m)^{-1} \alpha_m Q_m. \end{aligned} \quad (7.148)$$

(b)

By 2.26,

$$\mathbf{C}^{-1} = \beta \mathbf{I} - \beta^2 \Phi \Sigma \Phi^\top, \quad (7.149)$$

where

$$\Sigma = (\mathbf{A} + \beta \Phi^\top \Phi)^{-1}. \quad (7.150)$$

Therefore,

$$\begin{aligned} S_m &= \beta \|\varphi_m\|^2 - \beta^2 \varphi_m^\top \Phi \Sigma \Phi^\top \varphi_m, \\ Q_m &= \beta \varphi_m^\top \mathbf{t} - \beta^2 \varphi_m^\top \Phi \Sigma \Phi^\top \mathbf{t}. \end{aligned} \quad (7.151)$$

7.18

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= y_n^{t_n}(1 - y_n)^{1-t_n}, \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}), \end{aligned} \quad (7.152)$$

where

$$\begin{aligned} y_n &= \sigma(\mathbf{w}^\top \boldsymbol{\phi}_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ \mathbf{A} &= \text{diag}(\alpha_m). \end{aligned} \quad (7.153)$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}). \quad (7.154)$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ and $\mathbf{t}$ can be written as

$$\sum_{n=1}^N (t_n \ln y_n + (1 - t_n) \ln(1 - y_n)) - \frac{1}{2} \mathbf{w}^\top \mathbf{A} \mathbf{w}. \quad (7.155)$$

Then,

$$\begin{aligned} \nabla \ln p(\mathbf{w}|\mathbf{t}) &= \sum_{n=1}^N (t_n - y_n) \boldsymbol{\phi}_n - \mathbf{A} \mathbf{w}, \\ \nabla \nabla \ln p(\mathbf{w}|\mathbf{t}) &= - \sum_{n=1}^N y_n(1 - y_n) \boldsymbol{\phi}_n \boldsymbol{\phi}_n^\top - \mathbf{A}. \end{aligned} \quad (7.156)$$

Therefore,

$$\begin{aligned} \nabla \ln p(\mathbf{w}|\mathbf{t}) &= \boldsymbol{\Phi}(\mathbf{t} - \mathbf{y}) - \mathbf{A} \mathbf{w}, \\ \nabla \nabla \ln p(\mathbf{w}|\mathbf{t}) &= -\boldsymbol{\Phi}^\top \mathbf{B} \boldsymbol{\Phi} - \mathbf{A}, \end{aligned} \quad (7.157)$$

where

$$\mathbf{B} = \text{diag}(y_n(1 - y_n)). \quad (7.158)$$

7.19

Let $t_1, \dots, t_N$ be conditionally independent binary variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= y_n^{t_n}(1 - y_n)^{1-t_n}, \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}), \end{aligned} \quad (7.159)$$

where $\mathbf{w}$ is in M dimensions and

$$\begin{aligned} y_n &= \sigma(\mathbf{w}^\top \boldsymbol{\phi}_n), \\ \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ \mathbf{A} &= \text{diag}(\alpha_m). \end{aligned} \quad (7.160)$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}. \quad (7.161)$$

The logarithm of the integrand of the right hand side can be written as

$$-\frac{M}{2} \ln(2\pi) + \frac{1}{2} \ln(\det \mathbf{A}) - E(\mathbf{w}), \quad (7.162)$$

where

$$E(\mathbf{w}) = - \sum_{n=1}^N (t_n \ln y_n + (1 - t_n) \ln(1 - y_n)) + \frac{1}{2} \mathbf{w}^\top \mathbf{A} \mathbf{w}. \quad (7.163)$$

We have

$$\begin{aligned} \nabla E(\mathbf{w}) &= -\boldsymbol{\Phi}^\top (\mathbf{t} - \mathbf{y}) + \mathbf{A} \mathbf{w}, \\ \nabla \nabla E(\mathbf{w}) &= \boldsymbol{\Phi}^\top \mathbf{B} \boldsymbol{\Phi} + \mathbf{A}, \end{aligned} \quad (7.164)$$

where

$$\mathbf{B} = \text{diag}(y_n(1 - y_n)). \quad (7.165)$$

Then, we have the Taylor series

$$\begin{aligned} E(\mathbf{w}) &= E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \boldsymbol{\Sigma}_{\text{MAP}}^{-1} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ &\quad + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3), \end{aligned} \quad (7.166)$$

where

$$\begin{aligned} \mathbf{w}_{\text{MAP}} &= \mathbf{A}^{-1} \boldsymbol{\Phi}^\top (\mathbf{t} - \mathbf{y}), \\ \boldsymbol{\Sigma}_{\text{MAP}} &= (\mathbf{A} + \boldsymbol{\Phi}^\top \mathbf{B}_{\text{MAP}} \boldsymbol{\Phi})^{-1}, \end{aligned} \quad (7.167)$$

and

$$\begin{aligned} \mathbf{B}_{\text{MAP}} &= \text{diag}(y_{\text{MAP}_n}(1 - y_{\text{MAP}_n})), \\ y_{\text{MAP}_n} &= \sigma(\mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}_n). \end{aligned} \quad (7.168)$$

Then,

$$p(\mathbf{t}) \simeq (2\pi)^{-\frac{M}{2}} (\det \mathbf{A})^{\frac{1}{2}} \exp(-E(\mathbf{w}_{\text{MAP}})) I, \quad (7.169)$$

where

$$I = \int \exp \left(-\frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{MAP}})^{\text{T}} \boldsymbol{\Sigma}_{\text{MAP}}^{-1} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \right) d\mathbf{w}. \quad (7.170)$$

We have

$$I = (2\pi)^{\frac{M}{2}} (\det \boldsymbol{\Sigma}_{\text{MAP}})^{\frac{1}{2}}. \quad (7.171)$$

Then,

$$p(\mathbf{t}) \simeq (\det \mathbf{A})^{\frac{1}{2}} (\det \boldsymbol{\Sigma}_{\text{MAP}})^{\frac{1}{2}} \exp(-E(\mathbf{w}_{\text{MAP}})). \quad (7.172)$$

Then, $\ln p(\mathbf{t})$ can be approximated as

$$l(\mathbf{t}) = \frac{1}{2} \ln(\det \mathbf{A}) + \frac{1}{2} \ln(\det \boldsymbol{\Sigma}_{\text{MAP}}) - E(\mathbf{w}_{\text{MAP}}). \quad (7.173)$$

By 3.21, setting the derivative of $l(\mathbf{t})$ with respect to α_m to zero gives

$$0 = \frac{1}{2\alpha_m} + \frac{1}{2} \text{tr} \left(\boldsymbol{\Sigma}_{\text{MAP}}^{-1} \frac{\partial \boldsymbol{\Sigma}_{\text{MAP}}}{\partial \alpha_m} \right) - \frac{\partial E(\mathbf{w}_{\text{MAP}})}{\partial \alpha}. \quad (7.174)$$

We have

$$\mathbf{O} = \frac{\partial \boldsymbol{\Sigma}_{\text{MAP}}^{-1} \boldsymbol{\Sigma}_{\text{MAP}}}{\partial \alpha_m}. \quad (7.175)$$

The right hand side can be written as

$$\frac{\partial \boldsymbol{\Sigma}_{\text{MAP}}^{-1}}{\partial \alpha_m} \boldsymbol{\Sigma}_{\text{MAP}} + \boldsymbol{\Sigma}_{\text{MAP}}^{-1} \frac{\partial \boldsymbol{\Sigma}_{\text{MAP}}}{\partial \alpha_m} = \mathbf{e}_m \mathbf{e}_m^{\text{T}} \boldsymbol{\Sigma}_{\text{MAP}} + \boldsymbol{\Sigma}_{\text{MAP}}^{-1} \frac{\partial \boldsymbol{\Sigma}_{\text{MAP}}}{\partial \alpha_m}, \quad (7.176)$$

where $\mathbf{e}_m$ is the m -th vector of the standard basis in M dimensions. Then,

$$\boldsymbol{\Sigma}_{\text{MAP}}^{-1} \frac{\partial \boldsymbol{\Sigma}_{\text{MAP}}}{\partial \alpha_m} = -\mathbf{e}_m \mathbf{e}_m^{\text{T}} \boldsymbol{\Sigma}_{\text{MAP}}. \quad (7.177)$$

We have

$$\frac{\partial E(\mathbf{w}_{\text{MAP}})}{\partial \alpha} = \nabla E(\mathbf{w}_{\text{MAP}}) \frac{\partial \mathbf{w}_{\text{MAP}}}{\partial \alpha} + \frac{1}{2} w_{\text{MAP}_m}^2. \quad (7.178)$$

We have

$$\nabla E(\mathbf{w}_{\text{MAP}}) = 0. \quad (7.179)$$

Then,

$$0 = \frac{1}{2\alpha_m} - \frac{1}{2} \Sigma_{\text{MAP}_{mm}} - \frac{1}{2} w_{\text{MAP}_m}^2. \quad (7.180)$$

Therefore, the maximum likelihood solution to α_m satisfies

$$(\alpha_m)_{\text{ML}} \simeq (\Sigma_{\text{MAP}_{mm}} + w_{\text{MAP}_m}^2)^{-1}. \quad (7.181)$$

8 Graphical Models

8.1

Let $x_1, \dots, x_N$ be variables such that

$$p(\mathbf{x}) = \prod_{n=1}^N p(x_n | \text{pa}_n), \quad (8.1)$$

where pa_n is the set of parents of x_n . For $N = 1$, we have

$$\int p(x_1) dx_1 = 1. \quad (8.2)$$

Let us assume that

$$\int p(\mathbf{x}) d\mathbf{x} = 1. \quad (8.3)$$

Let $x_1, \dots, x_{N+1}$ be variables such that

$$p(\mathbf{x}') = \prod_{n=1}^{N+1} p(x_n | \text{pa}_n), \quad (8.4)$$

where

$$\mathbf{x}' = \begin{bmatrix} \mathbf{x} \\ x_{N+1} \end{bmatrix}. \quad (8.5)$$

Then,

$$\int p(\mathbf{x}') d\mathbf{x}' = \int p(\mathbf{x}) \left(\int p(x_{N+1} | \text{pa}_{N+1}) dx_{N+1} \right) d\mathbf{x}. \quad (8.6)$$

We have

$$\int p(x_{N+1} | \text{pa}_{N+1}) dx_{N+1} = 1. \quad (8.7)$$

Then,

$$\int p(\mathbf{x}') d\mathbf{x}' = \int p(\mathbf{x}) d\mathbf{x} = 1. \quad (8.8)$$

Therefore, the assumption is proved by induction on N .

8.2

If a directed graph has a topological ordering, then the graph is acyclic. On the other hand, if a directed graph is acyclic, then the graph has a topological ordering. Therefore, a graph having a topological ordering is equivalent to the graph being acyclic.

8.3

Let a , b and c be binary variables whose joint distribution is given in the table below:

a	b	c	$p(a, b, c)$
0	0	0	$\frac{24}{125}$
0	0	1	$\frac{18}{125}$
0	1	0	$\frac{6}{125}$
0	1	1	$\frac{27}{125}$
1	0	0	$\frac{24}{125}$
1	0	1	$\frac{8}{125}$
1	1	0	$\frac{6}{125}$
1	1	1	$\frac{12}{125}$

(a)

We have

$$\begin{aligned}
 p(a = 0, b = 0) &= \frac{42}{125}, \\
 p(a = 0) &= \frac{3}{5}, \\
 p(b = 0) &= \frac{74}{125}.
 \end{aligned} \tag{8.9}$$

Therefore,

$$p(a, b) \neq p(a)p(b). \tag{8.10}$$

(b)

By Bayes' theorem,

$$\begin{aligned}p(a = 0, b = 0|c = 0) &= \frac{p(a = 0, b = 0, c = 0)}{p(c = 0)}, \\p(a = 0|c = 0) &= \frac{p(a = 0, c = 0)}{p(c = 0)}, \\p(b = 0|c = 0) &= \frac{p(b = 0, c = 0)}{p(c = 0)}.\end{aligned}\tag{8.11}$$

The right hand sides are $\frac{2}{5}$, $\frac{1}{2}$ and $\frac{4}{5}$. Therefore,

$$p(a = 0, b = 0|c = 0) = p(a = 0|c = 0)p(b = 0|c = 0).\tag{8.12}$$

(c)

By Bayes' theorem,

$$\begin{aligned}p(a = 1, b = 0|c = 0) &= \frac{p(a = 1, b = 0, c = 0)}{p(c = 0)}, \\p(a = 1|c = 0) &= \frac{p(a = 1, c = 0)}{p(c = 0)}, \\p(b = 0|c = 0) &= \frac{p(b = 0, c = 0)}{p(c = 0)}.\end{aligned}\tag{8.13}$$

The right hand sides are $\frac{2}{5}$, $\frac{1}{2}$ and $\frac{4}{5}$. Therefore,

$$p(a = 1, b = 0|c = 0) = p(a = 1|c = 0)p(b = 0|c = 0).\tag{8.14}$$

(d)

By Bayes' theorem,

$$\begin{aligned}p(a = 0, b = 1|c = 0) &= \frac{p(a = 0, b = 1, c = 0)}{p(c = 0)}, \\p(a = 0|c = 0) &= \frac{p(a = 0, c = 0)}{p(c = 0)}, \\p(b = 1|c = 0) &= \frac{p(b = 1, c = 0)}{p(c = 0)}.\end{aligned}\tag{8.15}$$

The right hand sides are $\frac{1}{10}$, $\frac{1}{2}$ and $\frac{1}{5}$. Therefore,

$$p(a = 0, b = 1|c = 0) = p(a = 0|c = 0)p(b = 1|c = 0). \quad (8.16)$$

(e)

By Bayes' theorem,

$$\begin{aligned} p(a = 1, b = 1|c = 0) &= \frac{p(a = 1, b = 1, c = 0)}{p(c = 0)}, \\ p(a = 1|c = 0) &= \frac{p(a = 1, c = 0)}{p(c = 0)}, \\ p(b = 1|c = 0) &= \frac{p(b = 1, c = 0)}{p(c = 0)}. \end{aligned} \quad (8.17)$$

The right hand sides are $\frac{1}{10}$, $\frac{1}{2}$ and $\frac{1}{5}$. Therefore,

$$p(a = 1, b = 1|c = 0) = p(a = 1|c = 0)p(b = 1|c = 0). \quad (8.18)$$

(f)

By Bayes' theorem,

$$\begin{aligned} p(a = 0, b = 0|c = 1) &= \frac{p(a = 0, b = 0, c = 1)}{p(c = 1)}, \\ p(a = 0|c = 1) &= \frac{p(a = 0, c = 1)}{p(c = 1)}, \\ p(b = 0|c = 1) &= \frac{p(b = 0, c = 1)}{p(c = 1)}. \end{aligned} \quad (8.19)$$

The right hand sides are $\frac{18}{65}$, $\frac{9}{13}$ and $\frac{2}{5}$. Therefore,

$$p(a = 0, b = 0|c = 1) = p(a = 0|c = 1)p(b = 0|c = 1). \quad (8.20)$$

(g)

By Bayes' theorem,

$$\begin{aligned}p(a = 1, b = 0|c = 1) &= \frac{p(a = 1, b = 0, c = 1)}{p(c = 1)}, \\p(a = 1|c = 1) &= \frac{p(a = 1, c = 1)}{p(c = 1)}, \\p(b = 0|c = 1) &= \frac{p(b = 0, c = 1)}{p(c = 1)}.\end{aligned}\tag{8.21}$$

The right hand sides are $\frac{8}{65}$, $\frac{4}{13}$ and $\frac{2}{5}$. Therefore,

$$p(a = 1, b = 0|c = 1) = p(a = 1|c = 1)p(b = 0|c = 1).\tag{8.22}$$

(h)

By Bayes' theorem,

$$\begin{aligned}p(a = 0, b = 1|c = 1) &= \frac{p(a = 0, b = 1, c = 1)}{p(c = 1)}, \\p(a = 0|c = 1) &= \frac{p(a = 0, c = 1)}{p(c = 1)}, \\p(b = 1|c = 1) &= \frac{p(b = 1, c = 1)}{p(c = 1)}.\end{aligned}\tag{8.23}$$

The right hand sides are $\frac{27}{65}$, $\frac{9}{13}$ and $\frac{3}{5}$. Therefore,

$$p(a = 0, b = 1|c = 1) = p(a = 0|c = 1)p(b = 1|c = 1).\tag{8.24}$$

(i)

By Bayes' theorem,

$$\begin{aligned}p(a = 1, b = 1|c = 1) &= \frac{p(a = 1, b = 1, c = 1)}{p(c = 1)}, \\p(a = 1|c = 1) &= \frac{p(a = 1, c = 1)}{p(c = 1)}, \\p(b = 1|c = 1) &= \frac{p(b = 1, c = 1)}{p(c = 1)}.\end{aligned}\tag{8.25}$$

The right hand sides are $\frac{12}{65}$, $\frac{4}{13}$ and $\frac{3}{5}$. Therefore,

$$p(a = 1, b = 1|c = 1) = p(a = 1|c = 1)p(b = 1|c = 1). \quad (8.26)$$

(j)

By (b) to (i),

$$p(a, b|c) = p(a|c)p(b|c). \quad (8.27)$$

(k)

We have

$$\begin{aligned} p(a = 0) &= \frac{3}{5}, \\ p(a = 1) &= \frac{2}{5}. \end{aligned} \quad (8.28)$$

(l)

By Bayes' theorem,

$$\begin{aligned} p(c = 0|a = 0) &= \frac{p(c = 0, a = 0)}{p(a = 0)}, \\ p(c = 1|a = 0) &= \frac{p(c = 1, a = 0)}{p(a = 0)}, \\ p(c = 0|a = 1) &= \frac{p(c = 0, a = 1)}{p(a = 1)}, \\ p(c = 1|a = 1) &= \frac{p(c = 1, a = 1)}{p(a = 1)}, \end{aligned} \quad (8.29)$$

The right hand sides are $\frac{2}{5}$, $\frac{3}{5}$, $\frac{3}{5}$ and $\frac{2}{5}$.

(m)

By Bayes' theorem,

$$\begin{aligned}
p(b=0|c=0) &= \frac{p(b=0, c=0)}{p(c=0)}, \\
p(b=1|c=0) &= \frac{p(b=1, c=0)}{p(c=0)}, \\
p(b=0|c=1) &= \frac{p(b=0, c=1)}{p(c=1)}, \\
p(b=1|c=1) &= \frac{p(b=1, c=1)}{p(c=1)},
\end{aligned} \tag{8.30}$$

The right hand sides are $\frac{4}{5}$, $\frac{1}{5}$, $\frac{2}{5}$ and $\frac{3}{5}$.

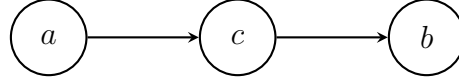
(n)

By (k) to (m),

$$p(a, b, c) = p(a)p(c|a)p(b|c). \tag{8.31}$$

(o)

By (n), the corresponding directed graph can be drawn as below:



8.4

Refer to 8.3.

8.5

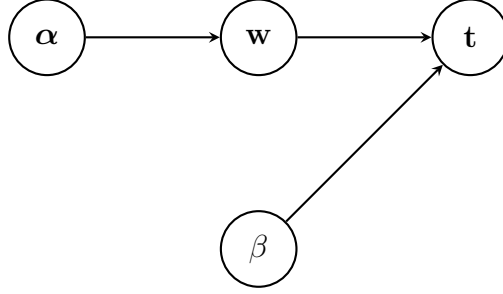
Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned}
p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\
p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{A}^{-1}),
\end{aligned} \tag{8.32}$$

where

$$\mathbf{A} = \text{diag}(\alpha_i). \tag{8.33}$$

Then, the corresponding directed graph can be drawn as below:



8.6 (Incomplete)

Let t be a binary variable such that

$$p(t|\phi) = \sigma(y)^t (1 - \sigma(y))^{1-t}, \quad (8.34)$$

where $\phi_1, \dots, \phi_M$ are binary variables and

$$\begin{aligned} \sigma(a) &= \frac{1}{1 + \exp(-a)}, \\ y &= \mathbf{w}'^\top \phi', \\ \mathbf{w}' &= \begin{bmatrix} w_0 \\ \mathbf{w} \end{bmatrix}, \phi' = \begin{bmatrix} 1 \\ \phi \end{bmatrix}. \end{aligned} \quad (8.35)$$

We have

$$1 - \sigma(y) = \sigma(-y). \quad (8.36)$$

Therefore,

$$p(t|\phi) = v^t (1 - v)^{1-t} \quad (8.37)$$

where

$$\begin{aligned} v &= 1 - (1 - \mu_0) \prod_{m=1}^M (1 - \mu_m)^{\phi_m}, \\ \mu_m &= p(\phi_m = 1). \end{aligned} \quad (8.38)$$

8.7

Let $x_1, \dots, x_N$ be variables such that

$$p(x_n | \text{pa}_n) = \mathcal{N} \left(\mathbf{x}_n | b_n + \sum_{n' \in \text{pa}_n} w_{nn'} x_{n'}, v_n \right), \quad (8.39)$$

where $\text{pa}_1, \dots, \text{pa}_n$ are parent nodes.

(a)

We have

$$\mathbb{E} x_n = \mathbb{E} (\mathbb{E} (x_n | \text{pa}_n)) . \quad (8.40)$$

The right hand side can be written as

$$\mathbb{E} \left(b_n + \sum_{n' \in \text{pa}_n} w_{nn'} x_{n'} \right) = b_n + \sum_{n' \in \text{pa}_n} w_{nn'} \mathbb{E} x_{n'} . \quad (8.41)$$

Therefore,

$$\mathbb{E} x_n = b_n + \sum_{n' \in \text{pa}_n} w_{nn'} \mathbb{E} x_{n'} . \quad (8.42)$$

(b)

We have

$$\text{cov}(x_n, x_{n'}) = \mathbb{E}(x_n - \mathbb{E} x_n)(x_{n'} - \mathbb{E} x_{n'}) . \quad (8.43)$$

We have

$$x_n = b_n + \sum_{n' \in \text{pa}_n} w_{nn'} x_{n'} + \sqrt{v_n} \epsilon_n , \quad (8.44)$$

where

$$p(\epsilon_n) = \mathcal{N}(\epsilon_n | 0, 1) . \quad (8.45)$$

Then, by (a),

$$x_{n'} - \mathbb{E} x_{n'} = \sum_{n'' \in \text{pa}_{n'}} w_{n'n''} (x_{n''} - \mathbb{E} x_{n''}) + \sqrt{v_{n'}} \epsilon_{n'} . \quad (8.46)$$

Then,

$$\begin{aligned} & \mathbb{E}(x_n - \mathbb{E} x_n)(x_{n'} - \mathbb{E} x_{n'}) \\ &= \mathbb{E}(x_n - \mathbb{E} x_n) \left(\sum_{n'' \in \text{pa}_{n'}} w_{n'n''} (x_{n''} - \mathbb{E} x_{n''}) + \sqrt{v_{n'}} \epsilon_{n'} \right) . \end{aligned} \quad (8.47)$$

The right hand side can be written as

$$\sum_{n'' \in \text{pa}_{n'}} w_{n'n''} \text{cov}(x_n, x_{n''}) + \sqrt{v_{n'}} \mathbb{E}(x_n - \mathbb{E} x_n) \epsilon_{n'} . \quad (8.48)$$

We have

$$\mathbb{E}(x_n - \mathbb{E} x_n) \epsilon_{n'} = \mathbb{E} \left(\sum_{n' \in \text{pa}_n} w_{nn'} (x_{n'} - \mathbb{E} x_{n'}) + \sqrt{v_n} \epsilon_n \right) \epsilon_{n'}. \quad (8.49)$$

The right hand side can be written as

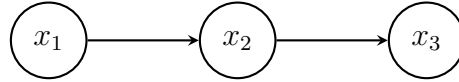
$$\sum_{n' \in \text{pa}_n} w_{nn'} \mathbb{E}(x_{n'} - \mathbb{E} x_{n'}) + \sqrt{v_n} \mathbb{E} \epsilon_n \epsilon_{n'} = \sqrt{v_n} I_{nn'}. \quad (8.50)$$

Therefore,

$$\text{cov}(x_n, x_{n'}) = \sum_{n'' \in \text{pa}_{n'}} w_{n'n''} \text{cov}(x_n, x_{n''}) + v_n I_{nn'}. \quad (8.51)$$

(c)

Let $N = 3$ and let the corresponding directed graph be as below:



By (a),

$$\mathbb{E} \mathbf{x} = \begin{bmatrix} b_1 \\ b_2 + w_{12} b_1 \\ b_3 + w_{23} (b_2 + w_{12} b_1) \end{bmatrix}. \quad (8.52)$$

By (b),

$$\text{var}(\mathbf{x}) = \begin{bmatrix} v_1 & w_{12} v_1 & w_{12} w_{23} v_1 \\ w_{12} v_1 & v_2 + w_{12}^2 v_1 & w_{23} (v_2 + w_{12}^2 v_1) \\ w_{12} w_{23} v_1 & w_{23} (v_2 + w_{12}^2 v_1) & v_3 + w_{23}^2 (v_2 + w_{12}^2 v_1) \end{bmatrix}. \quad (8.53)$$

8.8

Let a, b, c and d be variables such that

$$a \perp\!\!\!\perp b, c \mid d. \quad (8.54)$$

Then,

$$p(a, b, c \mid d) = p(a \mid d) p(b, c \mid d). \quad (8.55)$$

Then,

$$\int p(a, b, c|d)dc = p(a|d) \int p(b, c|d)dc, \quad (8.56)$$

so that

$$p(a, b|d) = p(a|d)p(b|d). \quad (8.57)$$

Therefore,

$$a \perp\!\!\!\perp b \mid d. \quad (8.58)$$

8.9 (Incomplete)

Let $\mathbf{x}_1, \dots, \mathbf{x}_N$ be variables such that

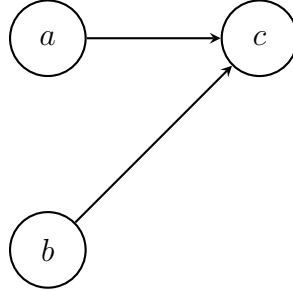
$$p(\mathbf{x}_n | \mathbf{x}_{n'}) = \frac{\prod_{n''=1}^N p(\mathbf{x}_{n''} | \text{pa}_{n''})}{\int \prod_{n''=1}^N p(\mathbf{x}_{n''} | \text{pa}_{n''}) d\mathbf{x}_{n'}}, \quad (8.59)$$

for $n' \neq n$, where $\text{pa}_1, \dots, \text{pa}_N$ are the parent nodes.

8.10

(a)

Let a, b and c be variables whose directed graph can be drawn as below:



Then,

$$p(a, b, c) = p(a)p(b)p(c|a, b), \quad (8.60)$$

so that

$$\int p(a, b, c)dc = p(a)p(b) \int p(c|a, b)dc. \quad (8.61)$$

We have

$$\int p(c|a, b)dc = 1. \quad (8.62)$$

Then,

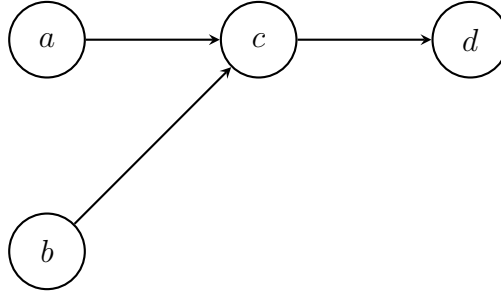
$$p(a, b) = p(a)p(b). \quad (8.63)$$

Therefore,

$$a \perp\!\!\!\perp b \mid \emptyset. \quad (8.64)$$

(b)

Let a , b , c and d be variables whose directed graph can be drawn as below:



By Bayes' theorem,

$$\begin{aligned} p(a, b|d) &= \frac{p(a, b, d)}{p(d)}, \\ p(a|d) &= \frac{p(a, d)}{p(d)}, \\ p(b|d) &= \frac{p(b, d)}{p(d)}. \end{aligned} \quad (8.65)$$

Then,

$$\frac{p(a, b|d)}{p(a|d)p(b|d)} = \frac{p(a, b, d)p(d)}{p(a, d)p(b, d)}. \quad (8.66)$$

If there do not exist q_1 and q_2 such that

$$p(a, b, d) = q_1(a, d)q_2(b, d), \quad (8.67)$$

we have

$$\frac{p(a, b, d)p(d)}{p(a, d)p(b, d)} \neq 1. \quad (8.68)$$

The graph gives

$$p(a, b, c, d) = p(a)p(b)p(c|a, b)p(d|c), \quad (8.69)$$

so that

$$\int p(a, b, c, d)dc = p(a)p(b) \int p(c|a, b)p(d|c)dc. \quad (8.70)$$

Then,

$$p(a, b, d) = p(a)p(b)p(d|a, b), \quad (8.71)$$

which does not guarantee the existence of such q_1 and q_2 . Then, in general,

$$p(a, b|d) \neq p(a|d)p(b|d). \quad (8.72)$$

Therefore,

$$a \not\perp b \mid d. \quad (8.73)$$

8.11

Let B , F , G and D be binary variables such that

$$\begin{aligned} p(D = 1|G = 1) &= \frac{9}{10}, p(D = 0|G = 0) = \frac{9}{10}, \\ p(G = 1|B = 1, F = 1) &= \frac{4}{5}, p(G = 1|B = 1, F = 0) = \frac{1}{5}, \\ p(G = 1|B = 0, F = 1) &= \frac{1}{5}, p(G = 1|B = 0, F = 0) = \frac{1}{10}, \\ p(B = 1) &= \frac{9}{10}, p(F = 1) = \frac{9}{10}, \end{aligned} \quad (8.74)$$

where B and F are independent.

(a)

By Bayes' theorem,

$$p(F = 0|D = 0) = \frac{p(D = 0, F = 0)}{p(D = 0)}. \quad (8.75)$$

By marginalisation,

$$\begin{aligned}
& p(D = 0, F = 0) \\
&= \sum_{B=0,1} \sum_{G=0,1} p(D = 0|G)p(G|B, F = 0)p(B)p(F = 0), \\
& p(D = 0, F = 1) \\
&= \sum_{B=0,1} \sum_{G=0,1} p(D = 0|G)p(G|B, F = 1)p(B)p(F = 1), \\
& p(D = 0) \\
&= \sum_{F=0,1} p(D = 0, F).
\end{aligned} \tag{8.76}$$

The right hand sides can be written as

$$\begin{aligned}
& \frac{9}{10} \frac{9}{10} \frac{1}{10} \frac{1}{10} + \frac{1}{10} \frac{1}{10} \frac{1}{10} \frac{1}{10} + \frac{9}{10} \frac{4}{5} \frac{9}{10} \frac{1}{10} + \frac{1}{10} \frac{1}{5} \frac{9}{10} \frac{1}{10} = \frac{187}{2500}, \\
& \frac{9}{10} \frac{4}{5} \frac{1}{10} \frac{9}{10} + \frac{1}{10} \frac{1}{5} \frac{1}{10} \frac{9}{10} + \frac{9}{10} \frac{1}{5} \frac{9}{10} \frac{9}{10} + \frac{1}{10} \frac{4}{5} \frac{9}{10} \frac{9}{10} = \frac{693}{2500}, \\
& \frac{187}{2500} + \frac{693}{2500} = \frac{44}{125}.
\end{aligned} \tag{8.77}$$

Therefore,

$$p(F = 0|D = 0) = \frac{17}{80}. \tag{8.78}$$

(b)

By Bayes' theorem,

$$p(F = 0|B = 0, D = 0) = \frac{p(D = 0, B = 0, F = 0)}{p(D = 0, B = 0)}. \tag{8.79}$$

By marginalisation,

$$\begin{aligned}
& p(D = 0, B = 0, F = 0) \\
&= \sum_{G=0,1} p(D = 0|G)p(G|B = 0, F = 0)p(B = 0)p(F = 0), \\
& p(D = 0, B = 0, F = 1) \\
&= \sum_{G=0,1} p(D = 0|G)p(G|B = 0, F = 1)p(B = 0)p(F = 1), \quad (8.80) \\
& p(D = 0, B = 0) \\
&= \sum_{F=0,1} p(D = 0, B = 0, F).
\end{aligned}$$

The right hands can be written as

$$\begin{aligned}
& \frac{9}{10} \frac{9}{10} \frac{1}{10} \frac{1}{10} + \frac{1}{10} \frac{1}{10} \frac{1}{10} \frac{1}{10} = \frac{41}{500}, \\
& \frac{9}{10} \frac{4}{5} \frac{1}{10} \frac{9}{10} + \frac{1}{10} \frac{1}{5} \frac{1}{10} \frac{9}{10} = \frac{333}{5000}, \quad (8.81) \\
& \frac{41}{5000} + \frac{333}{5000} = \frac{187}{2500}.
\end{aligned}$$

Therefore,

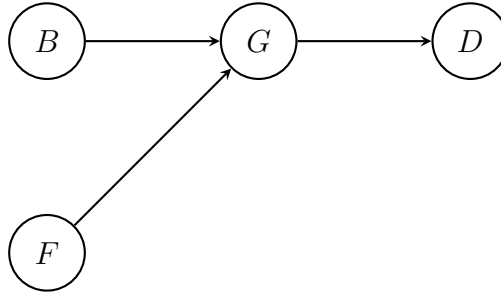
$$p(F = 0|B = 0, D = 0) = \frac{41}{374}. \quad (8.82)$$

(c)

By (a) and (b),

$$p(F = 0|D = 0) > p(F = 0|D = 0, B = 0). \quad (8.83)$$

The corresponding directed graph can be drawn as below:



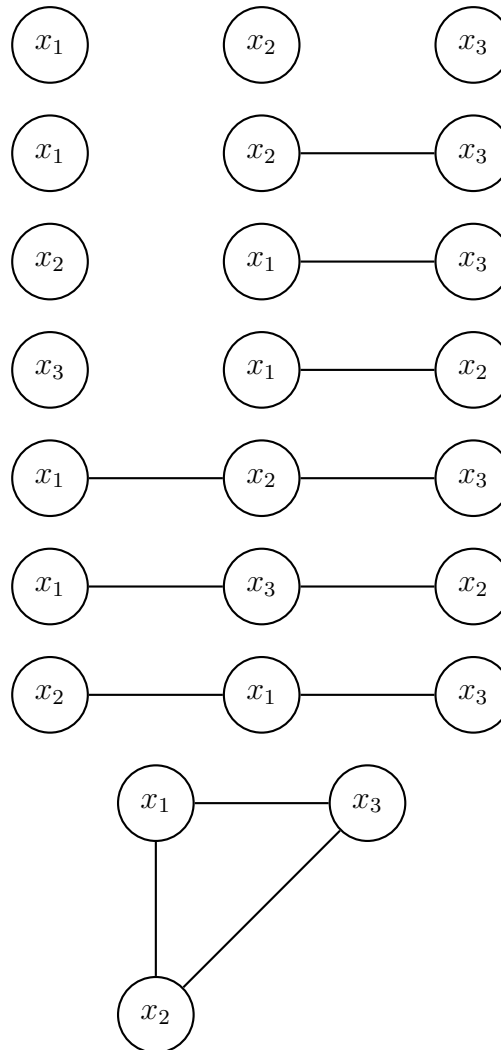
8.12

(a)

For N nodes, there are $\binom{N}{2}$ possibilities to choose a pair of 2 nodes. For each pair, the nodes are connected or isolated. Therefore, there are $2^{\binom{N}{2}}$ distinct undirected graphs for N nodes.

(b)

Let $N = 3$. Then, there are 8 distinct undirected graphs below:



8.13

Let $x_1, \dots, x_N$ and $y_1, \dots, y_N$ be variables from $\{-1, 1\}$ such that

$$p(\mathbf{x}, \mathbf{y}) = \frac{1}{Z} \exp(-E(\mathbf{x}, \mathbf{y})), \quad (8.84)$$

where

$$E(\mathbf{x}, \mathbf{y}) = h \sum_n x_n - \beta \sum_{\{n, n'\}} x_n x_{n'} - \eta \sum_n x_n y_n, \quad (8.85)$$

and $\{x_n, x_{n'}\}$ is a set of neighboring nodes. Then,

$$E(\mathbf{x}, \mathbf{y})|_{x_n=1} - E(\mathbf{x}, \mathbf{y})|_{x_n=-1} = 2 \left(h - \beta \sum_{\{n, n'\}} x_{n'} - \eta y_n \right). \quad (8.86)$$

8.14

Let $x_1, \dots, x_N$ and $y_1, \dots, y_N$ be variables from $\{-1, 1\}$ such that

$$p(\mathbf{x}, \mathbf{y}) = \frac{1}{Z} \exp(-E(\mathbf{x}, \mathbf{y})), \quad (8.87)$$

where

$$E(\mathbf{x}, \mathbf{y}) = -\eta \sum_{n=1}^N x_n y_n, \quad (8.88)$$

and $\eta > 0$. We have

$$x_n y_n = \begin{cases} 1, & x_n = y_n, \\ -1, & \text{otherwise.} \end{cases} \quad (8.89)$$

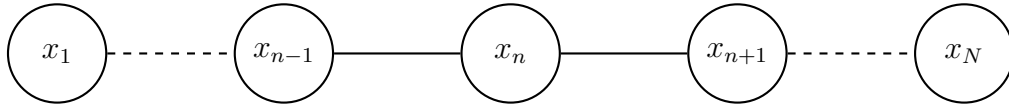
Therefore, $p(\mathbf{x}, \mathbf{y})$ will be maximised if $\mathbf{x} = \mathbf{y}$.

8.15

Let $x_1, \dots, x_N$ be variables such that

$$p(\mathbf{x}) = \frac{1}{Z} \prod_{n=1}^{N-1} \psi_{n, n+1}(x_n, x_{n+1}), \quad (8.90)$$

whose undirected graph can be drawn as below:



(a)

By marginalisation,

$$p(x_{n-1}, x_n) = \int p(\mathbf{x}) \prod_{n' \neq n-1, n} dx_{n'}. \quad (8.91)$$

The right hand side can be written as

$$\frac{1}{Z} \psi_{n-1, n}(x_{n-1}, x_n) \int \prod_{n' \neq n-1, n} \psi_{n', n'+1}(x_{n'}, x_{n'+1}) \prod_{n' \neq n-1, n} dx_{n'}. \quad (8.92)$$

The integral of the right hand side can be written as

$$\mu_\alpha(x_{n-1}) \mu_\beta(x_n), \quad (8.93)$$

where

$$\begin{aligned} \mu_\alpha(x_{n-1}) &= \int \prod_{n'=1}^{n-2} \psi_{n', n'+1}(x_{n'}, x_{n'+1}) \prod_{n'=1}^{n-2} dx_{n'}, \\ \mu_\beta(x_n) &= \int \prod_{n'=n}^{N-1} \psi_{n', n'+1}(x_{n'}, x_{n'+1}) \prod_{n'=n+1}^N dx_{n'}. \end{aligned} \quad (8.94)$$

Therefore,

$$p(x_{n-1}, x_n) = \frac{1}{Z} \mu_\alpha(x_{n-1}) \psi_{n-1, n}(x_{n-1}, x_n) \mu_\beta(x_n). \quad (8.95)$$

(b)

By Bayes' theorem,

$$p(x_n | x_N) = \frac{p(x_n, x_N)}{p(x_N)}. \quad (8.96)$$

By marginalisation,

$$p(x_n, x_N) = \int p(\mathbf{x}) \prod_{n' \neq n, N} dx_{n'}. \quad (8.97)$$

The right hand side can be written as

$$\frac{1}{Z} \int \prod_{n'=1}^{N-1} \psi_{n', n'+1}(x_{n'}, x_{n'+1}) \prod_{n' \neq n, N} dx_{n'}. \quad (8.98)$$

The integral of the right hand side can be written as

$$\int \prod_{n'=1}^{N-1} \psi_{n',n'+1}(x_{n'}, x_{n'+1}) \prod_{n' \neq n, N} dx_{n'} = \mu_{\alpha}(x_n) m_{\beta}(x_n, x_N), \quad (8.99)$$

where

$$\begin{aligned} \mu_{\alpha}(x_n) &= \int \prod_{n'=1}^{n-1} \psi_{n',n'+1}(x_{n'}, x_{n'+1}) \prod_{n'=1}^{n-1} dx_{n'}, \\ m_{\beta}(x_n, x_N) &= \int \prod_{n'=n}^{N-1} \psi_{n',n'+1}(x_{n'}, x_{n'+1}) \prod_{n'=n+1}^{N-1} dx_{n'}. \end{aligned} \quad (8.100)$$

By marginalisation,

$$p(x_N) = \int p(\mathbf{x}) \prod_{n' \neq N} dx_{n'}. \quad (8.101)$$

The right hand side can be written as

$$\frac{1}{Z} \int \prod_{n'=1}^{N-1} \psi_{n',n'+1}(x_{n'}, x_{n'+1}) \prod_{n'=1}^{N-1} dx_{n'}. \quad (8.102)$$

The integral of the right hand side can be written as $\mu_{\alpha}(x_N)$. Therefore,

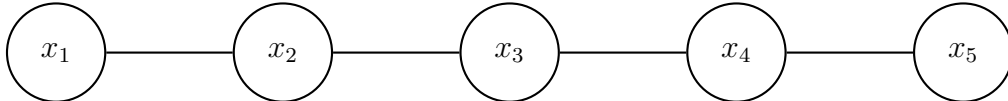
$$p(x_n | x_N) = \frac{\mu_{\alpha}(x_n)}{\mu_{\alpha}(x_N)} m_{\beta}(x_n, x_N). \quad (8.103)$$

8.16

Refer to 8.15(b).

8.17

Let $x_1, \dots, x_5$ be variables whose undirected graph can be drawn as below:



Let us assume that x_3 and x_5 are observed.

(a)

The path from x_2 to x_5 is blocked by x_3 , as it is observed. Therefore,

$$x_2 \perp\!\!\!\perp x_5 \mid x_3. \quad (8.104)$$

(b)

By Bayes' theorem,

$$p(x_2|x_3, x_5) = \frac{p(x_2, x_3, x_5)}{p(x_3, x_5)}. \quad (8.105)$$

By (a),

$$p(x_2, x_5|x_3) = p(x_2|x_3)p(x_5|x_3), \quad (8.106)$$

so that

$$p(x_2, x_5|x_3)p(x_3) = p(x_2|x_3)p(x_5|x_3)p(x_3). \quad (8.107)$$

By Bayes' theorem,

$$\begin{aligned} p(x_2, x_5|x_3)p(x_3) &= p(x_2, x_3, x_5), \\ p(x_5|x_3)p(x_3) &= p(x_3, x_5). \end{aligned} \quad (8.108)$$

Then,

$$\frac{p(x_2, x_3, x_5)}{p(x_3, x_5)} = p(x_2|x_3). \quad (8.109)$$

Therefore,

$$p(x_2|x_3, x_5) = p(x_2|x_3). \quad (8.110)$$

8.18 (Incomplete)

8.19 (Incomplete)

Let $\mathbf{x}$ be a variable such that

$$p(\mathbf{x}) = \prod_{s \in \text{ne}(x)} F_s(x, X_s)? \quad (8.111)$$

where

$$F_s(x, X_s) = f_s(x, x_1, \dots, x_N)G_1(x_1, X_{s1}) \cdots G_N(x_N, X_{sN}), \quad (8.112)$$

$\text{ne}(x)$ and X_s are the set of factor nodes which are neighbour of x and the set of all variables in the subtree connected to the variable node x via the factor node f_s . Then,

$$p(x) = \prod_{s \in \text{ne}(x)} \mu_{f_s \rightarrow x}(x), \quad (8.113)$$

where

$$\mu_{f_s \rightarrow x}(x) = \sum_{X_s} F_s(x, X_s). \quad (8.114)$$

We have

$$\sum_{X_s} F_s(x, X_s) = . \quad (8.115)$$

8.20 (Incomplete)

8.21 (Incomplete)

8.22 (Incomplete)

8.23 (Incomplete)

8.24 (Incomplete)

8.25 (Incomplete)

8.26 (Incomplete)

8.27

Let x and y be variables from $\{0, 1, 2\}$. The table below presents a joint distribution such that

$$p(\hat{x}, \hat{y}) = 0, \quad (8.116)$$

where $\hat{x}$ and $\hat{y}$ are the values maximising $p(x)$ and $p(y)$:

x	y	$p(x, y)$
0	0	0
0	1	$\frac{1}{4}$
0	2	$\frac{1}{4}$
1	0	$\frac{1}{4}$
1	1	0
1	2	0
2	0	$\frac{1}{4}$
2	1	0
2	2	0

9 Mixture Models and EM

10 Approximate Inference

11 Sampling Methods

12 Continuous Latent Variables

13 Sequential Data

14 Combining Models

15 Appendix

15.1

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ &= -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2}\mathbf{m}_0^\top \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\top\boldsymbol{\Phi} & -\beta\boldsymbol{\Phi}^\top \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}),$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^\top\mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\top\boldsymbol{\Phi})^{-1}. \end{aligned}$$

15.2

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned}$$

By Bayes' theorem,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}.$$

The logarithm of the integrand of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ &= -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2}\mathbf{m}_0^\top \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\top\boldsymbol{\Phi} & -\beta\boldsymbol{\Phi}^\top \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S}_0 & \mathbf{S}_0\boldsymbol{\Phi}^\top \\ \boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top \end{bmatrix},$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix} \mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0 \end{bmatrix}.$$

Therefore,

$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top).$$

15.3

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}.$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ &= -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{1}{2}\mathbf{m}_0^\top \mathbf{S}_0^{-1}\mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\top\boldsymbol{\Phi} & -\beta\boldsymbol{\Phi}^\top \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

Then,

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}),$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S}(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\boldsymbol{\Phi}^\top\mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \beta\boldsymbol{\Phi}^\top\boldsymbol{\Phi})^{-1}. \end{aligned}$$

Then, the logarithm of the integrand of the right hand side except the terms independent of t_{N+1} and $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}(t_{N+1} - \mathbf{w}^\top \boldsymbol{\phi}_{N+1})^2 - \frac{1}{2}(\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}) \\ &= -\frac{1}{2}\mathbf{v}'^\top \mathbf{M}'\mathbf{v}' + \mathbf{v}'^\top \mathbf{u}' - \frac{1}{2}\mathbf{m}^\top \mathbf{S}^{-1}\mathbf{m}, \end{aligned}$$

where

$$\mathbf{v}' = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M}' = \begin{bmatrix} \mathbf{S}^{-1} + \beta \boldsymbol{\phi}_{N+1} \boldsymbol{\phi}_{N+1}^\top & -\beta \boldsymbol{\phi}_{N+1} \\ -\beta \boldsymbol{\phi}_{N+1}^\top & \beta \end{bmatrix}, \mathbf{u}' = \begin{bmatrix} \mathbf{S}^{-1} \mathbf{m} \\ 0 \end{bmatrix}.$$

We have

$$\mathbf{M}'^{-1} = \begin{bmatrix} \mathbf{S} & \mathbf{S} \boldsymbol{\phi}_{N+1} \\ \boldsymbol{\phi}_{N+1}^\top \mathbf{S} & \beta^{-1} + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1} \end{bmatrix},$$

so that

$$\mathbf{M}'^{-1} \mathbf{u}' = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\top \boldsymbol{\phi}_{N+1} \end{bmatrix}.$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \mathcal{N}(t_{N+1} | \mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \beta^{-1} + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1}).$$

15.4

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}, \beta) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}|\beta) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0), \\ p(\beta) &= \text{Gam}(\beta|a_0, b_0), \end{aligned}$$

where $\mathbf{w}$ is a vector in M dimensions.

(a)

By marginalisation,

$$p(\mathbf{t}|\beta) = \int p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta)d\mathbf{w}.$$

The logarithm of $p(\mathbf{t}|\mathbf{w}, \beta)p(\mathbf{w}|\beta)$ except the terms independent of $\mathbf{w}$ can be written as

$$-\frac{\beta}{2}\|\mathbf{t} - \boldsymbol{\Phi}\mathbf{w}\|^2 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) = -\frac{1}{2}\mathbf{v}^\top \mathbf{M}\mathbf{v} + \mathbf{v}^\top \mathbf{u},$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta(\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\top \boldsymbol{\Phi}) & -\beta\boldsymbol{\Phi}^\top \\ -\beta\boldsymbol{\Phi} & \beta\mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \mathbf{S}_0^{-1}\mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\boldsymbol{\Phi}^\top \\ \beta^{-1}\boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}(\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top) \end{bmatrix},$$

so that

$$\mathbf{M}^{-1}\mathbf{v} = \begin{bmatrix} \mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0 \end{bmatrix}.$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}(\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top)).$$

(b)

By marginalisation,

$$p(\mathbf{t}) = \int_0^\infty p(\mathbf{t}|\beta)p(\beta)d\beta.$$

The logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of β can be written as

$$\begin{aligned} & -\frac{N}{2} \ln \beta^{-1} - \frac{\beta}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0) \\ & + (a_0 - 1) \ln \beta - b_0 \beta \\ & = (a - 1) \ln \beta - b \beta, \end{aligned}$$

where

$$\begin{aligned} a &= a_0 + \frac{N}{2}, \\ b &= b_0 + \frac{1}{2} (\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0). \end{aligned}$$

We have

$$\int_0^\infty \beta^{a-1} \exp(-b\beta) d\beta = b^{-a} \Gamma(a).$$

We have

$$b^{-a} = \left(1 + (\mathbf{t} - \boldsymbol{\mu})^\top (\nu \boldsymbol{\Sigma})^{-1} (\mathbf{t} - \boldsymbol{\mu}) \right)^{-\frac{\nu+N}{2}} b_0^{-a},$$

where

$$\begin{aligned} \boldsymbol{\mu} &= \Phi \mathbf{m}_0, \\ \boldsymbol{\Sigma} &= \frac{b_0}{a_0} (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top), \\ \nu &= 2a_0. \end{aligned}$$

Therefore,

$$p(\mathbf{t}) = \text{St}(\mathbf{t}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu).$$

15.5

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}, \beta) &= \mathcal{N}(t_n|\mathbf{w}^\top \boldsymbol{\phi}_n, \beta^{-1}), \\ p(\mathbf{w}|\beta) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \beta^{-1}\mathbf{S}_0), \\ p(\beta) &= \text{Gam}(\beta|a_0, b_0), \end{aligned}$$

where $\mathbf{w}$ is a vector in M dimensions.

(a)

By marginalisation,

$$p(t_{N+1}|\beta, \mathbf{t}) = \int p(t_{N+1}|\mathbf{w}, \beta) p(\mathbf{w}|\beta, \mathbf{t}) d\mathbf{w}.$$

By Bayes' theorem,

$$p(\mathbf{w}|\beta, \mathbf{t}) p(\mathbf{t}|\beta) = p(\mathbf{t}|\mathbf{w}, \beta) p(\mathbf{w}|\beta).$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2} \|\mathbf{t} - \boldsymbol{\Phi} \mathbf{w}\|^2 - \frac{\beta}{2} (\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1} (\mathbf{w} - \mathbf{m}_0) \\ &= -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} + \mathbf{v}^\top \mathbf{u} - \frac{\beta}{2} \mathbf{m}_0^\top \mathbf{S}_0^{-1} \mathbf{m}_0, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} \\ \mathbf{t} \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \beta (\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\top \boldsymbol{\Phi}) & -\beta \boldsymbol{\Phi}^\top \\ -\beta \boldsymbol{\Phi} & \beta \mathbf{I} \end{bmatrix}, \mathbf{u} = \begin{bmatrix} \beta \mathbf{S}_0^{-1} \mathbf{m}_0 \\ \mathbf{0} \end{bmatrix}.$$

Then,

$$p(\mathbf{w}|\beta, \mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \beta^{-1}\mathbf{S}),$$

where

$$\begin{aligned} \mathbf{m} &= \mathbf{S} (\mathbf{S}_0^{-1} \mathbf{m}_0 + \boldsymbol{\Phi}^\top \mathbf{t}), \\ \mathbf{S} &= (\mathbf{S}_0^{-1} + \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}. \end{aligned}$$

Then, the logarithm of $p(t_{N+1}|\mathbf{w}, \beta)p(\mathbf{w}|\beta, \mathbf{t})$ except the terms independent of $\mathbf{w}$ can be written as

$$\begin{aligned} & -\frac{\beta}{2}(t_{N+1} - \mathbf{w}^\top \boldsymbol{\phi}_{N+1})^2 - \frac{\beta}{2}(\mathbf{w} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}) \\ & = -\frac{1}{2}\mathbf{v}'^\top \mathbf{M}'\mathbf{v}' + \mathbf{v}'^\top \mathbf{u}' - \frac{\beta}{2}\mathbf{m}^\top \mathbf{S}^{-1}\mathbf{m}, \end{aligned}$$

where

$$\mathbf{v}' = \begin{bmatrix} \mathbf{w} \\ t_{N+1} \end{bmatrix}, \mathbf{M}' = \begin{bmatrix} \beta(\mathbf{S}^{-1} + \boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^\top) & -\beta\boldsymbol{\phi}_{N+1} \\ -\beta\boldsymbol{\phi}_{N+1}^\top & \beta \end{bmatrix}, \mathbf{u}' = \begin{bmatrix} \beta\mathbf{S}^{-1}\mathbf{m} \\ 0 \end{bmatrix}.$$

We have

$$\mathbf{M}'^{-1} = \begin{bmatrix} \beta^{-1}\mathbf{S} & \beta^{-1}\mathbf{S}\boldsymbol{\phi}_{N+1} \\ \beta^{-1}\boldsymbol{\phi}_{N+1}^\top \mathbf{S} & \beta^{-1}(1 + \boldsymbol{\phi}_{N+1}^\top \mathbf{S}\boldsymbol{\phi}_{N+1}) \end{bmatrix},$$

so that

$$\mathbf{M}'^{-1}\mathbf{u}' = \begin{bmatrix} \mathbf{m} \\ \mathbf{m}^\top \boldsymbol{\phi}_{N+1} \end{bmatrix}.$$

Therefore,

$$p(t_{N+1}|\beta, \mathbf{t}) = \mathcal{N}(t_{N+1}|\mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \beta^{-1}(1 + \boldsymbol{\phi}_{N+1}^\top \mathbf{S}\boldsymbol{\phi}_{N+1})).$$

(b)

By Bayes' theorem,

$$p(\beta|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\beta)p(\beta).$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \beta^{-1}\mathbf{S}_0 & \beta^{-1}\mathbf{S}_0\boldsymbol{\Phi}^\top \\ \beta^{-1}\boldsymbol{\Phi}\mathbf{S}_0 & \beta^{-1}(\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top) \end{bmatrix},$$

so that

$$\mathbf{M}^{-1}\mathbf{u} = \begin{bmatrix} \mathbf{m}_0 \\ \boldsymbol{\Phi}\mathbf{m}_0 \end{bmatrix}.$$

Then,

$$p(\mathbf{t}|\beta) = \mathcal{N}(\mathbf{t}|\boldsymbol{\Phi}\mathbf{m}_0, \beta^{-1}(\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top)).$$

The logarithm of the right hand side except the terms independent of β can be written as

$$-\frac{N}{2}\ln \beta^{-1} - \frac{\beta}{2}(\mathbf{t} - \boldsymbol{\Phi}\mathbf{m}_0)^\top (\mathbf{I} + \boldsymbol{\Phi}\mathbf{S}_0\boldsymbol{\Phi}^\top)^{-1}(\mathbf{t} - \boldsymbol{\Phi}\mathbf{m}_0).$$

Then the logarithm of $p(\mathbf{t}|\beta)p(\beta)$ except the terms independent of β can be written as

$$(a - 1) \ln \beta - b\beta,$$

where

$$a = a_0 + \frac{N}{2},$$

$$b = b_0 + \frac{1}{2}(\mathbf{t} - \Phi \mathbf{m}_0)^\top (\mathbf{I} + \Phi \mathbf{S}_0 \Phi^\top)^{-1} (\mathbf{t} - \Phi \mathbf{m}_0).$$

Therefore,

$$p(\beta|\mathbf{t}) = \text{Gam}(\beta|a, b).$$

(c)

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int_0^\infty p(t_{N+1}|\beta, \mathbf{t})p(\beta|\mathbf{t})d\beta,$$

By (a) and (b), the logarithm of $p(t_{N+1}|\beta, \mathbf{t})p(\beta|\mathbf{t})$ except the terms independent of β can be written as

$$-\frac{1}{2} \ln \beta^{-1} - \frac{(t_{N+1} - \mathbf{m}^\top \phi_{N+1})^2}{2(1 + \phi_{N+1}^\top \mathbf{S} \phi_{N+1})} \beta + (a - 1) \ln \beta - b\beta$$

$$= (a' - 1) \ln \beta - b' \ln \beta,$$

where

$$a' = a + \frac{1}{2},$$

$$b' = b + \frac{(t_{N+1} - \mathbf{m}^\top \phi_{N+1})^2}{2(1 + \phi_{N+1}^\top \mathbf{S} \phi_{N+1})}.$$

We have

$$\int_0^\infty \beta^{a'-1} \exp(-b'\beta) d\beta = \Gamma(a') b'^{-a'}.$$

We have

$$b'^{-a'} = \left(1 + \frac{(t_{N+1} - \mu)^2}{\nu \sigma^2}\right)^{-\frac{\nu+1}{2}} b^{-a'},$$

where

$$\begin{aligned}\mu &= \mathbf{m}^\top \boldsymbol{\phi}_{N+1}, \\ \sigma^2 &= \frac{b}{a}(1 + \boldsymbol{\phi}_{N+1}^\top \mathbf{S} \boldsymbol{\phi}_{N+1}), \\ \nu &= 2a.\end{aligned}$$

Therefore,

$$p(t_{N+1}|\mathbf{t}) = \text{St}(t_{N+1}|\mu, \sigma^2, \nu).$$

15.6

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned}$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^N (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0).$$

We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3),$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}).$$

Therefore,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}).$$

15.7

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned}$$

By marginalisation,

$$p(\mathbf{t}) = \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w})d\mathbf{w}.$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^N (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0).$$

We have the Taylor series

$$y(\mathbf{x}_n, \mathbf{w}) = y_{0n} + \mathbf{g}_{0n}^\top (\mathbf{w} - \mathbf{m}_0) + O(\|\mathbf{w} - \mathbf{m}_0\|^2),$$

where

$$\begin{aligned} y_{0n} &= y(\mathbf{x}_n, \mathbf{m}_0). \\ \mathbf{g}_{0n} &= \nabla_{\mathbf{w}} y(\mathbf{x}_n, \mathbf{m}_0). \end{aligned}$$

Then, $E(\mathbf{w})$ can be approximated as

$$\begin{aligned} & \frac{\beta}{2} \sum_{n=1}^N (t_n - y_{0n} - \mathbf{g}_{0n}^\top (\mathbf{w} - \mathbf{m}_0))^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0) \\ &= \frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v} - \mathbf{v}^\top \mathbf{u}, \end{aligned}$$

where

$$\mathbf{v} = \begin{bmatrix} \mathbf{w} - \mathbf{m}_0 \\ \mathbf{t} - \mathbf{y}_0 \end{bmatrix}, \mathbf{M} = \begin{bmatrix} \mathbf{S}_0^{-1} + \beta \mathbf{G}_0^\top \mathbf{G}_0 & -\beta \mathbf{G}_0^\top \\ -\beta \mathbf{G}_0 & \beta \mathbf{I} \end{bmatrix}.$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{S}_0 & \mathbf{S}_0 \mathbf{G}_0^\top \\ \mathbf{G}_0 \mathbf{S}_0 & \beta^{-1} \mathbf{I} + \mathbf{G}_0 \mathbf{S}_0 \mathbf{G}_0^\top \end{bmatrix}.$$

Therefore,

$$p(\mathbf{t}) \simeq \mathcal{N}(\mathbf{t}|\mathbf{y}_0, \beta^{-1} \mathbf{I} + \mathbf{G}_0 \mathbf{S}_0 \mathbf{G}_0^\top).$$

15.8

Let $t_1, \dots, t_N$ be conditionally independent variables such that

$$\begin{aligned} p(t_n|\mathbf{w}) &= \mathcal{N}(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}), \\ p(\mathbf{w}) &= \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0). \end{aligned}$$

By marginalisation,

$$p(t_{N+1}|\mathbf{t}) = \int p(t_{N+1}|\mathbf{w})p(\mathbf{w}|\mathbf{t})d\mathbf{w}.$$

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{t})p(\mathbf{t}) = p(\mathbf{t}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{t}$ and $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = \frac{\beta}{2} \sum_{n=1}^N (t_n - y(\mathbf{x}_n, \mathbf{w}))^2 + \frac{1}{2}(\mathbf{w} - \mathbf{m}_0)^\top \mathbf{S}_0^{-1}(\mathbf{w} - \mathbf{m}_0).$$

We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3),$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}).$$

Then,

$$p(\mathbf{w}|\mathbf{t}) \simeq \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}).$$

We have the Taylor series

$$\begin{aligned} y(\mathbf{x}_{N+1}, \mathbf{w}) &= y_{\text{MAP}N+1} + \mathbf{g}_{\text{MAP}N+1}^\top (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\ &\quad + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^2), \end{aligned}$$

where

$$\begin{aligned} y_{\text{MAP}N+1} &= y(\mathbf{x}_{N+1}, \mathbf{w}_{\text{MAP}}), \\ \mathbf{g}_{\text{MAP}N+1} &= \nabla_{\mathbf{w}} y(\mathbf{x}_{N+1}, \mathbf{w}_{\text{MAP}}). \end{aligned}$$

Then, the logarithm of the integrand except the terms independent of $\mathbf{w}$ can be approximated as

$$\begin{aligned}
& -\frac{\beta}{2} (t_{N+1} - y_{\text{MAP}_{N+1}} - \mathbf{g}_{\text{MAP}_{N+1}}^\top (\mathbf{w} - \mathbf{w}_{\text{MAP}}))^2 \\
& -\frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{MAP}})^\top \mathbf{H}_{\text{MAP}} (\mathbf{w} - \mathbf{w}_{\text{MAP}}) \\
& = -\frac{1}{2} \mathbf{v}^\top \mathbf{M} \mathbf{v},
\end{aligned}$$

where

$$\begin{aligned}
\mathbf{v} &= \begin{bmatrix} \mathbf{w} - \mathbf{w}_{\text{MAP}} \\ t_{N+1} - y_{\text{MAP}_{N+1}} \end{bmatrix}, \\
\mathbf{M} &= \begin{bmatrix} \mathbf{H}_{\text{MAP}} + \beta \mathbf{g}_{\text{MAP}_{N+1}} \mathbf{g}_{\text{MAP}_{N+1}}^\top & -\beta \mathbf{g}_{\text{MAP}_{N+1}} \\ -\beta \mathbf{g}_{\text{MAP}_{N+1}}^\top & \beta \end{bmatrix}.
\end{aligned}$$

We have

$$\mathbf{M}^{-1} = \begin{bmatrix} \mathbf{H}_{\text{MAP}}^{-1} & \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}_{N+1}} \\ \mathbf{g}_{\text{MAP}_{N+1}}^\top \mathbf{H}_{\text{MAP}}^{-1} & \beta^{-1} + \mathbf{g}_{\text{MAP}_{N+1}}^\top \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}_{N+1}} \end{bmatrix}.$$

Therefore,

$$p(t_{N+1} | \mathbf{t}) \simeq \mathcal{N}(t_{N+1} | y_{\text{MAP}_{N+1}}, \beta^{-1} + \mathbf{g}_{\text{MAP}_{N+1}}^\top \mathbf{H}_{\text{MAP}}^{-1} \mathbf{g}_{\text{MAP}_{N+1}}).$$

15.9

Let $\mathbf{t}_1, \dots, \mathbf{t}_N$ be conditionally independent variables from the standard basis in K dimensions such that

$$p(\mathbf{t}_n|\mathbf{w}) = \prod_{k=1}^K y_{nk}^{t_{nk}},$$

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}),$$

where

$$y_{nk} = y_k(\mathbf{a}_n),$$

$$y_k(\mathbf{a}) = \frac{\exp(a_k)}{\sum_{k'=1}^K \exp(a_{k'})},$$

$$\mathbf{a}_n = \mathbf{a}(\mathbf{w}, \phi_n).$$

(a)

By Bayes' theorem,

$$p(\mathbf{w}|\mathbf{T})p(\mathbf{T}) = p(\mathbf{T}|\mathbf{w})p(\mathbf{w}).$$

The logarithm of the right hand side except the terms independent of $\mathbf{w}$ can be written as $-E(\mathbf{w})$ where

$$E(\mathbf{w}) = -\sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_{nk} + \frac{\alpha}{2} \|\mathbf{w}\|^2.$$

We have the Taylor series

$$E(\mathbf{w}) = E(\mathbf{w}_{\text{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^T \mathbf{H}_{\text{MAP}}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^3),$$

where $\mathbf{w}_{\text{MAP}}$ is a stationary point of E and

$$\mathbf{H}_{\text{MAP}} = \nabla \nabla E(\mathbf{w}_{\text{MAP}}).$$

Then,

$$p(\mathbf{w}|\mathbf{T}) \simeq \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{H}_{\text{MAP}}^{-1}).$$

(b)

We have

$$p(\mathbf{t}_{N+1}|\mathbf{a}_{N+1}) = \prod_{k=1}^K y_k(\mathbf{a}_{N+1})^{t_{N+1,k}}.$$

We have the Taylor series

$$\mathbf{a}_{N+1} = \mathbf{a}_{\text{MAP},N+1} + \mathbf{J}_{\text{MAP},N+1}(\mathbf{w} - \mathbf{w}_{\text{MAP}}) + O(\|\mathbf{w} - \mathbf{w}_{\text{MAP}}\|^2),$$

where

$$\begin{aligned}\mathbf{a}_{\text{MAP},N+1} &= \mathbf{a}(\mathbf{w}_{\text{MAP}}, \phi_{N+1}), \\ \mathbf{J}_{\text{MAP},N+1} &= \nabla_{\mathbf{w}} \mathbf{a}(\mathbf{w}_{\text{MAP}}, \phi_{N+1}).\end{aligned}$$

Therefore,

$$p(\mathbf{t}_{N+1}|\mathbf{a}_{N+1}) \simeq \prod_{k=1}^K y_k(\alpha_{N+1})^{t_{N+1,k}},$$

where

$$\alpha_{N+1} = \mathbf{a}_{\text{MAP},N+1} + \mathbf{J}_{\text{MAP},N+1}(\mathbf{w} - \mathbf{w}_{\text{MAP}}).$$

(c)

By (a) and (b),

$$\begin{aligned}\mathbb{E}(\alpha_{N+1}|\mathbf{T}) &= \mathbf{a}_{\text{MAP},N+1}, \\ \text{var}(\alpha_{N+1}|\mathbf{T}) &= \mathbf{J}_{\text{MAP},N+1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{J}_{\text{MAP},N+1}^{\top}.\end{aligned}$$

Therefore,

$$p(\mathbf{a}_{N+1}|\mathbf{T}) \simeq \mathcal{N}(\mathbf{a}_{N+1}|\mathbf{a}_{\text{MAP},N+1}, \mathbf{J}_{\text{MAP},N+1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{J}_{\text{MAP},N+1}^{\top}).$$

(d)

By marginalisation,

$$p(\mathbf{t}_{N+1}|\mathbf{T}) = \int p(\mathbf{t}_{N+1}|\mathbf{a}_{N+1})p(\mathbf{a}_{N+1}|\mathbf{T})d\mathbf{a}_{N+1}.$$

Therefore, by (b) and (c),

$$p(\mathbf{t}_{N+1}|\mathbf{T}) \simeq \prod_{k=1}^K v_{N+1,k}^{t_{N+1,k}},$$

where

$$v_{N+1,k} = \int y_k(\mathbf{a}) \mathcal{N}(\mathbf{a} | \mathbf{a}_{\text{MAP } N+1}, \mathbf{J}_{\text{MAP } N+1} \mathbf{H}_{\text{MAP}}^{-1} \mathbf{J}_{\text{MAP } N+1}^{\top}) d\mathbf{a},$$

since $\mathbf{t}_{N+1}$ is from the standard basis in K dimensions.